



Power BI Administration

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0101: Welcome to the Course

Welcome to the Microsoft Power BI Administrator Course!

Most Power BI training is geared towards report developers, focusing on creating visualizations and dashboards. But what if your role goes beyond that? What if you're the one responsible for overseeing how Power BI is managed across your organization—ensuring governance, security, and seamless integration within the Microsoft ecosystem?



This course is designed specifically for Fabric administrators responsible for Power BI administration, as well as Power Platform and Microsoft 365 admins with related responsibilities. In today's data-driven landscape, effective administration and governance are essential to ensuring a well-managed and scalable Power BI environment.

Throughout this course, you'll gain a structured framework for integrating Power BI into your corporate environment. We'll walk through key tenant settings, permission management, and best practices to help you maintain security, compliance, and efficiency. More than just technical configurations, this course will show you how these administrative decisions impact your organization's data strategy and user productivity.

While there are no formal prerequisites, a basic understanding of Power BI fundamentals will be helpful.

It's important to note that this course focuses exclusively on Power BI administration within Microsoft Fabric. Administration of other Fabric services—such as Data Factory, Synapse Data Engineering, and Real-Time Analytics—will be covered separately.

By the end of this course, you'll have the knowledge and tools to confidently manage Power BI within your organization, ensuring a secure, well-governed, and efficient deployment.

0102: Power BI Administration Overview

Power BI administration refers to the governance and management of Power BI within a Microsoft 365 tenant. Historically, Microsoft used the term "Power BI administrator" to describe the role responsible for overseeing Power BI settings, security, and governance. However, with the introduction of Microsoft Fabric, this terminology has changed. Microsoft now refers to all tenant administrators simply as "Fabric administrators."

In this course, we make an important distinction. A Power BI administrator, as we define it, is responsible for managing Power BI within a tenant that does not have Fabric. This role sits within the broader Microsoft 365 administration structure, alongside other key roles:

Microsoft 365 Administration Roles



At the highest level, we have the Global Administrator. This role has full control over all Microsoft 365 services, including Power BI and Fabric. A Global Admin can manage users, security settings, licensing, and compliance policies across the entire tenant.

Next is the Power Platform Administrator. This role governs Power Apps and Power Automate but does not have control over Microsoft 365 user accounts or broader security settings outside the Power Platform.

Then we come to the role most relevant to this course: the Power BI Administrator. This is someone responsible for managing Power BI in a tenant without Fabric. Their focus is on Power BI service governance, security settings, and workspace management, without the added complexity of Fabric workloads such as OneLake or data engineering.

Finally, there's the Fabric Administrator, the official Microsoft term for tenant administrators today. While this role encompasses everything a Power BI Administrator does, it also includes additional

responsibilities, such as managing OneLake storage policies, Fabric capacity, and integration with Fabric-specific workloads.

Focus of This Course

Since this course is designed for Power BI Administrators (not Fabric Administrators), we will focus exclusively on managing Power BI in a non-Fabric environment. Our discussions will center on:

- Power BI Service administration
- Tenant settings and security policies
- Workspace management
- Governance best practices

Even though Microsoft now refers to all admins as Fabric Administrators, in practice, the Power BI Administrator role still exists for those managing Power BI-only tenants. For this reason, we will continue to use the term "Power BI Administrator" throughout the course where relevant.

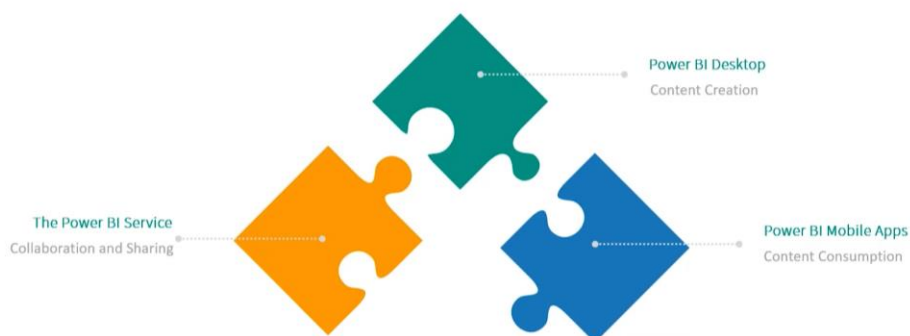
Power BI Administration in Context

Power BI administration can be understood in both a narrow and broader sense. In the narrow sense, it refers specifically to managing key settings and policies that control how Power BI is used within a given tenant. In the broader sense, it now exists within the context of Microsoft Fabric, which integrates Power BI with other services such as data engineering, real-time analytics, and data science.

Throughout this course, we will use the broader definition of administration, which includes:

- Governance of Power BI's adoption and use across the organization
- Data governance and workspace management
- Enforcing best practices in content creation and sharing

Power BI Components



Power BI consists of three core components:

Power BI Desktop – A Windows application for creating reports and data models

Power BI Service – An online Software-as-a-Service (SaaS) platform for sharing and managing reports

Power BI Mobile Apps – Available for Windows, Android, and iOS devices

As a Power BI Administrator, you will be responsible for managing all three components, though your primary area of activity will be the Power BI Service, now integrated into the broader Microsoft Fabric platform.

The Shift to Self-Service BI



Power BI is designed as a self-service business intelligence (BI) tool, enabling business users to gain insights from their data quickly. In traditional BI systems, the IT department controlled data access and reporting, requiring business users to submit requests for reports and data extracts. Power BI disrupts this model by giving users:

- Quick access to data
- Intuitive reporting tools
- Flexible options for sharing insights

Creating reports in Power BI requires less technical knowledge than traditional BI tools. This means users can work more independently, reducing their reliance on IT teams.

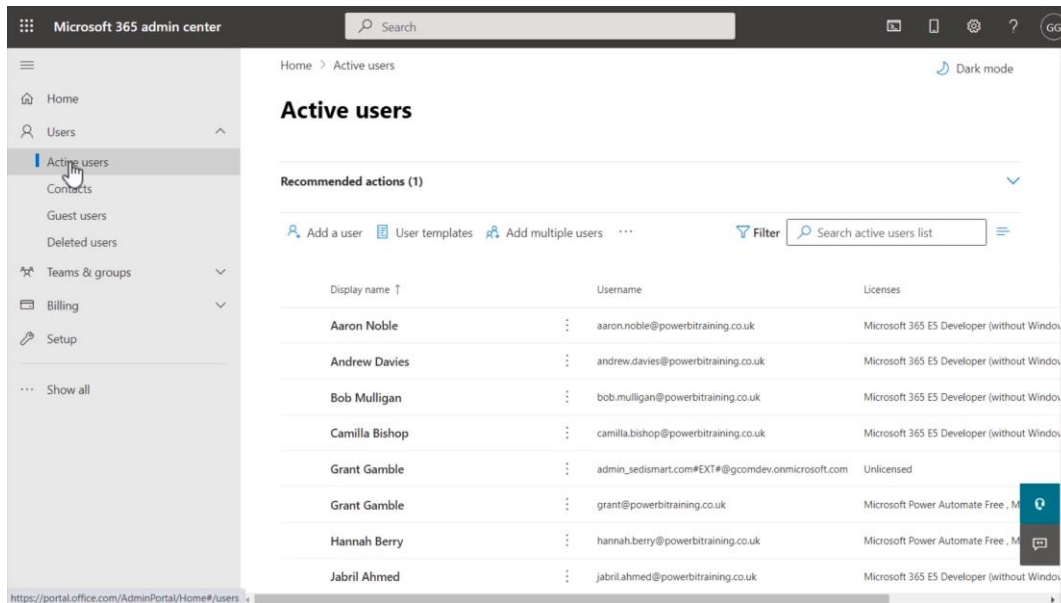
Challenges of Self-Service BI

While self-service BI provides greater flexibility, it also introduces challenges for administrators. These challenges vary by organization but typically include:

- Maintaining data quality and consistency
- Ensuring a single version of the truth
- Avoiding data silos and data anarchy
- Striking the right balance between user freedom and governance

As we progress through the course, we will explore these challenges in more detail, along with best practices for managing them effectively within your organization.

0103: Assigning Admin Roles in the Microsoft 365 Admin Centre



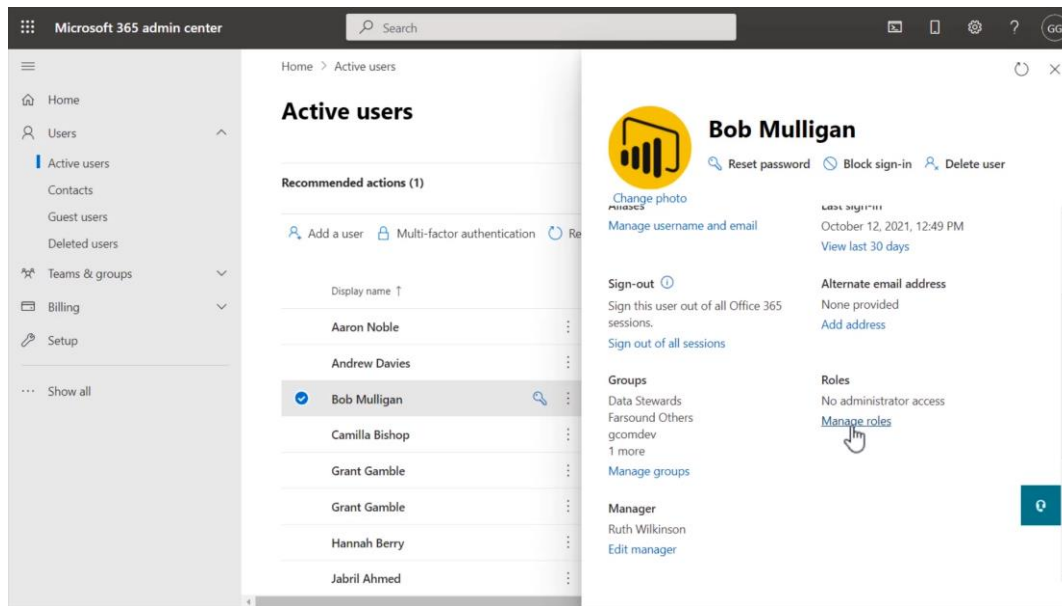
To assign someone as a Fabric Administrator or a Global Administrator, you must already hold the Global Administrator role yourself. There are two ways to assign the Fabric Administrator role: through the Microsoft 365 Admin Centre or Microsoft Entra ID. While Entra ID offers additional benefits, it requires all participants to have a Microsoft Entra ID Premium P2 license, which may not be available in all organizations.

Accessing the Microsoft 365 Admin Centre

In this section, we'll focus on assigning the admin role using the Microsoft 365 Admin Centre. You can access it directly from your Power BI tenant by clicking on Settings and selecting Admin Portal. From there, navigating to the Users tab will provide a quick link to the Microsoft 365 Admin Centre.

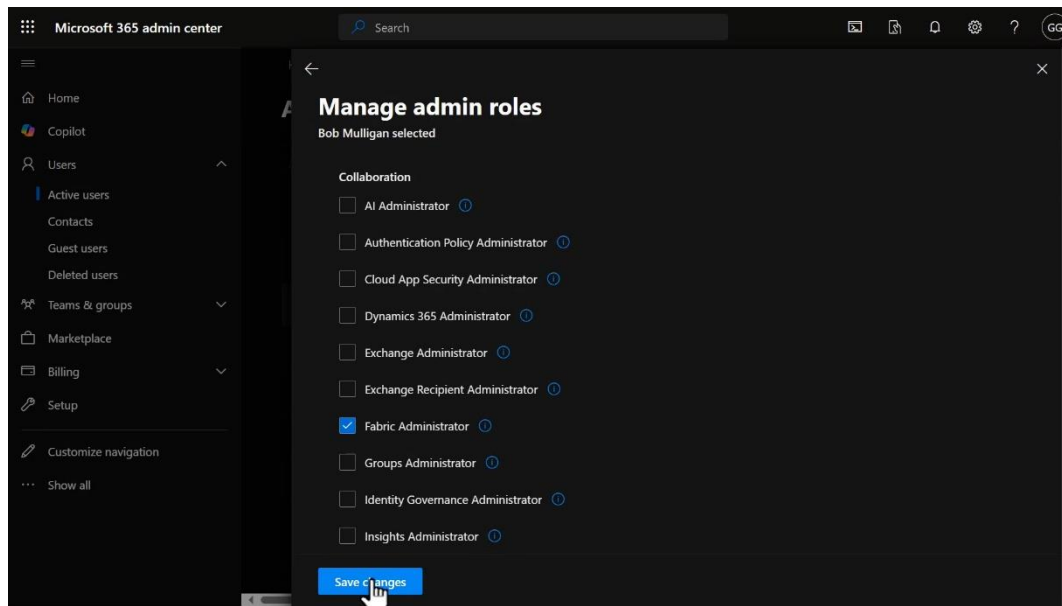
Once inside the Admin Centre, clicking on Active Users brings up a list of all users in the organization. Here, you can edit individual user settings, including their administrative roles.

Assigning the Fabric Administrator Role



Let's take an example: assigning Bob Mulligan as a Fabric Administrator.

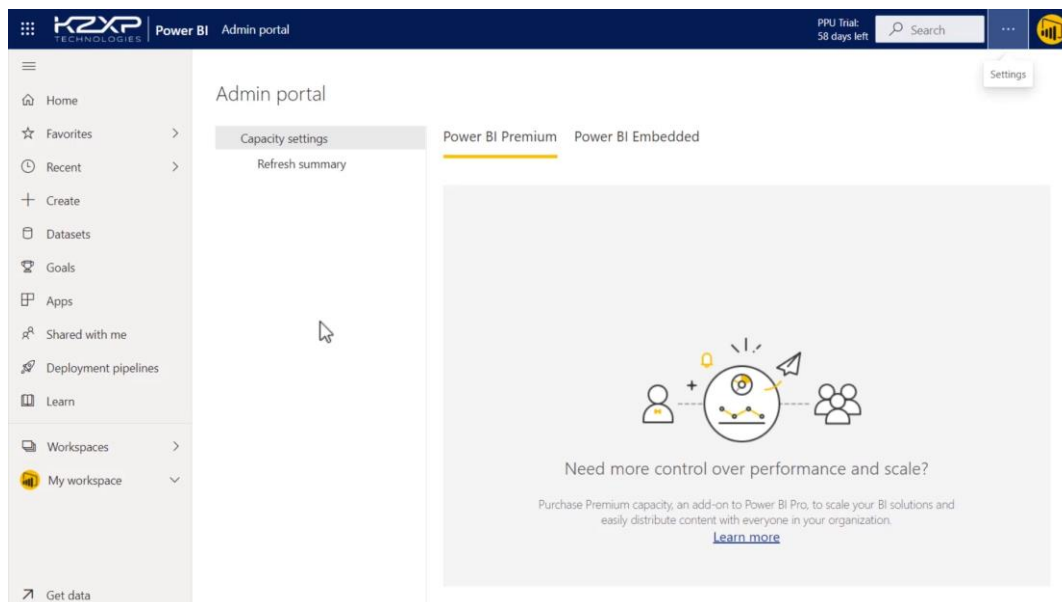
1. Locate Bob Mulligan in the Active Users list and click on his name.
2. This opens the User Settings window—scroll down to the Roles section and select Manage Roles.
3. By default, Bob is just a standard user. To assign admin privileges, we first need to enable Admin Centre Access.
4. Once admin access is enabled, a full list of administrator roles becomes available. Unlike radio buttons, these roles use checkboxes, meaning multiple roles can be assigned if needed.
5. Since we only want to assign the Fabric Administrator role (not Global Administrator), we expand the role categories by selecting Show All by Category.
6. Under the Collaboration section, we find and select Fabric Administrator.



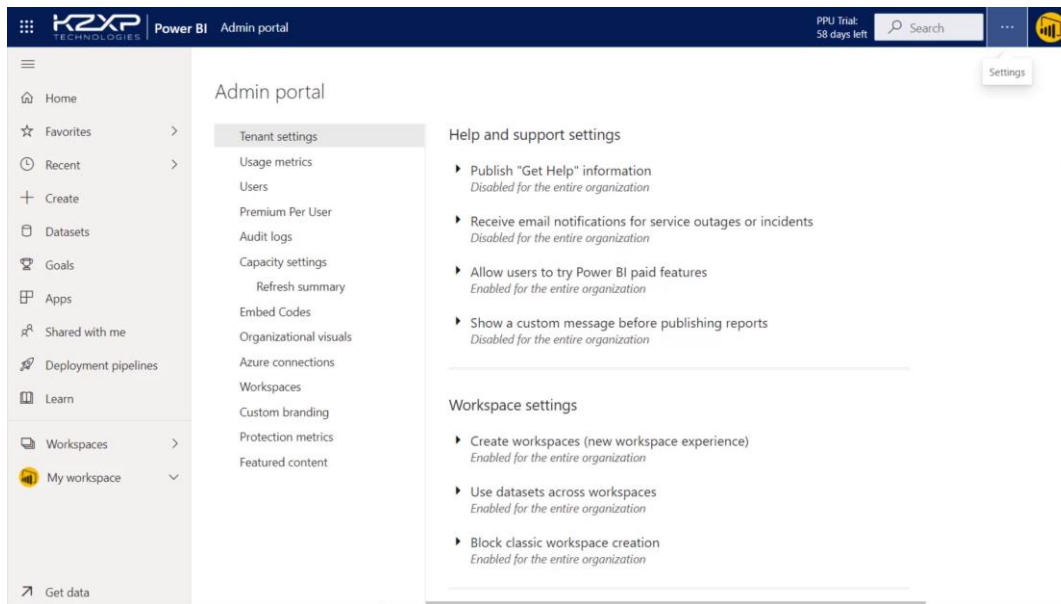
7. Click Save to apply the changes.

Verifying the Role Change

Before making the change, let's check what Bob currently sees in the Admin Portal. When logged in as Bob Mulligan, navigating to Settings > Admin Portal shows only the standard user view—he does not yet have access to admin portal settings.



After saving the changes and allowing a few minutes for the update to take effect, logging back in as Bob now shows a fully enabled Fabric Administrator experience. He can now manage Power BI and Fabric settings within the tenant.

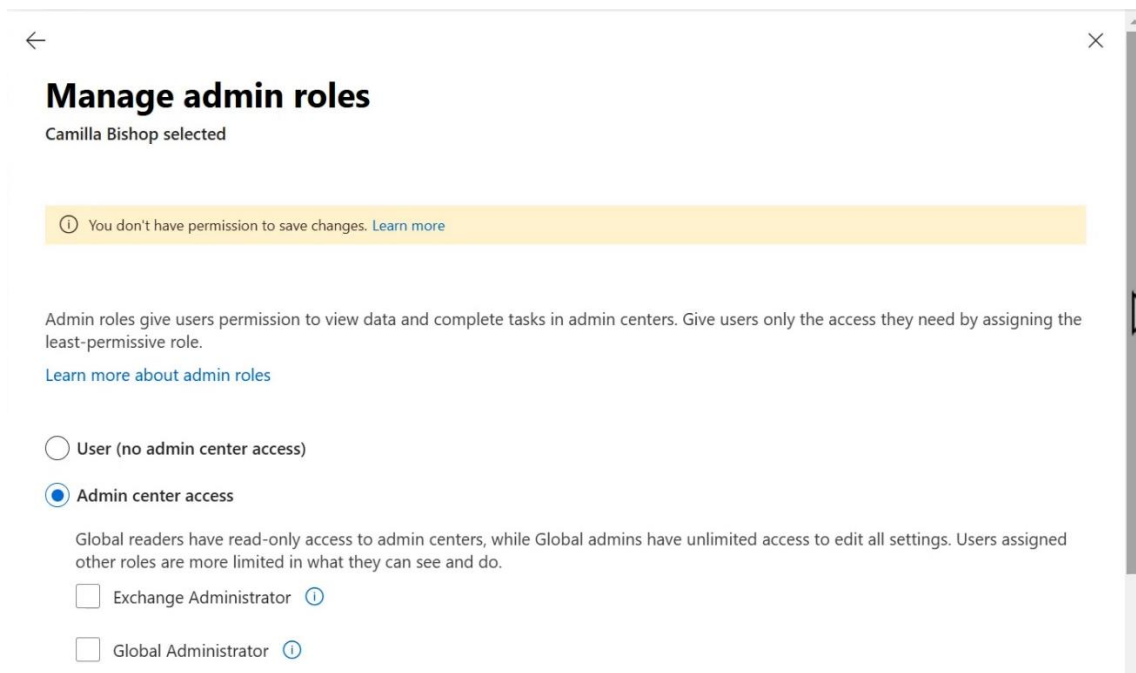


Fabric Administrator vs Global Administrator

While Bob now has Fabric Administrator privileges, he does not have Global Administrator rights. This means that:

- He can manage Power BI and Fabric settings.
- He cannot modify user roles, assign licenses, or edit account details for other users.

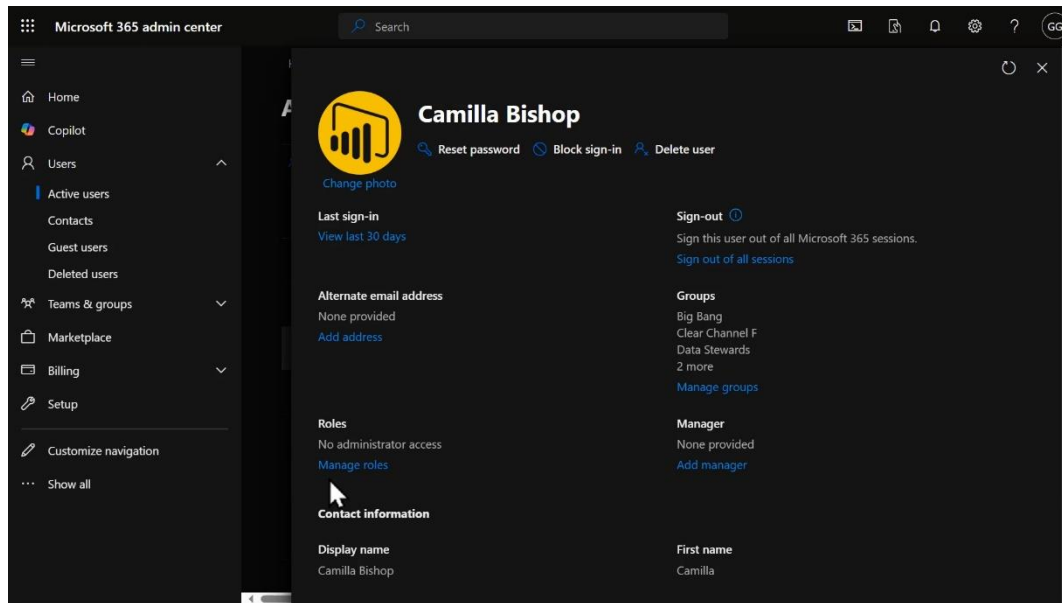
For example, if Bob attempts to edit Camilla Bishop's user settings, he can view her details but is unable to make changes. A button may appear suggesting that he can modify roles, but a warning message confirms he lacks the necessary permissions.



0103-assigning Admin Roles In The Microsoft 365 Admin Centre

By contrast, a Global Administrator has full control over the entire Microsoft 365 environment. If we switch back to an account with Global Admin rights, we can:

- Edit user contact information.
- Assign or revoke roles.
- Manage licenses and group memberships.

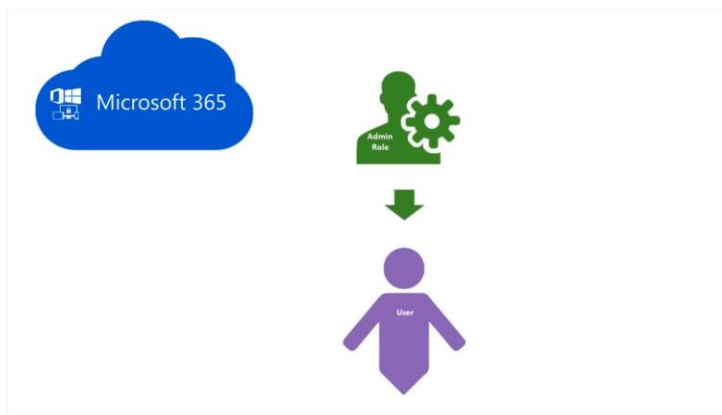


Summary

A Fabric Administrator has control over Power BI and Fabric settings, but a Global Administrator has full tenant-wide control over Microsoft 365. When assigning admin roles, always consider the level of access required and ensure the correct permissions are applied.

0104: Assigning Admin Roles in Microsoft Entra ID

In the previous section, we explored how to assign the Power BI Administrator role to a user within the Microsoft 365 Admin Center. When assigned this way, the role is permanent, which is not always ideal from a security perspective. For instance, if a user only needs admin privileges to adjust tenant settings in line with corporate policy, granting permanent admin access is unnecessary and introduces potential security risks. This is where Microsoft Entra ID provides a more controlled approach to admin assignments.

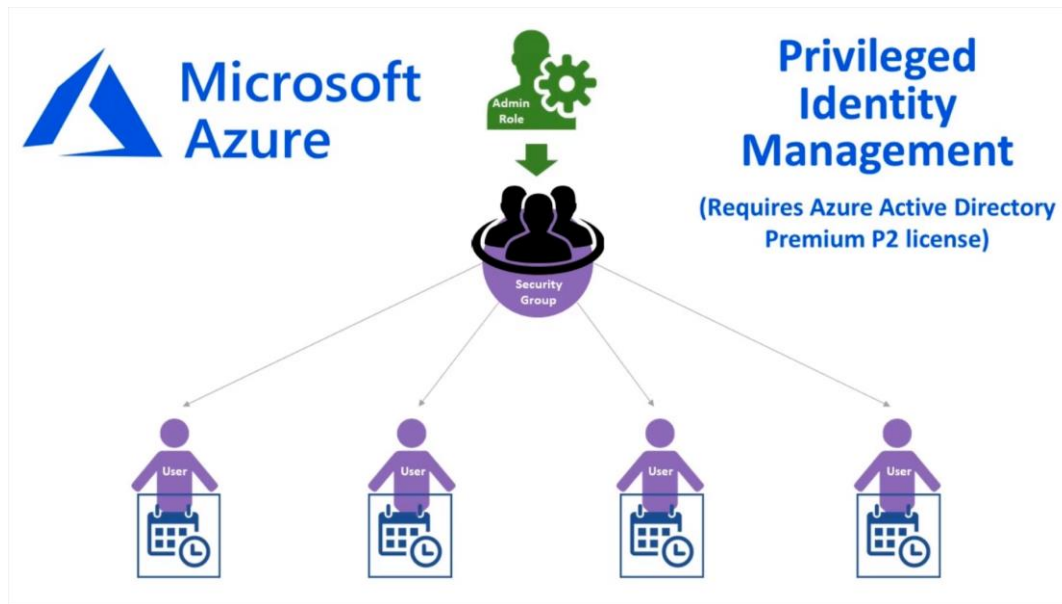


Why Use Microsoft Entra ID for Admin Role Assignments?

Microsoft Entra ID offers two key features that enhance security when assigning admin roles:

- **Group-Level Role Assignments** – Instead of assigning roles to individuals, you can assign admin privileges to a group, and then manage group membership separately.

Privileged Identity Management (PIM) – PIM allows you to grant admin privileges on a temporary or conditional basis rather than permanently.



However, it's important to note that both of these features require a Microsoft Entra ID Premium P2 license for each involved user.

Assigning Roles in Microsoft Entra ID

Let's start by removing the Power BI Administrator role from a user previously assigned via the Microsoft 365 Admin Center. In Entra ID, navigate to Users, select the user (e.g., Bob Mulligan), and review their assigned roles. Here, you have the option to remove the role. While standard user and group management can be done in both Microsoft 365 Admin Center and Entra ID, the features we are about to explore are unique to Entra ID.

Creating a Group for Admin Assignments

To assign admin privileges at the group level:

Go to Groups in Entra ID and create a new group.

Select Security as the group type.

Name the group (e.g., Fabric Administrators).

Add a description (e.g., Designed for users with the Fabric Administrator role).

Microsoft Azure Search resources, services, and docs (G+/)

Home > gcomdev > Groups >

New Group ...

Group type * ⓘ
Security

Group name * ⓘ
Power BI Administrators

Group description ⓘ
Designed specifically for users to whom the Power BI Administrator role has been assigned

Azure AD roles can be assigned to the group ⓘ
☐ Yes ☒ No

Membership type * ⓘ
Assigned

Owners
No owners selected

Members
No members selected

Create

At this point, you can add Owners and Members to the group. More importantly, if you enable the option "Microsoft Entra roles can be assigned to the group", a new Roles section appears.

Here, you can search for the Fabric Administrator role, select it, and assign it to the group. From this point forward, anyone added to this group automatically receives Fabric Administrator privileges.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > gcomdev | Groups > Groups | Overview >

New Group

Got feedback?

security

Group name *

Fabric Administrators

Group description

Designed specifically for users to whom the Fabric administrator role

Microsoft Entra roles can be assigned to the group

Yes

No

Membership type

Assigned

Owners

No owners selected

Members

No members selected

Roles

No roles selected

Create

Directory roles

Choose admin roles that you want to assign to this use

fabric

Role

Fabric Administrator

Select

Note that enabling role assignments for a group is a permanent setting that cannot be changed later.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > gcomdev | Groups > Groups | Overview >

New Group

Got feedback?

Creating a group to which Microsoft Entra roles can be assigned is a setting that cannot be changed later. Are you sure you want to add this capability? [Learn More.](#)

Yes

No

Designed specifically for users to whom the Fabric administrator role has been assigned.

Microsoft Entra roles can be assigned to the group

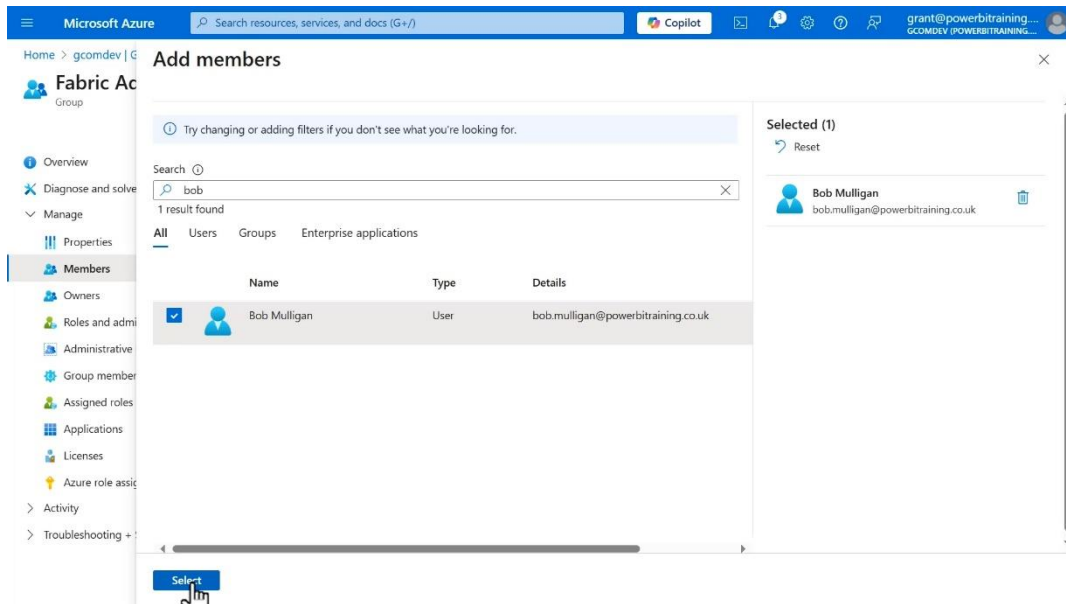
Yes

No

Membership type

Assigned

Once the group is created, you can add users. For instance, to reinstate Bob Mulligan as an administrator, go to Members, select Add Members, find Bob, and confirm.



Using Privileged Identity Management (PIM)

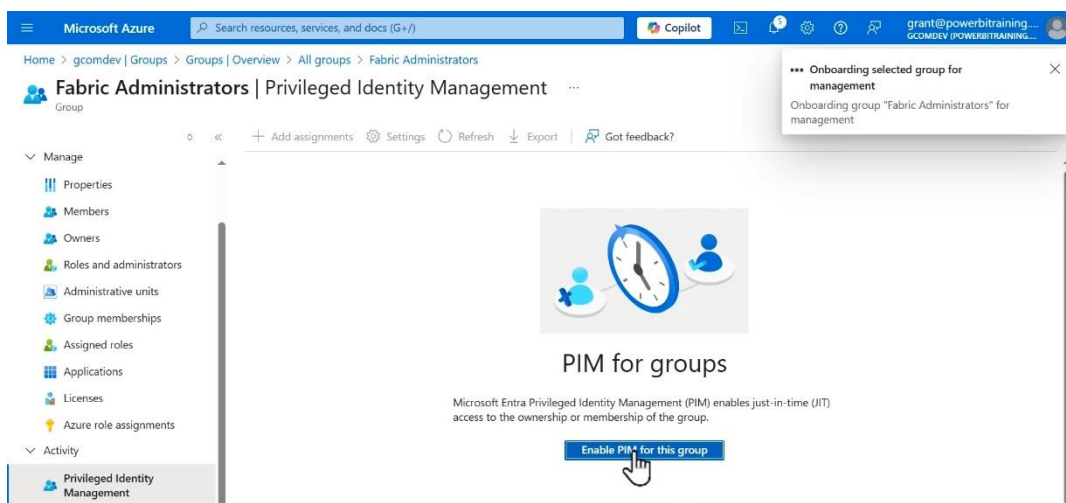
Microsoft Entra ID's Privileged Identity Management (PIM) feature introduces just-in-time access control for admin roles. Instead of granting permanent admin rights, PIM allows you to specify Active or Eligible assignments:

Active Assignments: Users immediately gain admin privileges but for a limited duration.

Eligible Assignments: Users must manually activate their privileges when needed, and each activation lasts for a defined period.

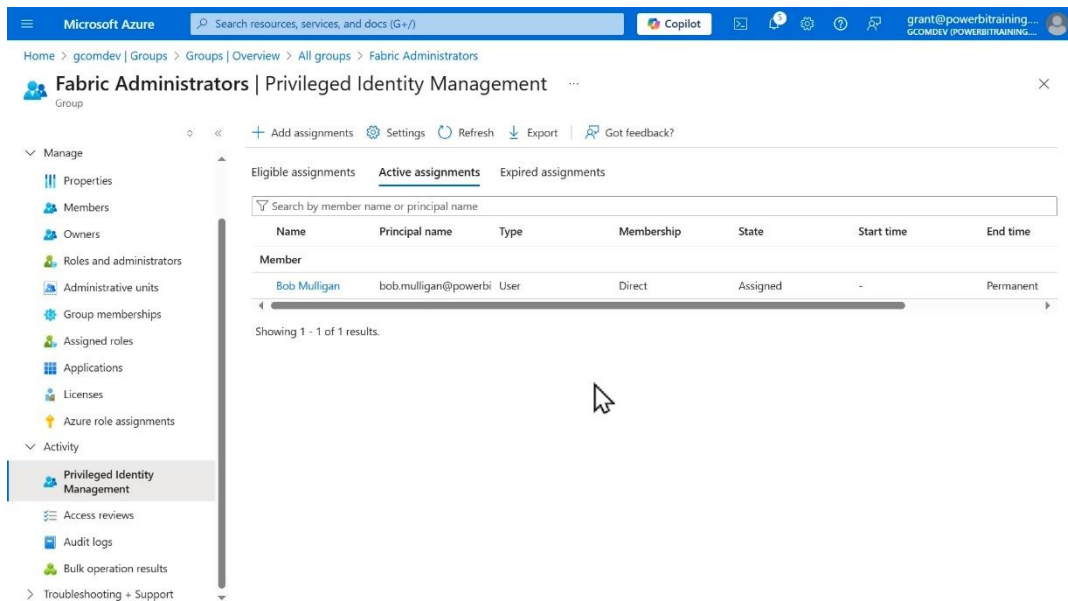
To enable PIM for the Fabric Administrators group:

- Go to Privileged Identity Management within Entra ID.
- Select Enable PIM for this group.



0104-assigning Admin Roles In Microsoft Entra Id

- Review existing assignments (e.g., Bob Mulligan, assigned permanently).

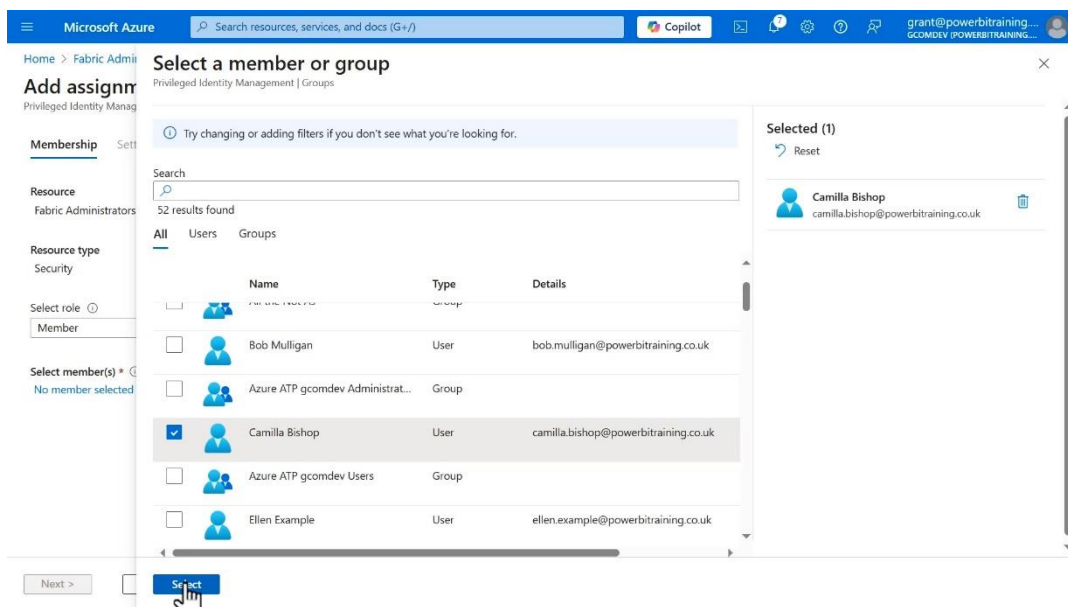


Now, let's assign admin privileges to two more users:

Assigning an Active Role (e.g., Camilla Bishop)

Click Add Assignments > Select Role and Member.

Choose Camilla Bishop, then click Select.



Set the Assignment Type to Active.

Specify a time frame (e.g., from today until August 31).

Add a justification (e.g., Participating in a Power BI proof of concept).

Click Assign.

The screenshot shows the 'Add assignments' page in the Microsoft Azure portal. The breadcrumb navigation is: Home > gcomdev | Groups > Groups | Overview > All groups > Fabric Administrators | Privileged Identity Management >. The page title is 'Add assignments' with a three-dot menu icon. Below the title is the breadcrumb 'Privileged Identity Management | Groups'. The 'Assignment type' section has two radio buttons: 'Eligible' and 'Active', with 'Active' selected. A note states 'Maximum allowed assignment duration is 6 month(s)'. The 'Assignment starts' section has a date field set to '03/14/2025' and a time field set to '9:37:15 AM'. The 'Assignment ends' section has a date field set to '08/31/2025' and a time field set to '10:37:15 AM'. The 'Enter justification' section has a text area containing 'Power BI POC'. At the bottom, there are three buttons: 'Assign' (highlighted with a mouse cursor), '< Prev', and 'Cancel'.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > gcomdev | Groups > Groups | Overview > All groups > Fabric Administrators | Privileged Identity Management >

Add assignments

Privileged Identity Management | Groups

Assignment type ⓘ

☐ Eligible

☒ Active

Maximum allowed assignment duration is 6 month(s).

Assignment starts *

03/14/2025 9:37:15 AM

Assignment ends *

08/31/2025 10:37:15 AM

Enter justification *

Power BI POC

Assign < Prev Cancel

Camilla is now an active administrator for the defined period.

Assigning an Eligible Role (e.g., Ben Carter)

Click Add Assignments > Select Role and Member.

Choose Ben Carter, then click Next.

Set the Assignment Type to Eligible.

Define the validity period (e.g., until August 31).

Click Assign.

Unlike Camilla, Ben does not automatically become an admin. Instead, he must activate his role each time he requires access.

Microsoft Azure

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Co

[Home](#) > [gcomdev | Groups](#) > [Groups | Overview](#) > [All groups](#) > [Fabric Administrators | Privileged Identity Management](#) >

Add assignments

Privileged Identity Management | Groups

Membership

Setting

Assignment type ⓘ

☒ Eligible

☐ Active

Maximum allowed eligible duration is 1 year(s).

Assignment starts *

03/14/2025

9:42:28 AM

Assignment ends *

08/31/2025

9:42:28 AM

Assign

< Prev

Cancel

The User Experience for Active vs. Eligible Assignments

Camilla Bishop (Active Assignment)

Microsoft Azure

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Copilot

grant@powerbitraining...
GCOMDEV | POWERBITRAINING...

[Home](#) > [gcomdev | Groups](#) > [Groups | Overview](#) > [All groups](#) > [Fabric Administrators](#)

Fabric Administrators | Privileged Identity Management

Group

+ Add assignments

Settings

Refresh

Export

Got feedback?

Manage

Properties

Members

Owners

Roles and administrators

Administrative units

Group memberships

Assigned roles

Applications

Licenses

Azure role assignments

Activity

Privileged Identity Management

Access reviews

Audit logs

Bulk operation results

Troubleshooting + Support

Eligible assignments

Active assignments

Expired assignments

Search by member name or principal name

	Principal name	Type	Membership	State	Start time	End time	Ac
ishop	camilla.bishop@powerbi	User	Direct	Assigned	3/14/2025, 9:40:55 AM	8/31/2025, 10:37:15 AM	Re
gan	bob.mulligan@powerbi	User	Direct	Assigned	-	Permanent	Re

Showing 1 - 2 of 2 results.

When Camilla logs into the Power BI Admin Portal, she immediately has full administrator privileges—no extra steps required. She can manage tenant settings and perform admin tasks from the moment of login.

Ben Carter (Eligible Assignment)

When Ben logs into the Power BI Admin Portal, he does not see admin options immediately. To activate his admin role:

He navigates to Azure > Manage Microsoft Entra ID > Fabric Administrators Group.

He clicks Privileged Identity Management and sees an option to Activate Now.

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > gcomdev | Groups > Groups | Overview > All groups > Fabric Administrators

Fabric Administrators | Privileged Identity Management

Group

+ Add assignments Settings Refresh Export Got feedback?

You have eligible administrative assignments for this resource. Activate now to manage PIM. →

Eligible assignments Active assignments Expired assignments

Search by member name or principal name

Name	Principal name	Type	Membership
------	----------------	------	------------

Overview

Diagnose and solve problems

Manage

Activity

Privileged Identity Management

Access reviews

Audit logs

Bulk operation results

Troubleshooting + Support

He clicks Activate, selects a duration (e.g., 4 hours), and provides a reason (e.g., Modify export and sharing settings).

Microsoft Azure Search resources, services, and docs (G+/) Copilot ben.carter@powerbitrai... GCOMDEV (POWERBITRAINING...)

Home > gcomdev | Groups > Groups | Overview > All groups > Fabric Administrators | Privileged Identity Management

My roles - Groups

Privileged Identity Management | Groups

Refresh Got feedback?

Eligible assignments Active assignments Expired assignments

Search by role or group

Role	Group	Group type	Membership
Member	Fabric Administrators	Security	Direct

Activate - Member

Privileged Identity Management | Groups

Roles Activate Status

☒ Custom activation start time

Start time * 14/03/2025 09:57:31

(UTC+00:00) Dublin, Edinburgh, Lisbon, London

Duration (hours) 4

The end time of activation would be 14/03/2025, 13:57:31

Reason (max 500 characters) * Modify export and sharing settings

Activate Cancel

After processing, the activation completes, and he gains admin privileges for the chosen time frame.

The system automatically removes his admin access when the period expires, ensuring security.

Key Benefits of Using PIM in Microsoft Entra ID

- Privileged Identity Management in Microsoft Entra ID provides several advantages:
- Improved Security – Reduces the risk of excessive admin access.
- Flexible Role Management – Supports group-level admin assignments.
- Time-Based Access Control – Allows for temporary and just-in-time admin rights.
- Auditability – Tracks when users activate admin roles and their justifications.

If your organization has Microsoft Entra ID Premium P2, possibly through a Microsoft 365 E5 subscription, leveraging these features is highly recommended.

0201: Creating a Power BI Tenant

To get started with Power BI, go to powerbi.microsoft.com. Here, you have two options: sign up for a free trial by selecting Try Free, or purchase a subscription by clicking Buy Now.

The sign-up process begins by entering your email address and clicking Set up account. If Microsoft does not recognise the domain associated with your email, you will be prompted to create a new account.



Thank you for choosing **Power BI Pro**

- 1 Let's set up your account
We can't finish signing you up because Power BI Pro isn't currently available for your work or school account associated with **norman.newcomer@newtenant.com**. If you want to sign up for Power BI Pro, you'll need to set up a new account.
[Create new account](#)
- 2 Tell us about yourself
- 3 Create your business identity
- 4 Payment & billing

Feedback

Choosing a Region

During setup, you will be asked to select a region. This decision is important, as the region determines where your Power BI tenant will be hosted—and once selected, it cannot be changed. If you ever need to check where your tenant is located, log into Power BI, click on Settings, then select Help and Support, followed by About Power BI. This section will display the location of your data.

First name: Norman

Last name: Newcomer

Business phone number: 0800 888 8888

Company name: New Tenant Limited

Country or region: United Kingdom

By clicking **Next**, you agree to our terms and conditions.

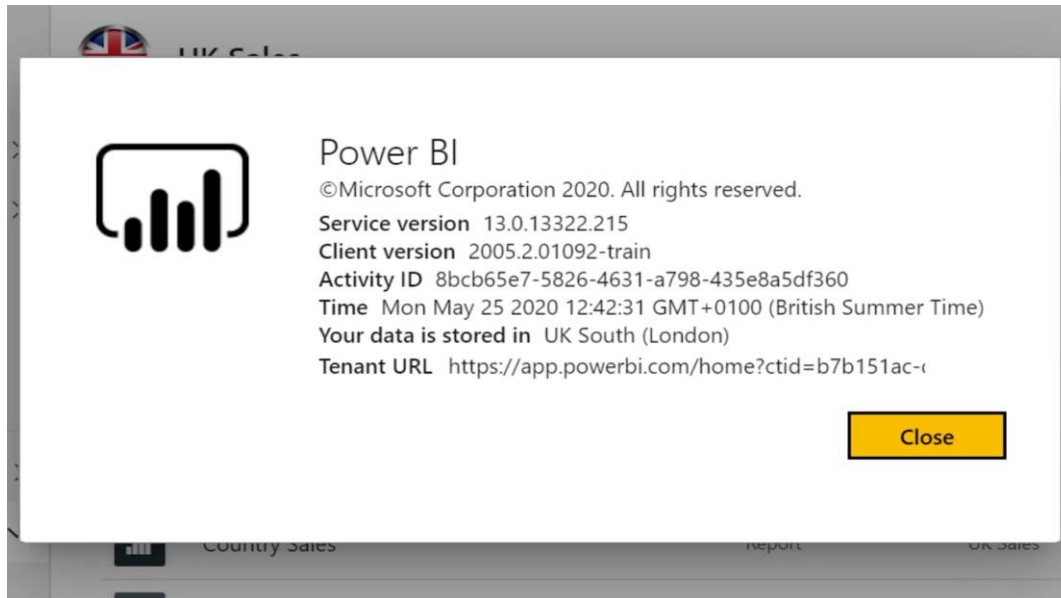
☐ I would like to receive information, tips, and offers from Microsoft about Microsoft Online Services, and for Microsoft to share my information with select partners so I can receive relevant information about their products and services. To learn more, or to unsubscribe at any time, view the [privacy Statement](#).

[Next](#)

What is a Power BI Tenant?

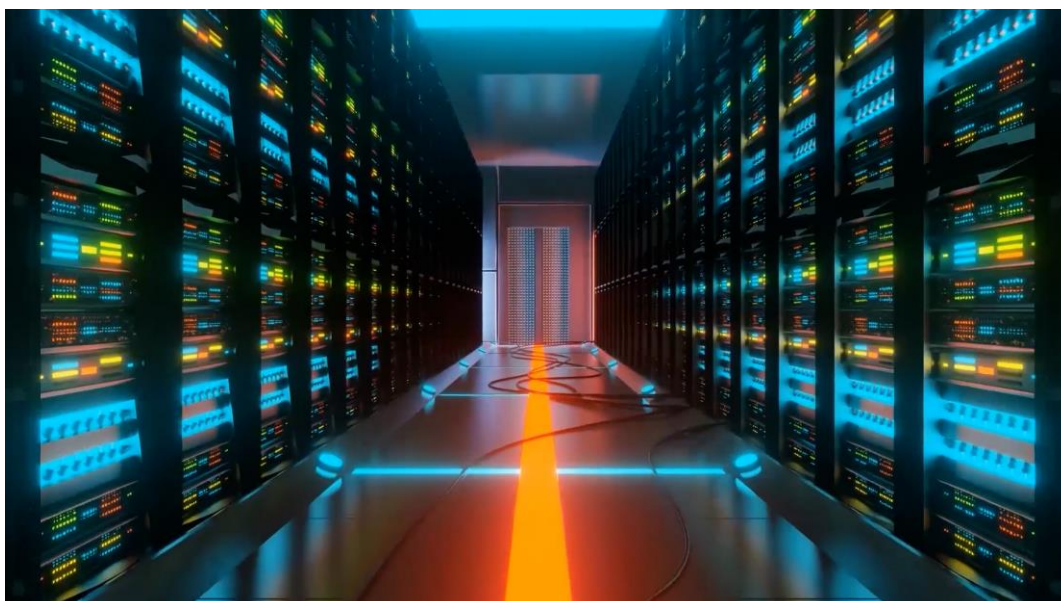
In Power BI, the tenant serves as the foundational organisational container, housing all your semantic models, reports, dashboards, and other assets. While the term "Power BI tenant" is commonly used, technically, all Power BI tenants exist within a broader Microsoft 365 (formerly Office 365) tenant.

How Power BI Fits into Microsoft 365



When an organisation's first user signs up for Power BI, they must use a corporate email address. If the associated domain already has an existing Microsoft 365 tenant, Power BI is simply added to that infrastructure. However, if no tenant exists, Microsoft automatically creates a new Microsoft 365 tenant—initially containing only Power BI but with the core Microsoft 365 infrastructure in place.

Shared vs. Dedicated Resources



0201-creating A Power BI Tenant

Unless an organisation purchases a Power BI Premium licence, its Power BI tenant shares cloud resources with other tenants, much like multiple businesses renting office space in the same building. However, in this cloud environment, you have no visibility into who else shares the infrastructure.

Microsoft

Get started with **Power BI Pro**
for \$9.99 user/month
Price shown in USD. Currency may vary by country.

- 1 Let's set up your account
Enter your work or school email address, we'll check if you need to create a new account.

By proceeding you acknowledge that if you use your organization's email, your organization may have rights to access and manage your data and account. [Learn More](#)
[Set up account](#)
- 2 Tell us about yourself
- 3 Create your business identity
- 4 Payment & billing
- 5 Start using Power BI Pro

Office 365

Privacy & cookies Terms of use Trademarks © 2020 Microsoft [Feedback](#)

A Premium licence provides dedicated capacity, ensuring more control over performance and resource allocation.

0202: Corporate Branding

Power BI offers a limited but effective set of custom branding options to help organisations maintain a consistent look and feel across their reports. These options include a logo, a cover image, and a theme colour. Without branding, users moving from an intranet environment to Power BI may feel like they have entered a generic Microsoft space. Applying these branding elements ensures that your Power BI environment remains aligned with your corporate identity.

Checking Your Admin Role

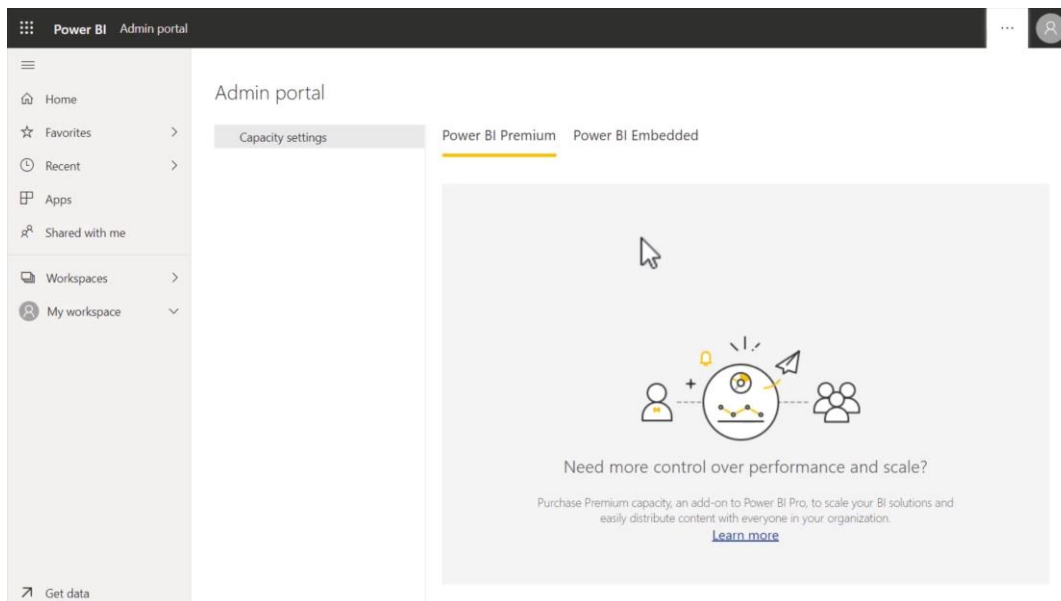
To configure custom branding, you need to be a Power BI administrator. If you're unsure whether you have the necessary permissions, follow these steps:

Log in to your Power BI tenant.

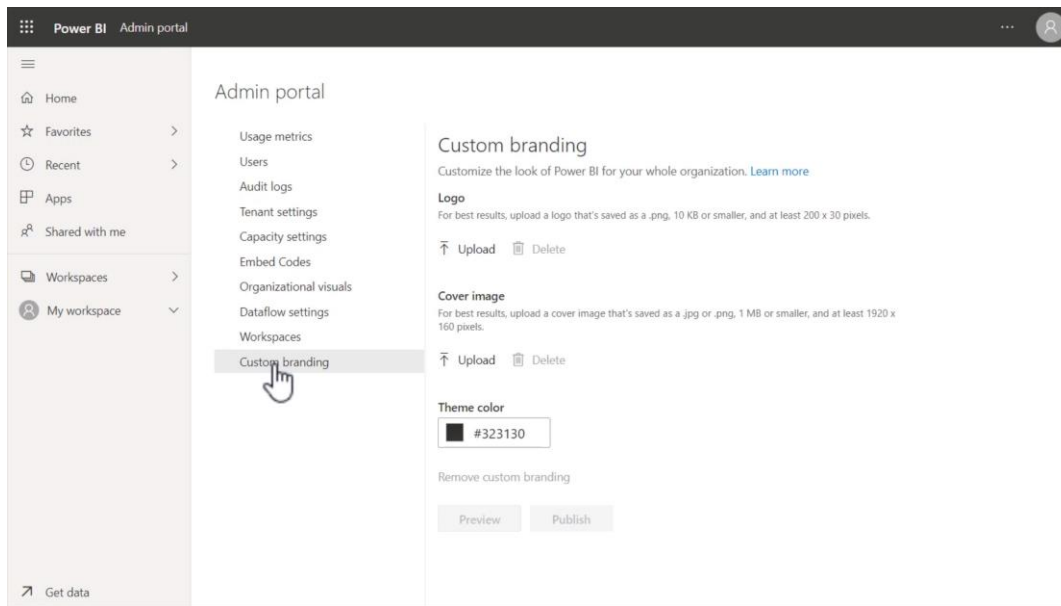
Click the ellipsis menu (top right of the screen).

Select Settings > Admin Portal.

If you only see Capacity Settings within the Admin Portal, this means you are not an administrator.



If you are an administrator, you will see a broader set of configuration options, including Custom Branding, which appears towards the bottom of the list.



Branding Elements and File Formats

The Custom Branding section consists of three elements:

Logo

Cover Image

Theme Colour

Power BI provides format guidelines for these elements. The logo must be in PNG or JPEG format:

PNG supports transparency, making it ideal for logos with non-rectangular shapes or text-based designs.

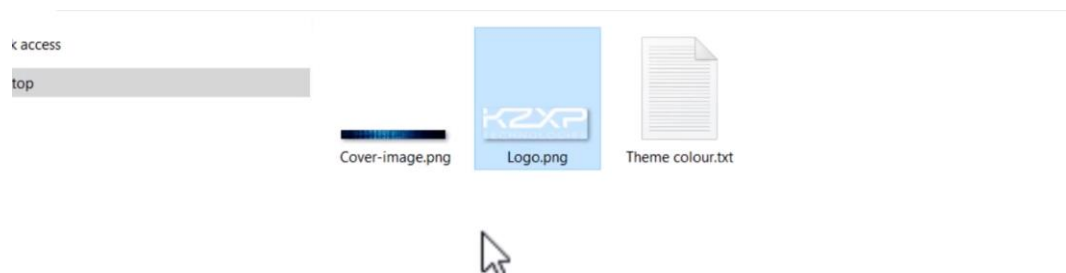
JPEG is always rectangular and better suited for photographic images.

When selecting a theme colour, consider its contrast with your logo. The theme colour applies to the top navigation bar, so if your logo is light, a darker theme colour is advisable for visibility.

Preparing and Uploading Branding Elements

If your logo is too large, you can resize it using simple tools like Microsoft Paint:

Right-click the image and select Open with Paint.



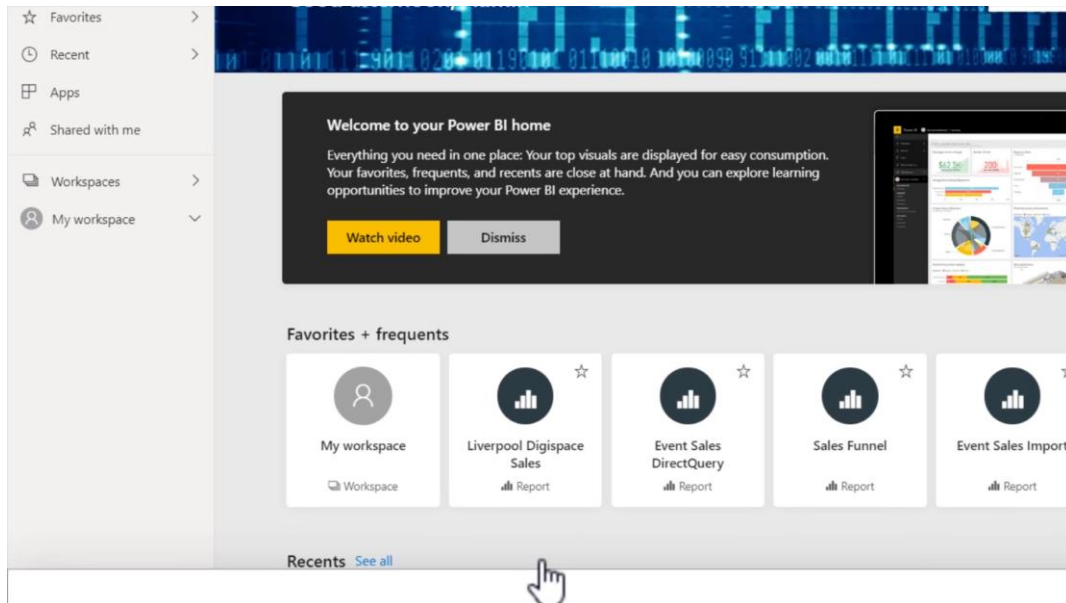
0202-corporate Branding

Click Resize and ensure Maintain Aspect Ratio is enabled.

Adjust the dimensions to a suitable size.

Once your branding elements are prepared:

Upload the logo – if it has a transparent background, it may not be immediately visible if the theme colour is similar.



Upload the cover image – ensure it is sized according to Power BI's recommendations.

Set the theme colour – enter the hex code and click to confirm.

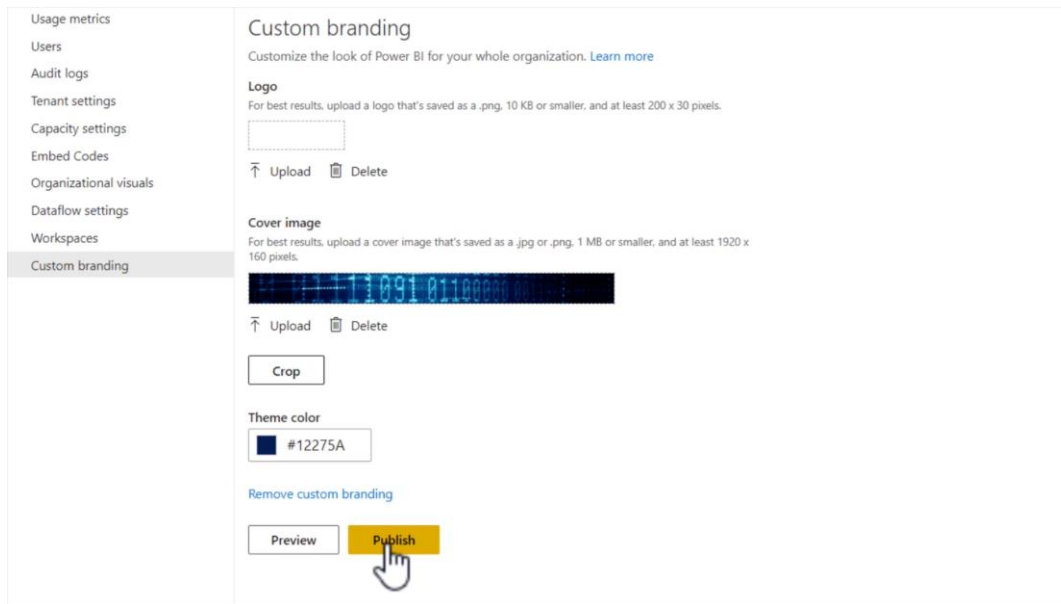
Previewing and Publishing

Before publishing, Power BI allows you to preview your changes. This lets you check how the logo, cover image, and theme colour appear together. For example, if you've uploaded a white logo on a dark blue background, the contrast should be clear.

The cover image appears only on the Power BI home page and is not visible when users navigate to other areas. If the preview looks correct:

Click Go Back to exit the preview.

Click Publish to apply the branding.



The screenshot shows the 'Custom branding' settings page in the Power BI Admin Portal. On the left is a navigation menu with options: Usage metrics, Users, Audit logs, Tenant settings, Capacity settings, Embed Codes, Organizational visuals, Dataflow settings, Workspaces, and Custom branding (which is highlighted). The main content area is titled 'Custom branding' and includes a 'Learn more' link. It has two sections: 'Logo' and 'Cover image'. The 'Logo' section has a text box for the logo and 'Upload' and 'Delete' buttons. The 'Cover image' section has a text box for the cover image, 'Upload' and 'Delete' buttons, and a 'Crop' button. Below these is a 'Theme color' section with a color picker set to '#12275A'. At the bottom are 'Remove custom branding', 'Preview', and 'Publish' buttons. A hand cursor is pointing at the 'Publish' button.

Reverting Branding Changes

If at any point you need to remove custom branding, Power BI provides a simple option:

Navigate to Custom Branding in the Admin Portal.

Click Remove Custom Branding.

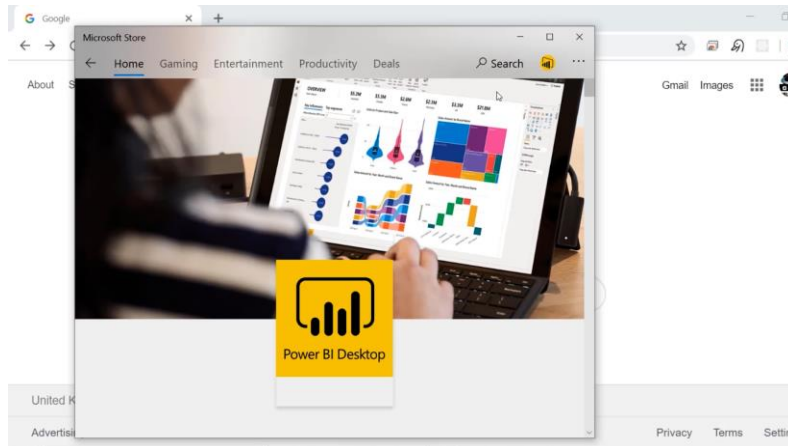
Click Publish to revert to the default Power BI appearance.



With these simple steps, you can ensure that your Power BI environment reflects your organisation's branding, making it a seamless part of your business ecosystem.

0203: Installing Power BI Desktop

Power BI Desktop is completely free to use. Microsoft's business model focuses on monetizing collaboration and content sharing rather than the standalone software itself. There are two ways to install Power BI Desktop: as a downloadable executable file or as a Windows 10 app from the Microsoft Store.

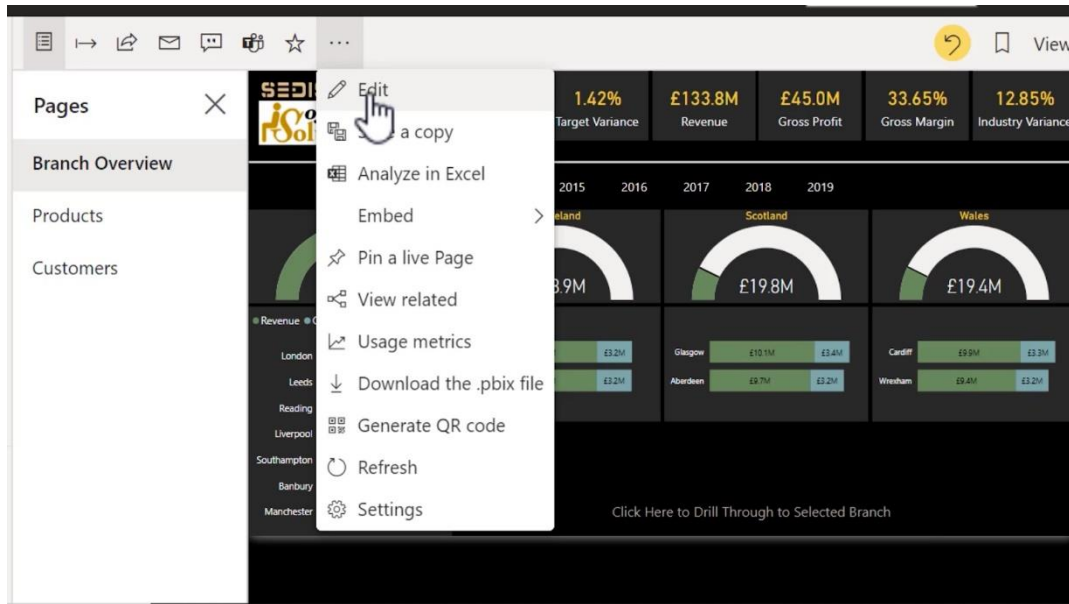


Choosing the Best Installation Method

Installing Power BI Desktop via the Microsoft Store has a couple of key advantages. First, it does not require administrator rights, making it more accessible for users in environments with restricted permissions. Second, it updates automatically, which is particularly beneficial since Microsoft releases a new version of Power BI Desktop every month.

However, not every user will necessarily need Power BI Desktop. This applies not only to report consumers but also to some report creators. If a team has already prepared the data, and certain users only need to build reports rather than work with the data itself, they can often do everything they need directly in the Power BI Service, without installing Power BI Desktop.

Editing Reports in Power BI Service



As the report owner, you can switch between consuming and editing reports in the Power BI Service. To edit a report, click the ellipsis menu (More Options) and choose Edit. This opens an interface similar to Power BI Desktop, allowing you to make changes directly in the cloud.

To maximize your workspace, you can:

- Close the navigation pane on the left by clicking Hide Navigation Pane.
- Put your browser into full-screen mode by pressing F11.

Close the fields pane if you need additional space on the page.

Working Without Power BI Desktop

Editing reports in the Power BI Service is particularly useful for Mac users. Power BI Desktop is only available on Windows, but by working in the Power BI Service, Mac users can still participate in report development. This flexibility ensures that all team members, regardless of their operating system, can contribute to the reporting process effectively.

Later, when we discuss Content Governance, we'll explore in more detail how to determine who needs Power BI Desktop and who can work entirely in the cloud.

0204: Power BI Licensing

Power BI Desktop is entirely free, but when it comes to the Power BI service, a subscription is required. There are three main licensing options: Power BI Free, Power BI Pro, and Power BI Premium. The Free and Pro licenses are assigned at an individual level, while the Premium license is applied at the tenant level.



Individual Licensing: Power BI Free vs. Power BI Pro

For individuals, the key decision is between Power BI Free and Power BI Pro, which is priced at \$10 (or £7.50) per user per month. The primary distinction between these two options is collaboration.



Power BI Free is intended for personal use, making it ideal for learning the product, creating reports, and experimenting. However, it does not support collaboration. If a user with a Free license attempts an operation that requires a Pro license—such as sharing content—a prompt appears, inviting them to upgrade to Pro or start a 60-day trial of Power BI Pro.

It's worth noting that this 60-day trial is not unlimited across a tenant. After approximately 50 users have been offered the trial within the same organisation, the option eventually disappears. At that point, users will simply be prompted to upgrade to Power BI Pro without the trial option.

Collaboration and Licensing Considerations

To enable collaboration within an organisation, every user who needs to interact with reports beyond their own personal workspace must have a Power BI Pro license. This applies to all participants, regardless of their level of involvement.

For example, a data analyst who actively develops reports every day requires the same Pro license as an executive who only logs in once a month to review a key report. This can make the Pro licensing model expensive for large organisations where many users only need occasional access.

Power BI Premium: A Scalable Enterprise Solution



For enterprises with hundreds or thousands of users, Power BI Premium provides an alternative to individual Pro licenses. Premium offers dedicated cloud resources, allowing an organisation to move away from the shared tenancy model of Free and Pro users. It also increases data usage limits and provides greater flexibility for scaling resources as demand grows.

However, even with Power BI Premium, not all licensing requirements are eliminated. Users who need to create and share reports must still have a Power BI Pro license. The main benefit of Premium is that users who only consume reports—meaning they access reports shared by others but do not create or distribute them—can do so with a Power BI Free license.

In summary, Power BI Free is suited for individual exploration, Power BI Pro is required for collaboration, and Power BI Premium is designed for large-scale deployments where report consumers can operate without incurring per-user licensing costs.

0205: Power BI Tenant Settings

Power BI tenant settings provide granular control over the features available to your organisation within the Power BI service, as well as to some extent in Power BI Desktop. These settings are particularly useful if you need to manage access to specific features due to data sensitivity concerns or if you want to tailor feature availability to different groups of users.

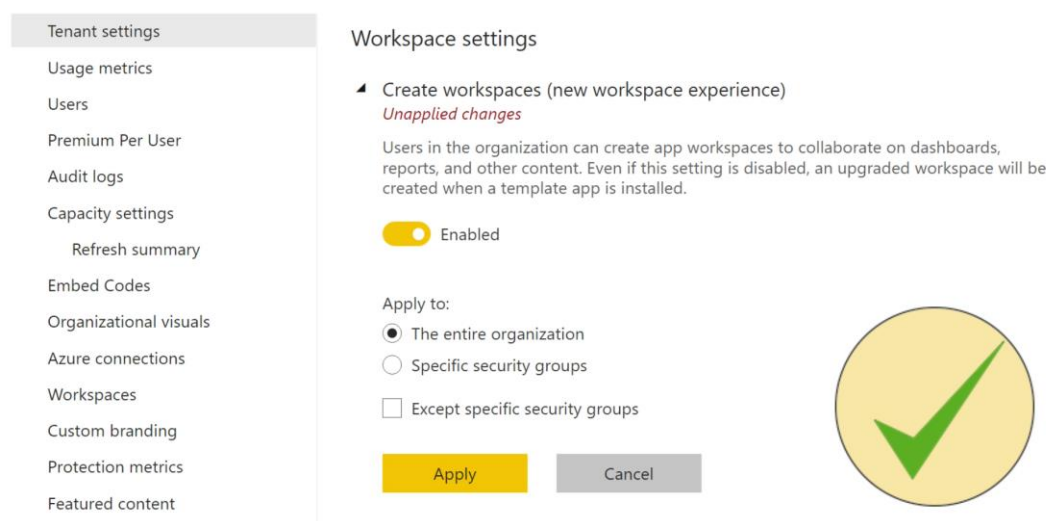
Within the Power BI admin centre, tenant settings allow you to dictate how users interact with Power BI objects within workspaces and define the operations they are permitted to perform. While these settings help establish governance policies, it's important to note that they are not security measures—they simply shape the governance approach that best suits your organisation's needs.

Understanding Tenant Setting Options

Most tenant settings follow a consistent pattern, offering three key options:

First, you can disable a setting for the entire organisation. This prevents all users from accessing the associated feature, ensuring it is completely unavailable across the organisation.

Admin portal



The screenshot displays the 'Admin portal' interface. On the left is a sidebar menu with 'Tenant settings' highlighted. The main content area is titled 'Workspace settings' and shows the configuration for 'Create workspaces (new workspace experience)'. A red text label 'Unapplied changes' is present. The description states: 'Users in the organization can create app workspaces to collaborate on dashboards, reports, and other content. Even if this setting is disabled, an upgraded workspace will be created when a template app is installed.' The setting is currently 'Enabled', indicated by a yellow toggle switch. Under 'Apply to:', the radio button for 'The entire organization' is selected. Below this, there is an unchecked checkbox for 'Except specific security groups'. At the bottom are 'Apply' and 'Cancel' buttons. A large yellow circle with a green checkmark is overlaid on the right side of the settings panel.

Second, you can enable a feature for everyone. This means that all users within the organisation will have access to the feature without restriction.

Third, and often the most practical option, you can enable a feature for only specific security groups. This allows you to grant access to particular teams while restricting it for others, providing a more flexible governance model.

Admin portal

Tenant settings
Usage metrics
Users
Premium Per User
Audit logs
Capacity settings
Refresh summary
Embed Codes
Organizational visuals
Azure connections
Workspaces
Custom branding
Protection metrics
Featured content

Workspace settings

Create workspaces (new workspace experience)
Unapplied changes

Users in the organization can create app workspaces to collaborate on dashboards, reports, and other content. Even if this setting is disabled, an upgraded workspace will be created when a template app is installed.

☒ Enabled

Apply to:

☐ The entire organization
☒ Specific security groups

☒ Except specific security groups

☒ Except specific security groups

Additionally, Power BI allows for more nuanced control. You can enable a feature for the entire organisation but exclude specific security groups. This means that while most users can access the feature, those in the excluded security groups will not. This flexibility allows for combinations of settings, ensuring that access can be fine-tuned—such as enabling a feature for one group while simultaneously disabling it for another.

Applying and Testing Tenant Settings

One key consideration when making changes to tenant settings is that they do not take effect immediately. It can take up to 15 minutes for a setting change to apply across the organisation, so you may need to allow for this delay when testing your configurations.

When configuring access, keep in mind that the most restrictive setting for a given user will always take precedence. For example, if a user is part of multiple security groups with conflicting settings, the setting that imposes the strictest restriction will apply.

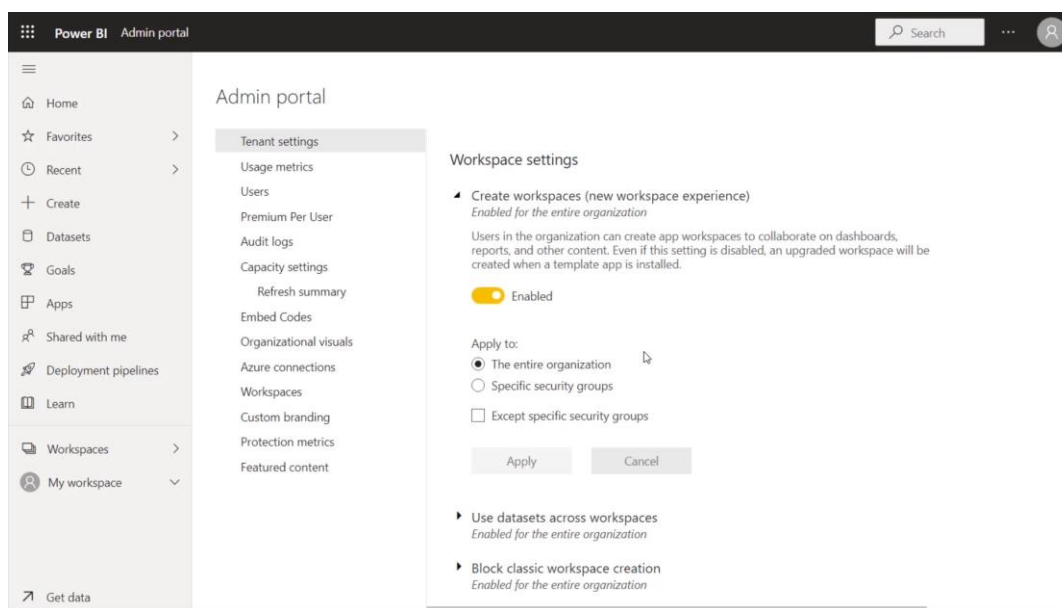
By carefully managing tenant settings, you can strike the right balance between enabling productivity and ensuring governance policies align with your organisation's needs.

0300: Overview of Workspace Administration

In a Power BI tenant, all content is stored within an object called a workspace. This section explores how workspaces are created and how access is granted to them.

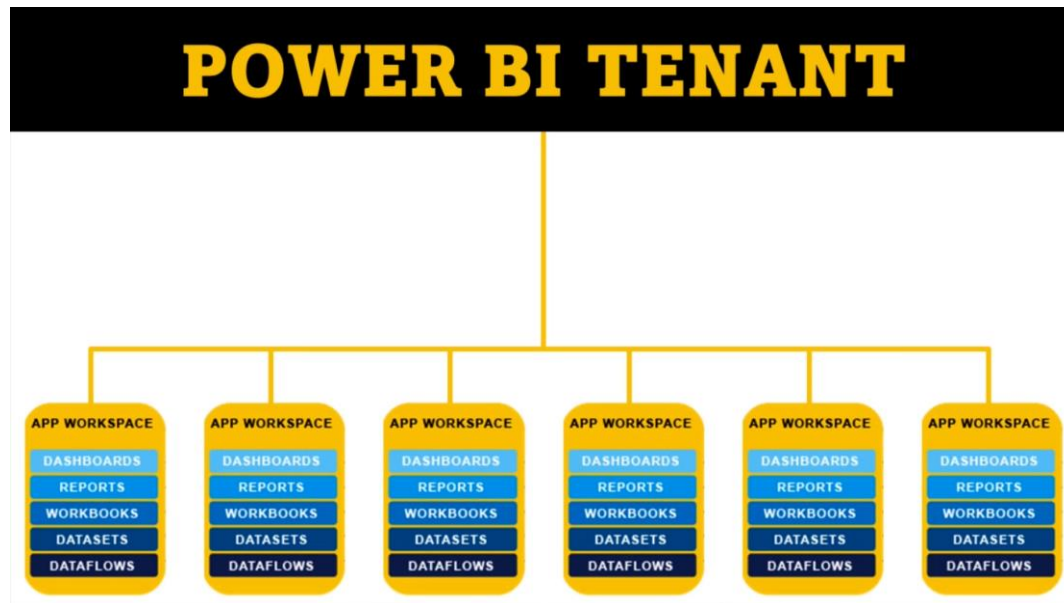
Managing Workspace Creation

One of the biggest challenges in workspace administration is preventing the uncontrolled proliferation of workspaces. It is easy to underestimate how quickly this can happen. In the early stages of Power BI adoption, when only a few workspaces exist, it may seem feasible to allow workspace creation to occur organically. However, as time passes, this organic growth can lead to an unmanageable and confusing environment. Without proper controls, an organisation may find itself with dozens or even hundreds of workspaces—many of which contain duplicate or conflicting information.



The key to avoiding this scenario is to establish clear policies from the outset. If the right structure is in place from day one, it becomes much easier to maintain a coherent and navigable workspace environment. A best practice is to restrict workspace creation to specific security groups, typically composed of administrators or designated individuals. This setting is controlled in the Power BI Admin Portal under Workspace Settings.

The Importance of a Coherent Workspace Structure



It is important to emphasise that restricting workspace creation is not about limiting functionality—it is about ensuring clarity and structure. A well-organised workspace structure allows users to understand where content should be placed and how it should be accessed. Therefore, it is crucial to design a logical framework for workspaces and ensure that everyone involved understands both the structure and the rationale behind it. Clear guidelines should be established to define what type of content belongs in each workspace.

Workspaces should be primarily used by content creators, with additional access provided for approval and testing. However, content consumers should not be given direct access to workspaces. Instead, they should consume reports and dashboards through read-only Power BI apps. The process of creating and managing these apps will be covered in Section 6: Sharing Administration.

Granting Access to Workspaces

Once workspaces have been created, the most effective way to manage access is through security groups. Granting access on an individual basis can quickly become difficult to manage and prone to inconsistencies. Security groups provide a scalable and controlled approach to access management.

Additionally, the ability to grant workspace access can be delegated. Assigning the Member role to a security group allows members of that group to grant access to other users—or preferably to other security groups. This delegation ensures that access management remains flexible while still being controlled within a structured framework.

By following these principles, organisations can maintain a well-structured and efficient Power BI environment, preventing unnecessary complexity while ensuring that the right users have appropriate access.

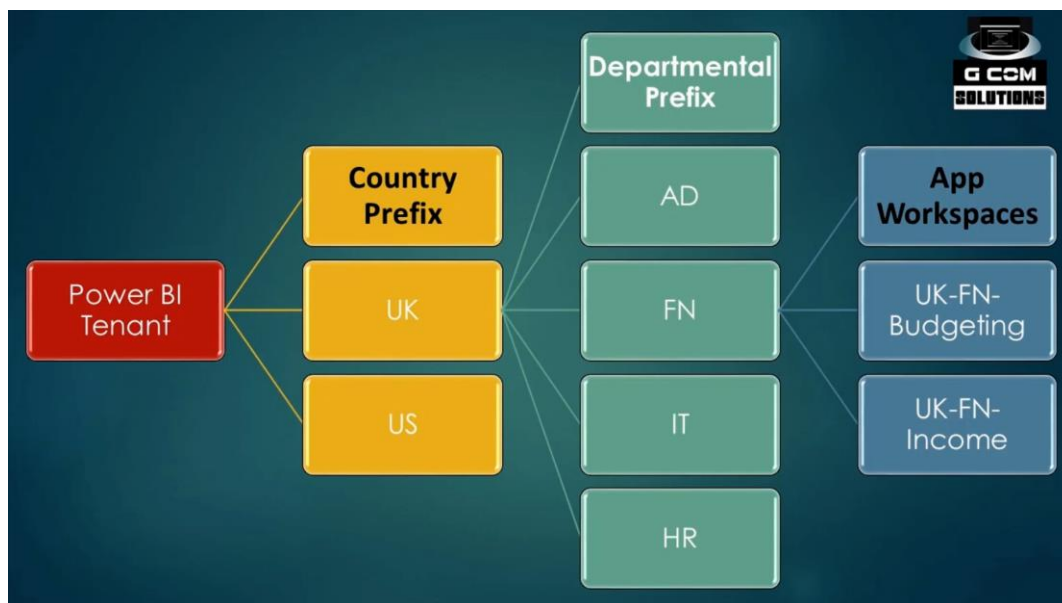
0301: Designing Workspace Structures

In Power BI, all content is stored within workspaces, sometimes referred to as App workspaces. These workspaces house three key types of content: reports, dashboards, and workbooks. Supporting this content are the underlying data structures—semantic models and dataflows. Once a workspace is prepared, it can be published as an app, which is typically a read-only version of the workspace. However, as we will explore later in the course, this distinction is not always rigid.

The Challenge of Workspace Organisation

Power BI does not provide built-in tools for structuring or grouping workspaces. From a technical standpoint, all workspaces are identical—there are no hierarchies, folders, or groupings available natively. As a Power BI administrator, it falls to you to design an effective workspace structure that aligns with your organisation's needs.

By default, workspaces are listed alphabetically. A practical way to introduce some structure is through naming conventions. For example, a workspace name can begin with a country prefix, followed by a departmental prefix, and finally, a descriptive theme. This approach helps keep related workspaces together when viewed in a list.



Deciding the Scope of Workspaces

Another important consideration is the level of granularity in your workspaces. Should you consolidate a broad set of related content into a single workspace, or create multiple, more focused workspaces? Power BI does not enforce a particular approach, so this decision will be guided by your organisation's broader information management practices.

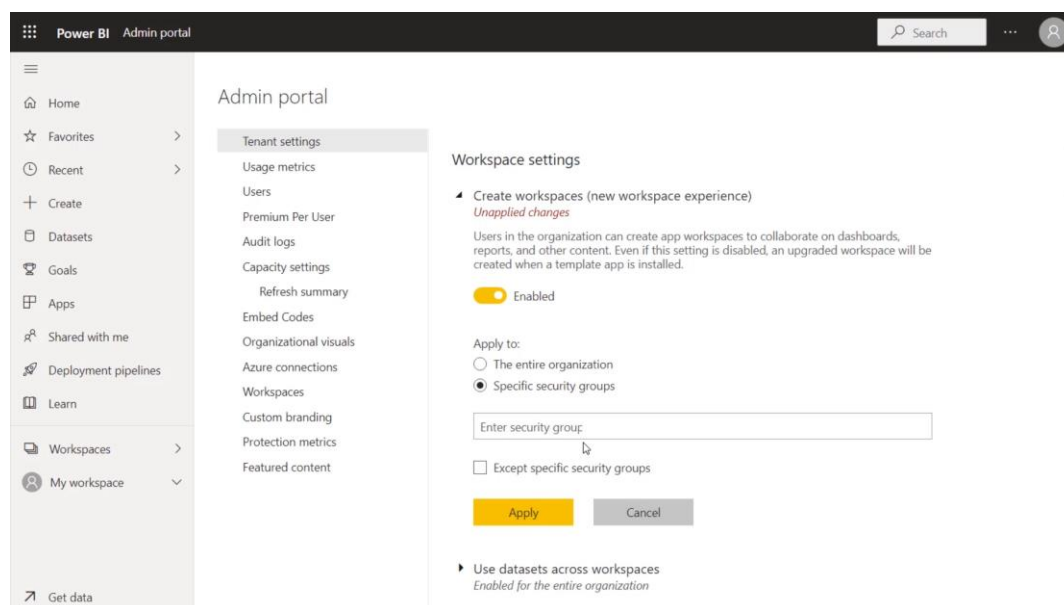
Many organisations follow existing naming and structuring conventions from other systems, ensuring consistency across platforms. Your chosen approach should align with how teams work and collaborate, making it easier for users to navigate and manage Power BI content effectively.

0302: Workspace Settings

The Workspace Settings section of the Power BI Admin Portal consists of three key options that help you manage workspace creation and semantic model usage across your organisation. These settings allow administrators to control who can create workspaces, how semantic models are shared, and whether classic workspaces should be used. Let's explore each setting in detail.

Controlling Workspace Creation

The Create Workspaces setting determines who within your Power BI tenant has permission to create new workspaces. As with many admin settings, this can be applied to the entire organisation or restricted to specific security groups.

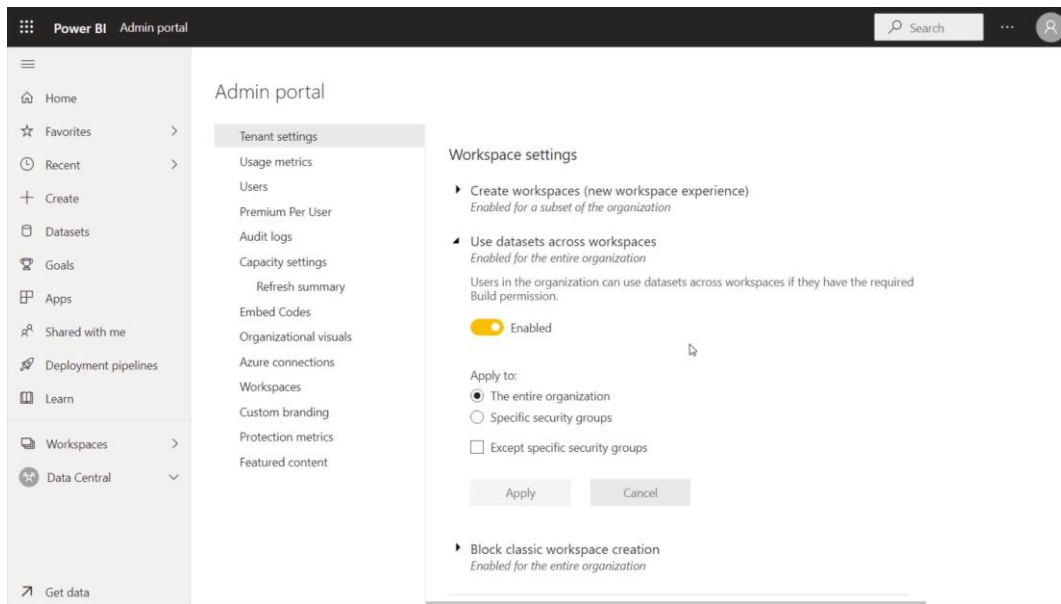


For example, if you want only members of a Workspace Administrators security group to create workspaces, you can configure the setting accordingly. Once in place, only users within this group will see the option to create a new workspace when they navigate to Workspaces in the Power BI Service. Users outside this group will receive an error message stating that App workspace creation is disabled. However, the message does not specify whether the restriction applies to the entire organisation or just the individual user.

Using Semantic models Across Workspaces

By default, Power BI workspaces function in a way where each report has its own accompanying semantic model. When a report is published from Power BI Desktop to the Power BI Service, you will typically see two separate items: the report and the semantic model.

This approach, however, can lead to semantic model duplication, where multiple reports are built on logically identical but physically separate semantic models. The Use Semantic models Across Workspaces setting helps to address this by allowing organisations to centralise semantic models.



With this enabled, reports can be built from a single, shared semantic model and then published to different workspaces.

For example, an organisation might have a dedicated Data Only workspace, where semantic models are stored without any reports, dashboards, or workbooks. Users can then access these centralised semantic models via the Manage Permissions feature, granting specific individuals or security groups the necessary access. They can then build reports using these shared semantic models while maintaining the flexibility to upload reports to other workspaces based on content and business needs.

This feature is particularly useful for organisations that want to maintain a single source of truth while reducing unnecessary duplication of semantic models across workspaces. We will return to this theme later when discussing Workspace Management in more depth.

Blocking Classic Workspace Creation

The final option, Block Classic Workspace Creation, controls whether users can create legacy Classic Workspaces—a feature from the earlier days of Power BI that was tightly integrated with Microsoft 365 Groups.

Previously, when a Classic Workspace was created, an underlying Microsoft 365 Group was also generated, and workspace administration was handled within Microsoft 365 rather than in Power BI itself. Many early adopters of Power BI still have a number of these Classic Workspaces.

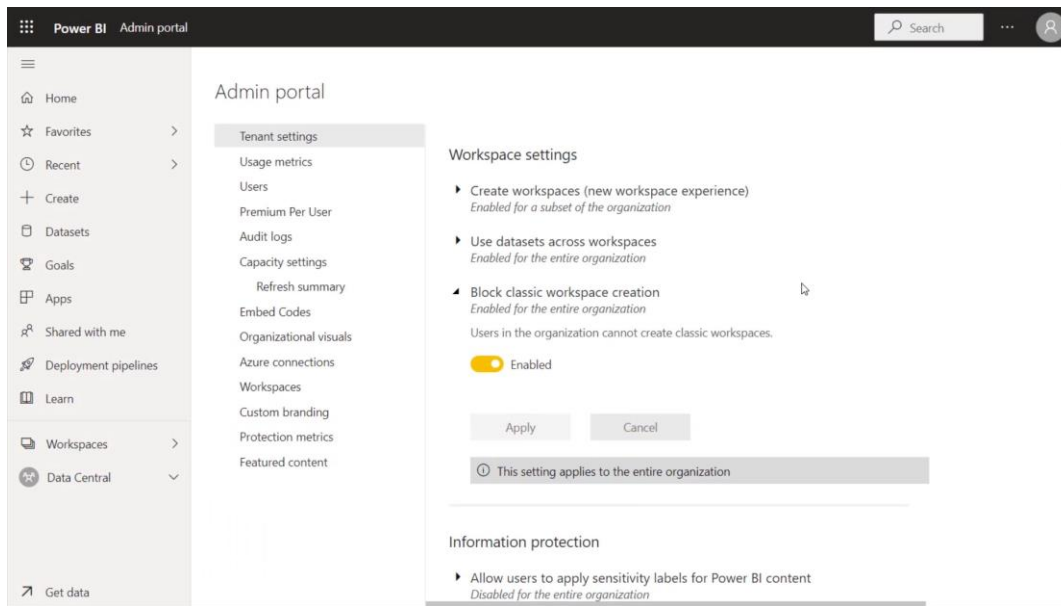
By contrast, the new workspace experience operates entirely within Power BI, without any dependency on Microsoft 365 Groups. This modern approach offers several advantages, including:

More granular role-based access control within Power BI.

Better governance over workspace administration.

Support for shared semantic models across multiple workspaces, a capability that is not available in Classic Workspaces.

0302-workspace Settings

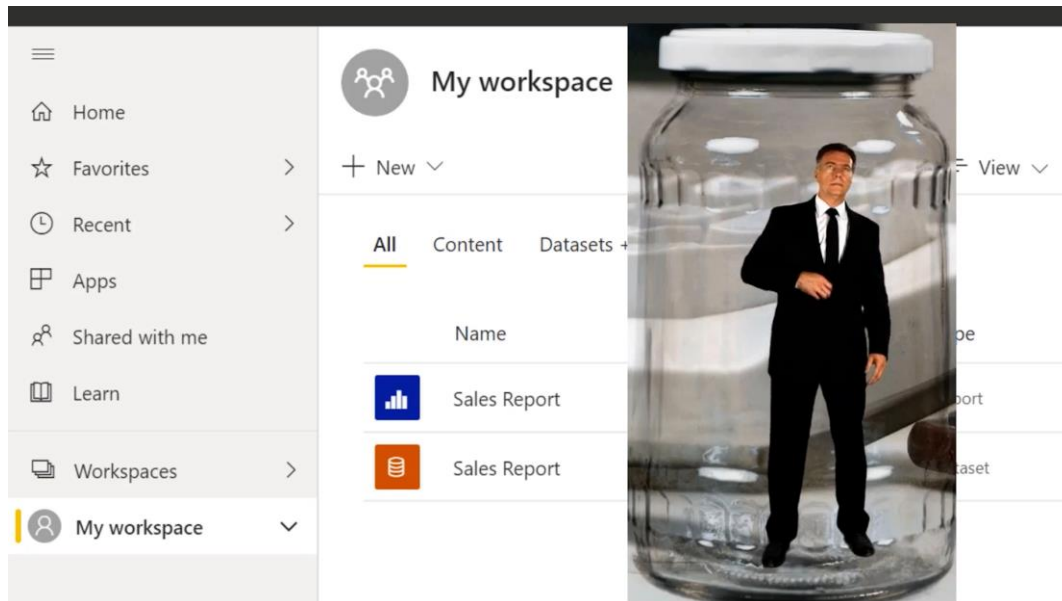


For most organisations, blocking Classic Workspace creation is the recommended approach. Even if Classic Workspaces already exist in your environment, there is little reason to allow users to create new ones. Enabling this setting ensures that all newly created workspaces use the modern workspace experience, providing a more robust and flexible environment for collaboration and semantic model management.

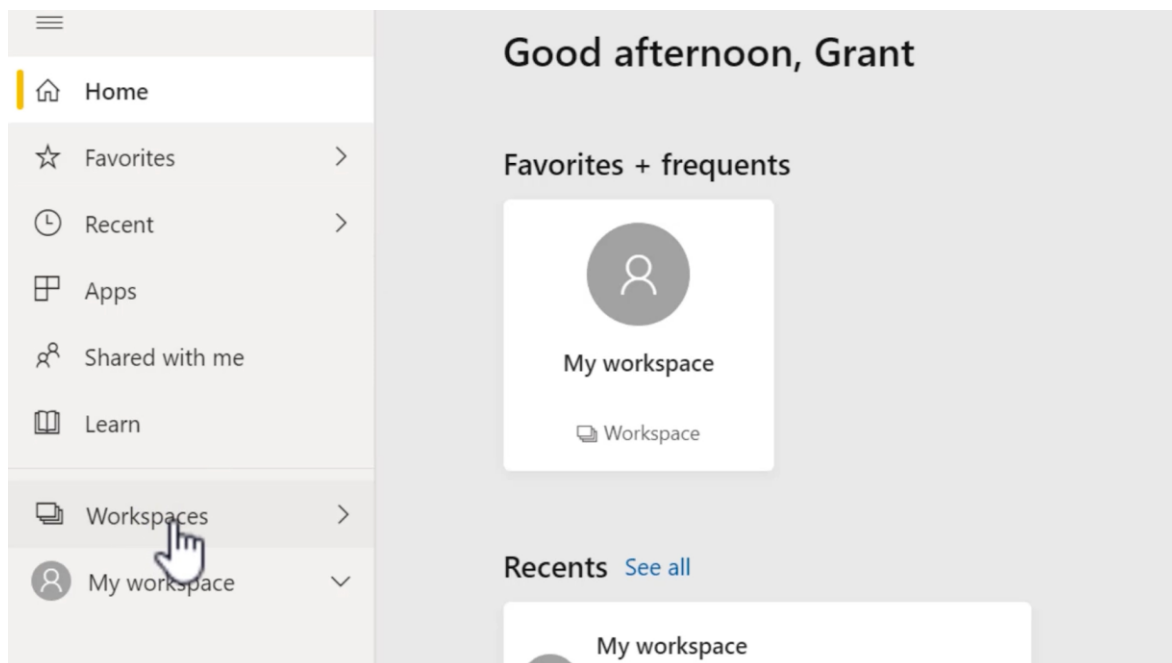
Retry

0303: My Workspace

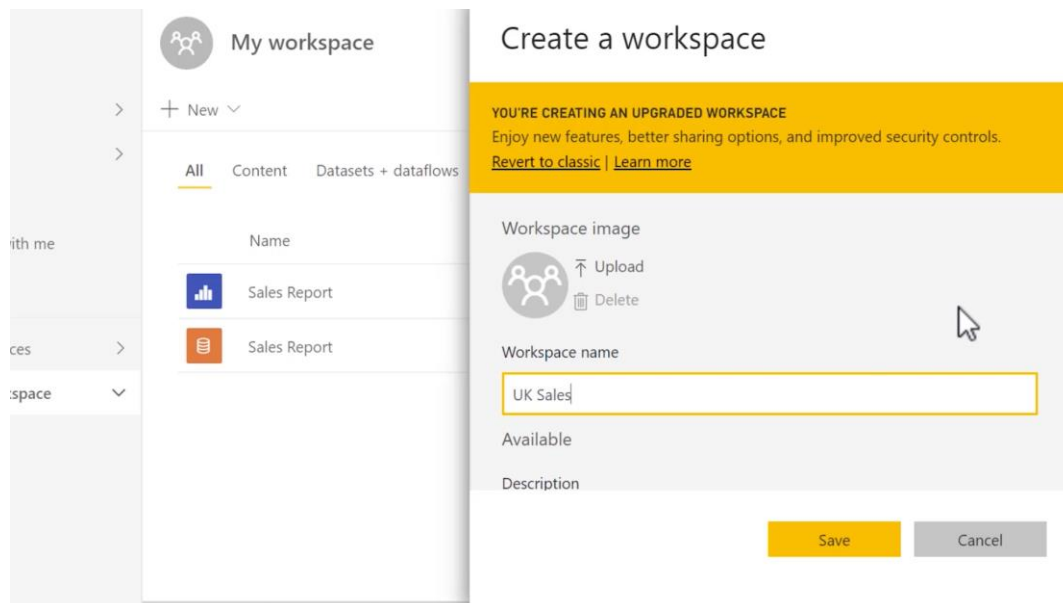
In Power BI, the creation of workspaces is managed by the tenant administrators. They determine whether workspaces should be centrally controlled or if users are allowed to create their own. Alongside these corporate workspaces, every user has a private space called My Workspace. This is an individual workspace linked to the user's account, meaning each user only has access to their own My Workspace. The only way to access another user's My Workspace would be to sign in using their credentials, which is not a standard practice.



In this example, a new account has been created, and currently, the only available workspace is My Workspace.

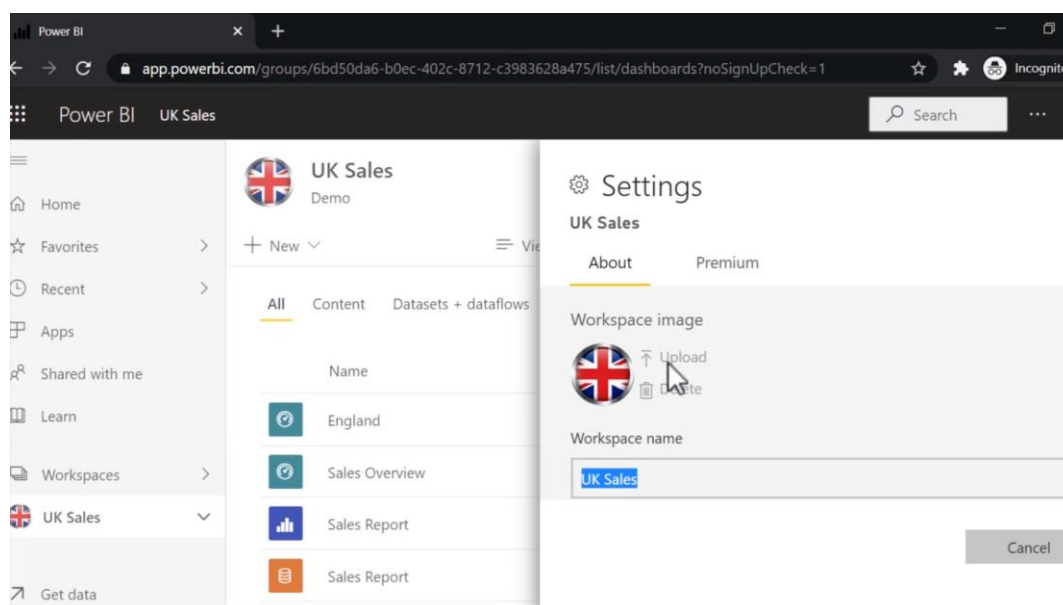


0304: Workspace Creation



Once you have the necessary permissions to create a workspace, the process is straightforward. Start by selecting Workspaces, then click the Create a workspace button. In this example, we'll create a workspace called UK Sales.

As the number of workspaces grows, it can be helpful to differentiate them visually. One way to do this is by assigning an image. To add an image, simply click the Upload workspace image button. Keep in mind that the image will be displayed in a circular frame. While you don't need to upload a circular image, be aware that any image you select will be cropped into this shape.



Next, enter a detailed description for the workspace. This should explain its purpose, the type of content that will be stored, and who will have access to it. For this demonstration, we'll keep it simple and enter "Demo." Once that's done, click Save.

0304-workspace Creation

With the workspace created, you now have a dedicated location to upload and manage content. From here, you'll have access to various sharing options, making it easier to distribute reports and insights across your organisation.

0305: Assigning Workspace Roles

One of the key differences between My Workspace and an App Workspace in Power BI is that an App Workspace allows multiple users to collaborate, whereas My Workspace is accessible only to its owner. When granting access to an App Workspace, there are four distinct roles: Viewer, Contributor, Member, and Admin. Each role determines the level of control a user has over content and workspace settings.

 App Workspace Permissions				
	Admin	Member	Contributor	Viewer
View content	Yes	Yes	Yes	Yes
Create, edit and delete content	Yes	Yes	Yes	No
Publish reports from Power BI Desktop	Yes	Yes	Yes	No
Publish/Update apps	Yes	Yes	No	No
Share individual reports and dashboards	Yes	Yes	No	No
Add members and contributors	Yes	Yes	No	No
Update and delete workspace	Yes	No	No	No
Delete users, members and other admins	Yes	No	No	No

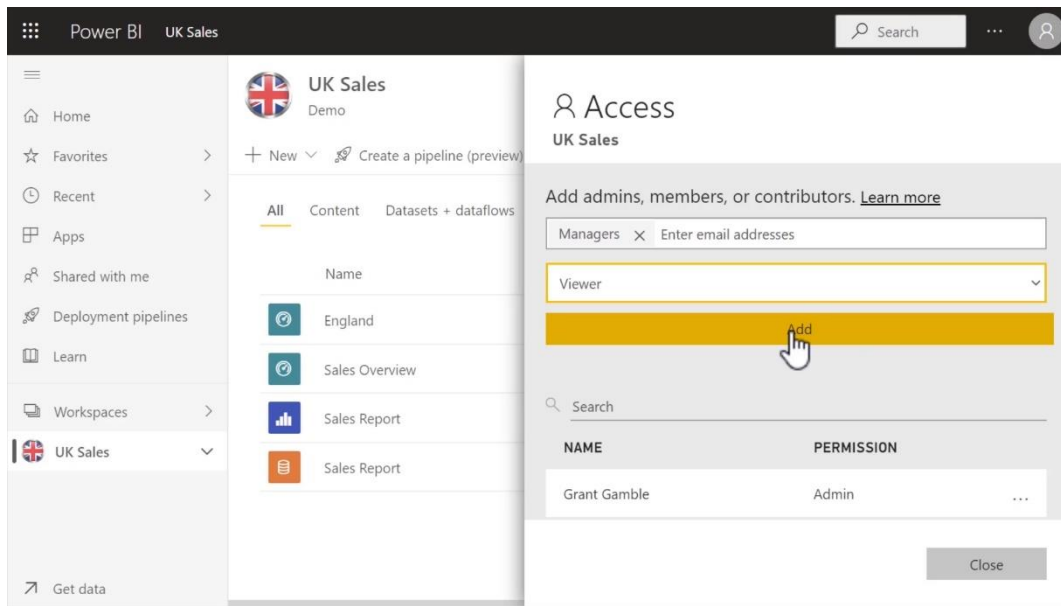
Viewer Role

The Viewer role is the most restricted. As the name suggests, viewers can only examine content—they cannot edit reports, dashboards, or semantic models. This role is typically assigned to managers, supervisors, or quality inspectors who need to review data but do not require editing privileges.

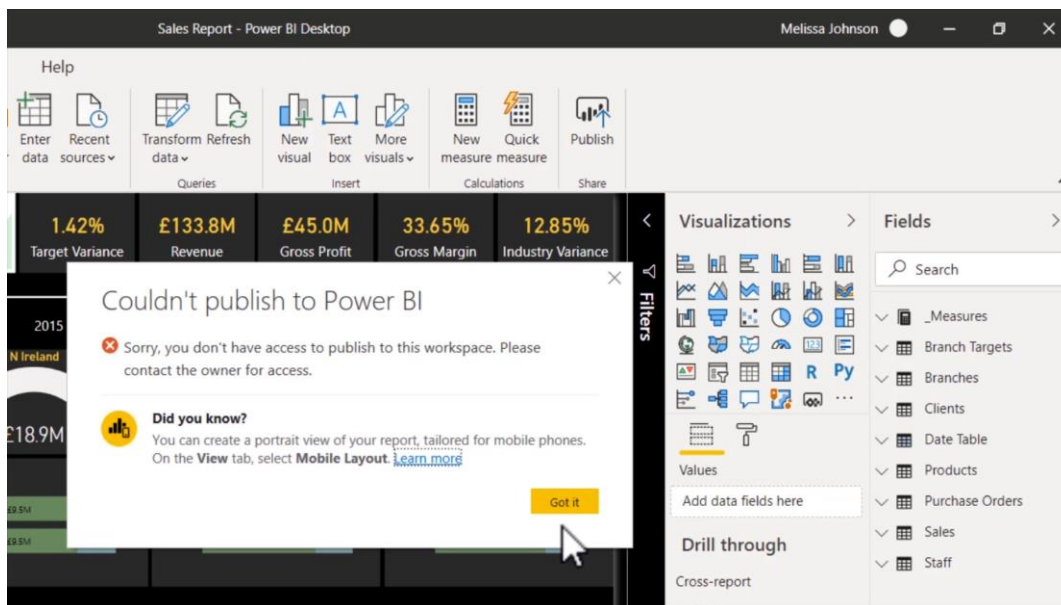
To assign a user the Viewer role:

- In the Power BI Service, go to Workspaces.
- Next to the relevant workspace, click the ellipsis (...) menu and select Workspace Access.
- Add an individual user or, preferably, a security group (e.g., "Managers") to simplify user management.
- Assign the Viewer role and click Add.

0305-assigning Workspace Roles



Once added, a user in the Viewer role can see the workspace and browse all available content but cannot edit reports, modify semantic models, or publish content to the workspace. If they attempt to publish from Power BI Desktop, they will receive an error message, indicating that they lack the necessary permissions.



Contributor Role

If a user needs the ability to create, edit, and delete content, they require Contributor access. Contributors can:

- Modify reports and dashboards.
- Create new content within the workspace.
- Delete reports, dashboards, and semantic models.

However, contributors cannot share reports or publish the workspace as an app. This role is ideal for report developers and content creators who manage reporting content but do not handle user permissions.

To assign the Contributor role:

Open Workspace Access for the relevant workspace.

Add a security group (e.g., "Report Developers").

Assign the Contributor role and click Add.

A contributor will now see all workspace content, including semantic models, and will have additional options when clicking the ellipsis (...) menu. Unlike viewers, contributors can delete semantic models but still lack the ability to share content.

Member Role

The Member role includes all Contributor permissions plus the ability to:

Share reports and dashboards.

Publish the workspace as an app.

Add and manage other users (except Admins).

Members are often team leads or senior analysts who need to distribute reports across the organization. They can add viewers, contributors, and other members, but they cannot assign Admin roles or modify workspace settings.

To assign the Member role:

Open Workspace Access for the workspace.

Add a security group (e.g., "Sales & Marketing").

Assign the Member role and click Add.

Once assigned, a member will see the Create App button, allowing them to publish reports as an app for wider distribution. Additionally, when they hover over reports, the Share button appears, enabling them to distribute reports and dashboards.

Admin Role

The Admin role grants full control over the workspace. Admins can:

- Manage all workspace content.
- Share and publish content.
- Add or remove any user, including other Admins.
- Modify workspace settings and, if necessary, delete the entire workspace.

Because of these powerful privileges, Admin roles are typically reserved for IT staff or Power BI administrators.

To assign the Admin role:

- Open Workspace Access.
- Add a security group (e.g., "Workspace Administrators").
- Assign the Admin role and click Add.

An Admin user will have the same level of control as the original workspace creator. This includes the ability to remove other admins, potentially including the person who originally created the workspace.

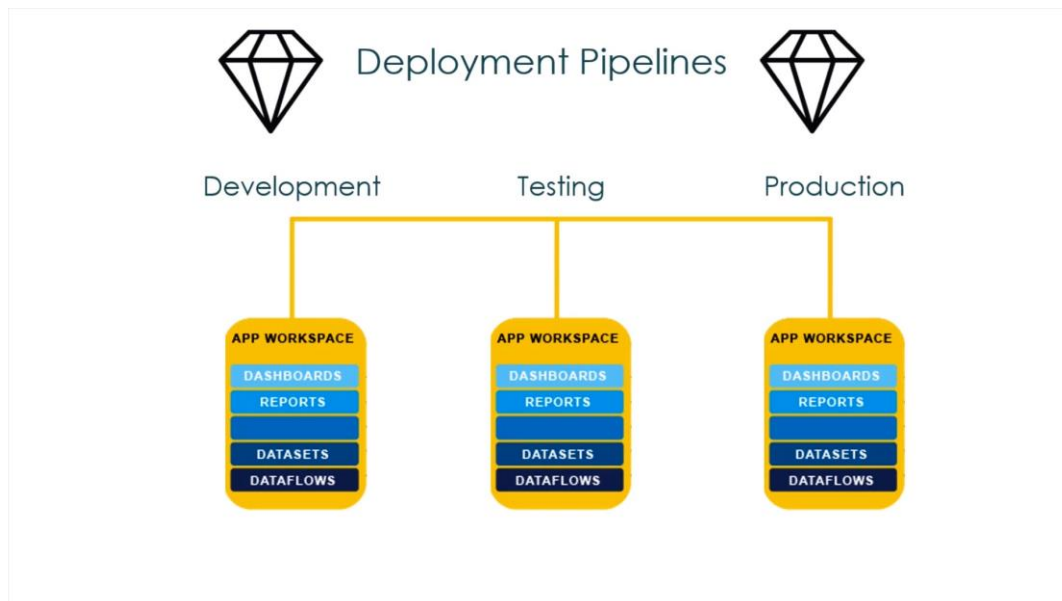
Summary of Workspace Roles

Power BI workspaces use four levels of permissions:

- Viewer – Can only view content.
- Contributor – Can edit and delete content but cannot share or publish.
- Member – Can share and publish but cannot modify workspace settings.
- Admin – Has full control, including user management and workspace deletion.

Assigning the right role ensures that users have appropriate access without unnecessary privileges. For efficient user management, security groups are preferable to individual user assignments, helping to simplify administration as team structures change.

0306: Using Deployment Pipelines



The standard process for creating and sharing content within the Power BI service involves developing reports in Power BI Desktop, publishing them to a workspace, and then making that workspace available as an app for users. Deployment pipelines extend this approach, introducing a structured, multi-tiered workflow that helps manage content across different environments—development, testing, and production—each corresponding to a separate workspace.

While it is possible to manually implement a similar process, deployment pipelines streamline the workflow, reducing inconsistencies and potential errors. However, to use this feature, you must meet specific requirements:

- A Power BI Pro licence

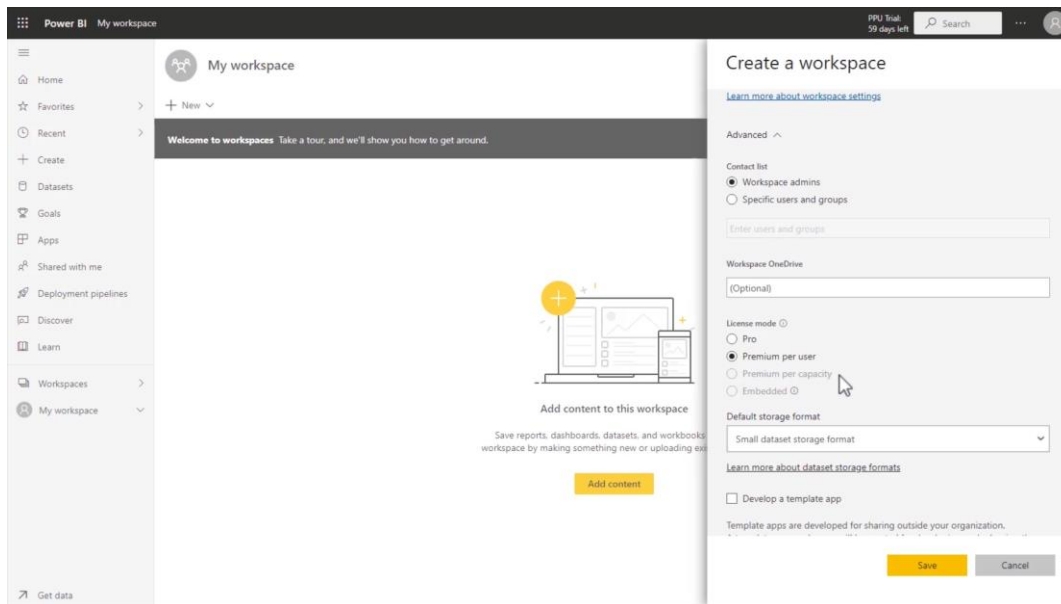
- Workspace admin rights

The workspace must be assigned to dedicated capacity, either via Power BI Premium, Power BI Premium Per User, or Power BI Embedded

Creating the Development Workspace

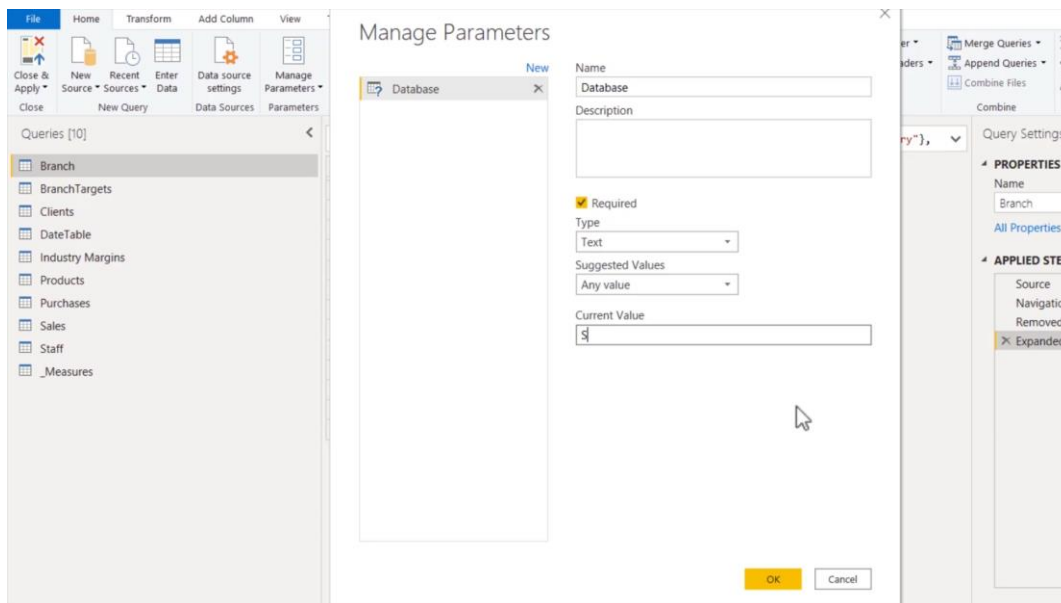
The first step in setting up a deployment pipeline is creating a workspace within the Power BI service. Navigate to Workspaces and select Create a workspace. This initial workspace will serve as the development stage. For example, if working with sales data, name the workspace Sales Dev.

Next, expand the Advanced section and assign the workspace to Premium Per User capacity, ensuring it meets the requirements for deployment pipelines. Once set, click Save.



Preparing Content with Parameters

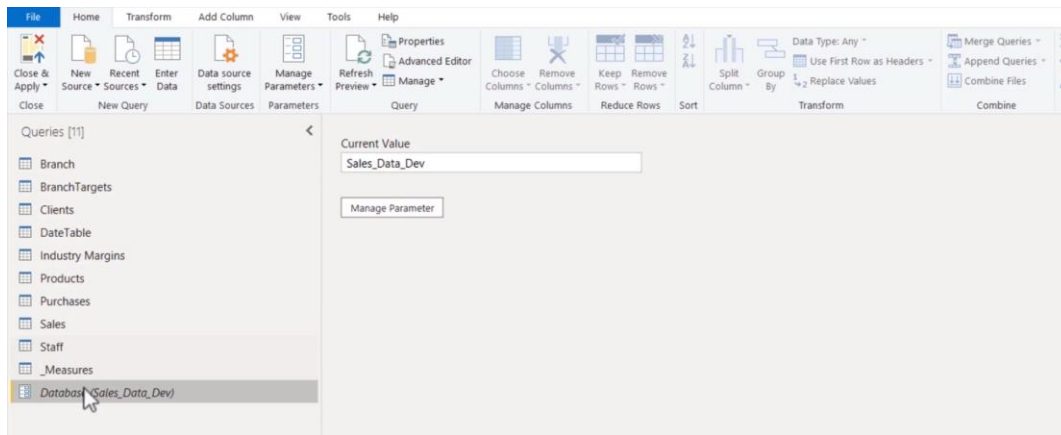
Before uploading content, it's useful to introduce parameters, which enable easy switching between data sources at different stages in the pipeline. This flexibility is particularly valuable when working with multiple environments.



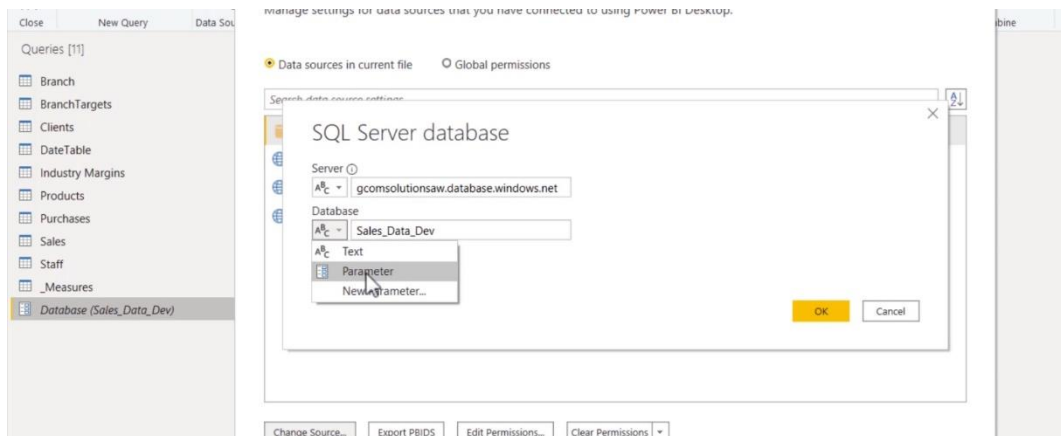
To create a parameter, go to Transform Data, then Manage Parameters > New Parameter. Name the parameter, e.g., Database, and set its type to Text—this is crucial, as leaving it as Any prevents modification when publishing to the Power BI service.

Enter an initial value for development, such as Sales_Data_Dev. In a real scenario, you might parameterize other aspects like the server name or file paths for CSV and Excel sources.

0306-using Deployment Pipelines

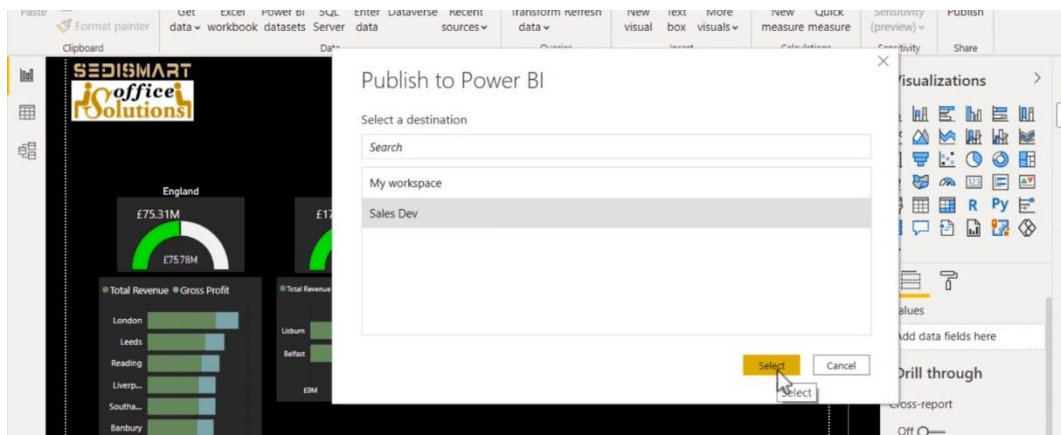


Once the parameter is created, it needs to be implemented. Open Data Source Settings, locate the database connection, and select Change Source. Instead of specifying the database as a text value, switch to the parameter, ensuring that all related tables will reference the appropriate database depending on the environment.



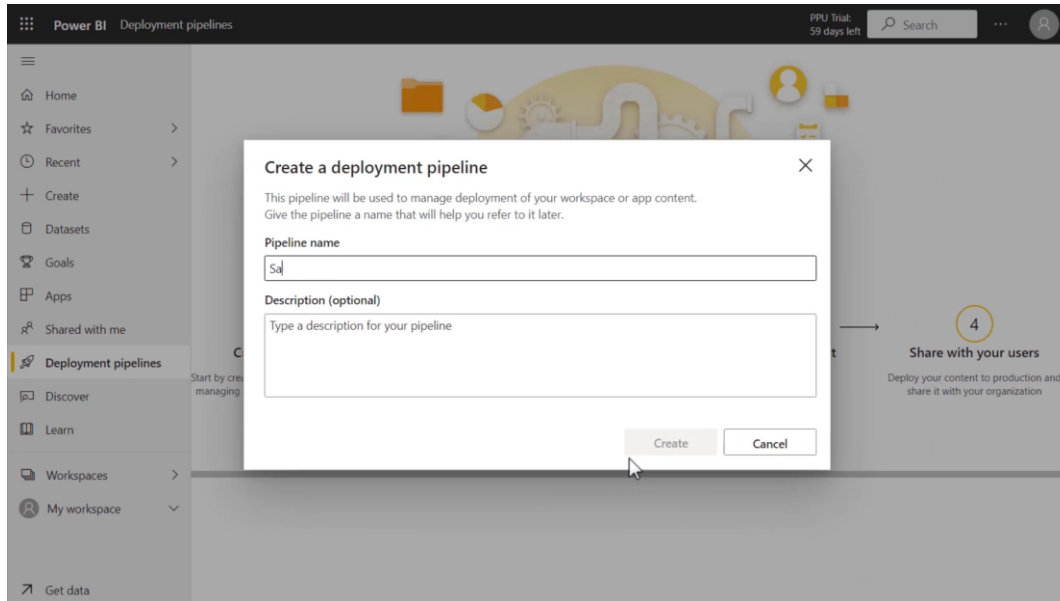
Publishing to the Power BI Service

With the parameter in place, click Close & Apply, then publish the report to the Power BI service. You will be prompted to save your work. Choose the development workspace (Sales Dev) and confirm the upload.

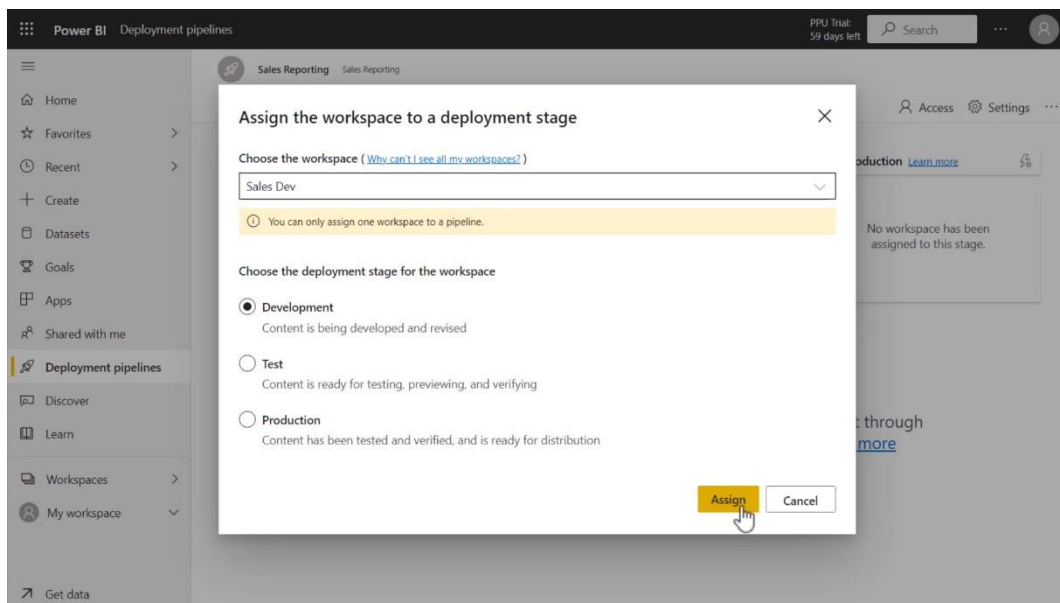


Creating and Assigning a Deployment Pipeline

Within the Power BI service, navigate to Deployment Pipelines and select Create a Pipeline. Name the pipeline appropriately, such as Sales Reporting, and provide a clear description outlining its purpose. Click Create to generate the pipeline.

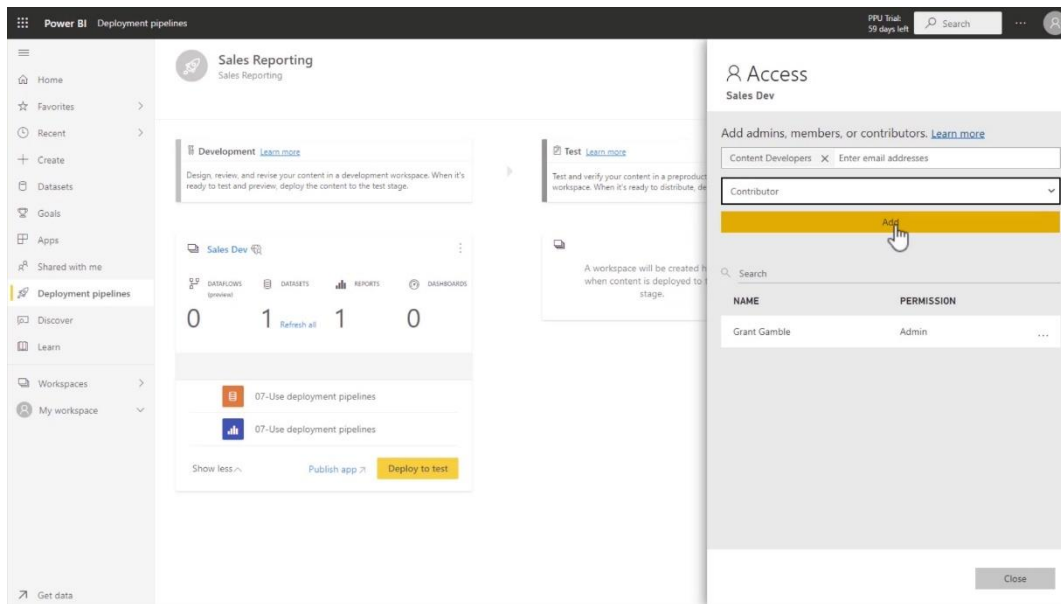


The pipeline consists of three stages—Development, Test, and Production. To begin, assign the Sales Dev workspace to the Development stage by clicking Assign a Workspace and selecting the appropriate workspace.



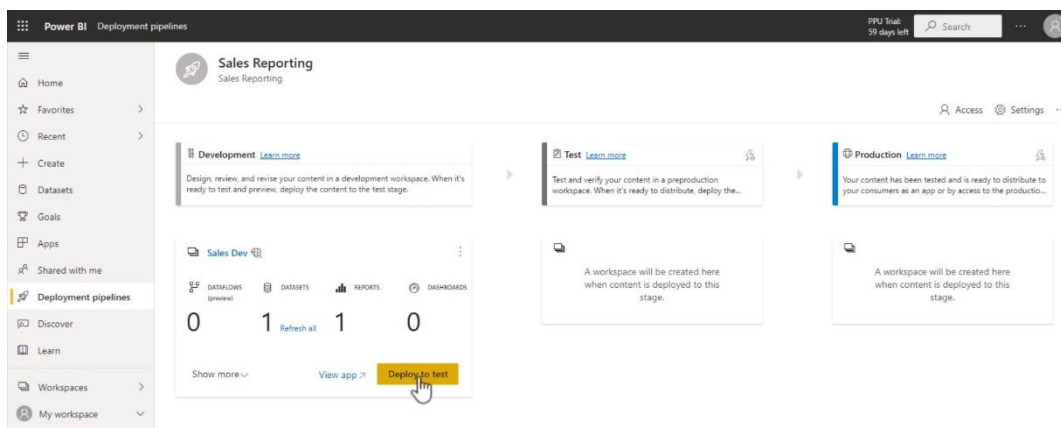
At this stage, you can review the workspace content and make adjustments. You can also manage workspace access, either by adding users via the workspace settings or directly through the deployment pipeline menu. For example, Power BI report developers might be assigned as Contributors.

0306-using Deployment Pipelines

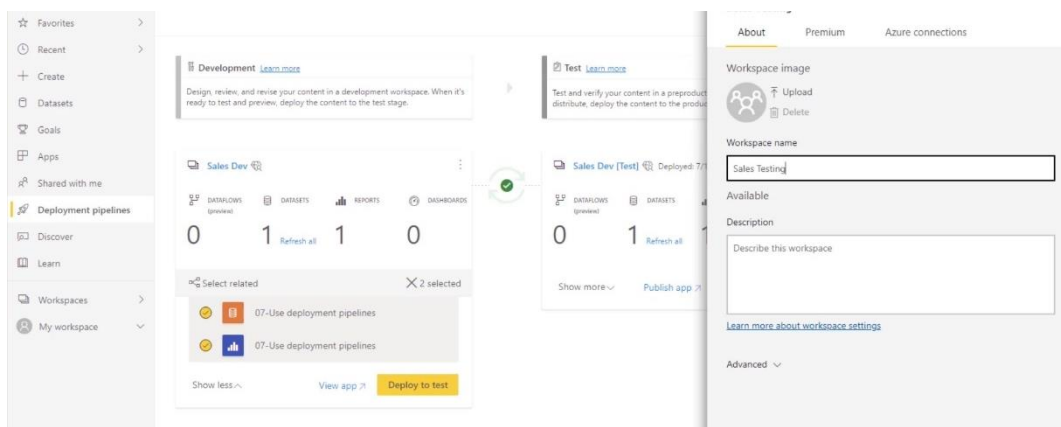


Deploying to the Testing Stage

Once the development phase is complete, deploy the content to the Test stage by clicking Deploy to Test. Power BI will automatically create a new workspace for testing, appending [Test] to the development workspace name.

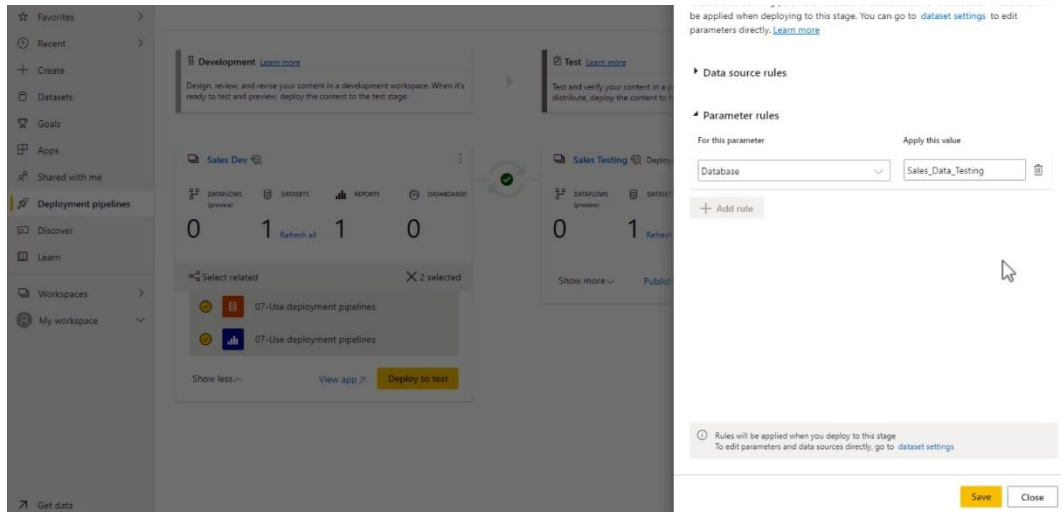


Since this default naming may not be ideal, update it in Workspace Settings, renaming it to something more meaningful, such as Sales Testing.

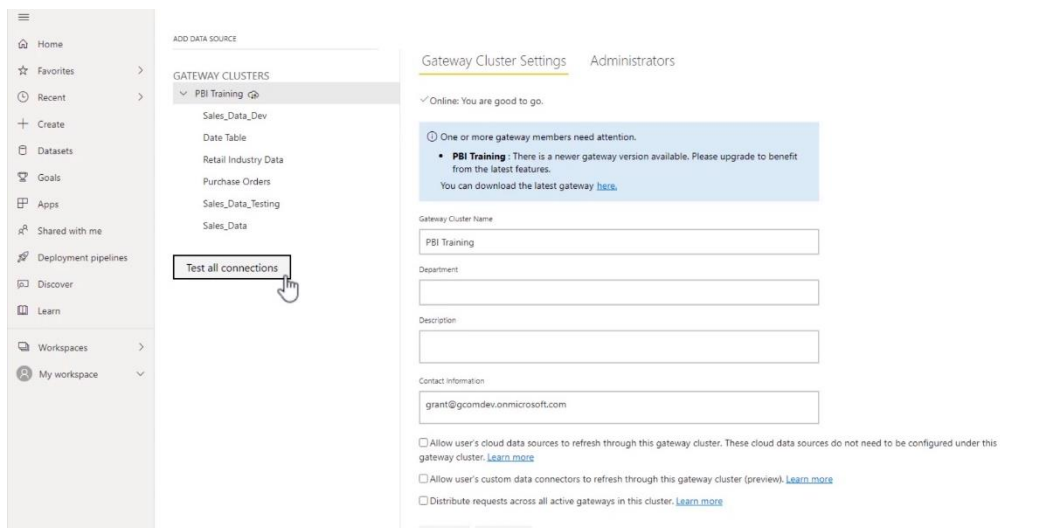


Configuring Deployment Settings

Next, update the parameter values to reflect the testing environment. Open Deployment Settings, select the semantic model, and under Parameter Rules, click Add Rule. Choose the Database parameter and change its value to Sales_Data_Testing. Save the changes and close the settings.

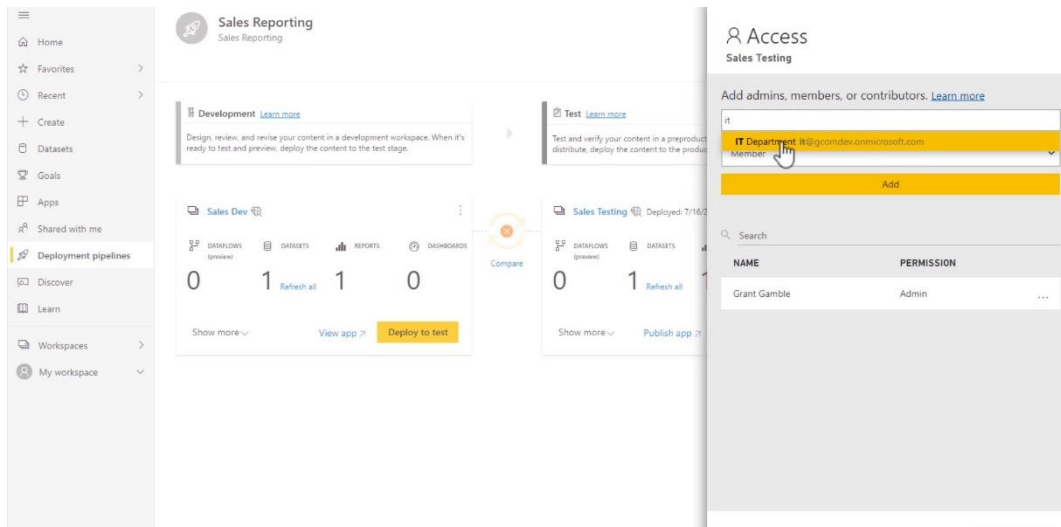


If using a data gateway, ensure that all necessary data sources have been added to the gateway cluster under Manage Gateways.



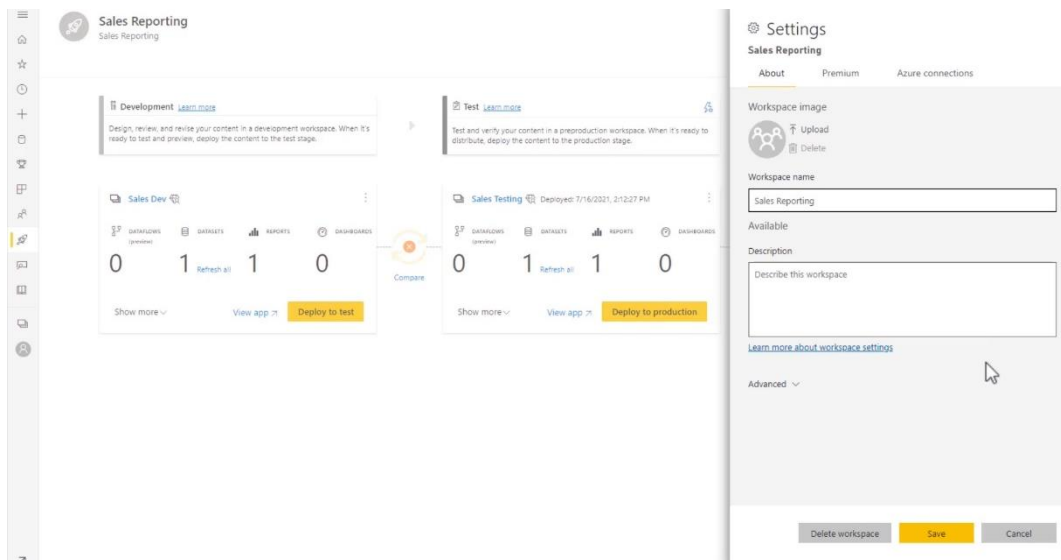
Access management is also crucial at this stage. For example, IT administrators might need Admin access to the workspace, while testers can be granted access via the Published App settings. Assign the Power BI Testing group to the test workspace and publish the app for them to review.

0306-using Deployment Pipelines



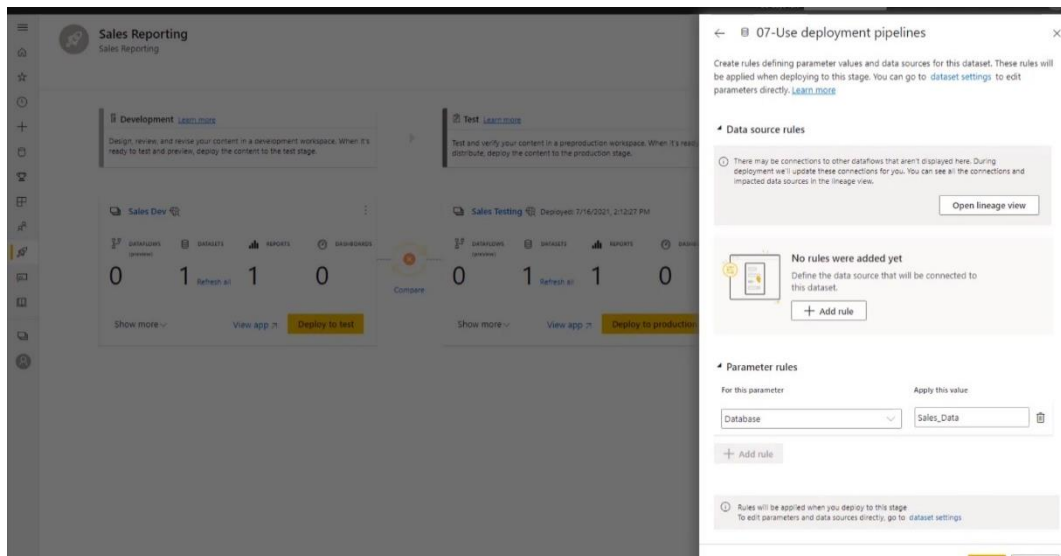
Deploying to Production

After thorough testing, deploy the content to the Production stage by selecting Deploy to Production. This process creates another new workspace. Rename it in Workspace Settings, e.g., Sales Reporting, to ensure consistency.

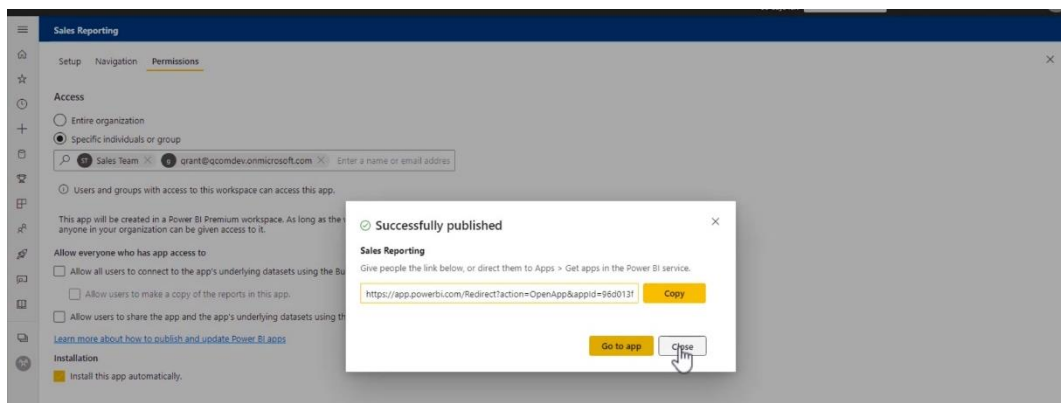


Update the parameter settings once again by modifying the Database parameter to reference the live production database (Sales_Data). Then, refresh the semantic model to pull in production data.

0306-using Deployment Pipelines



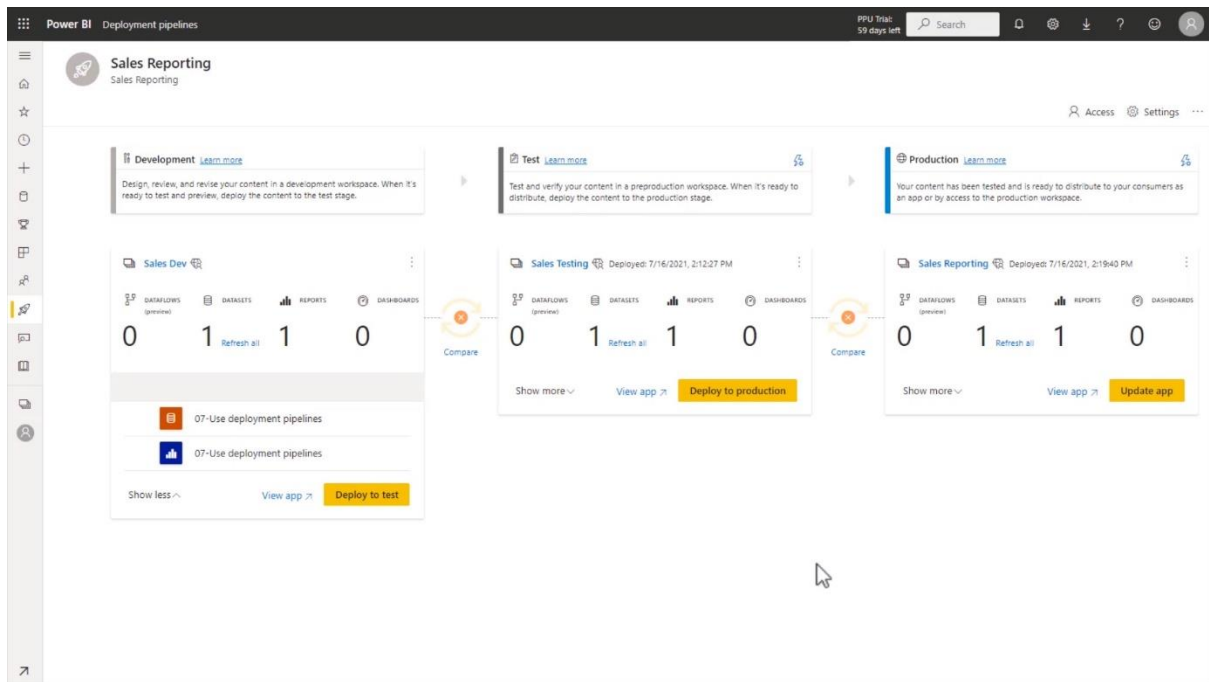
Finally, publish the app for end users. In the Publish App settings, assign the appropriate user group, such as Sales Team, ensuring they have access to the final reports.



Ongoing Content Updates

Deployment pipelines are not a one-time operation. As content evolves, new reports and updates can be added to the Development workspace, tested, and then deployed through the pipeline.

0306-using Deployment Pipelines



This structured approach ensures smooth transitions between environments, maintaining consistency and control over report distribution.

0400: Data Governance Overview

Data governance encompasses a range of themes related to the effective management of data assets. It involves the technologies and processes that ensure data within an organisation is trustworthy, accurate, complete, and accessible to those who need it. While data governance extends far beyond Power BI, this section focuses specifically on its role within the Power BI environment.

The Purpose of Data Governance in Power BI

The primary aim of data governance in Power BI is to help administrators and data stewards prevent the proliferation of multiple versions of the same data. Instead, the goal is to establish and maintain a single, authoritative version of the truth.



A key administrative setting in this regard is disabling the option to export data. When data is exported as an Excel or CSV file, it immediately falls out of sync with the original semantic model and will not be refreshed.

To mitigate this issue, Power BI provides the Analyse in Excel feature, which enables live connections to Power BI semantic models from Microsoft Excel. Another crucial aspect of governance is the creation of centralised shared semantic models, allowing multiple reports across different workspaces and apps to access the same underlying data. This ensures consistency and prevents data duplication.

Addressing Data Duplication in Power BI

Power BI is fundamentally a self-service BI tool, which can lead to the creation of data silos and duplicated semantic models. Since Power BI reports are built in Power BI Desktop and saved as .PBIX files, each report contains its own semantic model. If multiple reports draw from the same data sources independently, variations can emerge, making it difficult to keep semantic models aligned as they are updated.

The use of shared semantic models resolves this issue by creating a single, authoritative semantic model that all report developers reference. In this setup, each .PBIX file contains only the report layout and visualisations, while the semantic model itself remains centralised. This approach ensures that only one semantic model needs to be refreshed and updated.

Using PBIDS Files to Standardise Data Connections

Another method of enforcing consistency is through the use of .PBIDS files. These are simple text files that encapsulate connection information to data sources. By maintaining a library of standardised .PBIDS files, organisations can ensure that all report developers are connecting to the same databases, files, and SharePoint lists, reinforcing uniformity across reports.

Enforcing Data Governance with Dataflows

Dataflows are a key governance feature within Power BI that offer three primary benefits:

1. **Centralised Data Transformation** – Dataflows allow organisations to centralise data connections and transformations using Power Query Online within the Power BI service. This ensures that transformation logic is stored in a single, centralised location and made available to multiple report developers.
2. **Enhanced Security** – When dataflows are used, connections to original data sources become anonymised. Instead of accessing raw data sources directly, report developers connect to curated tables within dataflows, adding an extra layer of security.
3. **Optimised Data Refresh Management** – Semantic models often contain connections to multiple data sources with differing refresh schedules. Some sources may require daily refreshes, while others might only need weekly updates. Without dataflows, these conflicting requirements often lead to unnecessary refresh operations. Dataflows allow related data sources to be grouped according to their refresh needs, ensuring efficient and targeted refresh processes.

Managing Data Refresh with the On-Premises Data Gateway

Data refresh in Power BI is facilitated through the On-Premises Data Gateway, which serves as a bridge for securely transferring data between on-premises sources and Power BI semantic models or dataflows in the cloud. In this section, you will learn how to:

Set up the on-premises data gateway

Configure refresh schedules

Optimise refresh performance using incremental refresh

Semantic model Certification for Trustworthy Data

Another crucial aspect of data governance is semantic model certification. Power BI allows administrators to assign specific data specialists the ability to certify semantic models, ensuring they meet quality standards before being used across the organisation. However, to maintain integrity, this capability should be granted only to selected security groups rather than the entire organisation.

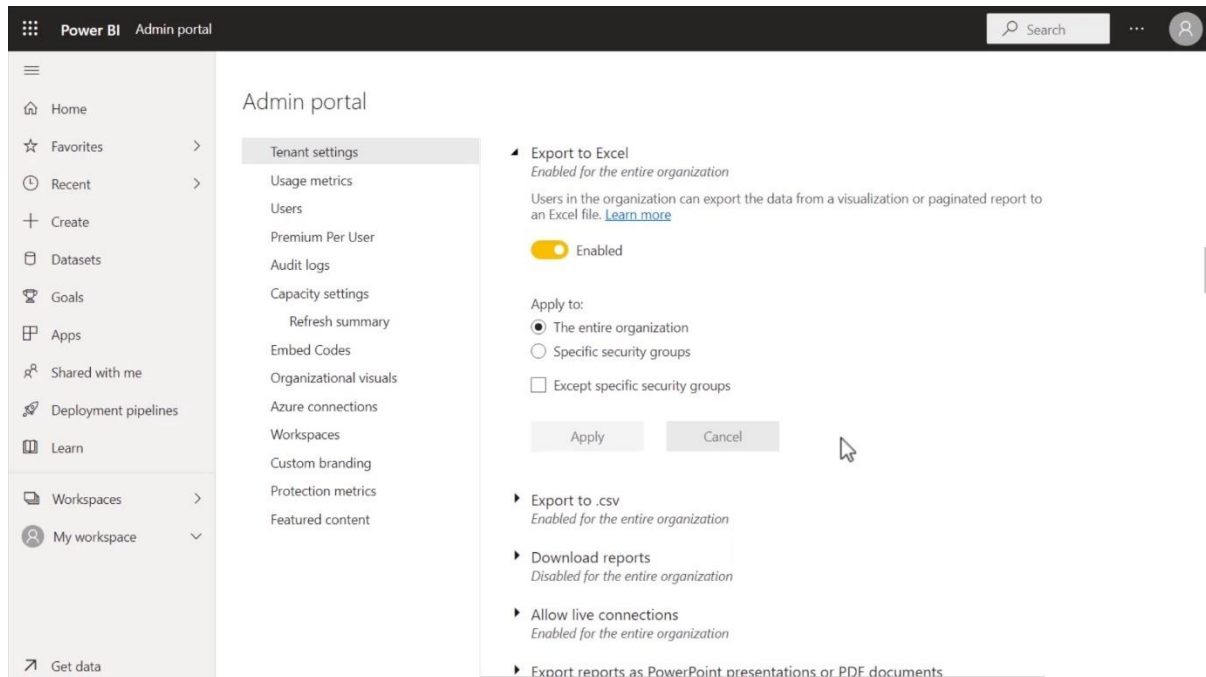
Beyond certification itself, administrators must also ensure that certified semantic models are easily discoverable. In the Admin Portal, under Discovery Settings, the Make certified content discoverable option should be enabled. This helps users identify and rely on certified semantic models rather than creating their own, reducing unnecessary duplication and reinforcing governance standards.

Summary

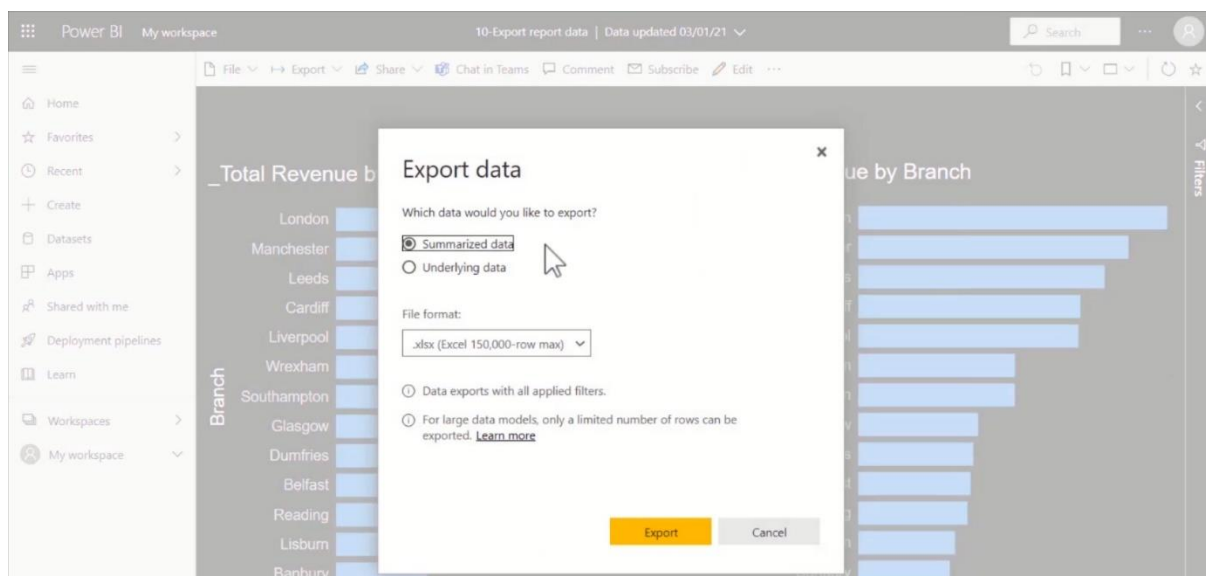
Effective data governance within Power BI focuses on centralisation, consistency, and security. By leveraging shared semantic models, PBIDS files, dataflows, and certification mechanisms, organisations can prevent data duplication, enhance trust in data, and streamline data refresh processes. These governance practices ensure that Power BI remains a powerful and reliable tool for business intelligence.

0401: Live Data Connections versus Exporting

Power BI provides options for exporting data from a visual, including Export to Excel and Export to CSV.



These features allow users to extract either the visible data from a visual or the underlying semantic model.



However, enabling these options is generally not recommended. The main concern is that exported data quickly becomes outdated as the live semantic model continues to evolve.

A more effective alternative is the Analyse in Excel feature, which ensures that users are always working with live data. Unlike static exports, this approach maintains a direct connection to the semantic model in the Power BI service, preventing users from developing separate versions of the data that may become misaligned with the authoritative source.

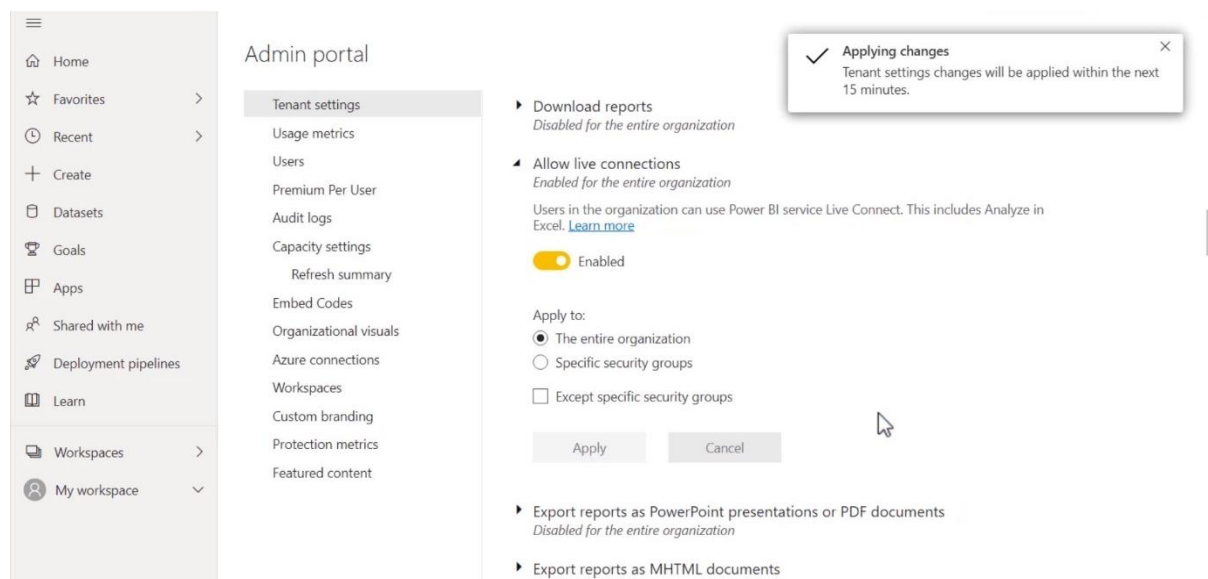
The Advantage of Live Connections

When granting access to Power BI semantic models, the live connection feature is the preferred approach. Live connections provide external clients with read-only access to semantic models stored in the Power BI service. This eliminates the need to export data and ensures that all users are always viewing the same, up-to-date version of the semantic model.

Live connections also play a crucial role in data centralization. By leveraging shared semantic models, organizations can enforce a single version of the truth, reducing discrepancies caused by multiple, manually maintained copies of the same data.

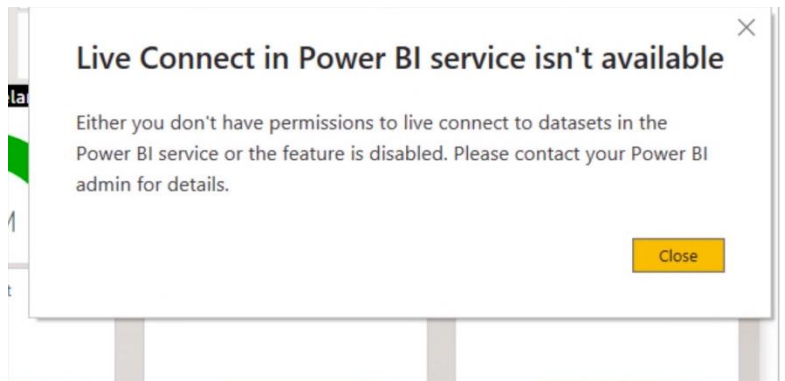
Enabling Live Connections and Analyse in Excel

To take full advantage of live connections, the Allow Live Connection setting should be activated in Power BI. This enables users to establish direct connections to shared semantic models and create multiple reports from the same data source, rather than relying on individual exports.



Additionally, enabling live connections activates the Analyse in Excel feature. This allows users to build PivotTables, PivotCharts, and dashboards powered by real-time data from a Power BI semantic model.

If the live connection feature is disabled, the Analyse in Excel option becomes unavailable (greyed out). Furthermore, in Power BI Desktop, attempting to use the Get Data > Power BI Semantic model command will result in an error message, indicating that live connections to the Power BI service are not available.



Summary

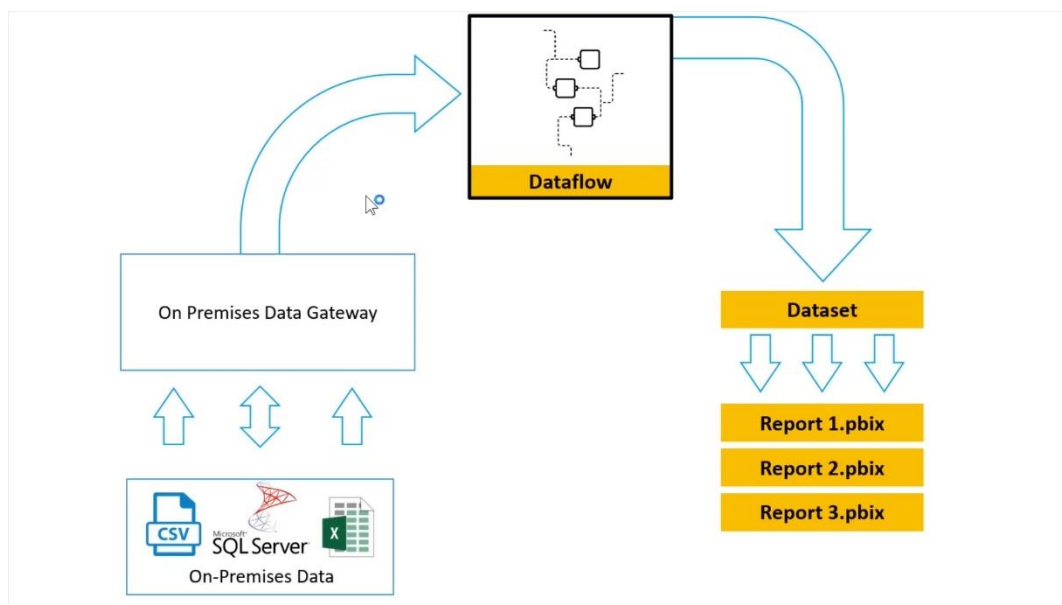
Rather than allowing users to export static Excel or CSV files, organizations should encourage the use of live connections and Analyse in Excel to ensure that all users work with the most up-to-date data. Live connections support data consistency, improve collaboration, and maintain the integrity of reports and dashboards within the Power BI environment.

0402: Centralizing Data Cleansing with Dataflows

Dataflows provide a centralized, cloud-based solution for managing data connections and transformations. Instead of repeating transformations across multiple reports, organizations can create a single dataflow in the cloud, where all transformation logic is stored in one place. This ensures consistency and allows different teams to use a unified data source.

The Role of Shared Semantic models

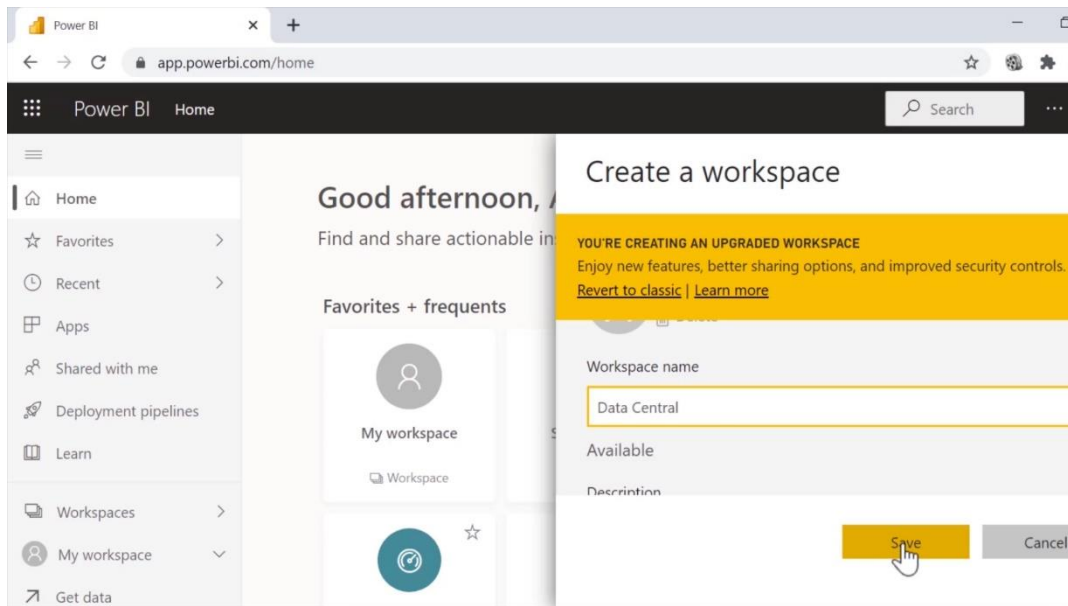
In addition to centralizing dataflows, another key aspect of data centralization is shared semantic models. These semantic models are built from one or more dataflows and serve as a common foundation for reporting. When constructing a semantic model, users can model the data, define relationships, and add calculated measures—without yet creating a report. The goal is to develop a robust data model and then publish it as a shared semantic model within a workspace. This approach prevents duplication and ensures that reports across the organization rely on a single, consistent version of the data.



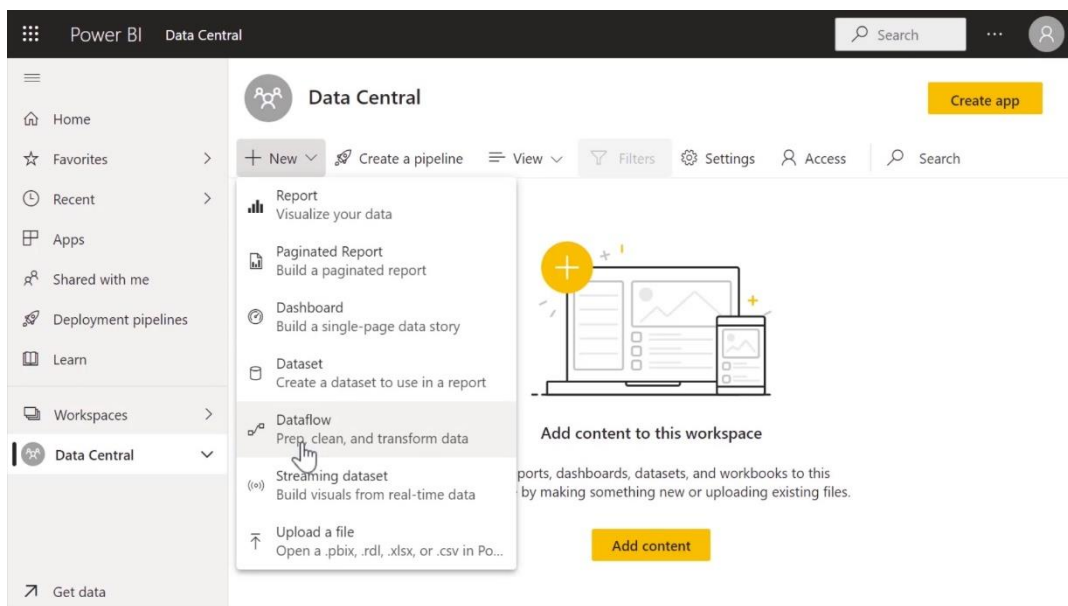
Setting Up a Dataflow in Power BI Service

The process of creating a dataflow begins in the Power BI service. First, a workspace needs to be created to store both dataflows and shared semantic models. In this example, we create a workspace named "Data Central" and then proceed to create a new dataflow within it.

0402-centralizing Data Cleansing With Dataflows

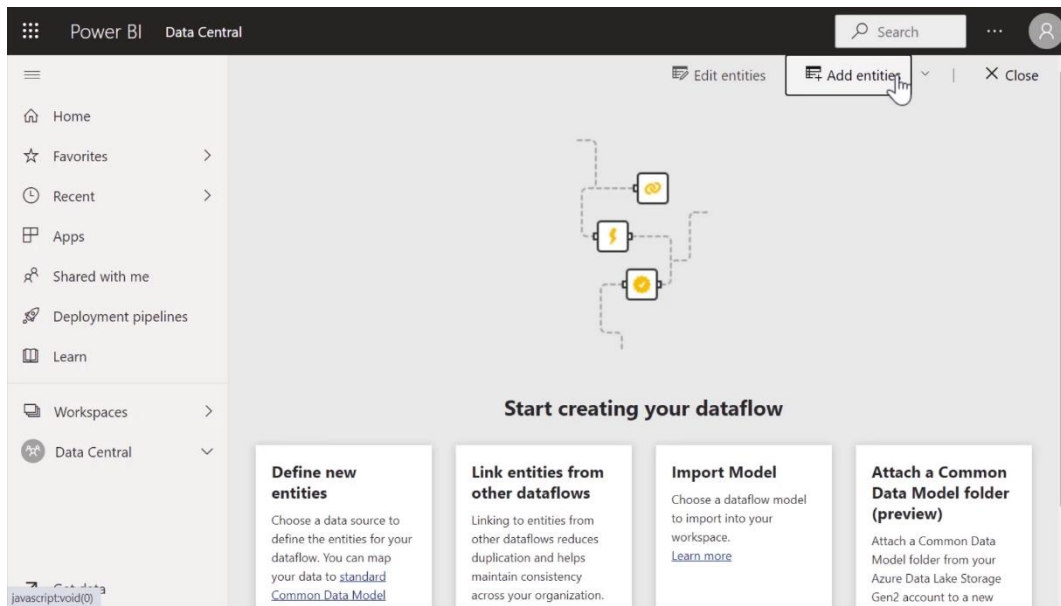


To create a dataflow, we navigate to the workspace, click New, and select Dataflow.



This generates a blank dataflow where we can define queries. In Power BI, these queries are called entities. To create an entity, we use the Add entities button, which provides access to the same data sources available in Power BI Desktop.

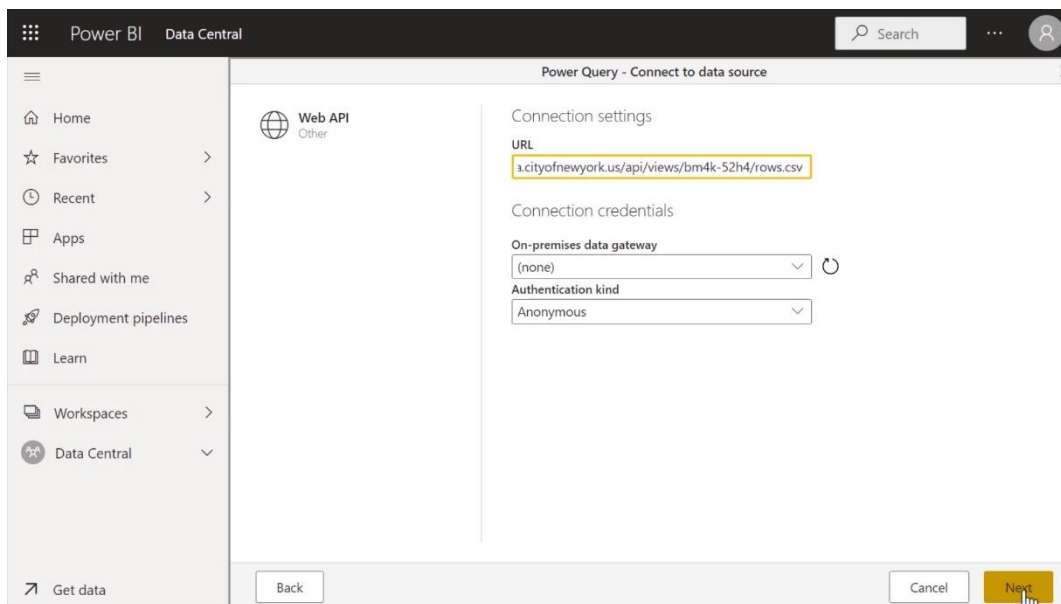
0402-centralizing Data Cleansing With Dataflows



Connecting to Data Sources

Since this tutorial focuses on web-based sources, we retrieve URLs from a shared semantic model folder and connect to them using the Web API connector. Although one of our data sources is a CSV file, we still use the Web API connector because the file is hosted online.

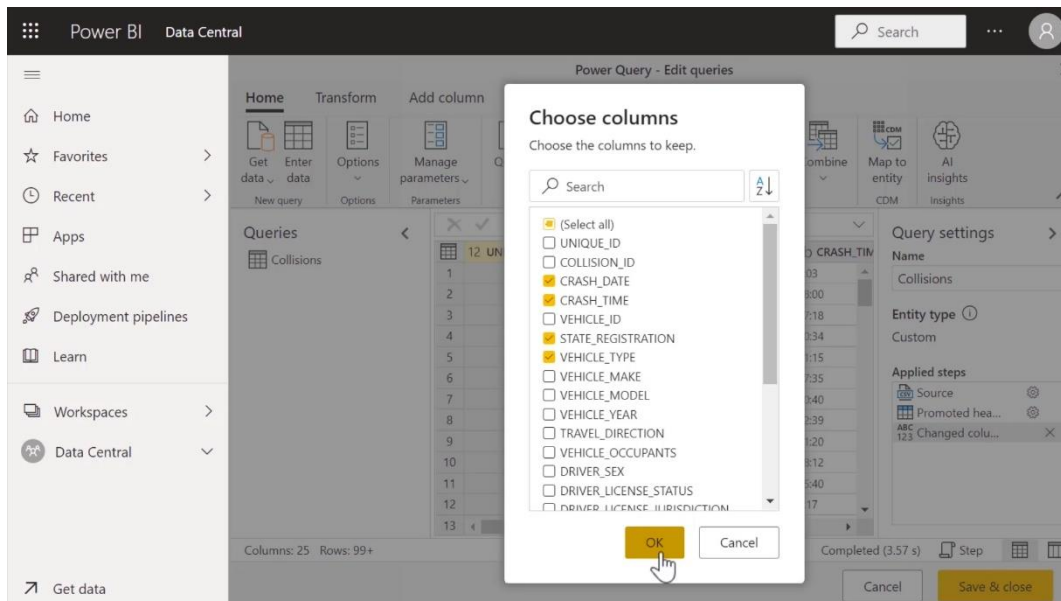
<https://data.cityofnewyork.us/api/views/bm4k-52h4/rows.csv>



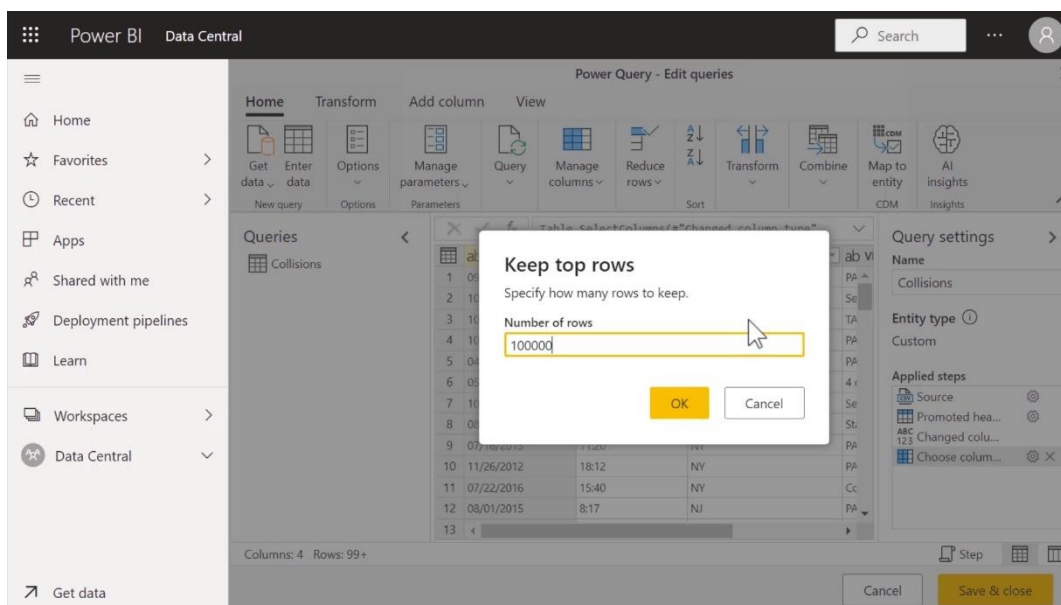
Authentication for these sources is set to Anonymous, meaning no credentials are required. However, note that loading data from an external source may take some time. Once the data is retrieved, we proceed with transformations.

Transforming Data in a Dataflow

The first semantic model contains accident data. We rename the query Collisions and refine it by selecting only the necessary columns: Crash Date, Crash Time, State Registration, and Vehicle Type.



Given the large size of the semantic model, we reduce the number of rows using the Keep Top Rows option, limiting it to the top 100,000 records.



Next, we adjust data types. For users outside the US, regional settings may affect date formats. To avoid issues, we go to Options > Project Options and change the locale to English (United States) before converting the Crash Date column to a date format. Finally, we convert Crash Time and verify the text columns.

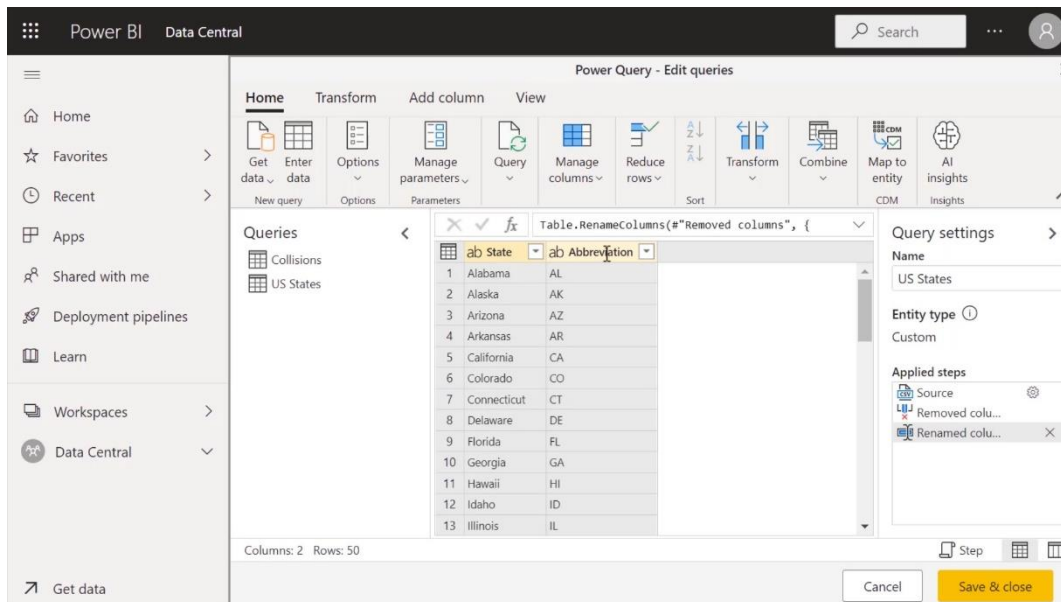
Adding a Second Data Source

0402-centralizing Data Cleansing With Dataflows

To add another data source, we use the Web API connector again, keeping authentication set to Anonymous.

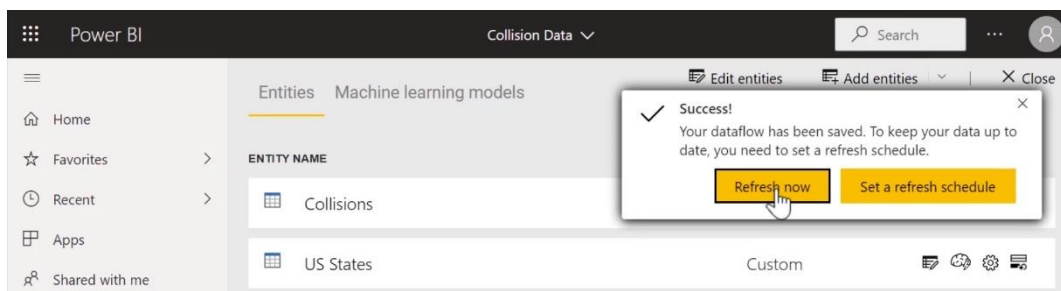
https://scottontechnology.com/wp-content/uploads/2014/08/50_us_states_all_data.csv

This dataset contains US state abbreviations. We rename the query US States, remove unnecessary columns, and rename the remaining two to State and Abbreviation.

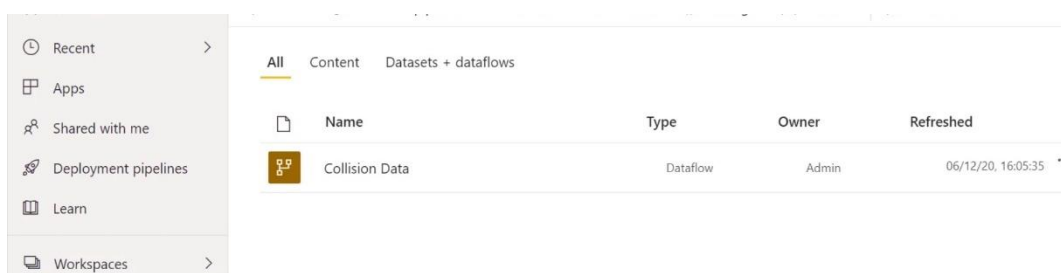


Finalizing and Refreshing the Dataflow

Once both entities are defined, we save and close the dataflow. After validation, we name the dataflow Collision Data and trigger a refresh. This ensures that data is available when connecting to the dataflow from Power BI Desktop.

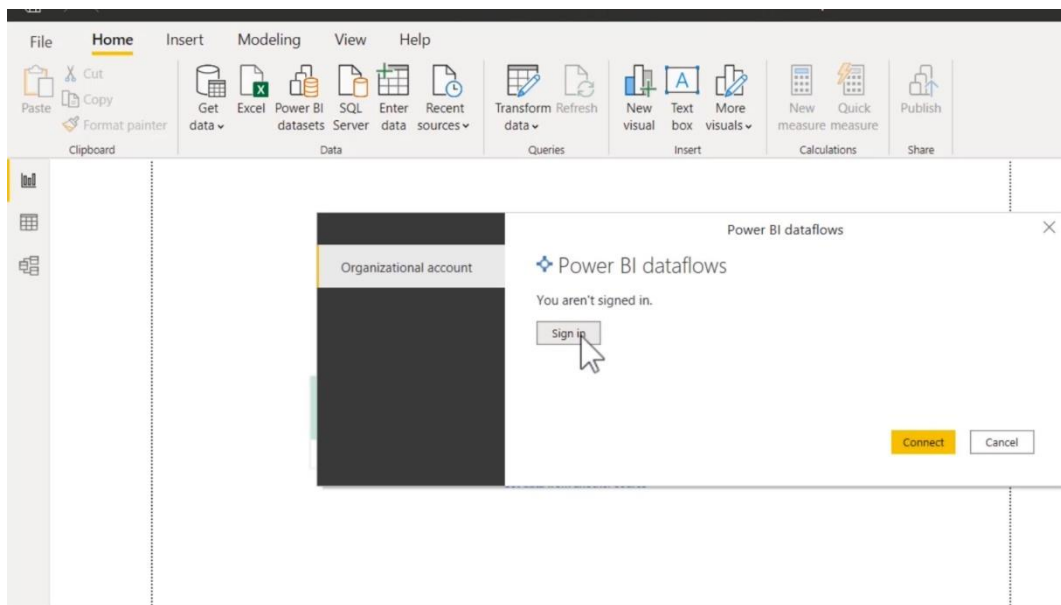


Returning to the Data Central workspace, we monitor the data load process. Once complete, the dataflow is ready for use in Power BI Desktop, where it can be connected to for further analysis and reporting.

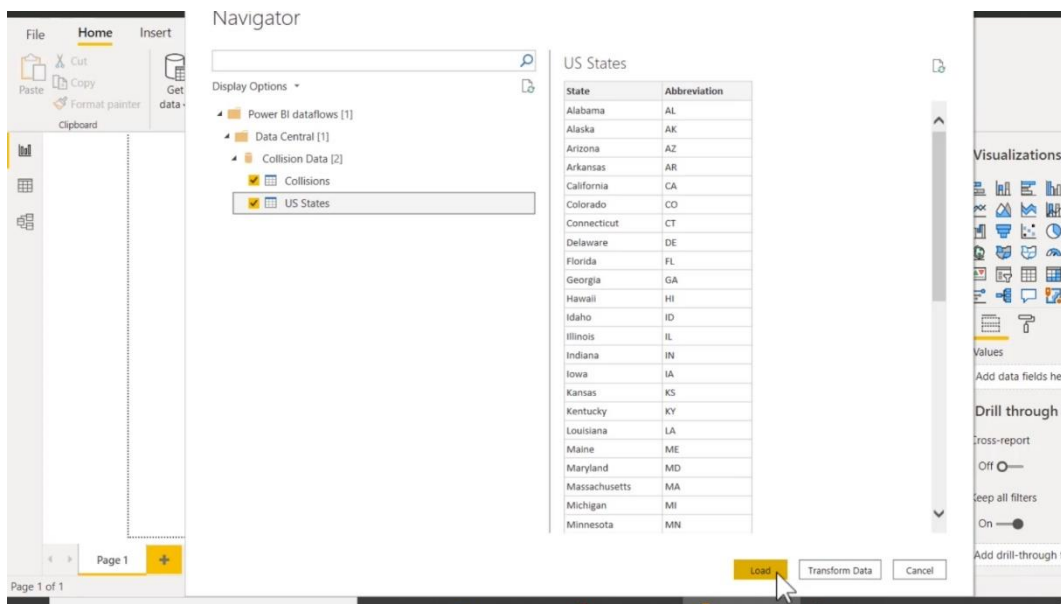


0403: Centralizing Data with Shared Semantic models

Now that we have created a data flow, the next step is to set up a shared semantic model based on it. To begin, go to Get Data and select Power BI Dataflows. You will need to sign in using your usual Power BI credentials. Once signed in, click Connect.



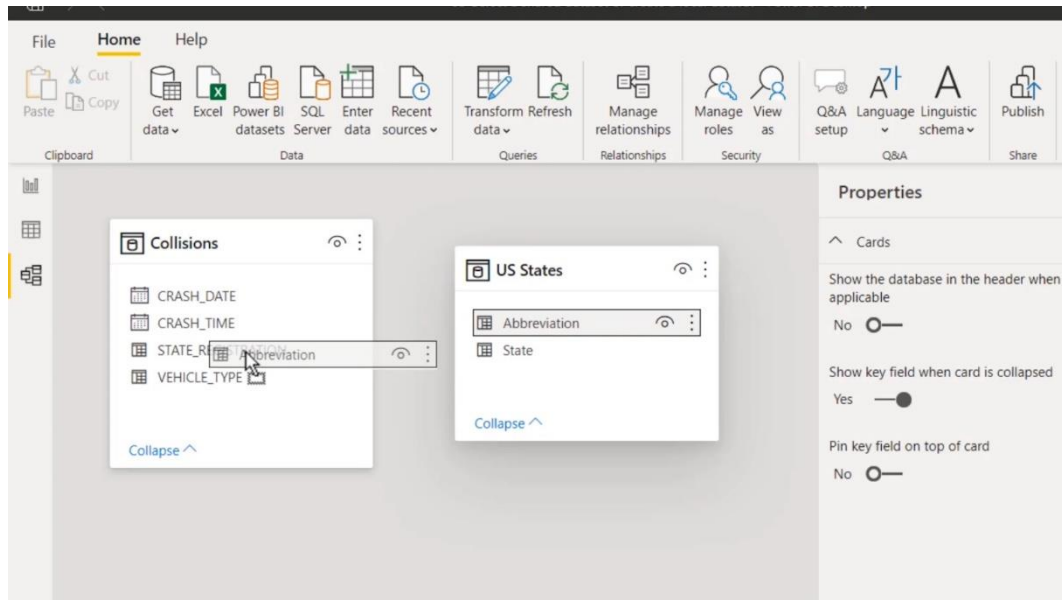
In the Data Central workspace, you'll see the newly created Collision Data dataflow, which contains two entities. Select both and click Load.



While it is possible to apply additional transformations at this stage, it is generally best practice to centralize all transformation logic within the dataflow itself rather than making further changes in the semantic model.

Creating Relationships and Measures

Once the data is loaded, switch to the Model View to establish relationships between tables. For example, you can create a relationship between US States[Abbreviation] and Collisions[State_Registration] to complete the semantic model structure.



To organize your measures, it is useful to create a dedicated table. Click Enter Data, name the table (a common convention is to call it `_Measures`), and create a single placeholder column. Once the table is created, right-click it and select New Measure.

As an example, let's create a measure to count the number of collisions. Use the `COUNTROWS` function on the Collisions table:

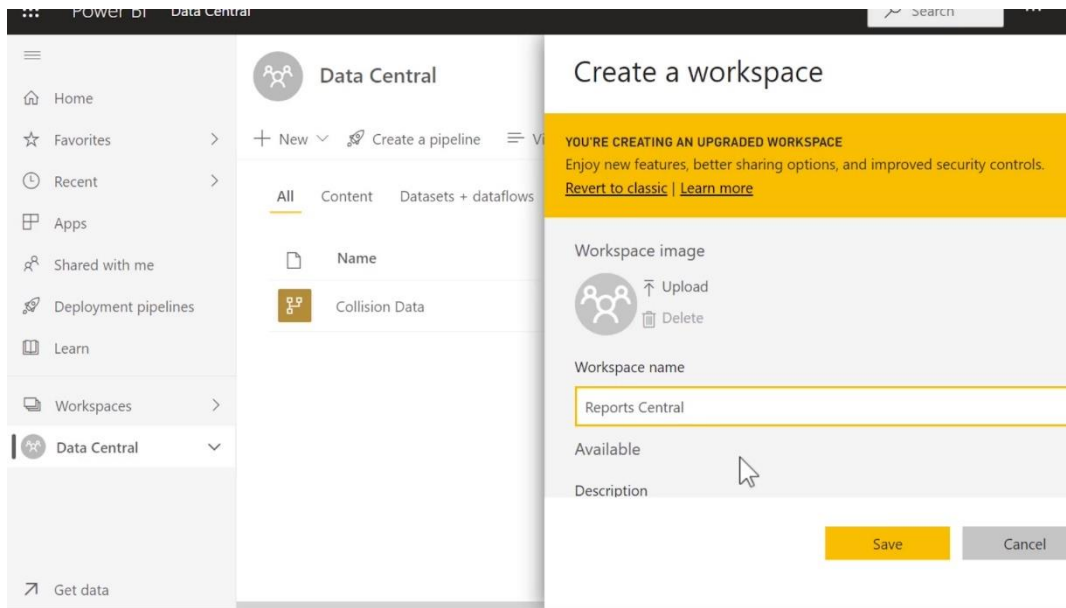
Number of Collisions = COUNTROWS(Collisions)

Once this measure is added, the placeholder column can be deleted. Refreshing the Fields pane will update the icon for the measures table.

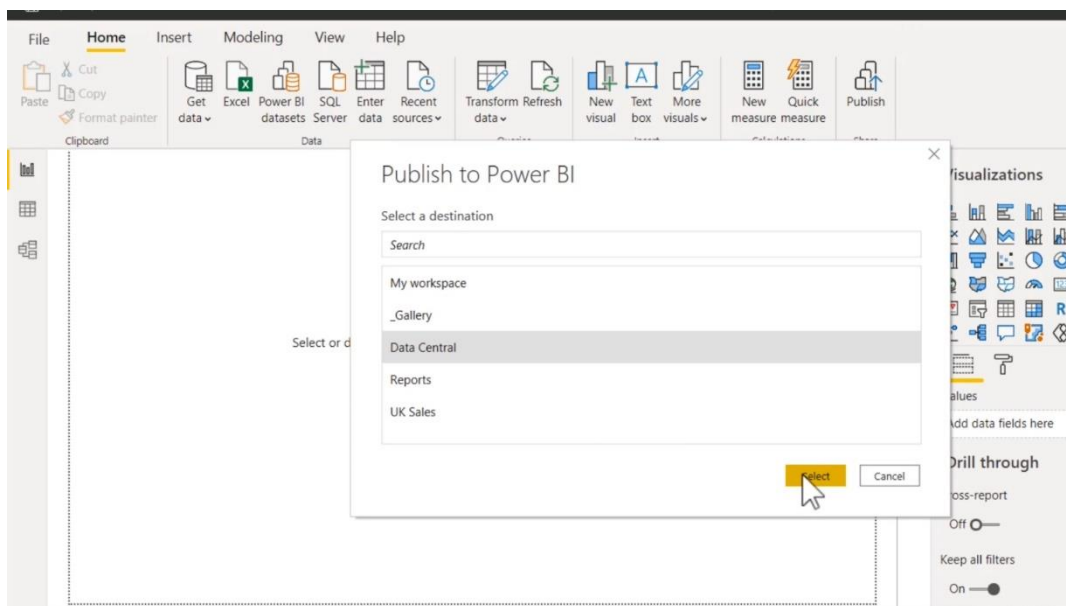
Publishing the Semantic model

To make this semantic model available for reporting, create a new workspace in Power BI Service.

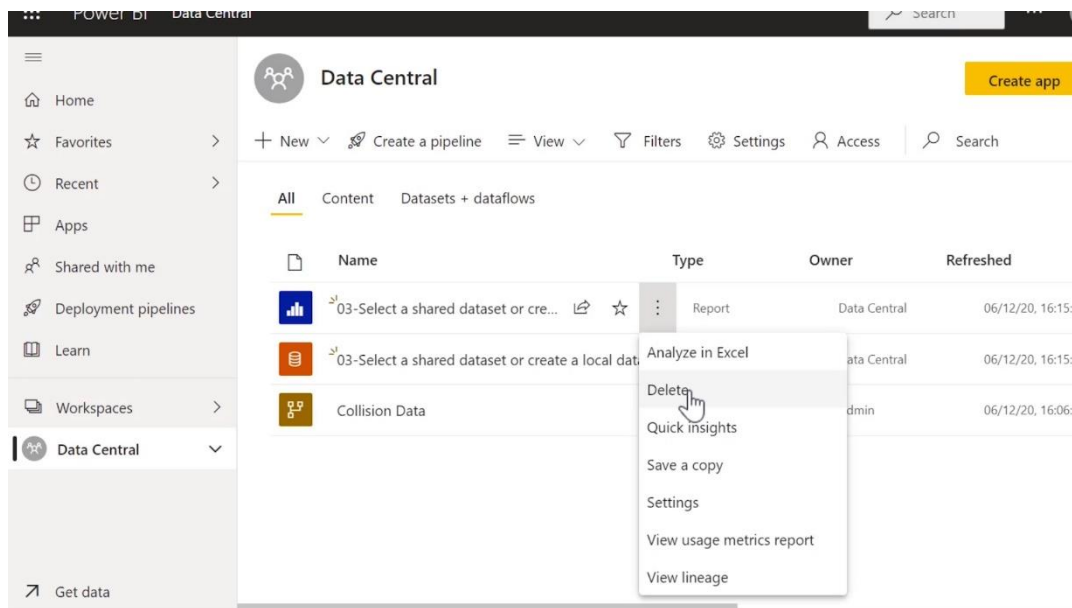
0403-centralizing Data With Shared Datasets



For clarity, let us now create a workspace for reports only (“Report Central”), to emphasize the demarcation between data focused and report focused workspaces.

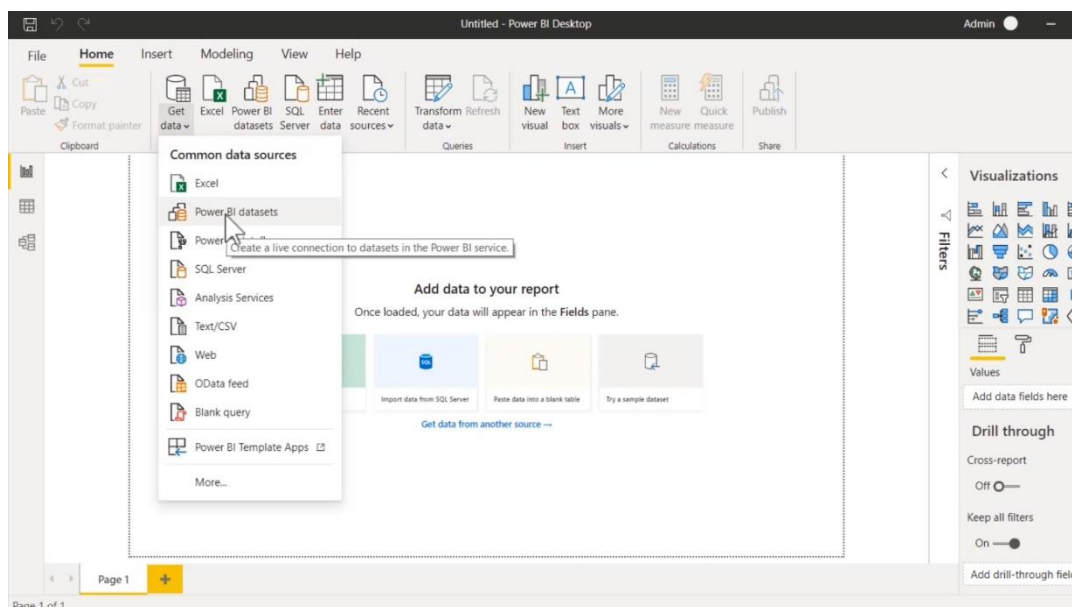


Each time a report is published from Power BI Desktop, both a Report and a Semantic model are created. Since we are primarily interested in the semantic model, we can delete the automatically generated report and retain only the semantic model.



Building Reports from a Shared Semantic model

To create a report using this shared semantic model, open Power BI Desktop and go to File > New. Then, connect to the semantic model stored in the Data Central workspace. You can do this by selecting Power BI Semantic models under Get Data.



Once signed in, search for the semantic model and click Create. This will provide access to the data model, but it will be read-only—you will not be able to modify relationships, rename fields, or make other structural changes. However, you can still create additional measures within the report.

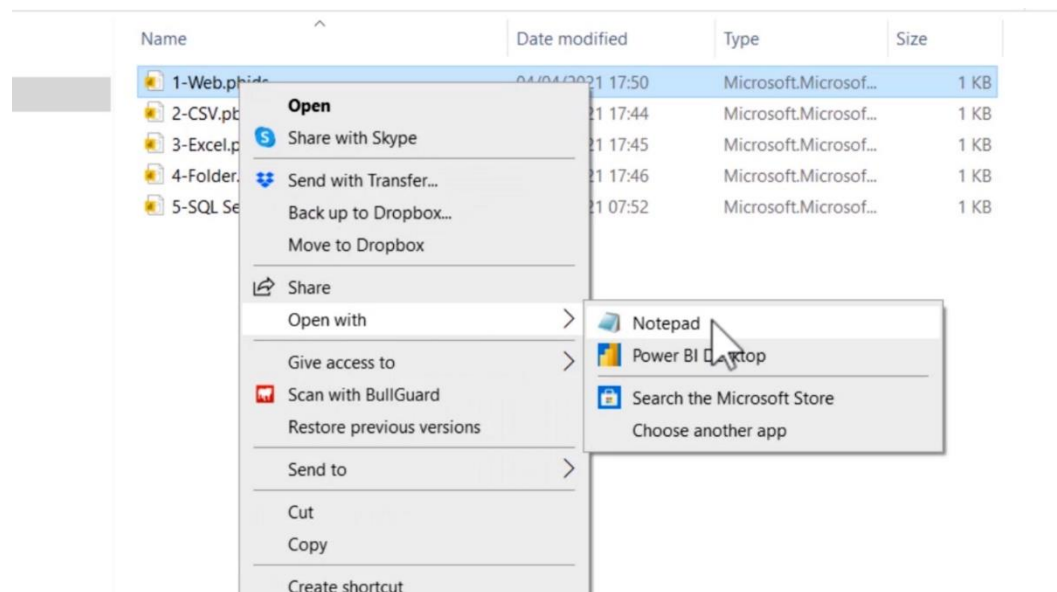
This approach ensures a centralized, consistent, and controlled way of managing semantic models while allowing reports to be developed independently based on approved data models.

0404: Using PBIDS Files

PBIDS files are used in Power BI to encapsulate connections to data sources, ensuring consistency in how users connect to databases, files, and web sources. They help standardize the connection process, particularly useful for new users and organizations that want to enforce best practices.

Understanding the PBIDS File Structure

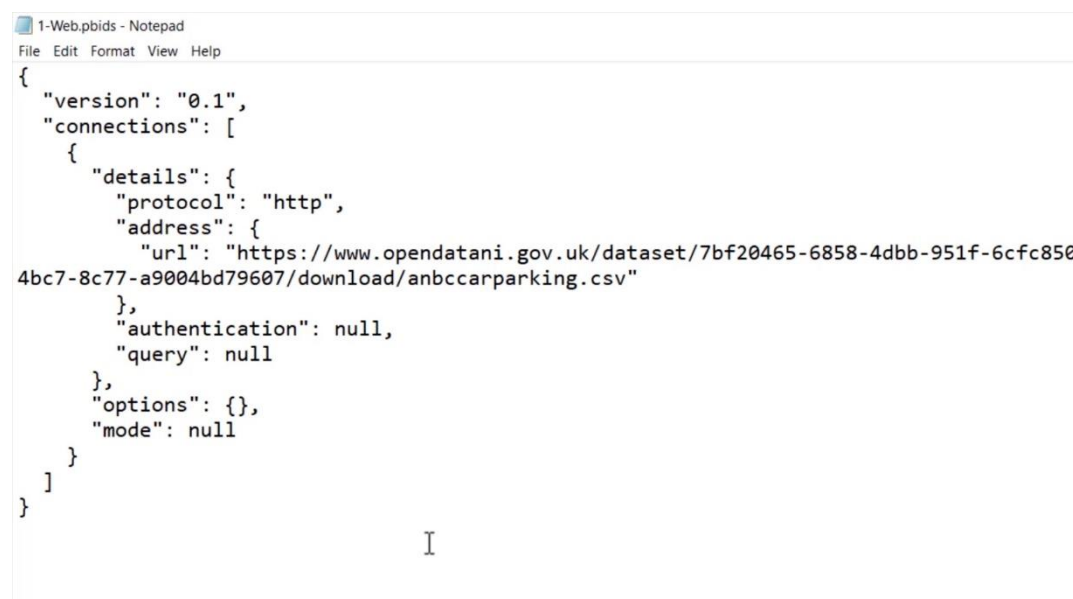
Despite their .pbids extension, PBIDS files are essentially JSON files. If we open one in a text editor rather than Power BI Desktop, we can see their structure.



Each PBIDS file contains two top-level elements:

Version – Specifies the PBIDS file version.

Connections – Defines how Power BI connects to the data source.



Inside the Connections section, we find:

Details – The key section containing information such as the protocol (e.g., http, file, tds), the address (URL, file path, or server name), authentication details, and any query parameters.

Options – Additional settings that can be defined for the connection.

Mode – Determines whether the connection uses Import mode, DirectQuery, or defaults to Import if set to null.

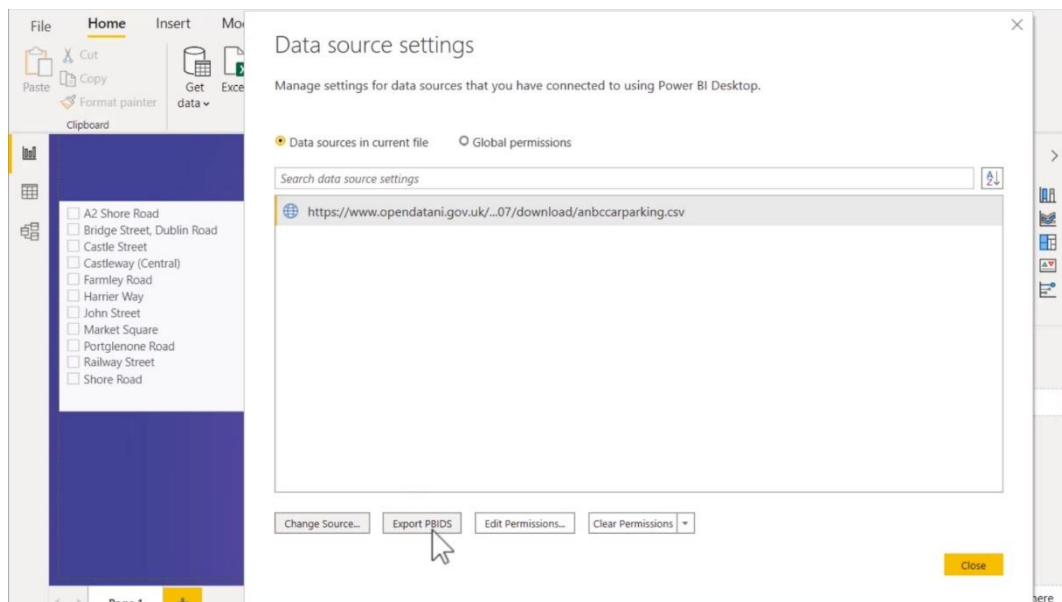
Creating a PBIDS File

Since PBIDS files are JSON, they can be written manually. However, a more efficient method is exporting a PBIDS file directly from Power BI Desktop:

Open a Power BI file that already contains a connection to the required data source.

Navigate to Transform Data > Data Source Settings.

Click Export PBIDS and save the file.



This method ensures accuracy by generating a file with the correct settings rather than creating one from scratch.

Exploring Different Data Source Connections

Web Connection

A PBIDS file for a web connection uses the http protocol. The key attributes in the Details section include:

Protocol: http

Address: The URL of the data source

Authentication: Set to null if no authentication is required

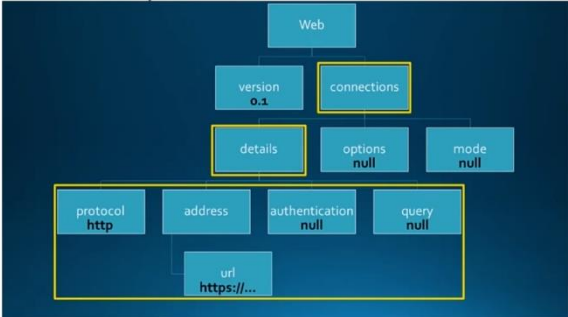
Query: Set to null if no specific query parameters are needed

1-Web.pbids - Notepad
File Edit Format View Help

```

{
  "version": "0.1",
  "connections": [
    {
      "details": {
        "protocol": "http",
        "address": {
          "url": "https://www.opendatani.gov.uk/dataset/7bf20465-6858-4dbb-951f-6cfc8504bc7-8c77-a9004bd79607/download/anbccarparking.csv"
        },
        "authentication": null,
        "query": null
      },
      "options": {},
      "mode": null
    }
  ]
}

```



The diagram illustrates the structure of a Web PBIDS file. It shows a root node 'Web' branching into 'version 0.1' and 'connections'. The 'connections' node further branches into 'details', 'options null', and 'mode null'. The 'details' node branches into 'protocol http', 'address', 'authentication null', and 'query null'. The 'address' node branches into 'url https://...'.

Connecting to a CSV File

A CSV file connection follows the same PBIDS format but with a few key differences:

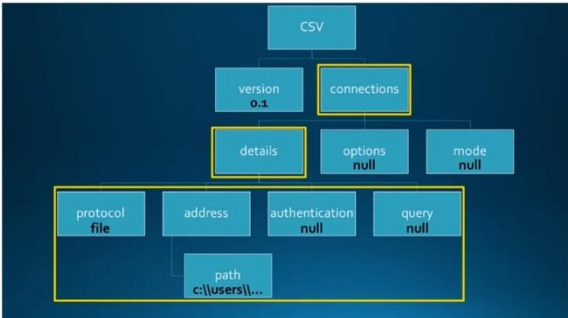
Protocol: file instead of http

Address: A file path instead of a URL

```

{
  "version": "0.1",
  "connections": [
    {
      "details": {
        "protocol": "file",
        "address": {
          "path": "ENTER FILE PATH"
        },
        "authentication": null,
        "query": null
      },
      "options": {},
      "mode": null
    }
  ]
}

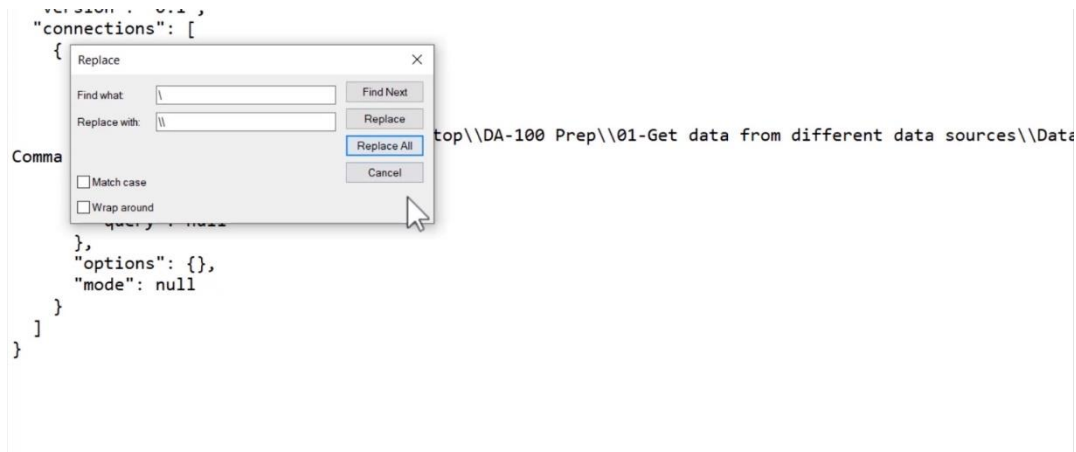
```



The diagram illustrates the structure of a CSV PBIDS file. It shows a root node 'CSV' branching into 'version 0.1' and 'connections'. The 'connections' node further branches into 'details', 'options null', and 'mode null'. The 'details' node branches into 'protocol file', 'address', 'authentication null', and 'query null'. The 'address' node branches into 'path c:\\users\\...'.

When entering the file path, Windows users need to escape backslashes by doubling them. This can be done efficiently using the Find and Replace function (Ctrl + H), replacing single backslashes (\) with double backslashes (\\).

0404-using Pbids Files



Connecting to an Excel File

The structure of a PBIDS file for an Excel connection is identical to that of a CSV file. The key difference is that Power BI treats Excel files as containers, meaning that once connected, users must select specific tables, worksheets, or named ranges within the workbook.



Connecting to a Folder

A folder connection is useful when working with multiple files in a directory. The structure of the PBIDS file remains largely the same, with the Protocol set to folder. The file path must be specified correctly, using the same double backslash convention as with CSV and Excel files.

When opened, Power BI will display the folder contents. Users can then select Combine & Transform, choose a sample file, and proceed with data transformations.

0404-using Pbids Files

4-Folder.pbids - Notepad

File Edit Format View Help

```
{
  "version": "0.1",
  "connections": [
    {
      "details": {
        "protocol": "folder",
        "address": {
          "path": "ENTER FOLDER PATH"
        },
        "authentication": null,
        "query": null
      },
      "options": {},
      "mode": null
    }
  ]
}
```

A hierarchical diagram showing the structure of a Folder PBIDS file. The root node is 'Folder'. It branches into 'version' (0.1) and 'connections'. 'connections' branches into 'details', 'options' (null), and 'mode' (null). 'details' branches into 'protocol' (folder), 'address' (path: c:\users\...), 'authentication' (null), and 'query' (null). 'options' and 'mode' are shown as separate nodes.

Connecting to SQL Server

A PBIDS file can also be used to connect to a SQL Server database. The key attributes include:

Protocol: tds (Tabular Data Stream)

Address: The server name and database name

Mode: Can be Import, DirectQuery, or left as null (defaults to Import mode)

Once the PBIDS file is configured, opening it in Power BI will prompt users to select tables or views from the database. If authentication credentials are already saved, users will be taken directly to the Navigator to select the required data.

File Edit Format View Help

```
{
  "version": "0.1",
  "connections": [
    {
      "details": {
        "protocol": "tds",
        "address": {
          "server": "ENTER SERVER NAME",
          "database": "Sales_Data"
        },
        "authentication": null,
        "query": null
      },
      "options": {
        "CreateNavigationProperties": false
      },
      "mode": "DirectQuery"
    }
  ]
}
```

A hierarchical diagram showing the structure of a SQL Server PBIDS file. The root node is 'SQL Server'. It branches into 'version' (0.1) and 'connections'. 'connections' branches into 'details', 'options', and 'mode' (DirectQuery). 'details' branches into 'protocol' (tds), 'address' (server: ENTER SERVER NAME, database: Sales_Data), 'authentication' (null), and 'query' (null). 'options' branches into 'CreateNavigationProperties' (false). 'mode' is shown as a separate node.

Summary

PBIDS files provide a structured, standardized way of defining data source connections in Power BI. They help maintain consistency across teams, simplify the connection process for new users, and ensure connections are established correctly. Whether connecting to web data, files, folders, or SQL Server, the process remains fundamentally the same—define the protocol, specify the address, and configure additional options as required.

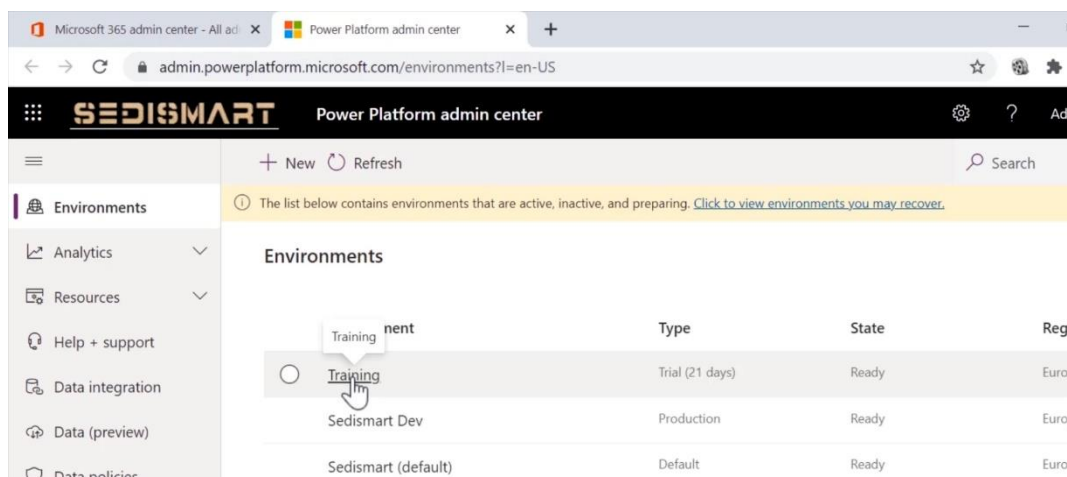
0405: Using Microsoft Dataverse

Microsoft Dataverse, formerly known as the Common Data Service, is a user-friendly database solution designed to store and manage business data efficiently. It provides a collection of pre-built tables, each tailored for a specific type of data, with appropriate columns already in place. Users can import their data directly into these tables or extend the structure by creating custom tables and modifying the built-in ones.

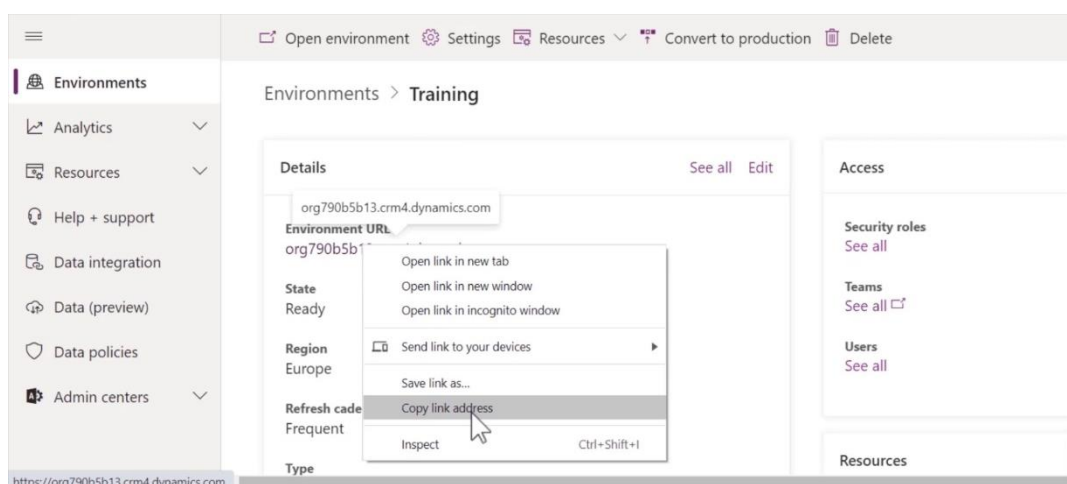
Accessing Dataverse in Power Apps Admin Centre

Dataverse tables are typically managed within the Power Apps Admin Centre. To access this, navigate to the Microsoft 365 Admin Centre, click on Show All, then select All Admin Centres and open Power Apps Admin Centre. From here, environments that store Dataverse databases can be managed.

To connect to a Dataverse database, an environment URL is required. In this example, we use a training environment containing sample data.

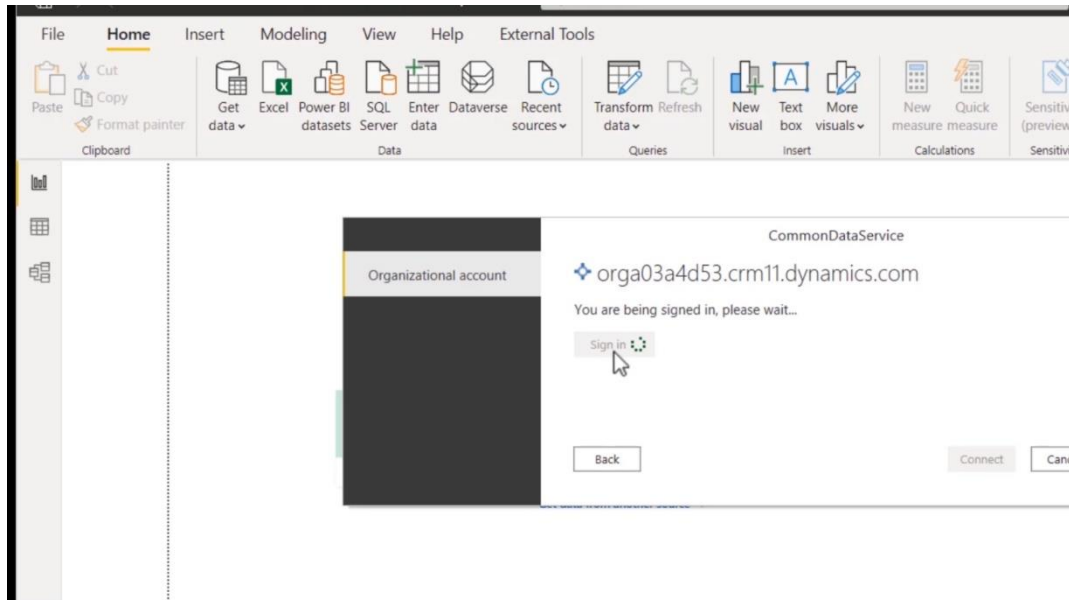


Clicking on the environment in the Admin Centre reveals its properties, including the environment URL.



Connecting Dataverse to Power BI

To connect Power BI to a Dataverse database, open Power BI Desktop and either click on the Database button or go to Get Data, then select Dataverse. The connection can be established using either Import Mode or DirectQuery Mode. DirectQuery Mode is useful when working with multiple data sources that need to be combined, but for this demonstration, we will proceed with Import Mode.

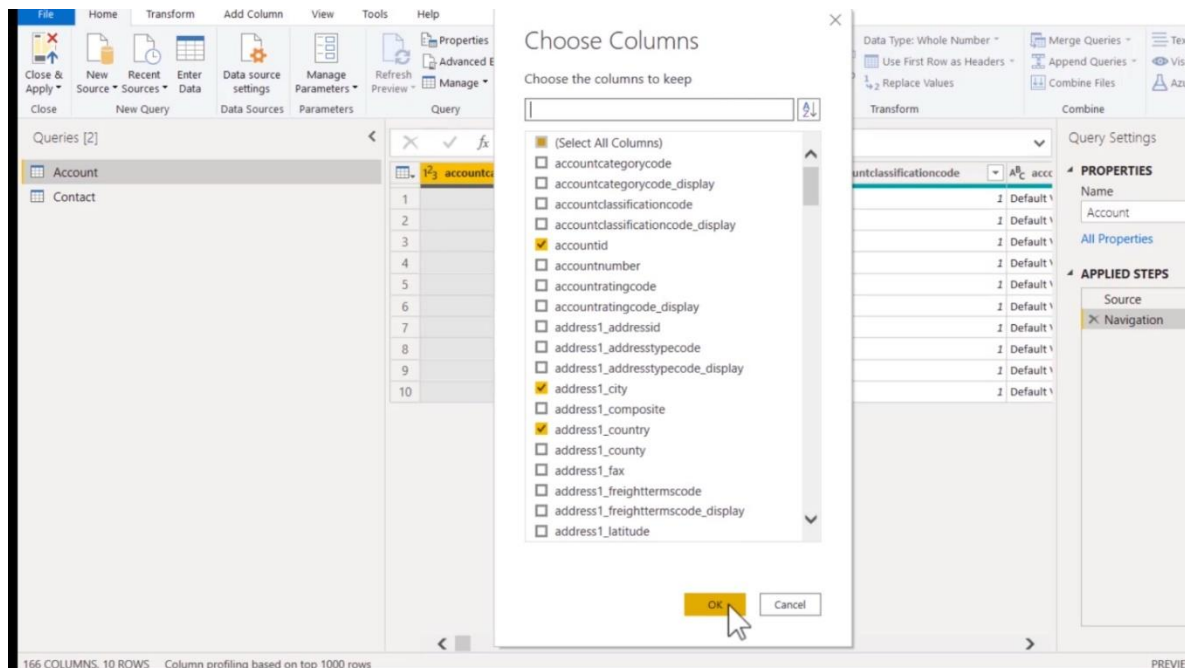


After pasting the environment URL into the connection dialog and clicking OK, sign in to the environment when prompted. Next, we select the tables we want to work with—Account and Contact—before transforming the data in Power Query Editor.

Transforming Data in Power Query Editor

Each Dataverse table contains numerous columns, but only a subset is typically needed for reporting. To refine our semantic model, we use the Choose Columns option:

In our example, for the Account table, we deselect all columns, then select Account ID, Name, Postal Code, City, State, and Country.

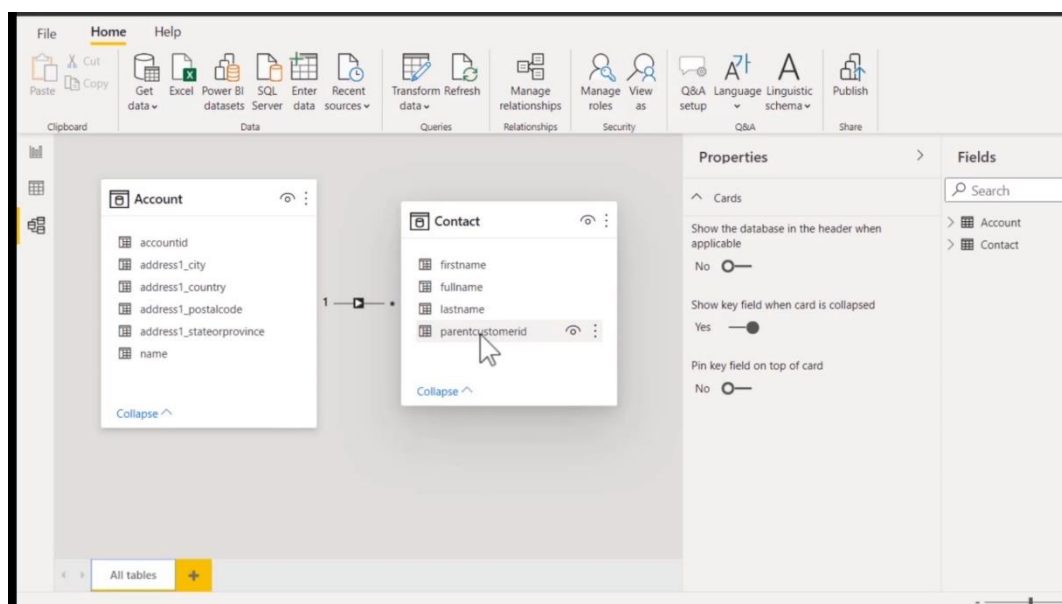


For the Contact table, we deselect all columns, then select Parent Customer ID (which links to Account ID), First Name, Full Name, and Last Name.

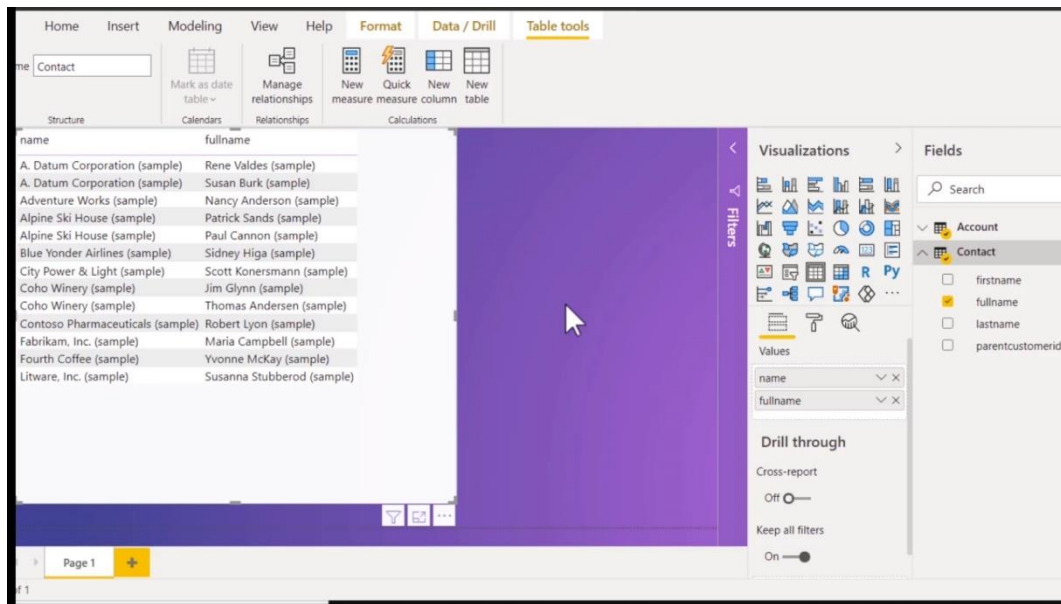
Once the required columns are selected, click Close & Apply to load the transformed data into Power BI.

Creating Relationships and Visualising Data

In Model Mode, a relationship needs to be established between the Account and Contact tables. This is done by linking Account ID in the Account table to Parent Customer ID in the Contact table.



To verify the relationship, we create a Table Visual in Power BI, adding Account Name and Contact Full Name. If the relationship is correctly configured, the data from both tables will align as expected.



Formatting and Saving the Report

To enhance readability, we adjust the theme and font size:

Navigate to View > Customize Current Theme.

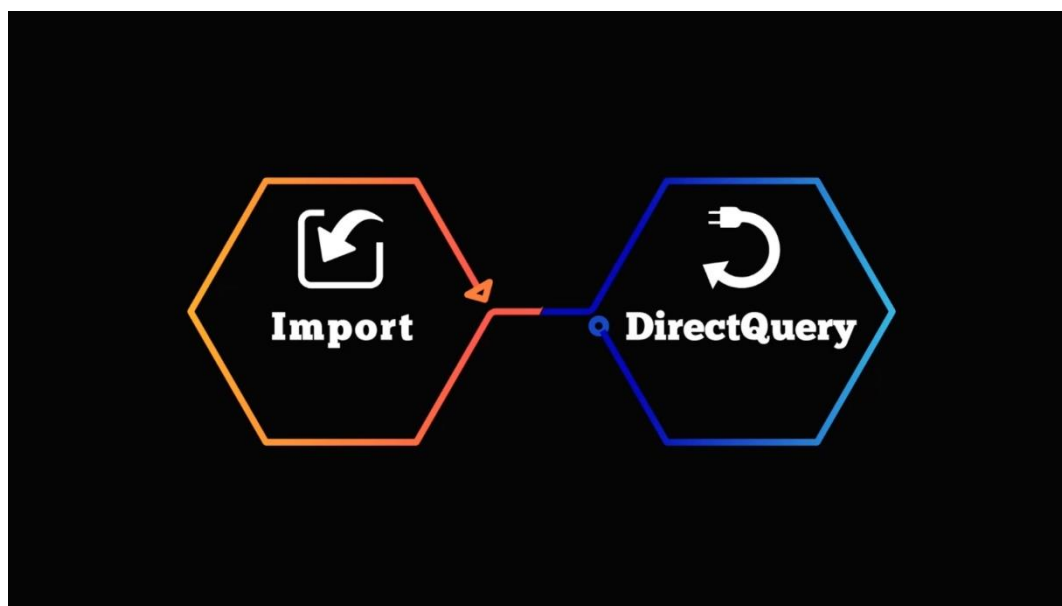
Increase Text Size to 16 and Title Size to 18.

Finally, after verifying the data and relationships, save the report to preserve the changes.

This concludes our introduction to connecting Microsoft Dataverse to Power BI, selecting and transforming data, creating relationships, and visualising results.

0406: Choosing a Storage Mode

When connecting to data sources, Power BI provides three storage modes: Import, DirectQuery, and Composite. Each mode has distinct implications for performance, flexibility, and data freshness. Understanding how they work will help you choose the best approach for your reports.



Import Mode

Import mode is the default and most commonly used option. Data is imported into the data model, where it is compressed, optimized, and stored on disk. When queried or refreshed, the data is loaded into memory, enabling fast performance.

This mode offers the most flexibility for report developers, as it supports the full range of Power Query (M language) transformations and DAX functions. It also enables features such as Q&A and Quick Insights. However, the main limitation is that imported data needs to be refreshed periodically, so it may not always reflect the most up-to-date information.

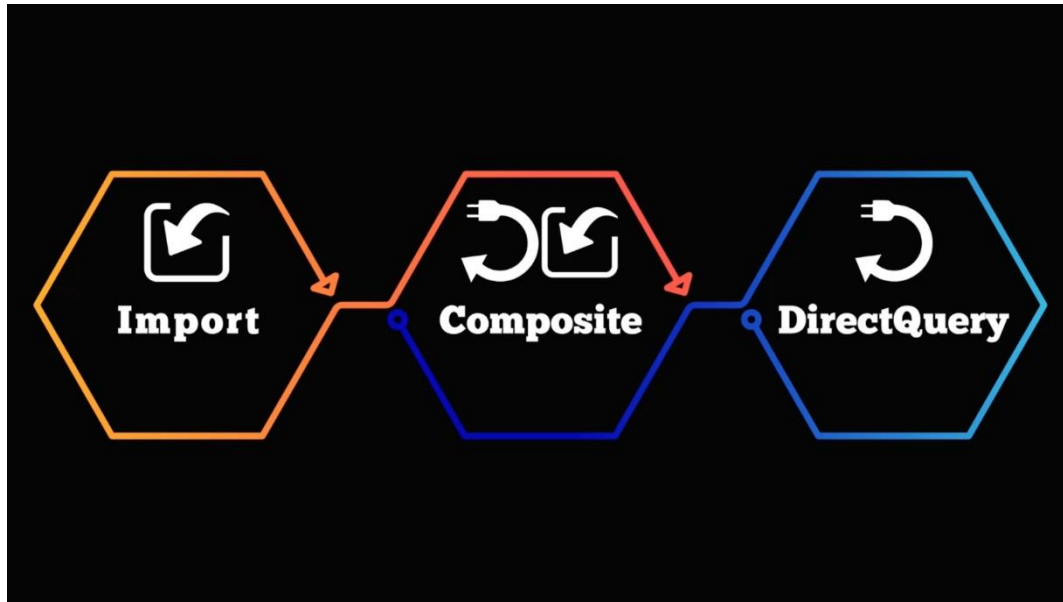
DirectQuery Mode

In DirectQuery mode, data is not imported into the report's data model. Instead, every time a visual is refreshed or interacted with, Power BI sends a query directly to the data source.

The advantage of this approach is that it provides access to real-time or near-real-time data without the need for scheduled refreshes. Additionally, it allows users to work with very large semantic models that would be impractical to import. However, DirectQuery performance depends on the speed of the underlying data source. If the data source is slow or under heavy load, reports may become sluggish.

Another key difference is that transformations in Power Query must be converted into native queries that the data source can understand (e.g., SQL for SQL Server). If a transformation cannot be translated, it will not be available unless the table is switched to Import mode.

Composite Mode



Composite mode provides the flexibility to combine Import and DirectQuery tables within the same model. It effectively extends DirectQuery mode by allowing additional tables to be included in Import mode.

There are two ways to transition a model from DirectQuery to Composite mode:

- Adding a connection to a different data source.
- Creating a DAX-calculated table.

For example, if a report is set up in DirectQuery mode but lacks a date table, adding a date table to the model will automatically switch the report to Composite mode. Similarly, importing a static table or creating a calculated table in DAX has the same effect.

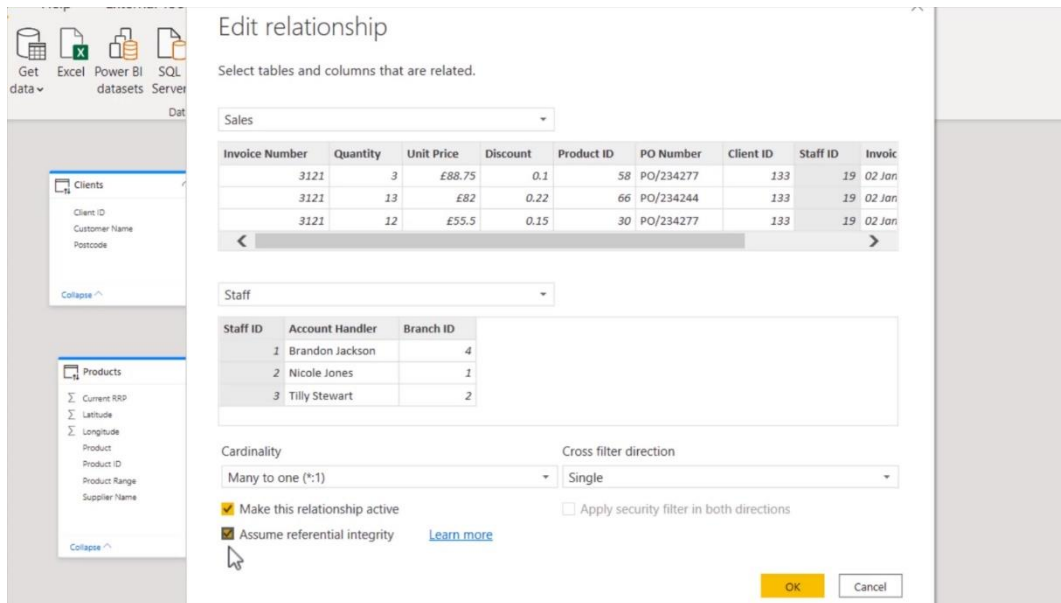
Additionally, Power BI allows individual tables to be converted from DirectQuery to Import mode on a case-by-case basis. However, once a table has been switched to Import mode, it cannot be reverted to DirectQuery. The only way to reintroduce it in DirectQuery mode would be to delete and re-add the table from the source.

Working with DirectQuery Mode

When using DirectQuery mode, transformations must be supported by the underlying database. For example, changing all text values in a column to uppercase can work if Power BI can convert it to a SQL statement. However, some transformations, such as capitalizing the first letter of each word, may not be possible unless the entire semantic model is switched to Import mode.

Another difference with DirectQuery is that the Data view is unavailable in Power BI Desktop since no data is stored locally. In Model view, tables in DirectQuery mode are marked with a blue bar at the top.

When editing relationships between tables in DirectQuery mode, an option called "Assume Referential Integrity" is available. This setting, which is typically disabled in Import mode, allows Power BI to optimize queries by using INNER JOIN instead of OUTER JOIN when it is safe to do so. This improves performance but requires that the data source enforces referential integrity (e.g., no null values in key columns).



Switching to Composite Mode

To transition from DirectQuery to Composite mode, a non-DirectQuery table must be introduced. This can be done by:

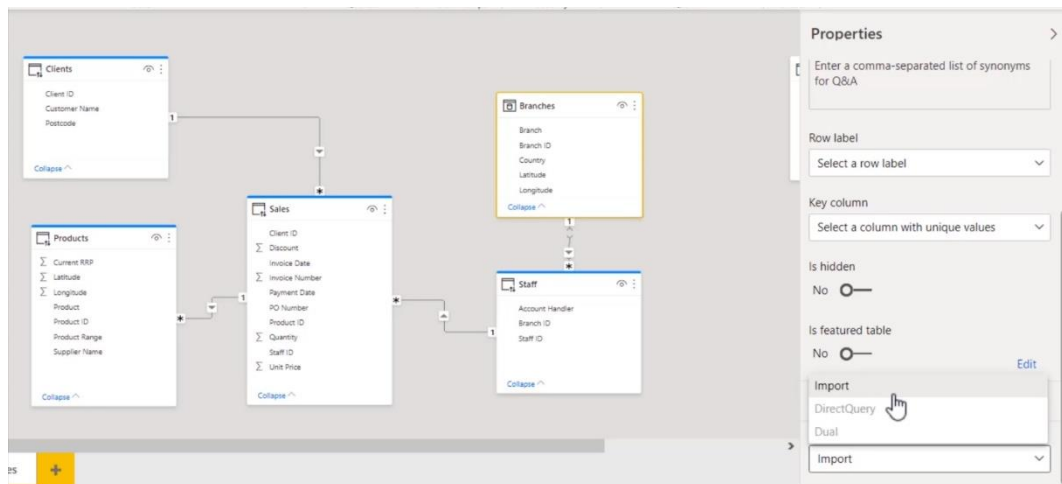
- Adding a DAX-calculated table (e.g., a simple date table using CALENDARAUTO()).
- Importing a table from the data source instead of using DirectQuery.

Once this change is made, two key differences will appear:

The Data view becomes available in Power BI Desktop.

The storage mode indicator in the bottom right changes from DirectQuery to Mixed.

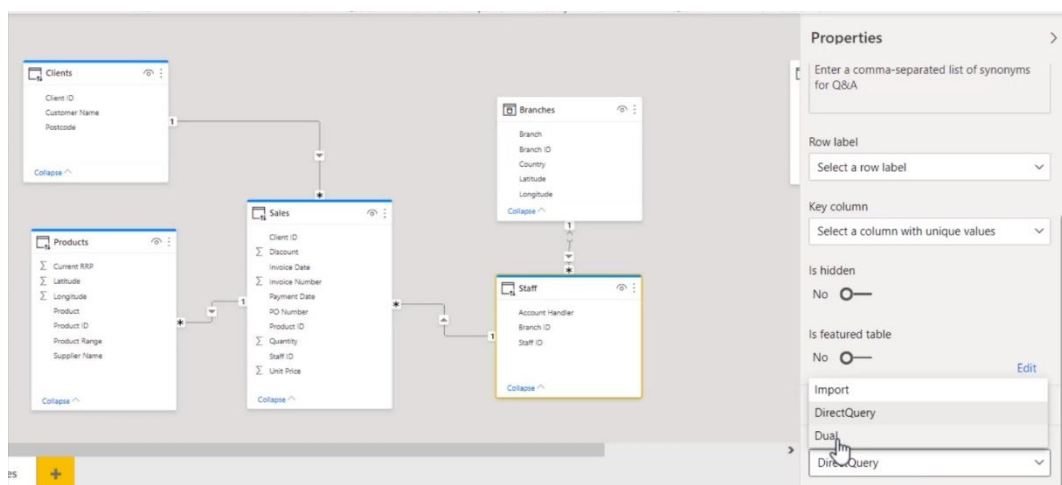
Another approach to modifying storage mode is to change an individual table's setting from DirectQuery to Import. This can be done in the Properties pane under the Advanced section. However, once a table has been switched to Import mode, it cannot be converted back to DirectQuery.



Dual Mode

A third storage option, Dual mode, allows a table to function as both Import and DirectQuery simultaneously. Power BI can decide at query time whether to use the imported data or to send a live query to the data source, depending on what will yield better performance.

For example, if the Staff table is switched to Dual mode, Power BI stores an imported copy while maintaining the ability to send live queries when needed. This mode is indicated by a dashed blue line at the top of the table in Model view.



Dual mode offers flexibility, as it can optimize performance while still allowing up-to-date queries when necessary. However, not all data sources or scenarios benefit from Dual mode, so careful consideration is needed when using it.

0407: Using Data Source Parameters

Power Query parameters have long been used to dynamically change data sources in Power BI. In this tutorial, we'll extend that concept to not only change the data source but also adjust the data type. The goal is to start with Excel-based data sources and seamlessly transition to SQL Server once database credentials become available.

Overview of the Solution

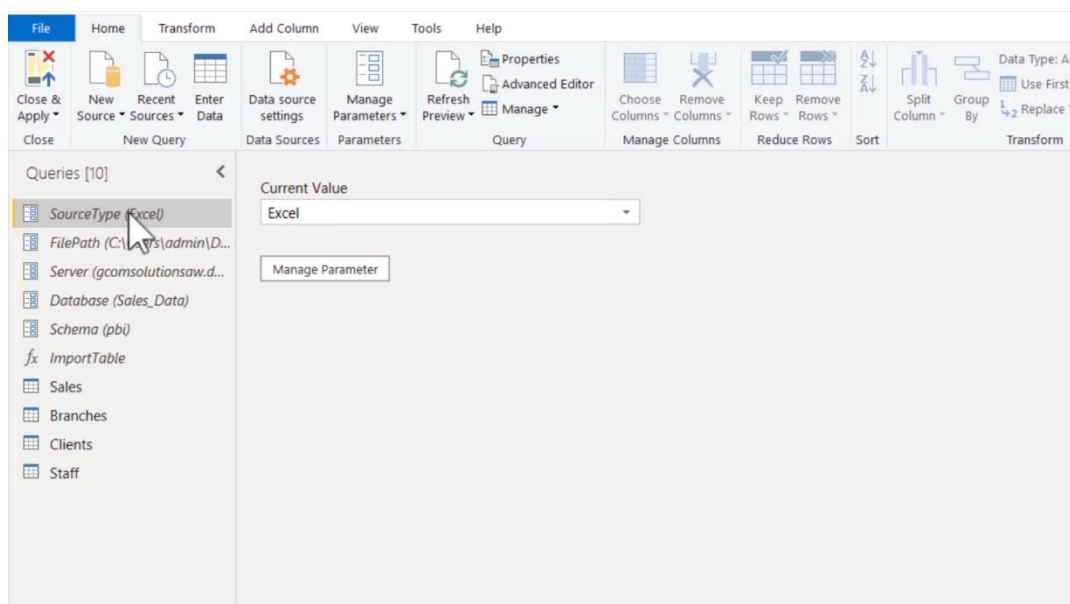
Our approach involves five parameters:

A source type parameter that allows switching between Excel and SQL Server.

A file path parameter for the Excel workbook location.

Three parameters for SQL Server: server name, database name, and schema name.

These parameters feed into a function that determines whether data should be retrieved from an Excel workbook or a SQL Server database. This ensures that changing the source is as simple as selecting an option.



Creating the Parameters

Let's begin by setting up the necessary parameters in Power Query.

Open Power BI Desktop and click Transform Data to launch Power Query Editor.

Navigate to Manage Parameters > New Parameter.

Create the Source Type parameter:

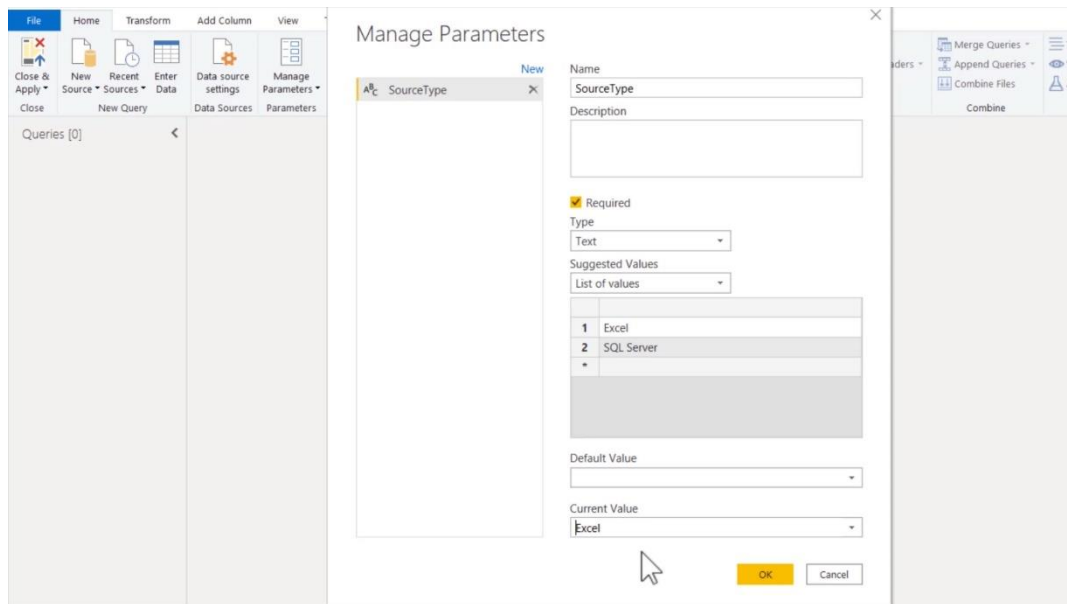
Name: SourceType

Type: Text

0407-using Data Source Parameters

Suggested Values: List of Values (Excel, SQL Server)

Current Value: Excel

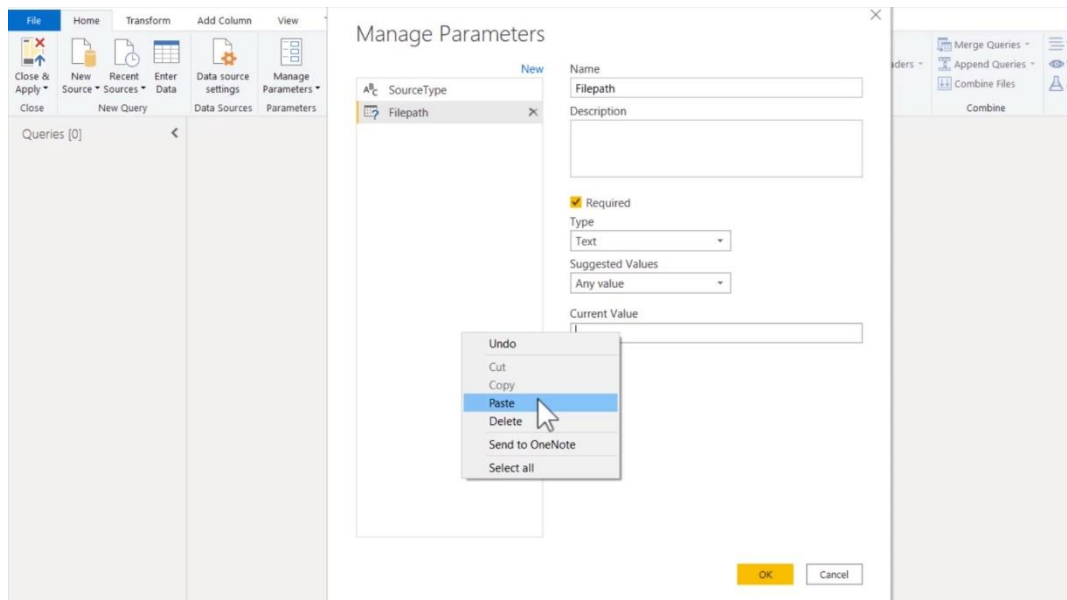


Create the File Path parameter:

Name: FilePath

Type: Text

Value: (Paste the Excel file path here)

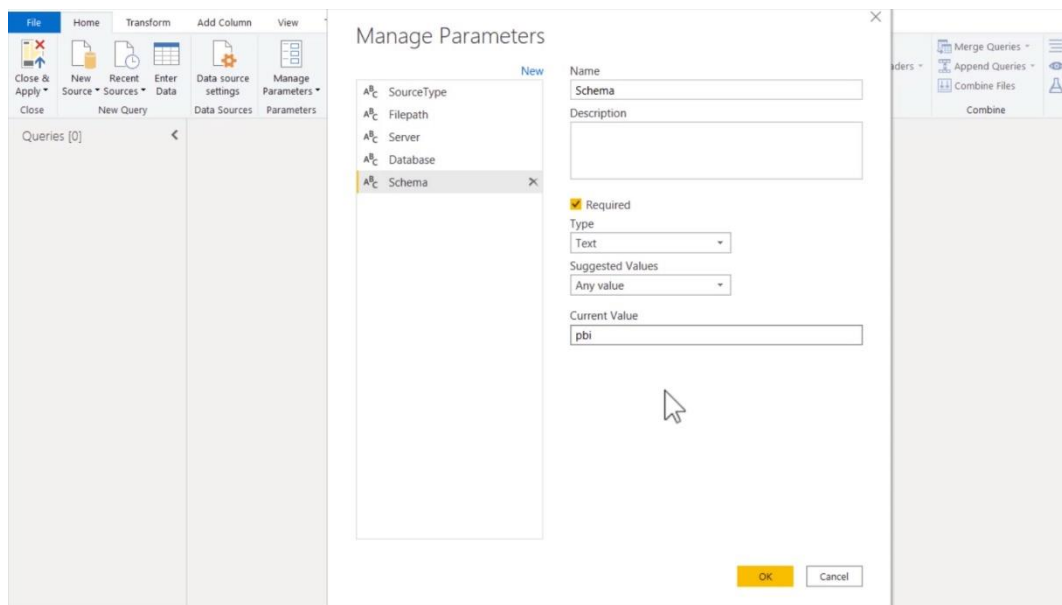


Create the three SQL Server parameters:

Server Name (Text)

Database Name (e.g., Sales_Data)

Schema Name (e.g., pbi)



With these five parameters set, we can proceed to creating the function.

Building the Function

To dynamically import tables, we'll create an M language function.

Create a blank query:

Go to New Source > Blank Query

Rename the query to ImportTable

Open Advanced Editor

Define the function header:

Name: GetData

Parameter: TableName (Text)

Return Type: Table

Implement logic to determine the data source:

Use an if statement to check the value of SourceType.

If Excel, load the data using Excel.Workbook(File.Contents(FilePath)).

If SQL Server, use Sql.Database(ServerName, DatabaseName).

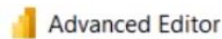
Navigate to the required table:

In Excel, locate the table within the metadata structure using Source{[Item=TableName, Kind="Table"]}[Data].

In SQL Server, use a similar approach, but filtering based on Schema and TableName.

Return the final table:

The function should return the processed table, allowing seamless switching between sources.



ImportTable

```

1  let
2      GetData = (TableName as text) as table =>
3          let
4              Source =
5                  if SourceType = "Excel"
6                  then Excel.Workbook(File.Contents(Filepath))
7                  else Sql.Database(Server, Database),
8              OutputTable =
9                  if SourceType = "Excel"
10                 then Source{[Item = TableName, Kind = "Table"]}[Data]
11                 else Source{[Item = TableName, Schema = Schema]}[Data]
12             in
13                 OutputTable
14 in
15     GetData

```

Get Data
 HexagonSmartApi.GetODataMetadata

✓ No syntax errors have been detected.

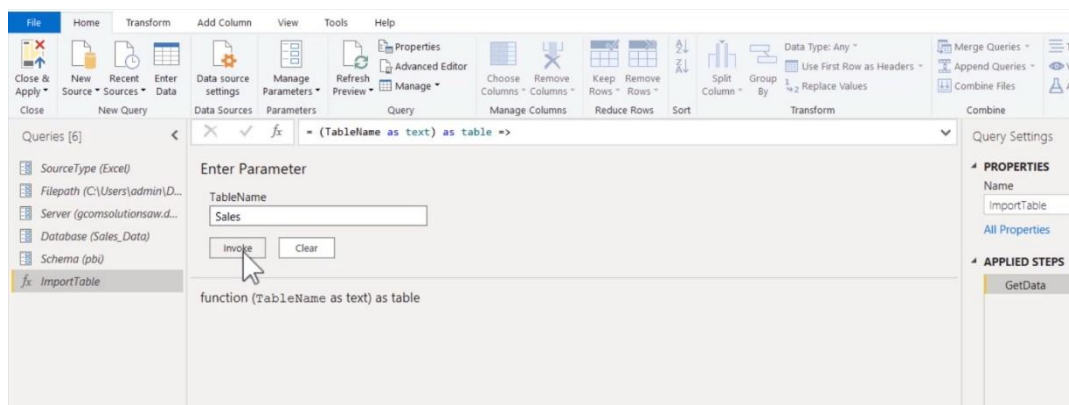
Applying the Function

Now that the function is ready, we can use it to import tables:

Invoke the function for each table:

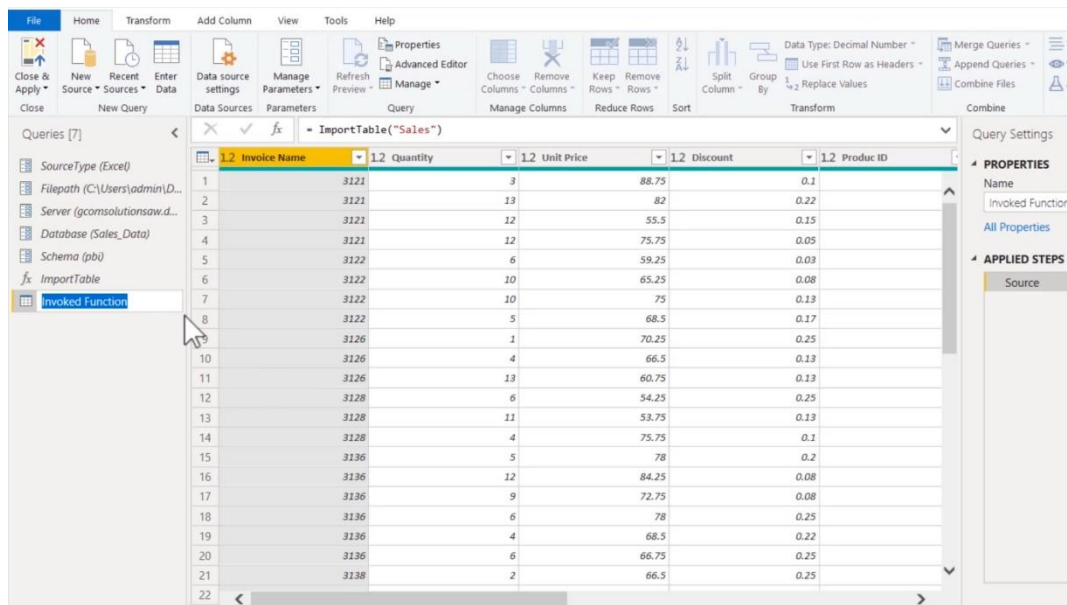
Select Invoke Custom Function and choose GetData.

Pass in the table names: Sales, Clients, Products, Staff, and Branches.



0407-using Data Source Parameters

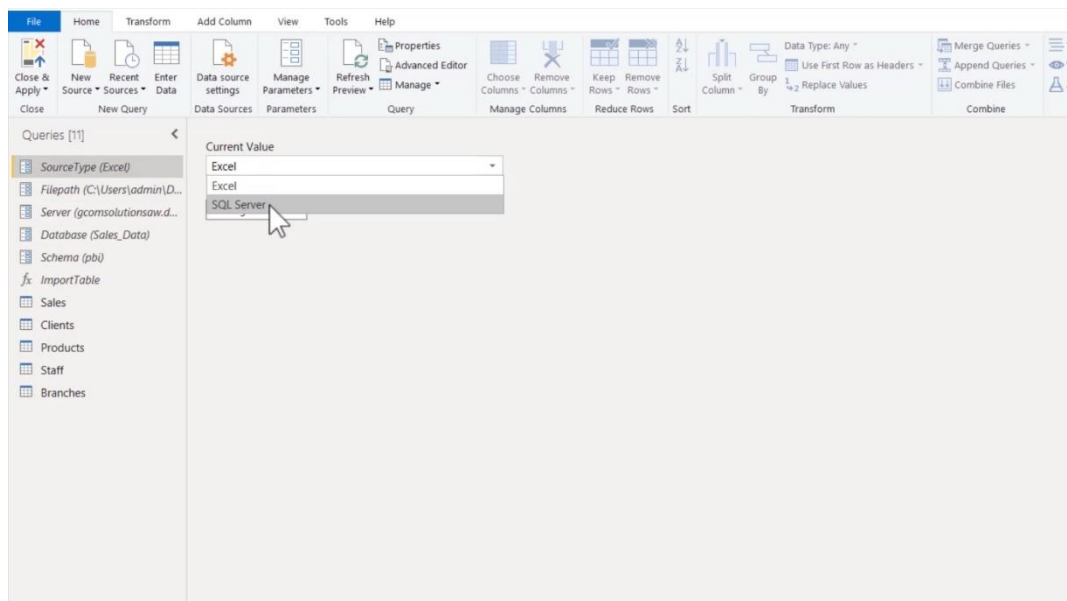
Rename tables as necessary.



	1.2 Invoice Name	1.2 Quantity	1.2 Unit Price	1.2 Discount	1.2 Product ID
1	3121	3	88.75	0.1	
2	3121	13	82	0.22	
3	3121	12	55.5	0.15	
4	3121	12	75.75	0.05	
5	3122	6	59.25	0.03	
6	3122	10	65.25	0.08	
7	3122	10	75	0.13	
8	3122	5	68.5	0.17	
9	3126	1	70.25	0.25	
10	3126	4	66.5	0.13	
11	3126	13	60.75	0.13	
12	3128	6	54.25	0.25	
13	3128	11	53.75	0.13	
14	3128	4	75.75	0.1	
15	3136	5	78	0.2	
16	3136	12	84.25	0.08	
17	3136	9	72.75	0.08	
18	3136	6	78	0.25	
19	3136	4	68.5	0.22	
20	3136	6	66.75	0.25	
21	3138	2	66.5		
22					

Test switching between data sources:

In the Source Type parameter, switch from Excel to SQL Server.



The imported tables will now come from SQL Server instead of Excel, assuming identical table structures.

Summary

By using Power Query parameters, we've created a flexible solution that allows seamless switching between Excel and SQL Server without modifying queries. This approach is particularly useful when starting with local files and later migrating to a central database.

0409: Granting Access to Semantic models

When you publish a report from Power BI Desktop, the corresponding workspace will contain both a report and a semantic model. As the creator, you automatically have access to both and can build additional reports from the semantic model. However, to allow others to do the same, you need to grant access. There are two primary ways to do this:

Granting direct access to the semantic model

Providing access via a workspace app

Let's explore both methods.

Granting Direct Access to a Semantic model

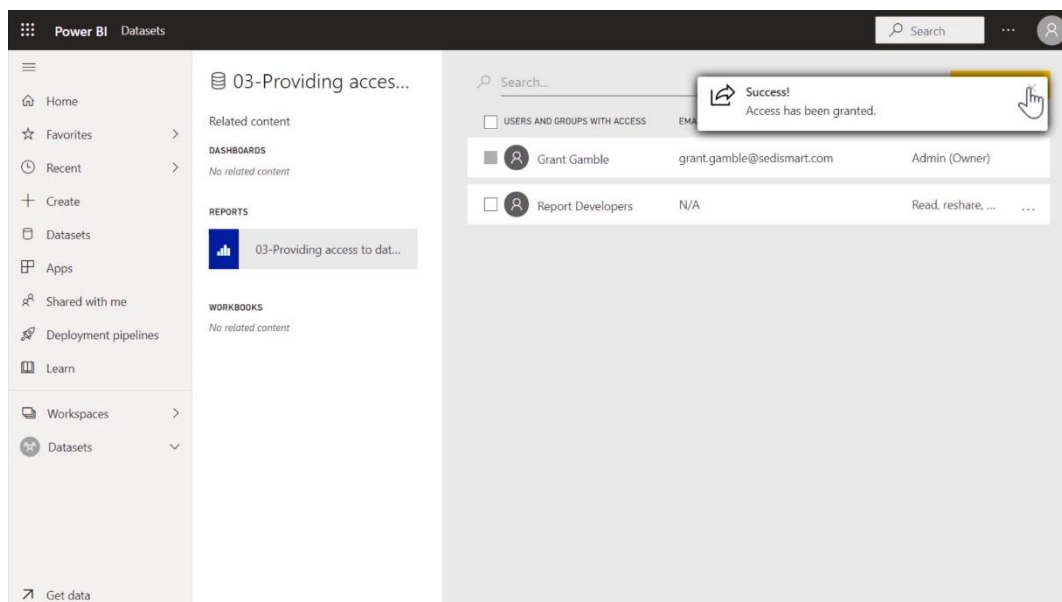
The first method is to grant users direct access to an individual semantic model. To do this:

Locate the semantic model within the workspace.

Click on the ellipsis (...) next to the semantic model name and select Manage permissions.

Click Add user and specify who should have access.

At this stage, you can enter individual usernames, but a better approach is to use security groups. For example, if you grant access to a security group called Report Developers, anyone in that group will be able to build reports from the semantic model.



Granting Access via a Workspace App

The second approach is to provide access to semantic models via a workspace app. The key difference here is that users will gain access to all semantic models included in the app, rather than a single semantic model.

0409-granting Access To Semantic models

To do this:

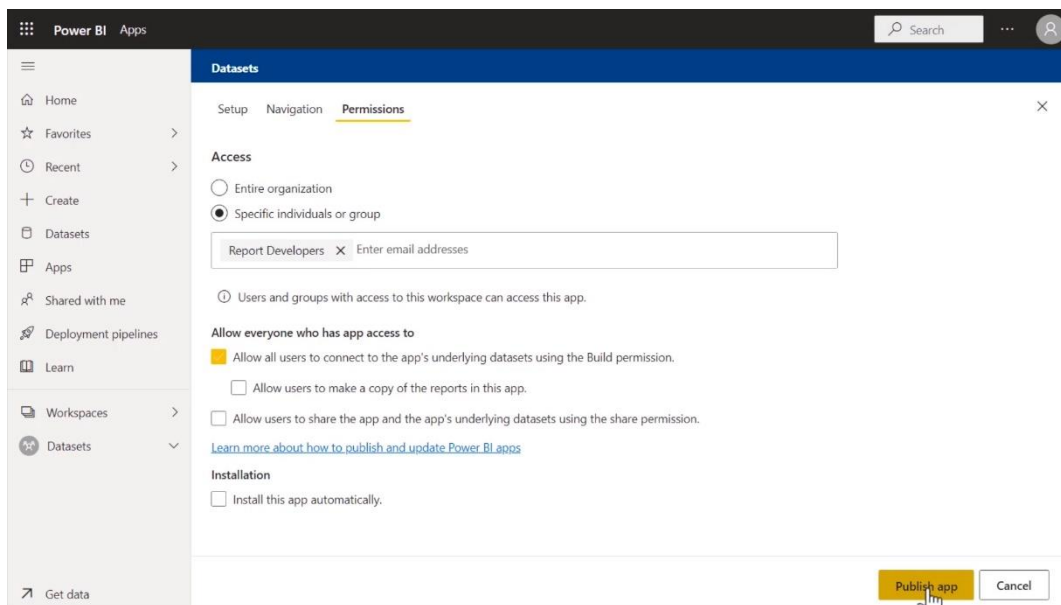
Click Create app within the workspace.

Follow the wizard, which consists of three sections: Setup, Navigation, and Permissions. In the Permissions section, locate the option to grant users the Build permission.

Enable Allow all users to connect to the app's underlying semantic models using the Build permission.

Add individual users or security groups as needed.

Publish the app.

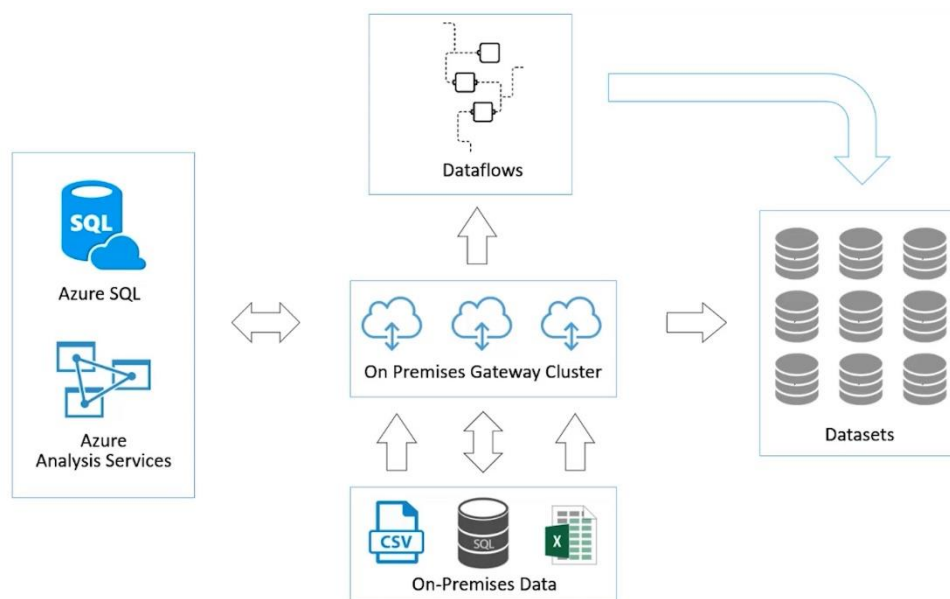


Once published, users with the necessary permissions can connect to the app's semantic models and build reports accordingly.

Choosing between these two methods depends on whether you want to grant access to a single semantic model or multiple semantic models at once. If users only need access to one semantic model, direct access is often preferable. However, if they require broader access across multiple semantic models, an app-based approach may be more efficient.

0410: On-Premises Data Gateway Setup

The on-premises data gateway in Power BI serves two key functions. When working in import mode, the gateway ensures that uploaded data remains in sync with the original source by enabling scheduled refreshes. This is typically used when importing data from files such as Excel or CSV. In DirectQuery mode, the gateway operates in a bi-directional manner, executing queries against a SQL Server database and sending results back to Power BI in real time for report visualization.



Downloading and Installing the On-Premises Data Gateway

To begin, the gateway software must be downloaded and installed. This can be done directly from the Power BI Service:

Click on the Download menu in the top-right corner.

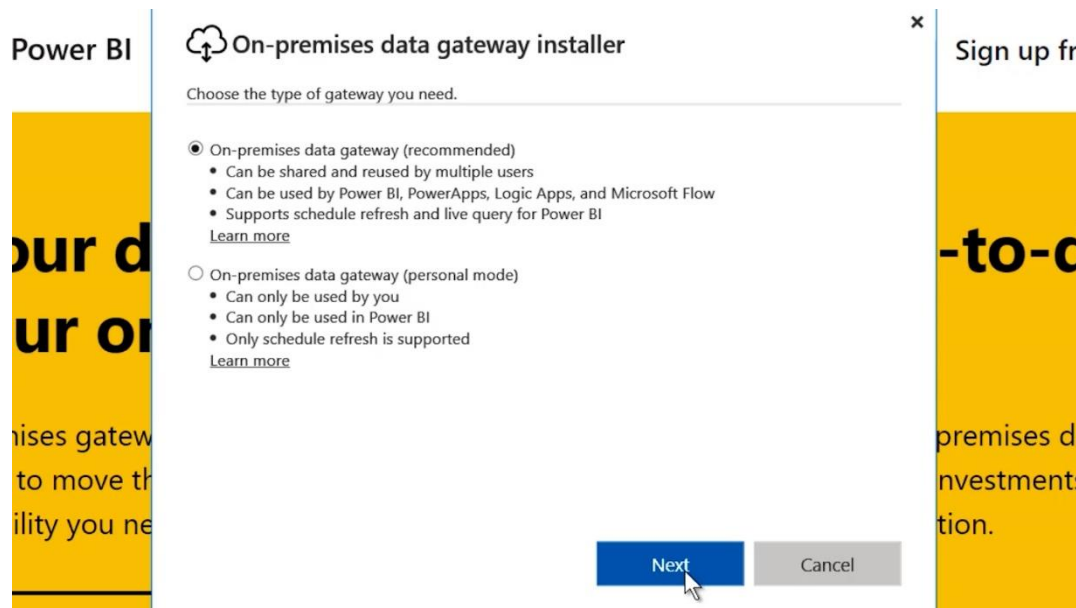
Select Data Gateway to navigate to the download page.

Click Download Gateway and run the downloaded executable file.

Choose the mode: Enterprise Mode is recommended for most business scenarios. Personal Mode is available as a fallback if an enterprise-wide gateway cannot be used.

After selecting Enterprise Mode (as opposed to personal), click Next to continue.

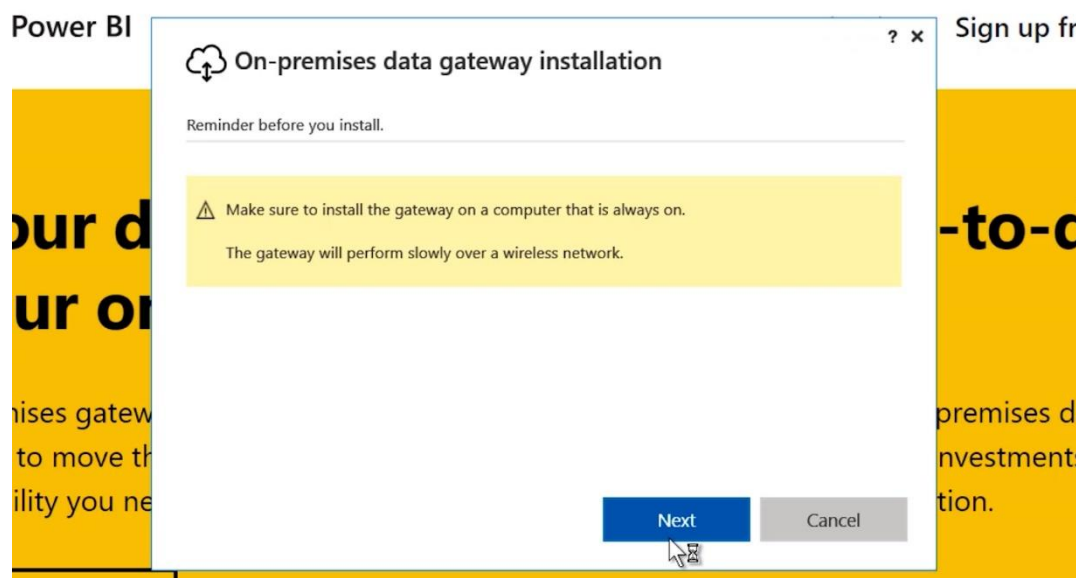
0410-on-premises Data Gateway Setup



Once the installer launches:

Click Next to proceed with installation.

The system will remind you that the gateway should be installed on a machine that is always on. After confirming this, proceed with the installation by accepting the terms of use and clicking Install.



0410-on-premises Data Gateway Setup

https://powerbi.microsoft.com

Power BI

On-premises data gateway

Almost done.

Installation was successful!

Email address to use with this gateway:

Next, you need to sign in to register your gateway.

[Sign in](#) [Cancel](#)

Power BI

On-premises data gateway

You are signed in as grantgamble@gcomsolutions.co.uk and are ready to register the gateway.

New on-premises data gateway name

☐ Add to an existing gateway cluster

Recovery key (8 character minimum)

ⓘ This key is ne it in a safe place.

Confirm recovery key

[Learn more about gateway clusters](#)

We'll use this region to connect the gateway to cloud services: UK South [Change Region](#)

[Configure](#) [Cancel](#)

0410-on-premises Data Gateway Setup

On-premises data gateway

You are signed in as grantgamble@gcomsolutions.co.uk and are ready to register the gateway.

New on-premises data gateway name

☐ Add to an existing gateway cluster

Recovery key (8 character minimum)

ⓘ This key is needed to restore the gateway and can't be changed. Record it in a safe place.

Confirm recovery key

[Learn more about gateway clusters](#)

We'll use this region to connect the gateway to cloud services: UK South [Change Region](#)

Configure **Cancel**

Registering the Gateway

Once installation is complete, the next step is to register the gateway. This requires signing in with your usual Power BI credentials.

During registration, you can either:

Create a new gateway cluster – This is typically the case when setting up a gateway for the first time.

Add to an existing cluster – If additional redundancy is needed, a new gateway can be added to an existing cluster by entering the recovery key used when setting up the first gateway in that cluster.

In this example, we are starting a new cluster. We assign a name (e.g., "Training"), do not need to enter a recovery key, and click Configure.

On-premises data gateway

You are signed in as grantgamble@gcomsolutions.co.uk and are ready to register the gateway.

New on-premises data gateway name

Training

☐ Add to an existing gateway cluster

Recovery key (8 character minimum)

ⓘ This key is needed to restore the gateway and can't be changed. Record it in a safe place.

Confirm recovery key

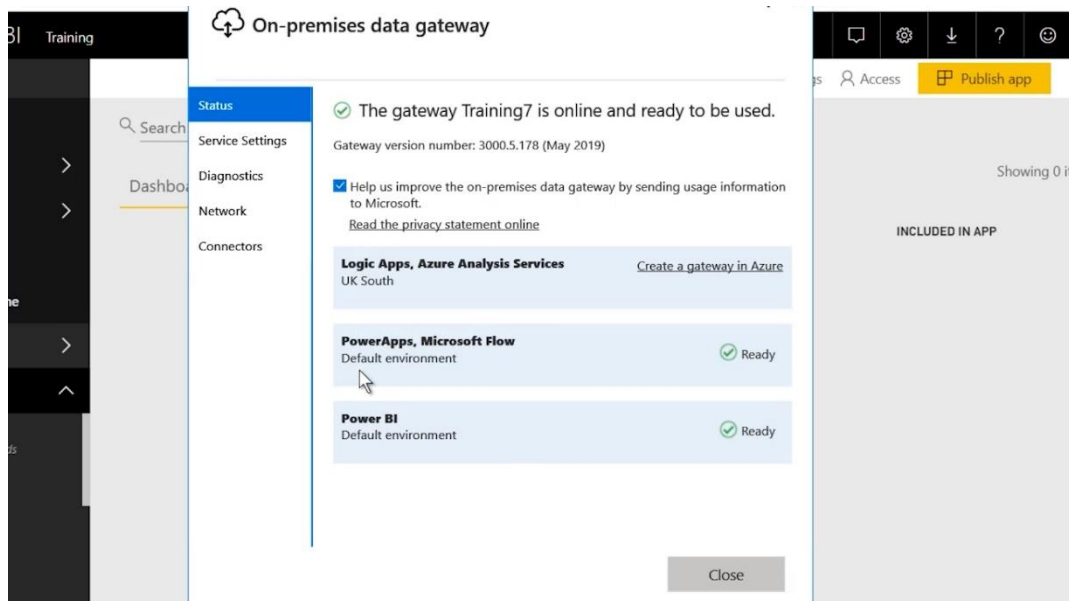
[Learn more about gateway clusters](#)

We'll use this region to connect the gateway to cloud services: UK South [Change Region](#)

Configure **Cancel**

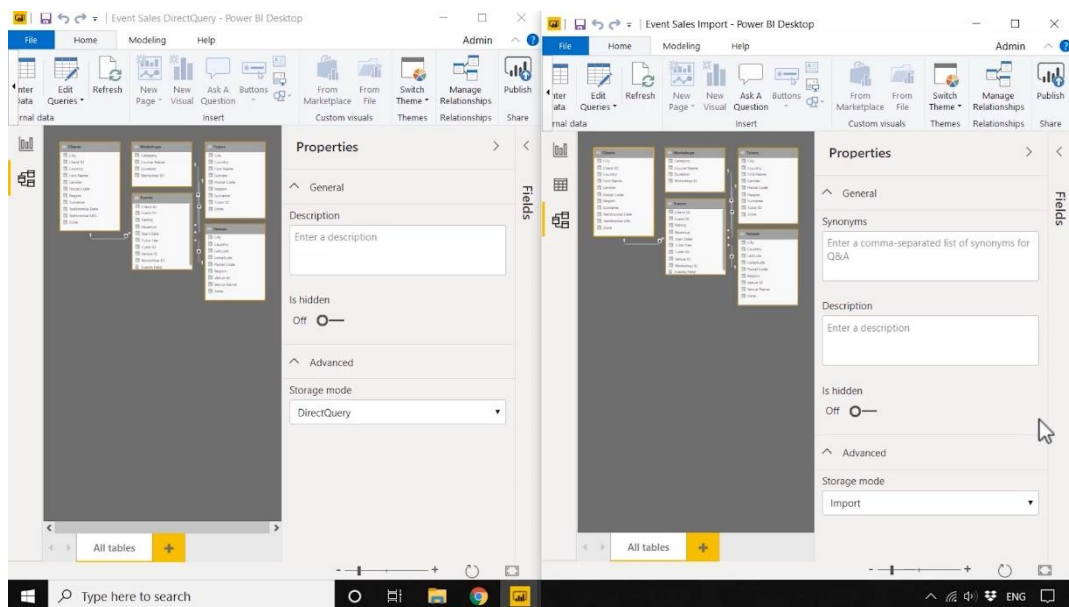
0410-on-premises Data Gateway Setup

Once registration is complete, a confirmation message appears. The gateway is now ready for use with Power BI, Power Apps, and Microsoft Flow.



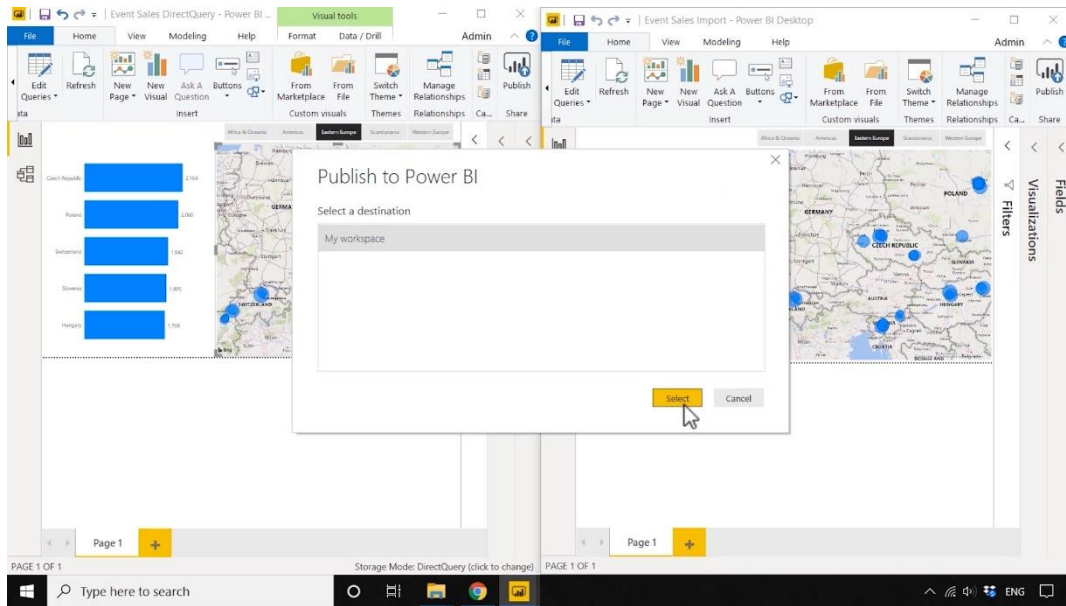
Publishing Reports and Configuring the Gateway

To illustrate how the gateway functions, consider two reports. Both reports are identical except for their storage mode: one uses import mode and the other DirectQuery.

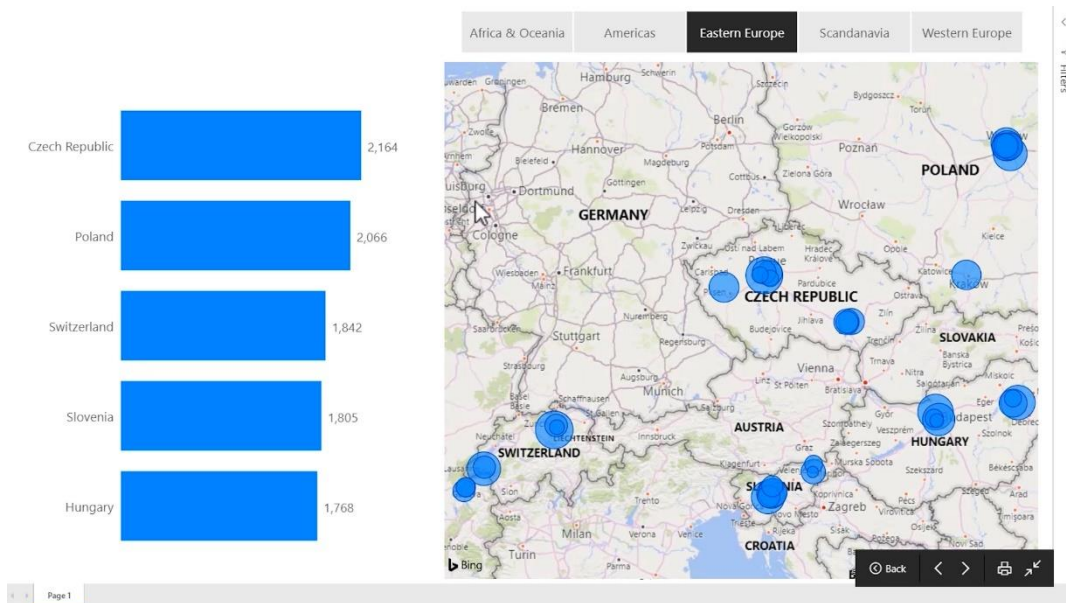


After confirming this in Power BI Desktop, we publish them to the Power BI Service.

0410-on-premises Data Gateway Setup



The Import mode report functions immediately, as all data is already embedded.



However, the DirectQuery mode report does not display because it requires a gateway to access the on-premises data source.

Configuring the Data Source in Power BI Service

To enable the DirectQuery report, we must configure the gateway in the Power BI Service:

Navigate to Settings and select Manage Gateways.

Click Add Data Source in the top-left corner.

Name the data source (e.g., "Event Sales").

0410-on-premises Data Gateway Setup

Choose SQL Server as the data source type.

Enter the server name and database name (e.g., "Event Sales").

Select the authentication method (typically Windows Authentication).

If using a Windows 10 machine, input login credentials in the format domain\username followed by the password.

Click Add to establish the connection.

My workspace > Gateways

New data source

Test all connections

Event Sales

Data Source Type

SQL Server

Server

DESKTOP-M900O1K

Database

EventSales

Authentication Method

Windows

The credentials are encrypted using the key stored on-premises on the gateway server. [Learn more](#)

Username

Password

☐ Skip Test Connection

Once the connection is successful, the next step is to link the semantic model to the gateway:

Go to Semantic models, locate Event Sales DirectQuery, and open its settings.

Under Gateway Connection, enable Use a Data Gateway.

The previously configured gateway should appear automatically.

General Alerts Subscriptions Dashboards **Datasets** Workbooks

Event Sales DirectQuery

Event Sales Import

SQL Azure Test

Settings for Event Sales DirectQuery

[Refresh history](#)

Gateway connection

To use a data gateway, make sure the computer is online and the data source is added in [Manage Gateways](#). If you're using an On-premises data gateway (standard mode), please select the corresponding data sources and then click apply.

Use a data gateway

☒ On

Gateway	Department	Contact information	Status	Actions
Training7		admin@sedismart.com	Running on DESKTOP-M900O1K	

Data sources included in this dataset:

SqlServer("server":"desktop-m900o1k","database":"eventsales")	Maps to:	Event Sales
---	----------	-------------

You have no personal gateways installed. [Learn more](#)

[Install now](#)

[Apply](#) [Discard](#)

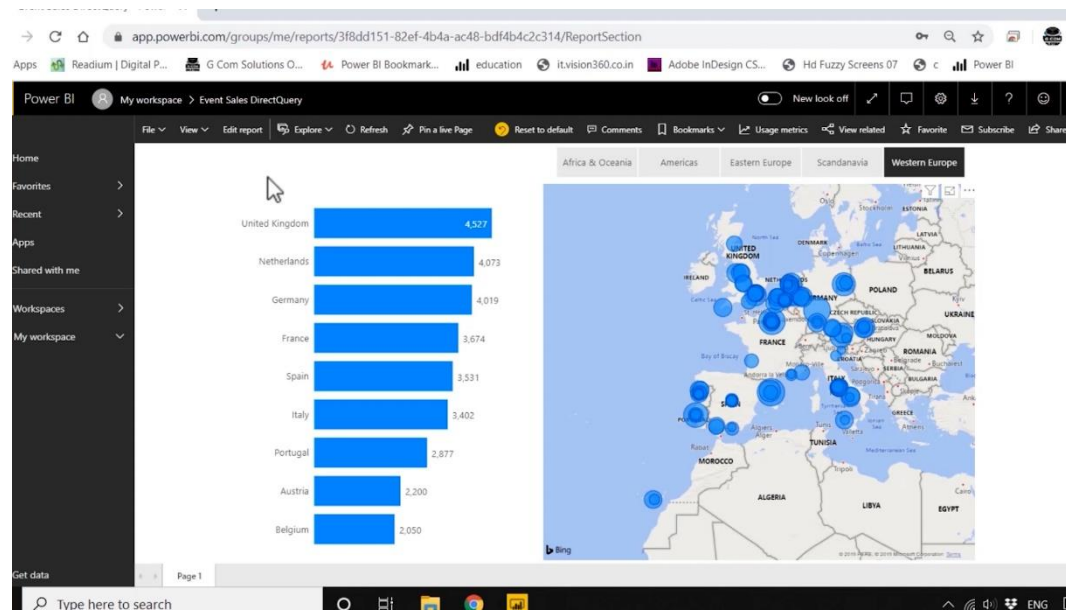
0410-on-premises Data Gateway Setup

Ensure the SQL Server connection maps to the newly created data source.

Click Apply to finalize the setup.

Verifying the Gateway Connection

Now that the gateway is in place, the DirectQuery report is accessible. As users interact with the report, the gateway queries the on-premises database in real time, ensuring live data is retrieved and displayed dynamically.

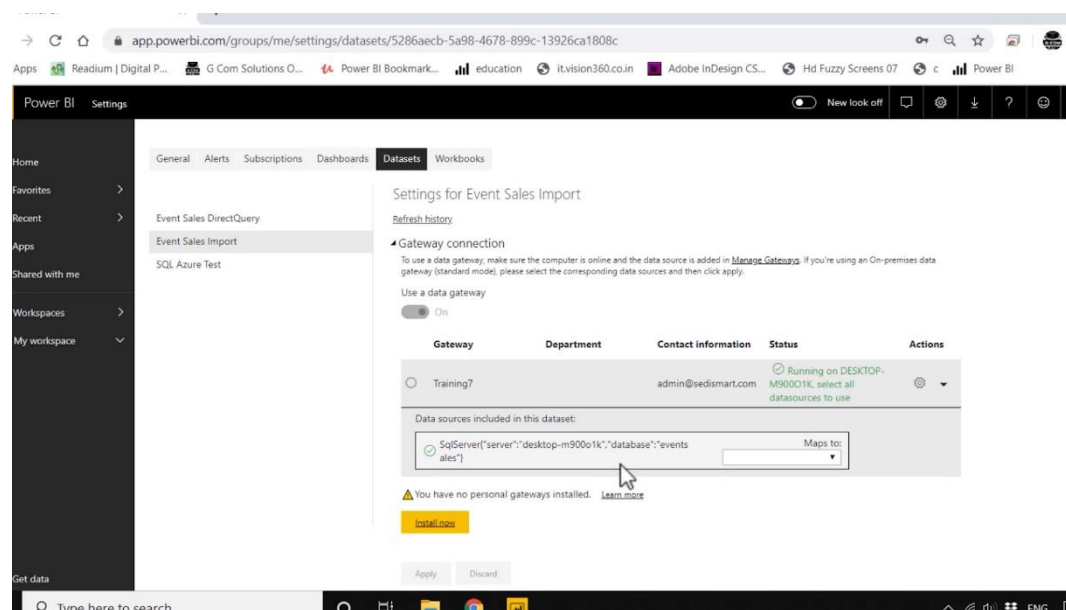


This setup allows seamless integration of on-premises data with Power BI Service, ensuring reports remain up-to-date and interactive while maintaining secure access to corporate databases.

0411: Creating Refresh Schedules

When working in Import Mode, data is refreshed using the Data Gateway, ensuring it stays in sync with the on-premises source. To configure this, we begin by accessing the settings of the relevant semantic model—in this case, the Event Sales Import semantic model.

The system automatically detects the Data Gateway, and we confirm that our SQL Server connection correctly maps to the Event Sales database. Once verified, we click Apply to save these settings.



Scheduling the Refresh

Next, we scroll down to the Scheduled Refresh section. Here, we define the refresh frequency—for example, setting it to daily.

We can then specify up to eight refresh times throughout the day. To add multiple refresh slots, we simply click Add Another Time for each additional refresh point.

If the semantic model is hosted in a Premium workspace, the scheduling limit increases significantly, allowing up to 48 refresh times per day.

Finally, we click Apply to confirm our schedule, ensuring that our semantic model stays current with the latest available data.

0411-creating Refresh Schedules

▼ Daily ▼

Time zone

(UTC) Dublin, Edinburgh, Lisbon, London ▼

Time

6 ▼ 30 ▼ AM ▼ X

10 ▼ 00 ▼ AM ▼ X

[Add another time](#)

☒ Send refresh failure notifications to the dataset owner

Email these users when the refresh fails

Enter email addresses

Apply Discard

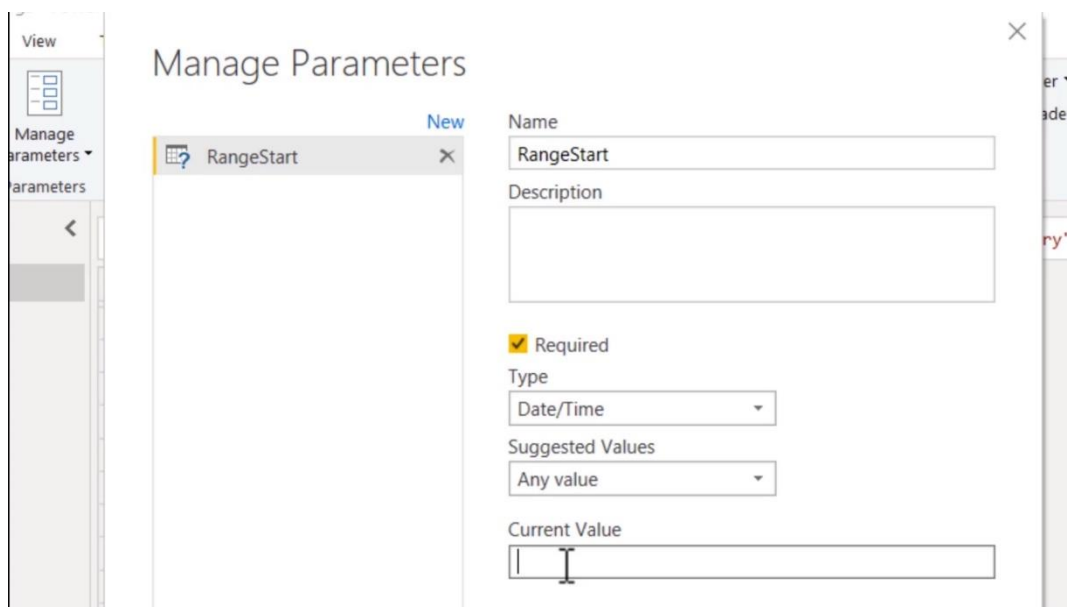
0412: Implementing Incremental Refresh

When working with extremely large semantic models, refresh performance can become a challenge. As the refresh duration increases, there is a risk that the scheduled refresh cannot be completed within the allocated time. The key issue is that by default, refresh is an all-or-nothing process—every time a refresh is triggered, the entire semantic model or dataflow is refreshed.

Incremental refresh addresses this problem by allowing only the data that has changed to be refreshed. Data that remains unchanged is left as is, significantly reducing the amount of data being processed and making the refresh more efficient.

Setting Up Incremental Refresh

To configure incremental refresh, the first step is to define two required parameters within Power Query. These parameters act as placeholders for defining the date range of the data to be refreshed.



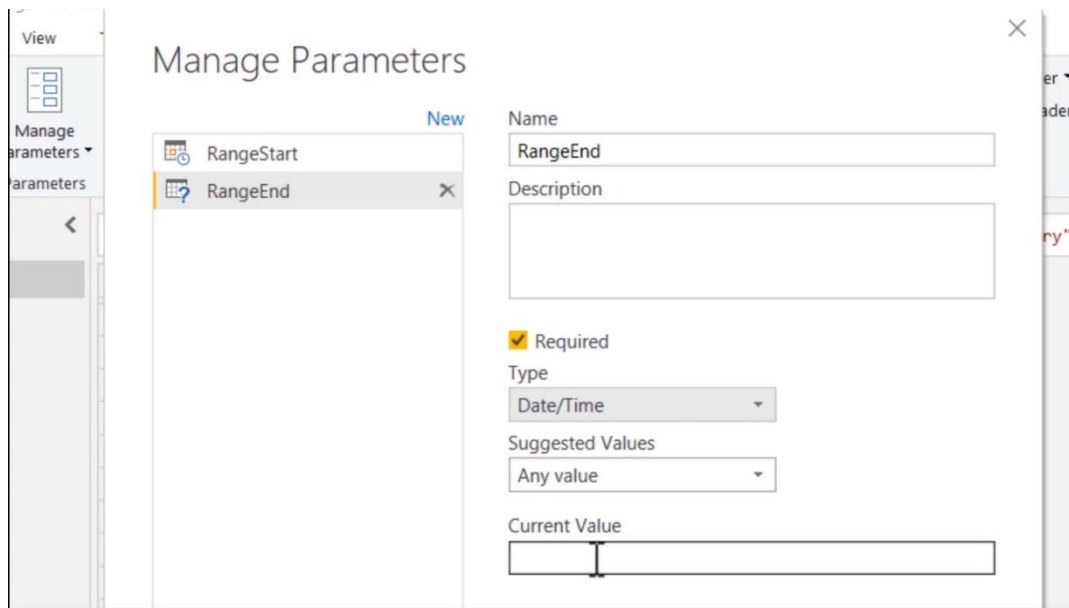
The screenshot shows the 'Manage Parameters' window in Power Query. On the left, a sidebar contains a 'View' button and a 'Manage parameters' dropdown. The main area has a list of parameters with 'RangeStart' selected. To the right of the list, the details for 'RangeStart' are shown: 'Name' is 'RangeStart', 'Description' is empty, 'Required' is checked, 'Type' is 'Date/Time', 'Suggested Values' is 'Any value', and 'Current Value' is empty.

Creating Parameters

Begin by opening Power Query and navigating to Manage Parameters. Click New Parameter, and create two parameters with specific names:

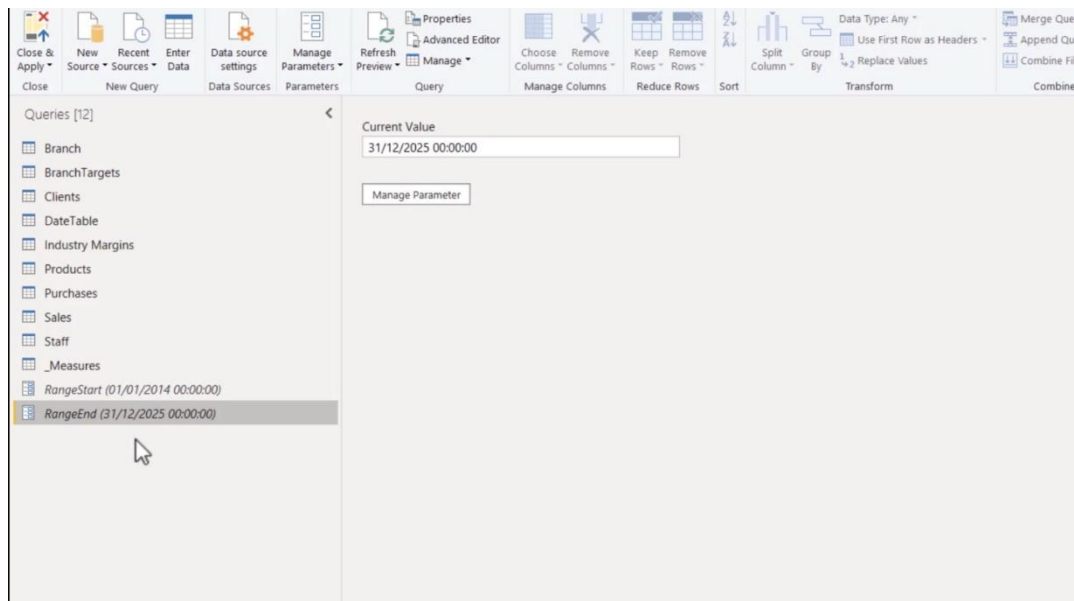
RangeStart (must be written exactly as shown, without spaces)

RangeEnd (must also be written exactly as shown, without spaces)



Once the first parameter is created, create the second parameter in the same way, ensuring both are correctly named and formatted.

Both parameters should have their data type set to Date/Time. To ensure that there is some initial data when the query is applied, provide temporary values.



Applying the Date Filters

With the parameters set up, the next step is to apply them to filter the semantic model. Typically, this filter is applied to the main fact table, which contains transactional data and is subject to frequent updates.

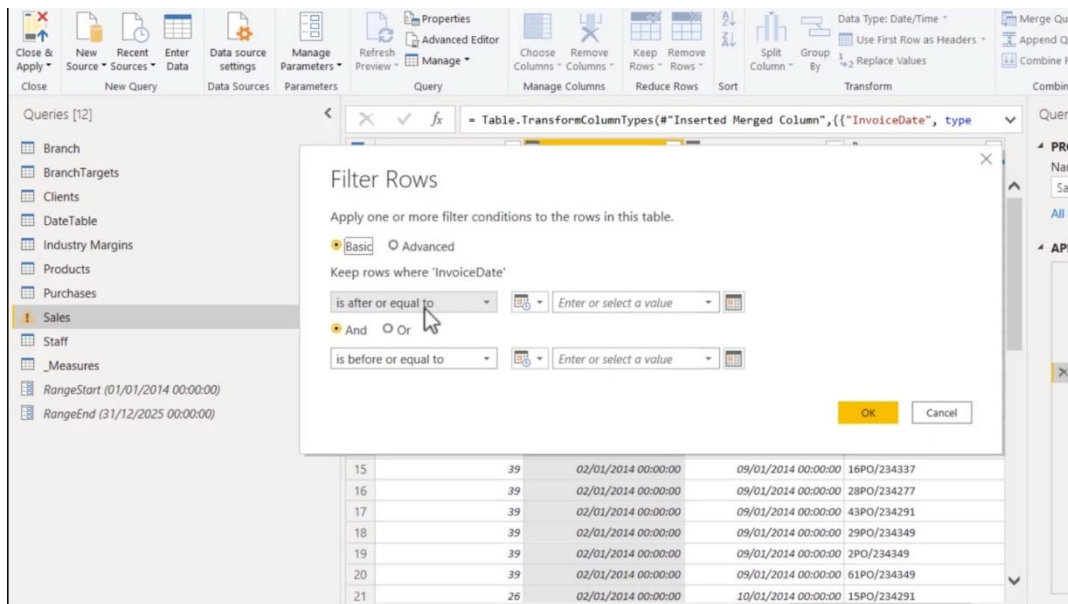
Identify the relevant Date/Time column—in our example Invoice Date—and apply a filter using Date/Time Filters → Between. Set the filter conditions as follows:

Keep rows where Invoice Date is greater than or equal to RangeStart

0412-implementing Incremental Refresh

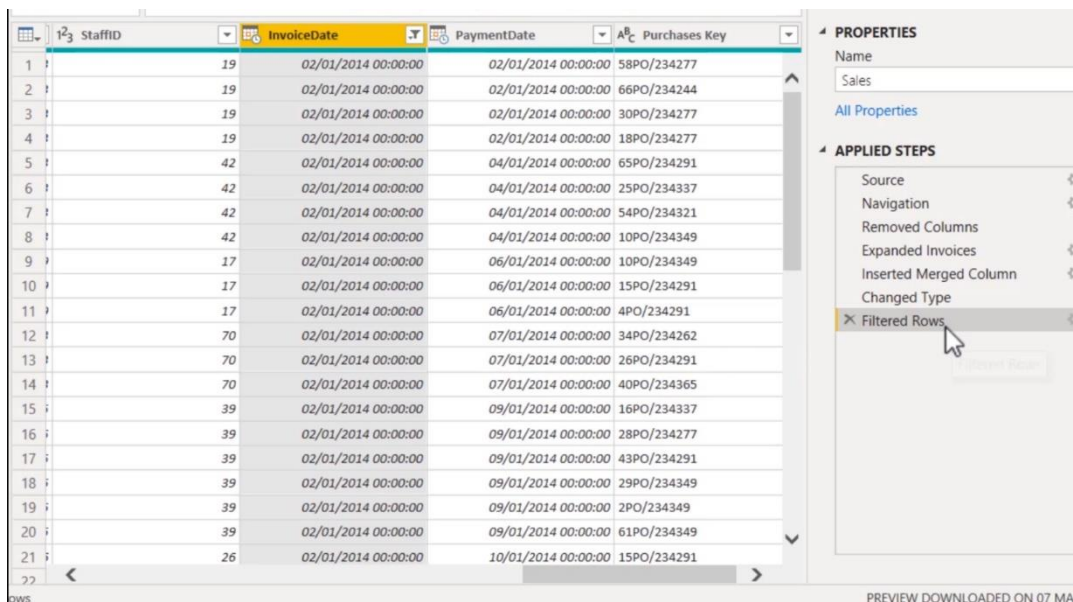
Keep rows where Invoice Date is less than or equal to RangeEnd

This step does not permanently restrict the data but instead establishes a mechanism that Power BI will later use when applying the incremental refresh policy.



Ensuring Query Folding

An important consideration when implementing incremental refresh is query folding. Query folding ensures that Power BI translates Power Query transformations into an efficient native query for the underlying data source, such as SQL Server.

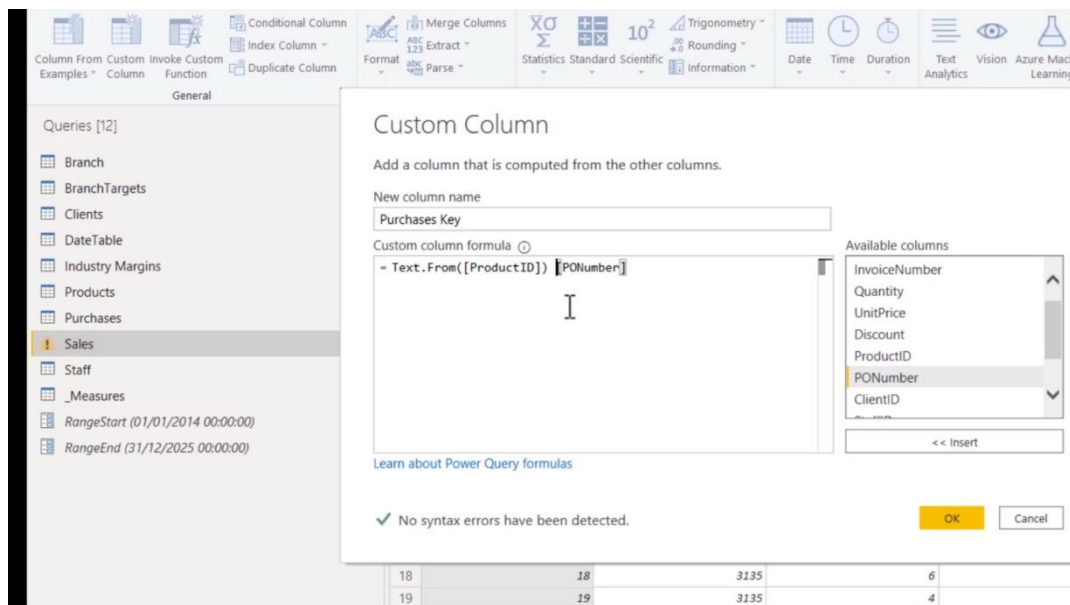


0412-implementing Incremental Refresh

	StaffID	InvoiceDate	PaymentDate	Purchases Key
1	19	02/01/2014	02/01/2014	58PO/234277
2	19	02/01/2014	02/01/2014	66PO/234244
3	19	02/01/2014	02/01/2014	30PO/234277
4	19	02/01/2014	02/01/2014	18PO/234277
5	42	02/01/2014	04/01/2014	65PO/234291
6	42	02/01/2014	04/01/2014	25PO/234337
7	42	02/01/2014	04/01/2014	54PO/234321
8	42	02/01/2014	04/01/2014	10PO/234349
9	17	02/01/2014	06/01/2014	10PO/234349
10	17	02/01/2014	06/01/2014	15PO/234291
11	17	02/01/2014	06/01/2014	4PO/234291
12	70	02/01/2014	07/01/2014	34PO/234262
13	70	02/01/2014	07/01/2014	26PO/234291
14	70	02/01/2014	07/01/2014	40PO/234365
15	39	02/01/2014	09/01/2014	16PO/234337
16	39	02/01/2014	09/01/2014	28PO/234277
17	39	02/01/2014	09/01/2014	43PO/234291
18	39	02/01/2014	09/01/2014	29PO/234349
19	39	02/01/2014	09/01/2014	2PO/234349
20	39	02/01/2014	09/01/2014	61PO/234349
21	26	02/01/2014	10/01/2014	15PO/234291

To check for query folding, right-click the final step in Power Query and select View Native Query. If this option is greyed out, it indicates that query folding has been broken at some point in the transformations.

If query folding is broken, backtrack through the steps to identify the cause. In many cases, certain transformations—such as complex column merges—can interrupt query folding. Where necessary, rework transformations to use simpler, native SQL-compatible operations, such as creating calculated columns using straightforward concatenation rather than complex custom formulas.



Configuring the Incremental Refresh Policy

Once the Power Query steps are complete, close and apply the changes, then set up the incremental refresh policy.

In the Fields pane, right-click the fact table (e.g., Sales table) and select Incremental Refresh. Toggle the feature on and note that, once published to the Power BI service, the semantic model cannot be downloaded back into Power BI Desktop.

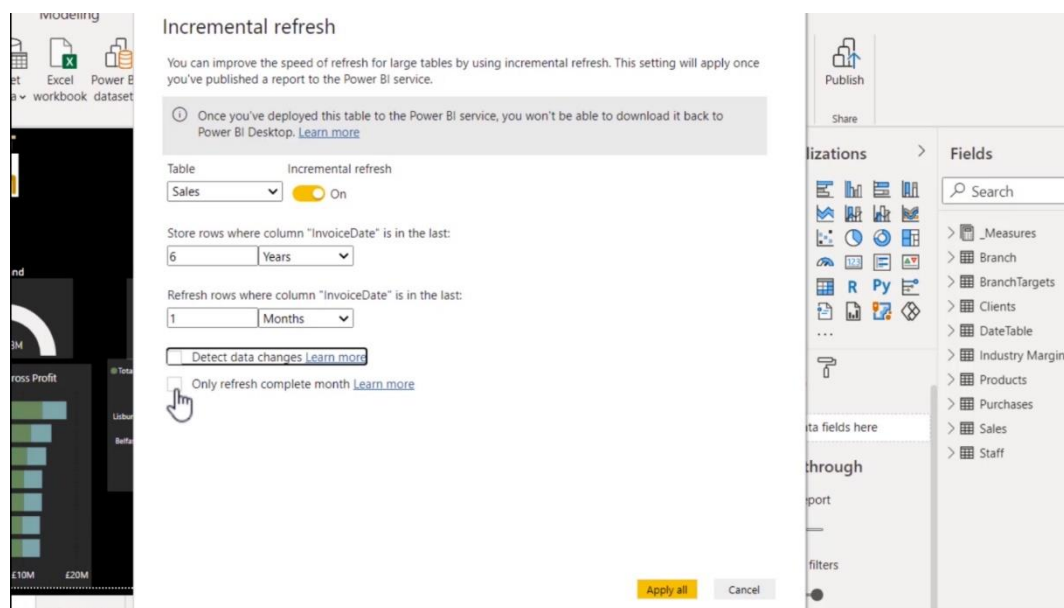
Defining the Storage and Refresh Period

The policy requires defining:

Data Retention Period – The amount of historical data to store in the semantic model (e.g., six years).

Refresh Period – The timeframe for which data will be refreshed (e.g., the last month).

The refresh period should align with how frequently new data is added. For example, if data is updated daily but historical records remain unchanged, refreshing only the past month may be an optimal balance.



Optional Refresh Settings

There are two optional settings to consider:

Detect data changes – If a column exists that reliably indicates when a row has been modified, it can be used to further refine the refresh process. In this example, the semantic model only contains Invoice Date and Payment Date, so this option is left off.

Only refresh complete periods – If partial data refreshes could lead to inconsistencies (e.g., incomplete monthly figures), enabling this option ensures that only fully completed periods are refreshed. If this is not a concern, leave it unchecked.

Once all settings are configured, click Apply to finalize the incremental refresh policy.

Conclusion

Configuring incremental refresh in Power BI Desktop can significantly improve refresh performance for large semantic models. By selectively updating only recent changes, the process becomes much

0412-implementing Incremental Refresh

more efficient, reducing load times and ensuring that reports remain up to date without excessive processing overhead.

0413: Large Semantic model Storage Format

When working with Power BI Desktop, it's important to understand the upload limits for semantic models. By default, when you upload a file to Power BI, the maximum size allowed is 10 gigabytes. However, if you're using a Power BI Pro licence and uploading to a shared workspace, that limit is significantly lower—just one gigabyte. Unfortunately, there's no way to increase this one-gigabyte limit for shared workspaces.

For premium workspaces, however, Power BI now provides a way to exceed the 10-gigabyte limit. This enables semantic models to scale significantly, potentially reaching the limits available in Azure Analysis Services—currently up to 400 gigabytes. The key to achieving this is the Large Semantic model Storage Format, a setting that can be applied either at the workspace level or the semantic model level.



Applying Large Semantic model Storage Format at the Workspace Level

To enable Large Semantic model Storage Format at the workspace level, follow these steps:

Navigate to Workspaces in the Power BI Service.

Click on the ellipsis menu next to the desired workspace.

Select Workspace Settings.

Go to the Premium tab.

At this stage, if a dedicated capacity has not yet been assigned, you'll need to do so. Power BI premium capacity can be provided by one of three products:

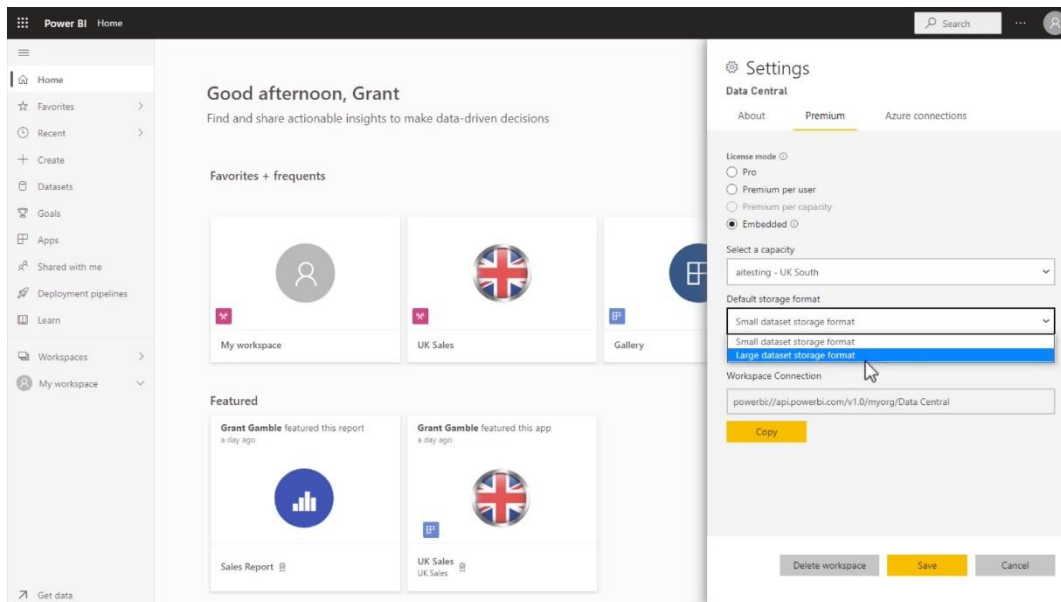
Premium Per User

0413-large Semantic model Storage Format

Premium Per Capacity

Power BI Embedded

In this example, we'll use Power BI Embedded, with a single capacity running called AI Testing. Once assigned, we now have the option to switch the default storage format from Small Semantic model Storage Format to Large Semantic model Storage Format.



It's important to note that switching to Large Semantic model Storage Format at the workspace level only affects new semantic models created from that point onward. Existing semantic models will not be impacted by this change. In this case, we'll leave the setting on Small Semantic model Storage Format and click Save.

Applying Large Semantic model Storage Format at the Semantic model Level

If you need to apply Large Semantic model Storage Format to an existing semantic model, you can do so at the semantic model level by following these steps:

Navigate to the workspace containing the semantic model (in this example, Data Central).

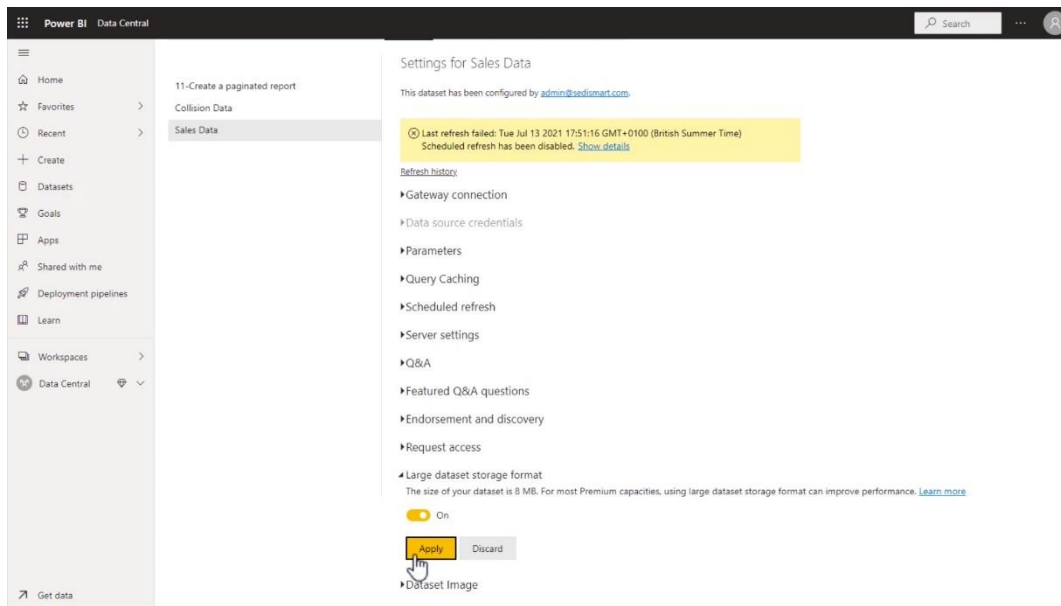
Locate the semantic model, click on the ellipsis menu, and select Settings.

Scroll down to the option for Large Semantic model Storage Format.

Since the semantic model is in a workspace with dedicated capacity, you'll be able to enable this option.

Click Apply and wait for the conversion to complete.

0413-large Semantic model Storage Format



Once the conversion is successful, you'll see a confirmation message. At this point, the semantic model will no longer be downloadable in its entirety. Instead, it will be able to expand dynamically by refreshing the data and appending new information over time.



Enabling Scheduled Refresh

After switching to Large Semantic model Storage Format, you can configure scheduled refresh to keep the semantic model up to date:

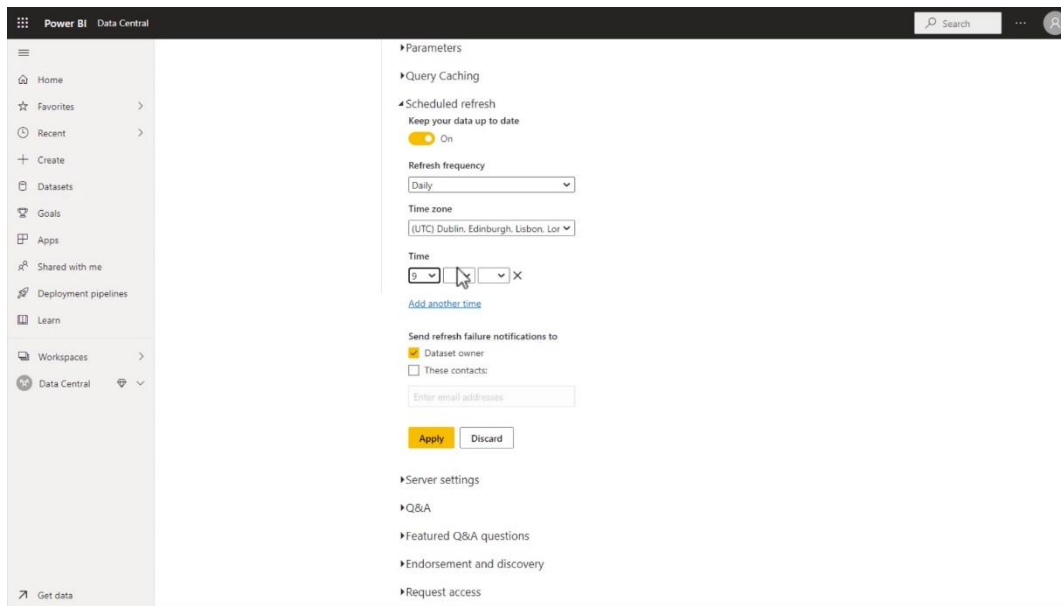
In the semantic model settings, locate the Scheduled Refresh section.

Click Add Another Time.

Select the desired refresh time for the semantic model.

In this example, we'll configure the semantic model to refresh once per day at a specific time. This ensures the semantic model remains up to date without requiring manual intervention.

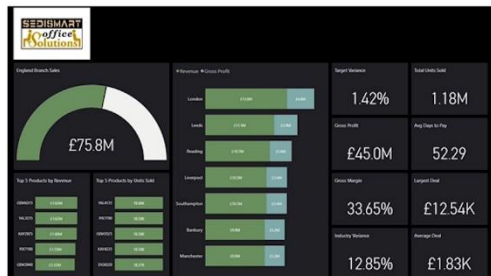
0413-large Semantic model Storage Format



By enabling Large Semantic model Storage Format and configuring scheduled refresh, you can effectively manage large-scale semantic models in Power BI, allowing for continuous data expansion while maintaining performance.

0500: Overview of Content Administration

Power BI content is organised within workspace apps and consists of three key elements: reports, dashboards, and Excel workbooks. Each of these serves a distinct purpose and has its own role in content administration.



Dashboards

- Created in Power BI service
- Single page
- Dependent on reports
- ~~No slicers~~
- ~~No cross-filtering or cross-highlighting~~
- ~~No drill-through and drill-down~~
- Additional tiles available
- Q&A natural language questions



Reports

- Created in Power BI Desktop
- Multipage
- Self-contained
- Slicers and filtering
- Cross-filtering and cross-highlighting
- Drill-through and drill-down

Reports, Dashboards, and Excel Workbooks

Name	Type	Owner	Refreshed
England	Dashboard	UK Sales	—
Expenses 2014-19	Workbook	UK Sales	10/08/20, 11:12:03
Sales Overview	Dashboard	UK Sales	—
Sales Report	Report	UK Sales	05/08/20, 16:03:44
Sales Report	Dataset	UK Sales	05/08/20, 16:03:44

Reports, created in Power BI Desktop, form the primary repository for Power BI content. They can be as detailed as required and may contain an unlimited number of pages. In contrast, dashboards are built exclusively in the Power BI service and are limited to a single page, though this can be scrollable and accommodate substantial content. Dashboards are often used to summarise key information

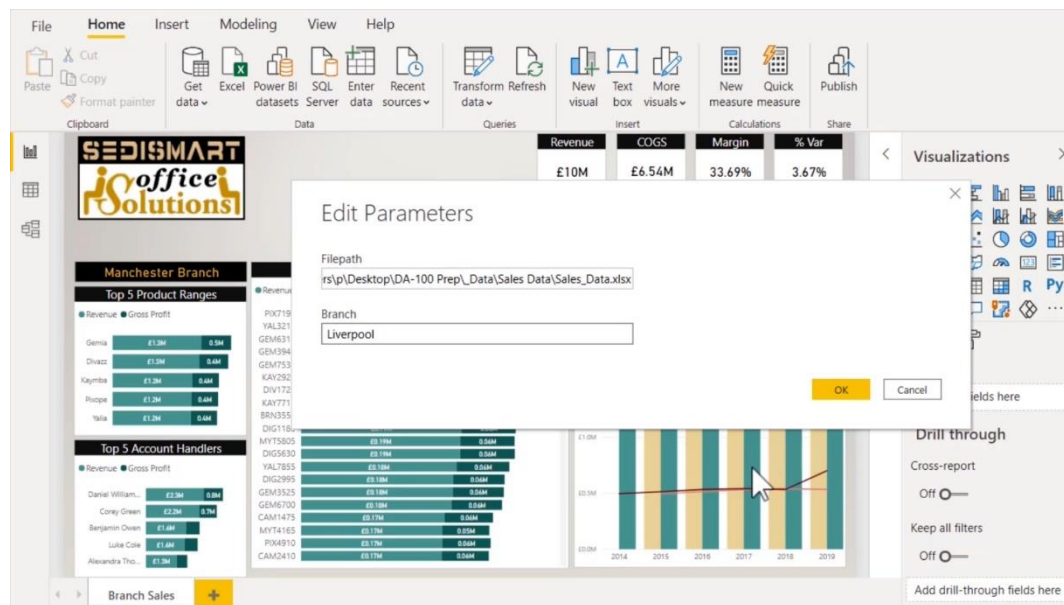
from multiple reports and can also display real-time streaming data, making them useful in scenarios where reports may not be sufficient.

The ability to upload Excel workbooks into Power BI workspaces is a somewhat unusual feature. Typically, Power BI adoption aims to move organisations away from Excel-based reporting, given Power BI's superior capabilities. However, if your organisation has an extensive library of Excel reports and assets, it may be practical to continue using them alongside Power BI content, at least in the short term.

Standardising Content with Templates

As an administrator, ensuring consistency in report design across your organisation is important. A straightforward way to achieve this is through the use of templates. Templates offer multiple benefits: they save time, prevent duplication of effort, and help avoid errors and omissions. More experienced report developers can create templates, while less experienced users can utilise them to maintain consistency in reporting.

Beyond design, templates can also help standardise data manipulation. Their usefulness can be further enhanced by implementing query parameters. Parameters allow a report's content to be dynamic, based on user-defined values. When a report built from a template contains parameters, developers must specify values when opening the template, ensuring that only relevant data is loaded.



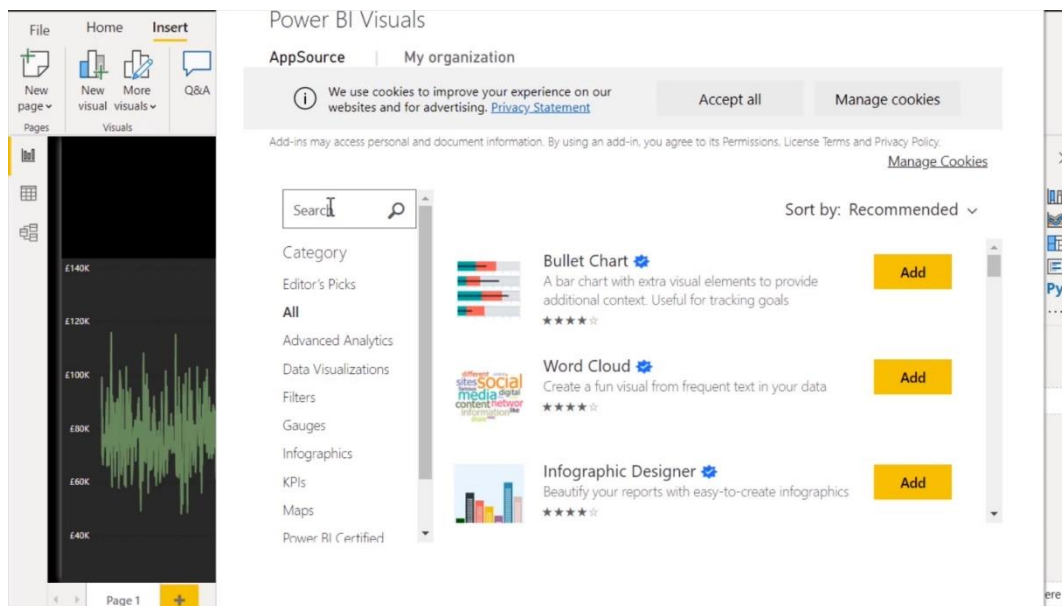
Enhancing Reports with Custom Visuals

Another way to improve Power BI content is through custom visuals. These visuals provide additional functionality beyond what is available in standard Power BI visualisations, while still behaving similarly in terms of formatting and interactions.

To maintain governance and consistency, administrators can manage the use of custom visuals through the Power BI Certified Visuals feature. This allows organisations to specify which custom visuals report developers are permitted to use, ensuring that only approved visuals are included in reports.

Ensuring Trustworthy Content

A crucial aspect of content administration is ensuring the trustworthiness of reports. Users need confidence that the content they rely on is accurate and validated. One way to achieve this is by enabling report certification. This feature, managed within export and sharing settings, allows a designated group—typically administrators and senior report developers—to certify reports. Certified reports display a badge, signalling to users that the content meets internal quality standards.



Another mechanism for maintaining control over content is the use of sensitivity labels. This feature, available within Microsoft 365's Information Protection settings, allows users to label reports according to their sensitivity level. By applying these labels, organisations can ensure that appropriate security measures are in place, protecting sensitive data while enabling effective collaboration.

0501: Reports and Dashboards Compared

The term "dashboard" is not unique to Power BI—it's a widely used concept across various platforms. For example, Excel dashboards have been around for years. However, in the Power BI environment, the term has a very specific meaning. One of the most common areas of confusion is the distinction between a report and a dashboard, as many people use the terms interchangeably. It's not uncommon to hear someone say they are creating a "Power BI dashboard" when they are actually building a Power BI report. So, let's take a structured look at the key differences between the two.

Where Dashboards and Reports Are Created

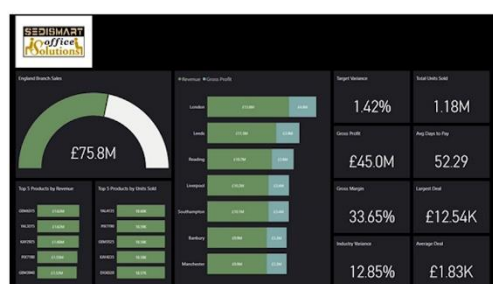
A fundamental difference lies in where dashboards and reports are built. Dashboards can only be created in the Power BI Service. By contrast, reports are primarily created in Power BI Desktop. This distinction is important because it influences how users interact with and manage these elements.

Purpose and Structure

Dashboards and reports serve different purposes. Dashboards provide a single-page overview designed for busy users who need to grasp key insights at a glance. Everything relevant is consolidated on a single screen, making dashboards ideal for high-level monitoring.

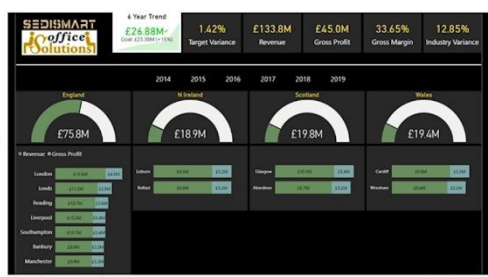
Reports, on the other hand, are much more detailed and typically span multiple pages. They allow for in-depth exploration, making them suitable for users who need to analyze data in more depth.

The Relationship Between Dashboards and Reports



Dashboards

- Created in Power BI service
- Single page
- Dependent on reports
- ~~No slicers~~
- ~~No cross-filtering or cross-highlighting~~
- ~~No drill-through and drill-down~~
- Additional tiles available
- Q&A natural language questions



Reports

- Created in Power BI Desktop
- Multipage
- Self-contained
- Slicers and filtering
- Cross-filtering and cross-highlighting
- Drill-through and drill-down

Dashboards are dependent on reports. Most of the content displayed on a dashboard comes from one or more reports. If a user interacts with a dashboard tile (such as clicking a visual), they will be taken to the corresponding report page from which that visual was pinned.

Reports, by contrast, are self-contained—they do not rely on dashboards and can function independently.



Interactivity and Filtering

Reports offer more interactivity than dashboards, particularly when it comes to filtering. While individual visuals can be pinned from a report to a dashboard, some elements—such as slicers—cannot be pinned. As a result, filtering options are only available within the report itself.

Similarly, features like cross-filtering and cross-highlighting can only be applied within a report page. Dashboards do not support these interactive filtering options.

The same restriction applies to drill-through and drill-down capabilities. These features, which allow users to explore data at different levels of detail, are exclusive to reports and are not available on dashboards.

Unique Features of Dashboards

Despite these limitations, dashboards have features that are not available in reports. For example, dashboards can include real-time data tiles, videos, and images, none of which can be created within a report.

Another dashboard-only feature is Q&A, which allows users to enter natural language questions and receive automatically generated visuals based on Power BI's built-in AI capabilities. This makes dashboards particularly useful for self-service analytics.

Blurring the Lines: The Pin Live Page Feature

There is one feature that blurs the distinction between dashboards and reports: the Pin Live Page feature. This allows an entire report page to be pinned to a dashboard rather than just individual visuals. When this is done, the resulting dashboard contains much of the interactivity of the original report page, including filtering, cross-filtering, and drill-down. However, some features—such as drill-through and tooltip pages—are not retained.

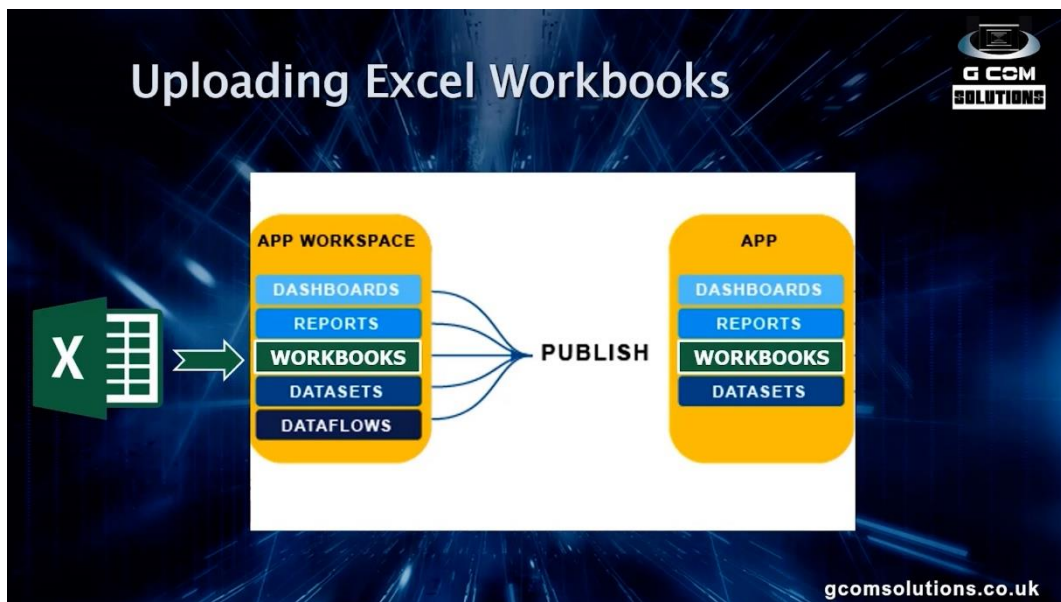


For users looking to create a dashboard in the traditional sense (i.e., a multi-page interactive experience rather than Power BI's specific definition), using the Pin Live Page feature provides the closest equivalent.

0502: Uploading Excel Workbooks

In Power BI, an app can include three types of content: dashboards, reports, and Excel workbooks. Importantly, when we talk about uploading an Excel workbook, we mean the entire file—worksheets, pivot tables, charts, and formatting—not just the raw data. However, there are a couple of limitations: VBA macros and Power Pivot data models are not included when a workbook is uploaded.

Uploading Excel workbooks is particularly valuable during the early stages of Power BI adoption. Many organisations encounter resistance from experienced Excel users who prefer to continue using their familiar tools for data analysis and reporting. Allowing these users to upload their Excel solutions into Power BI app workspaces helps maintain engagement and encourages them to transition gradually, rather than feeling forced into an entirely new way of working.

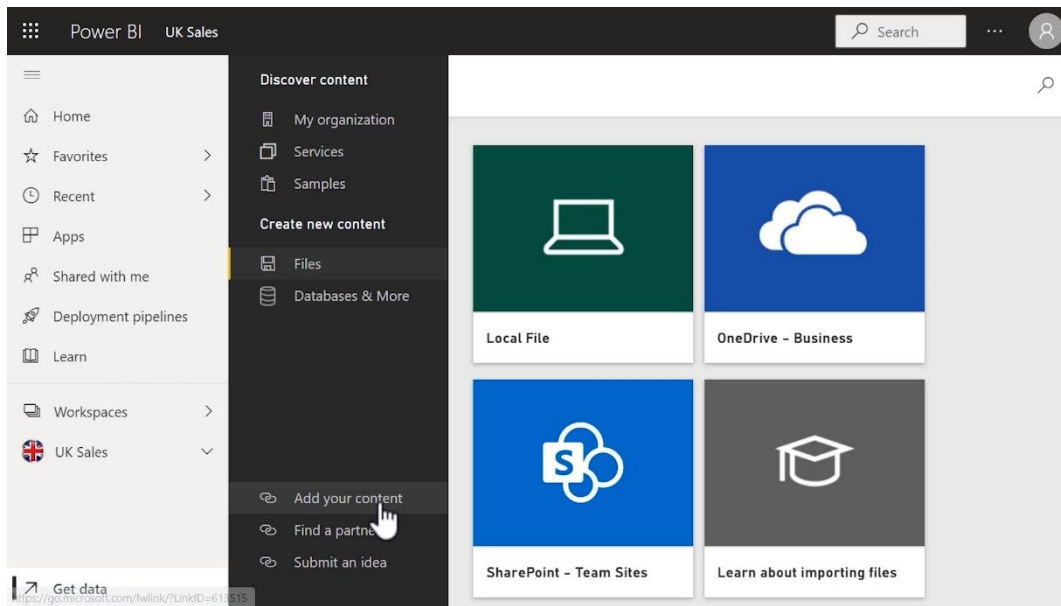


Uploading an Excel Workbook

To upload an Excel workbook into the Power BI service, start by navigating to the relevant workspace. In this example, we'll use the UK Sales workspace.

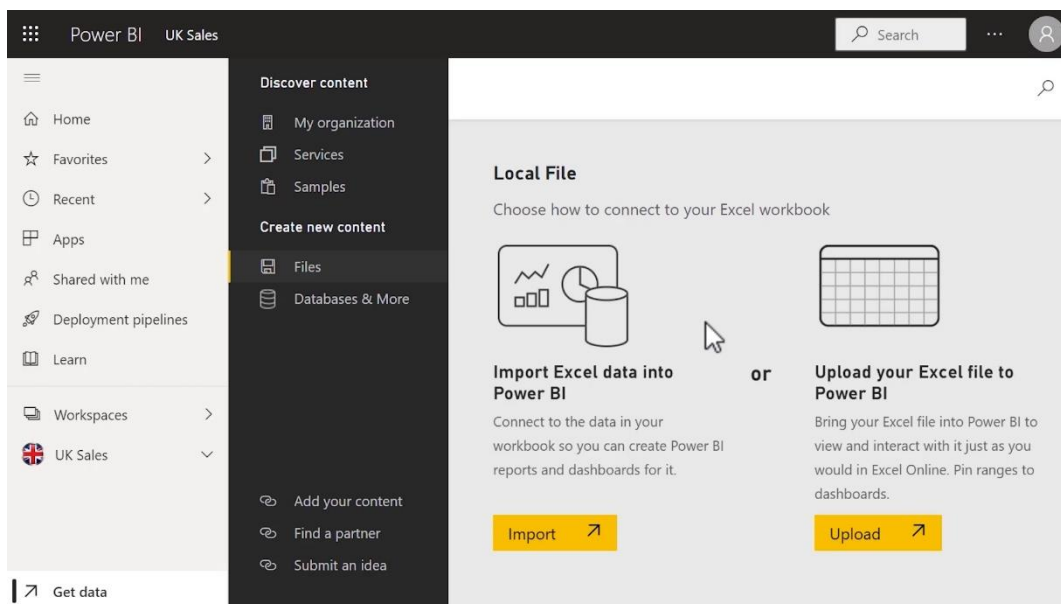
1. In the bottom left corner of the screen, click Get Data.
2. Scroll down to the Files option and click Get.
3. Choose the file location. Power BI allows you to upload from various sources, but in this case, we'll select Local File.
4. Navigate to the relevant folder and double-click on the desired workbook (e.g., expenses 2014 to 2019).

0502-uploading Excel Workbooks



5. At this stage, Power BI offers two options: Import (which extracts the data from the workbook) or Upload (which keeps the workbook intact). Since we want to retain the workbook structure, we choose Upload.

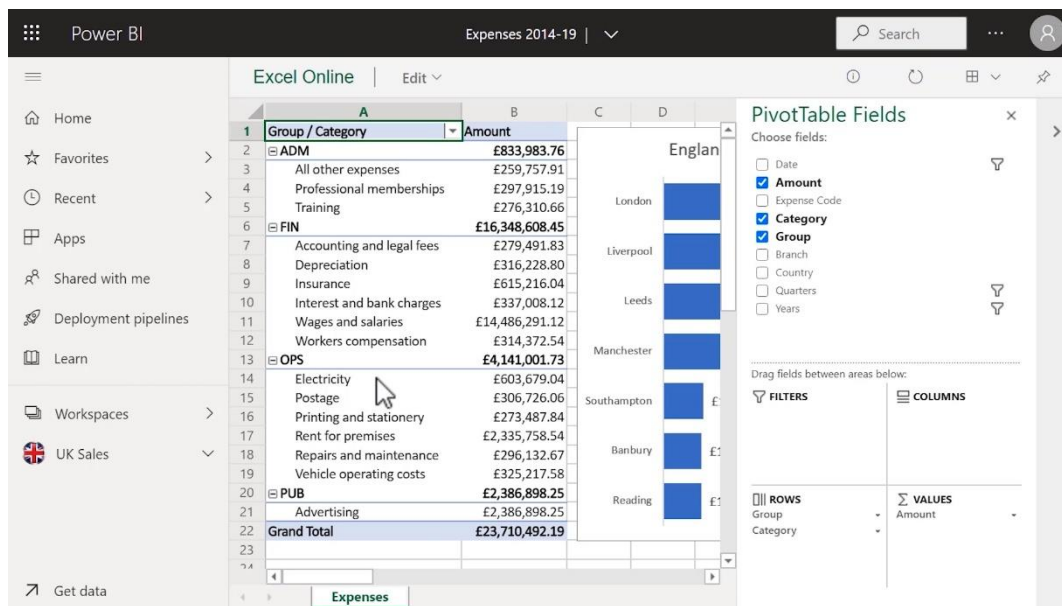
Interacting with an Uploaded Workbook



Once uploaded, the workbook appears in the workspace alongside dashboards and reports. Clicking on the workbook name opens it in Excel Online, where you can view its contents. Although it opens in read-only mode, users can make personal changes to the workbook during their session. These changes persist only for the duration of the session and do not affect the original file.

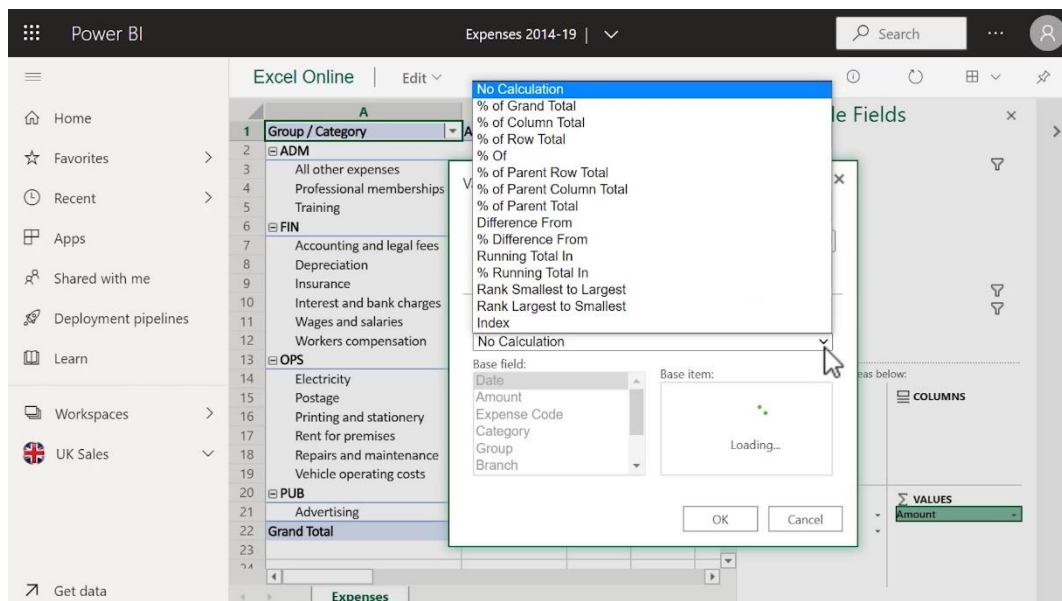
For example, let's say we want to display percentages instead of raw values in a pivot table:

0502-uploading Excel Workbooks



In the PivotTable Fields pane, double-click on the relevant field (e.g., Amount).

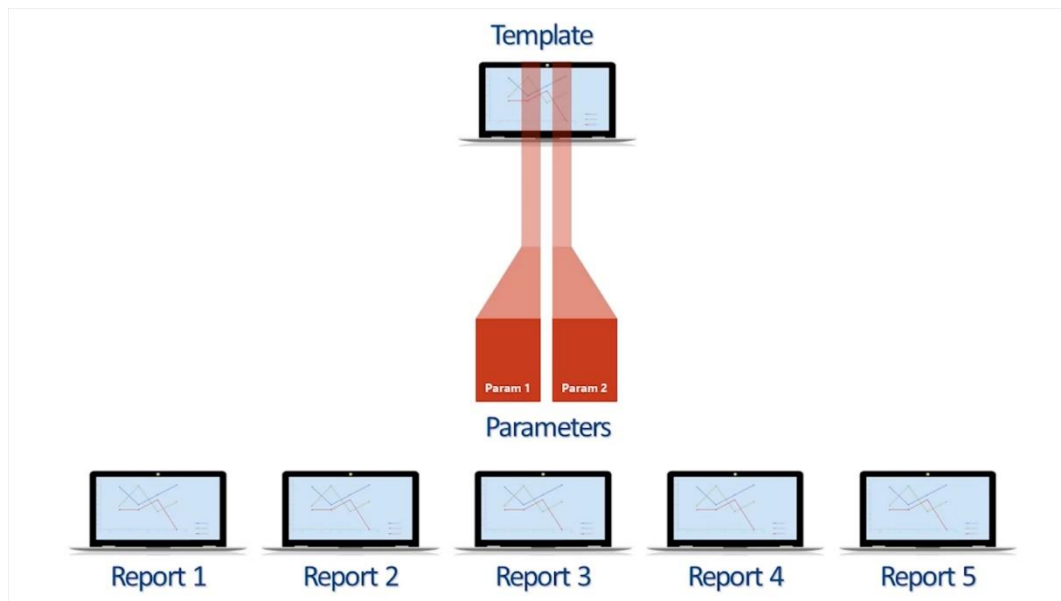
Navigate to Show Values As and select % of Column Total (or another relevant option).



Why This Matters

This feature is a great way to bridge the gap between Excel and Power BI. It provides a familiar environment for Excel users while also introducing them to Power BI's ecosystem. By allowing workbooks to be uploaded directly into an app workspace, organisations can ease the transition and build confidence among users who may be hesitant to move away from Excel-based reporting.

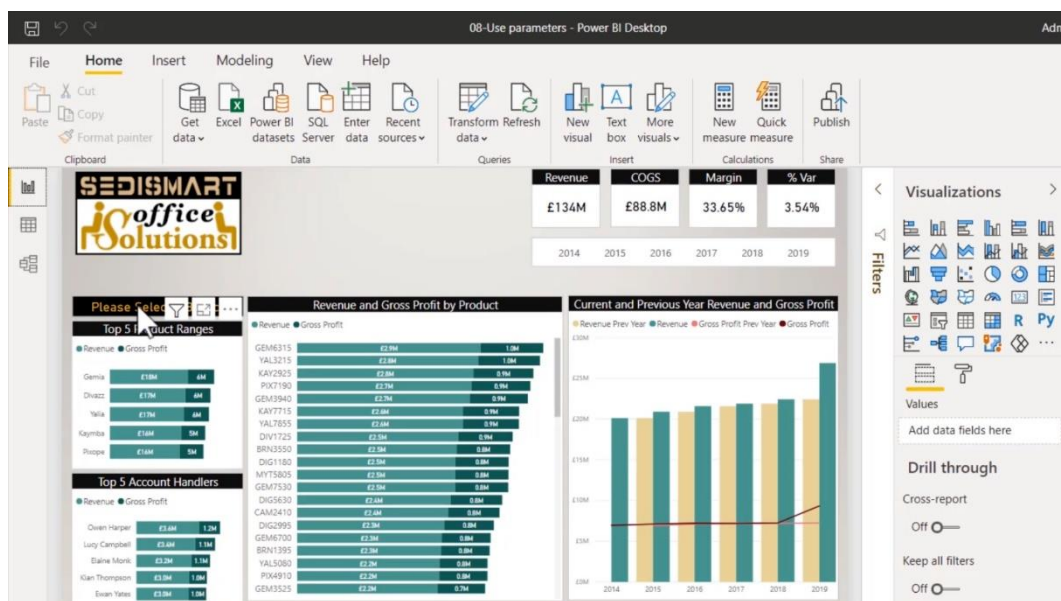
0503: Creating a Template with Parameters



Parameters allow you to control key aspects of your data dynamically—such as filtering rows or changing data sources—without modifying the underlying queries manually. This feature is particularly useful when working with templates, as it ensures that each time a template is used, users can input specific values for predefined parameters, dynamically shaping the loaded data.

Defining a Parameter for the Data Source

In our example, we have a sales report that focuses on a specific branch. To ensure that our queries work correctly across different environments, we need to parameterize the data source.



0503-creating A Template With Parameters

This involves creating a parameter that stores the file path of the data source, ensuring queries reference a dynamic value rather than a hardcoded path.

To create a parameter in Power Query:

Navigate to the Home tab and select Manage Parameters > New Parameter.

Name the parameter FilePath and set its type to Text.

Choose one of the three data entry methods:

Any Value (manual input)

List of Values (predefined options)

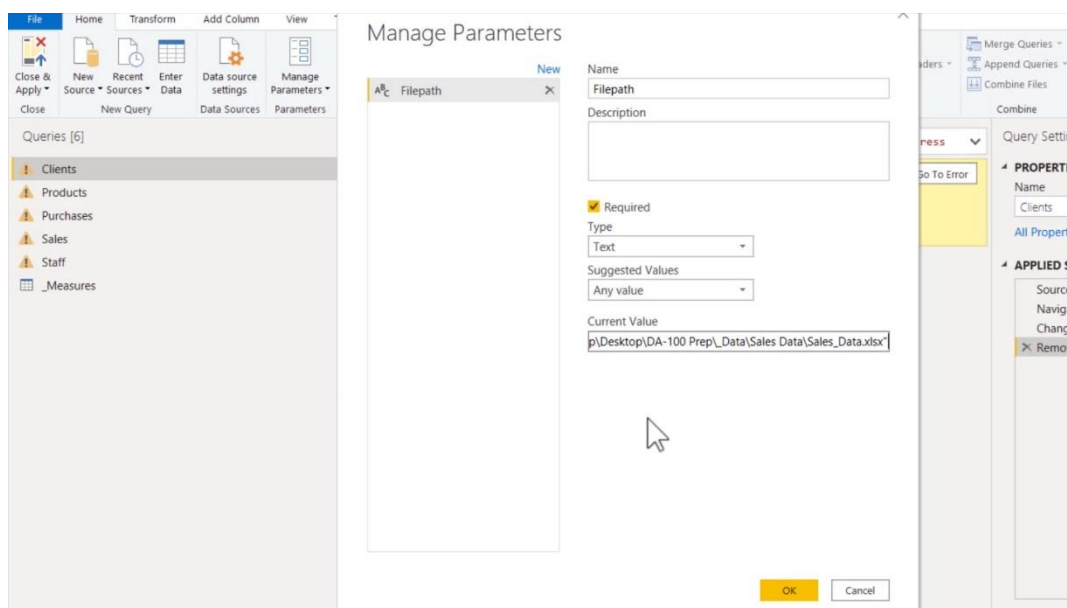
Query (a dynamic list based on another query)

For simplicity, select Any Value and enter the current file path.

To retrieve the correct file path:

Hold down the Shift key, right-click the data file (e.g., sales_data.xlsx), and choose Copy as path.

Paste the path into the Current Value field, ensuring you remove any automatically added double quotes.



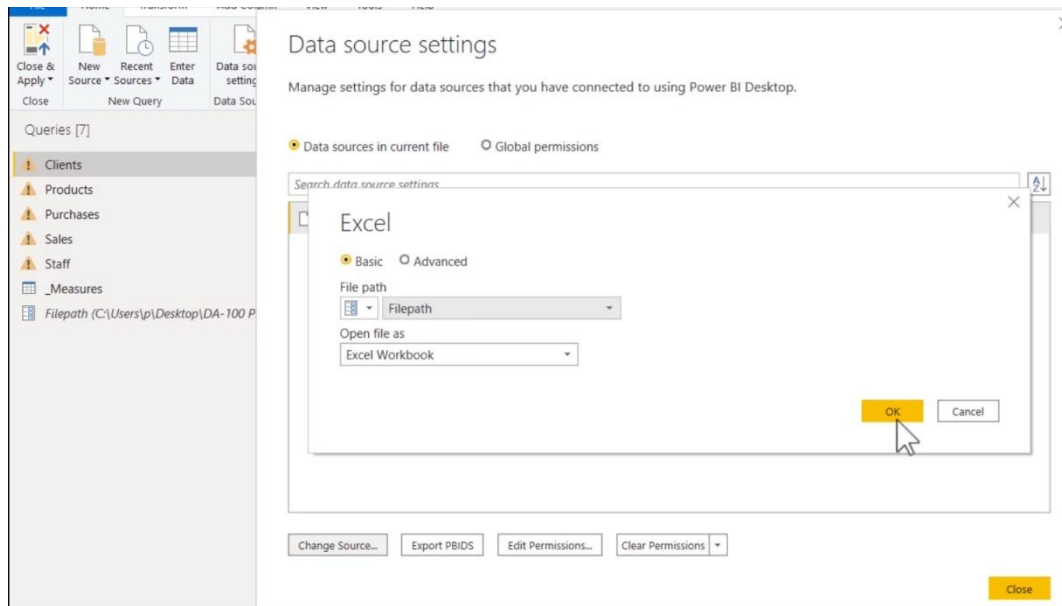
To apply the parameter globally:

Open Data Source Settings and modify the existing file path reference.

Instead of entering a static path, switch the reference type to Parameter and select FilePath.

This ensures all queries update dynamically based on the parameter's value.

0503-creating A Template With Parameters



Creating a Parameter for Branch Selection

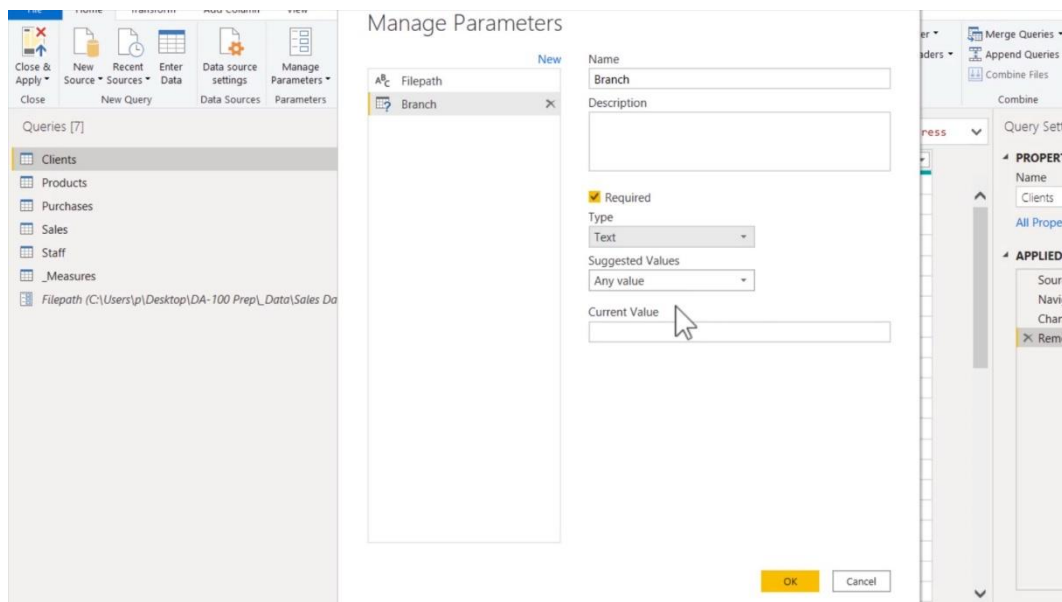
Now that our data source is parameterized, we can create another parameter to filter our report by branch.

Follow the same process to create a new parameter:

Go to Manage Parameters > New Parameter.

Name the parameter Branch and set the type to Text.

Select Any Value, then enter an initial value (e.g., "London").



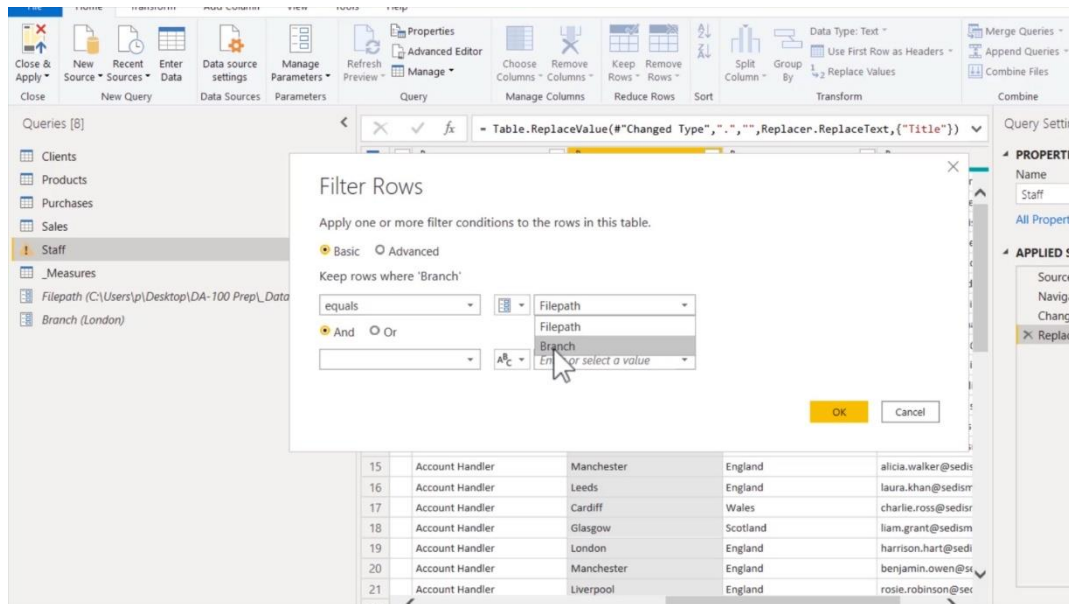
The branch information is stored in the Staff table, so we apply the parameter as a filter:

Navigate to the Branch column in the Staff table.

0503-creating A Template With Parameters

Apply a Text Filter and select Equals.

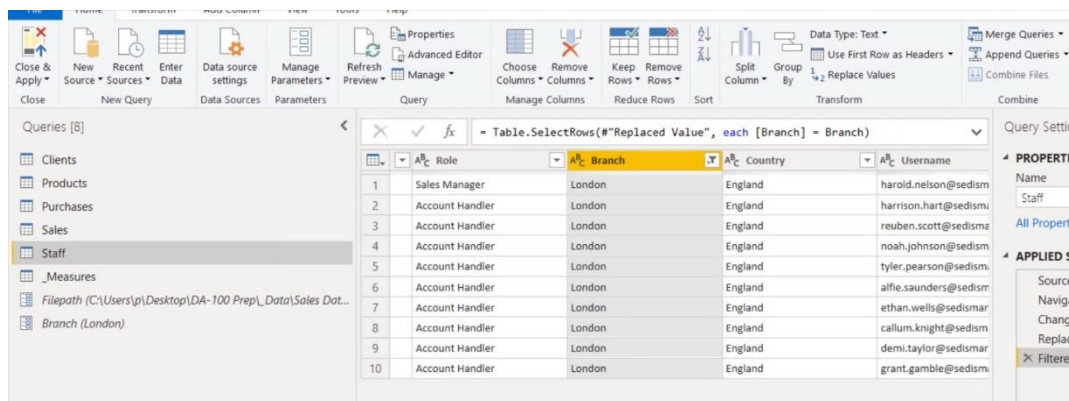
Instead of entering a static value, switch to Parameter and choose Branch.



Now, the report dynamically updates to show only staff from the selected branch. Changing the parameter value (e.g., from "London" to "Manchester") filters the semantic model accordingly.

Ensuring the Filter Affects All Relevant Data

Since the Staff table is a dimension table, applying a branch filter here doesn't automatically impact the Sales table. To ensure consistency, we need to apply the filter across related tables.



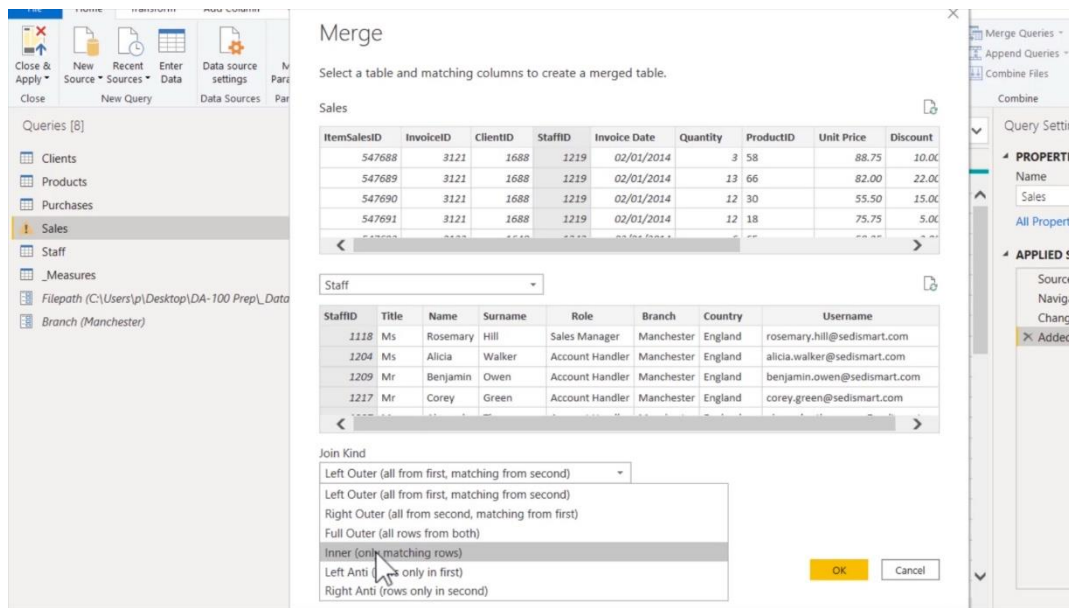
This is achieved using Merge Queries:

Select the Sales table and choose Merge Queries.

Merge it with the Staff table using Staff ID as the common column.

Choose the Inner Join option to keep only matching rows.

0503-creating A Template With Parameters



Click OK to finalize the merge.

This ensures that only sales records associated with staff from the selected branch are included in the report. Any non-matching sales rows are automatically excluded.

0504: Controlling the Use of Custom Visuals

Power BI includes a range of built-in visuals, but Microsoft has also opened the platform to third-party developers, allowing them to create additional custom visuals. These can be made available publicly, either for free or as paid editions. Some custom visuals have both a free version with basic functionality and a paid version with additional features.

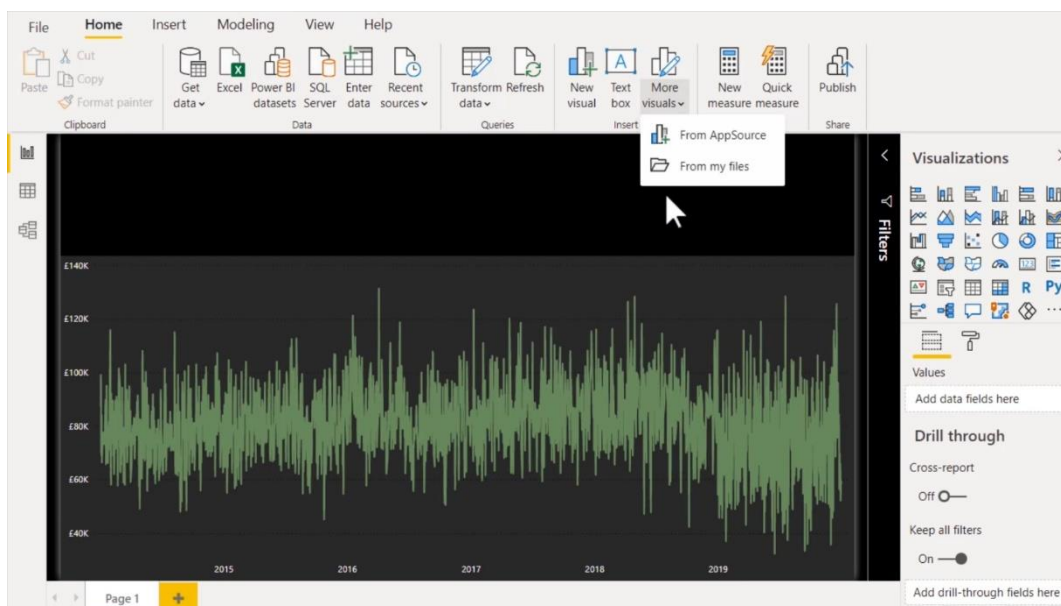
Adding a Custom Visual

Custom visuals can be added from the AppSource or from a local file. The recommended approach is to use AppSource, as this ensures access to the latest version. If an organisation has internal developers who create custom visuals, they can be uploaded from a file, but even in this case, it is advisable to first upload them to AppSource and retrieve them from there.

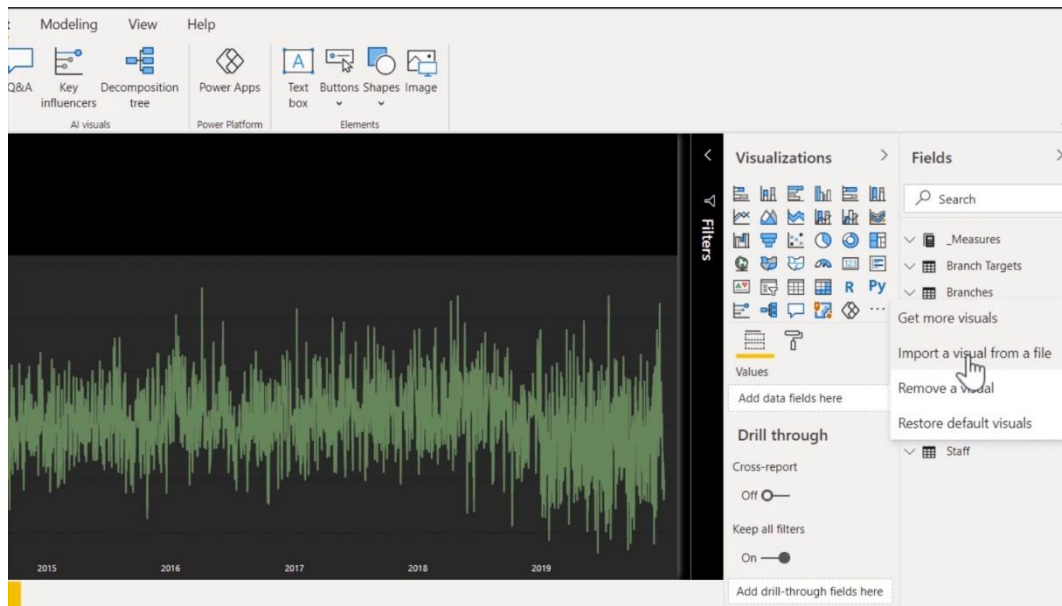
Custom visuals can be added via multiple locations in Power BI Desktop:

The Home tab contains the option More visuals → From AppSource or From My Files.

The Insert tab has the same command.



The Visualisations pane offers Get more visuals through its ellipsis menu, but this only allows importing from a file.



To add a visual, log in with a valid Power BI account, navigate to AppSource, and either browse categories or use the search function. Selecting a visual and clicking Add installs it. Custom visuals appear at the bottom of the Visualisations pane and can be right-clicked to be permanently pinned for easy access.

Configuring and Using Custom Visuals

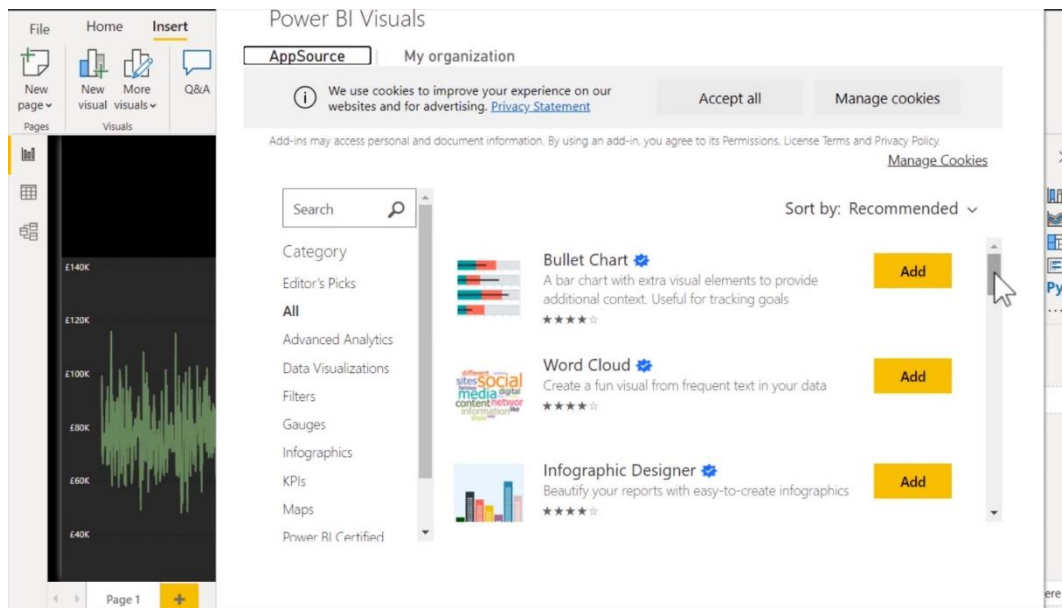
Once added, a custom visual can be placed on a report and formatted via the Format pane. In the case of a Timeline Slicer, for instance, assigning a Date field enables filtering over time, while formatting adjustments (such as colour and label changes) enhance readability.

Certified vs. Non-Certified Custom Visuals

Some custom visuals display a blue starburst icon, indicating that they are Power BI Certified. These visuals meet Microsoft's development standards, ensuring that they do not make external calls as part of their operation.

To filter AppSource results to show only certified visuals:

Use the category list on the left and select Power BI Certified.



Organisations that prioritise security may enforce policies permitting only certified visuals to be used in reports.

Controlling Custom Visuals in Power BI Service

Administrators can manage the use of custom visuals in the Power BI Admin Portal. To access these settings:

Click the ellipsis menu in Power BI Service.

Navigate to Settings → Admin Portal.

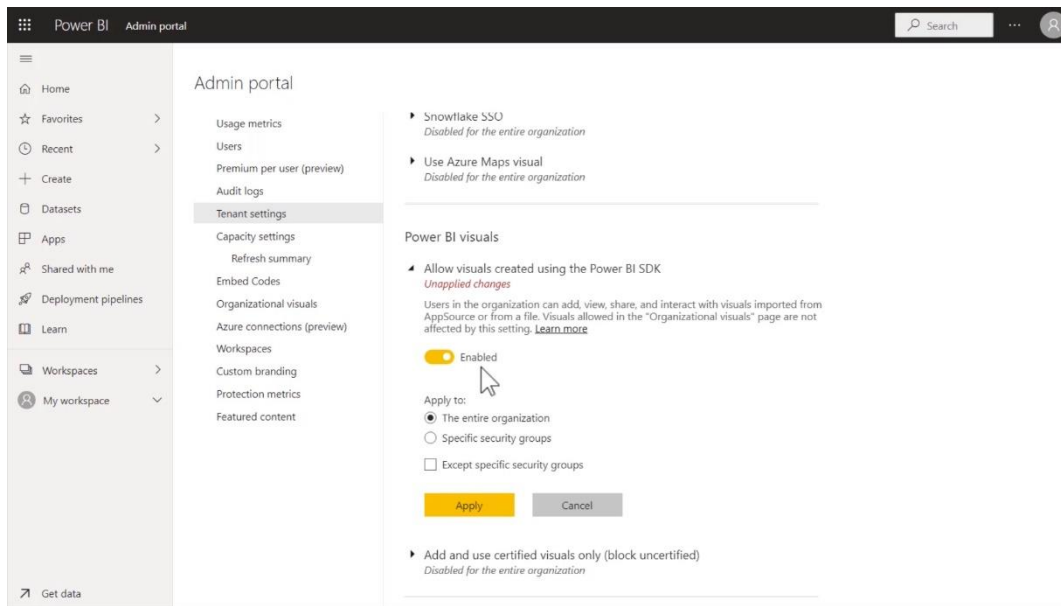
Open Tenant settings and scroll to Power BI visuals.

There are two key settings:

Disabling Custom Visuals – Prevents the use of any custom visuals across the organisation. This is the most restrictive security setting.

Allowing Only Certified Visuals – Ensures that only Power BI Certified visuals can be used. This restriction takes effect when reports are published; within Power BI Desktop, all custom visuals function normally.

0504-controlling The Use Of Custom Visuals

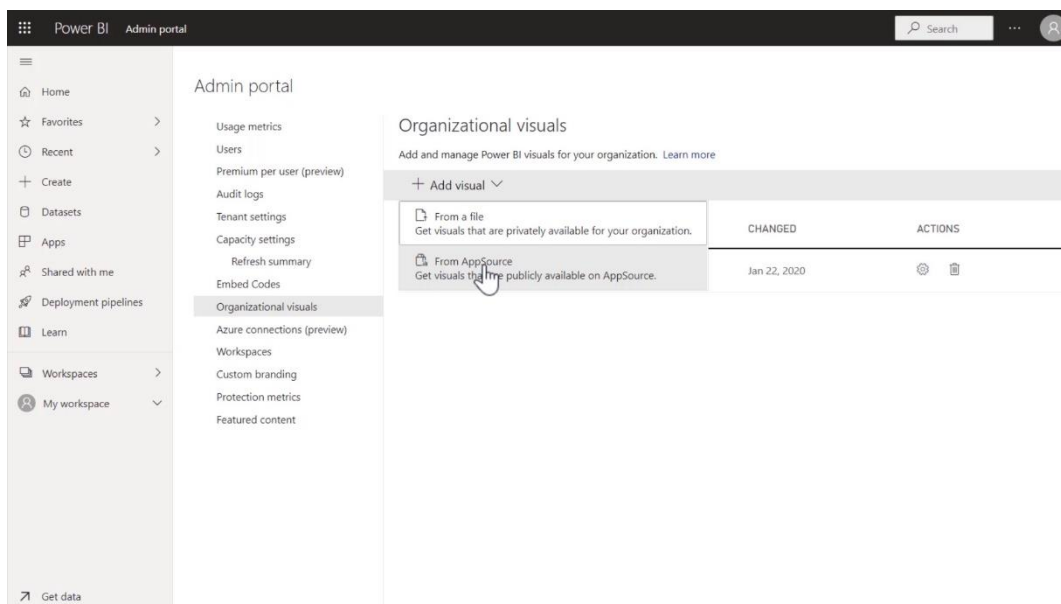


Organisational Custom Visuals

Even if custom visuals are disabled at the tenant level, Organisational Visuals can still be permitted. This allows an organisation to approve and distribute a specific set of custom visuals.

To configure Organisational Visuals:

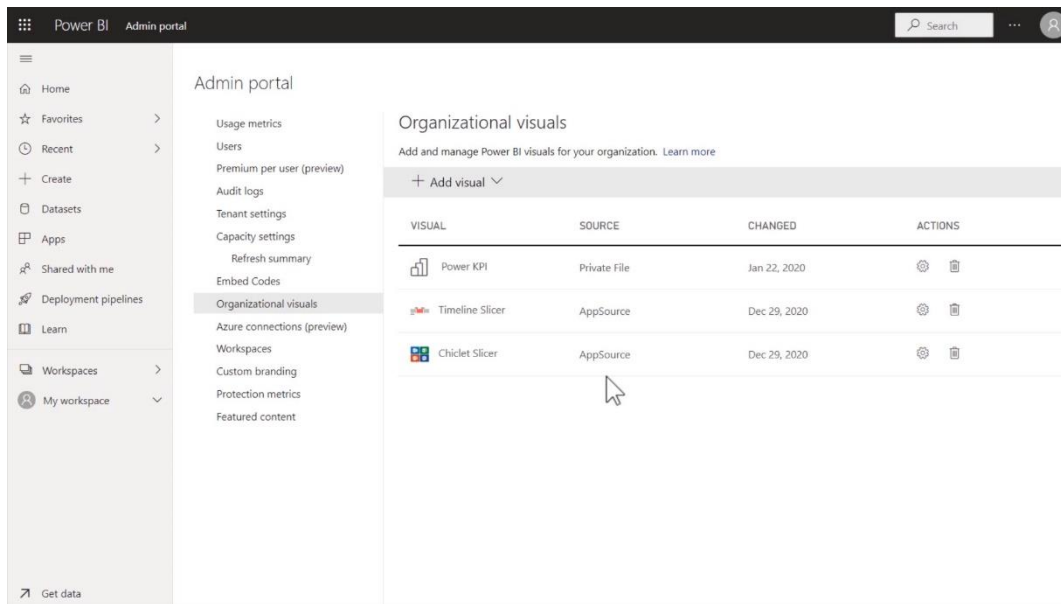
In the Power BI Admin Portal, go to the Organizational visuals section.



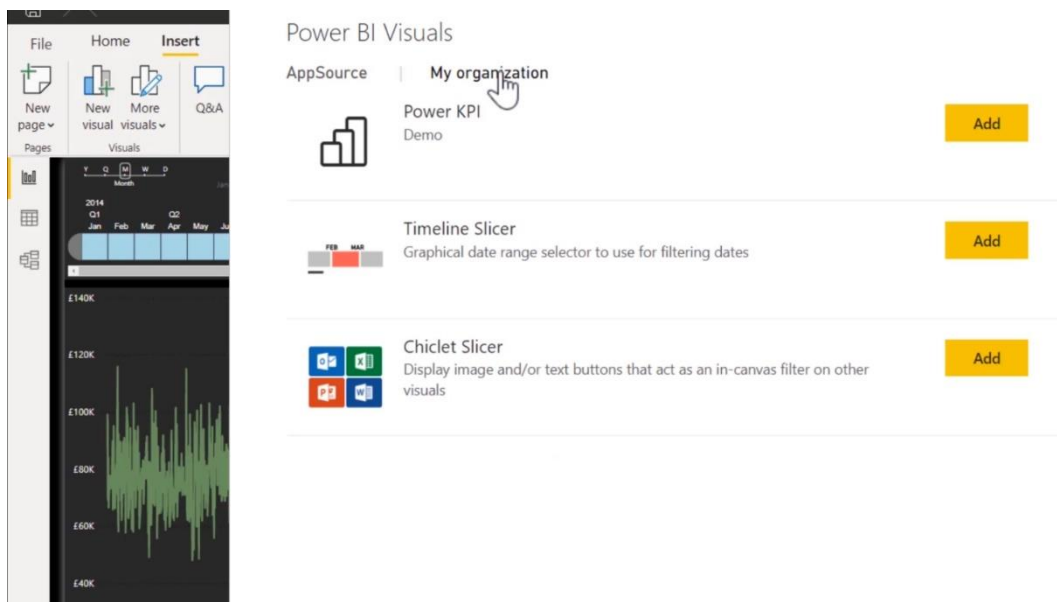
Click Add and choose a file to upload (typically for in-house custom visuals).

Alternatively, select a visual from AppSource.

0504-controlling The Use Of Custom Visuals



Once added, these visuals become accessible under My Organisation in AppSource.



Report developers should be instructed to use only visuals from this section to ensure compliance with organisational policies.

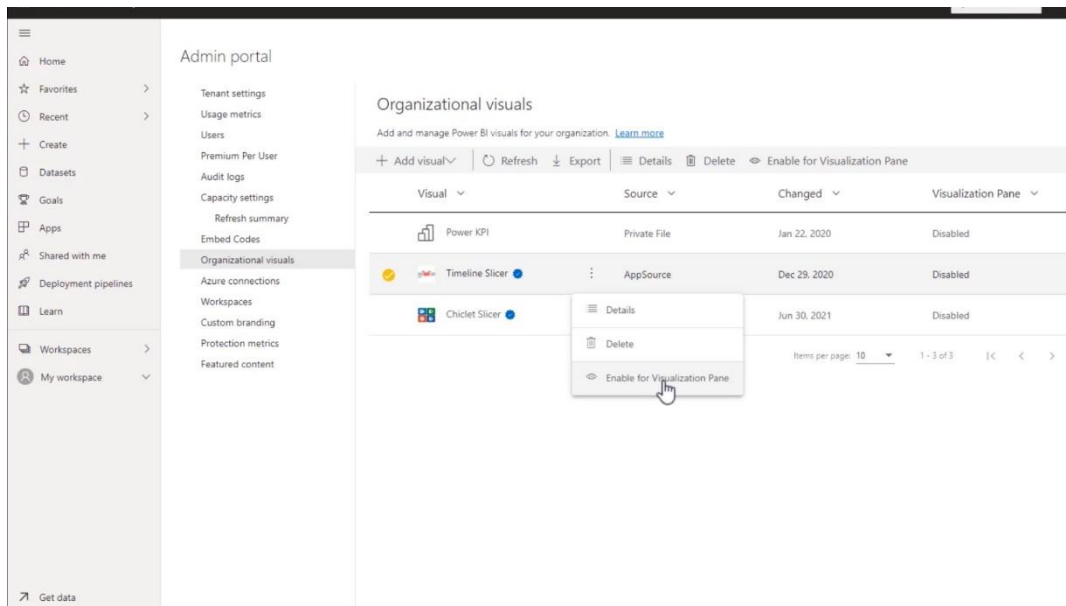
Enforcing Organisational Visuals for Users

Power BI Service allows administrators to automatically add approved organisational visuals to all users' Visualisations panes. To do this:

Go to Power BI Service → Settings for an organisational visual.

Activate the option Enable for Visualizations Pane.

0504-controlling The Use Of Custom Visuals



Once applied, these visuals appear as standard options in the Visualisations pane, eliminating the need for users to manually add them. This ensures consistency and security while maintaining ease of use for report developers.

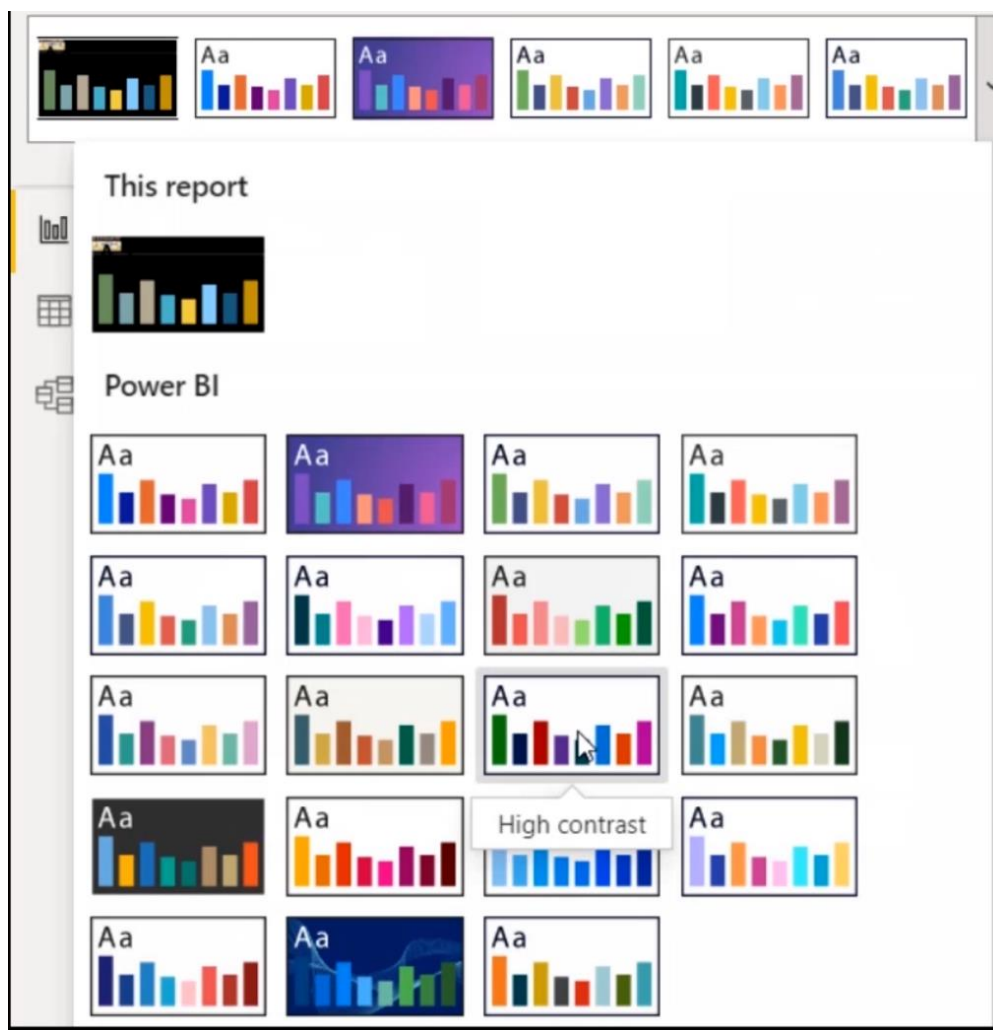


0505: Making Content Accessible

Ensuring Colour Accessibility

One of the key aspects of accessibility in Power BI is ensuring that your colour scheme enables all users to interpret your reports effectively. The best way to manage this is by using Power BI themes, which allow for global changes throughout your report. If you rely on custom colours outside the theme, making adjustments later becomes significantly more time-consuming.

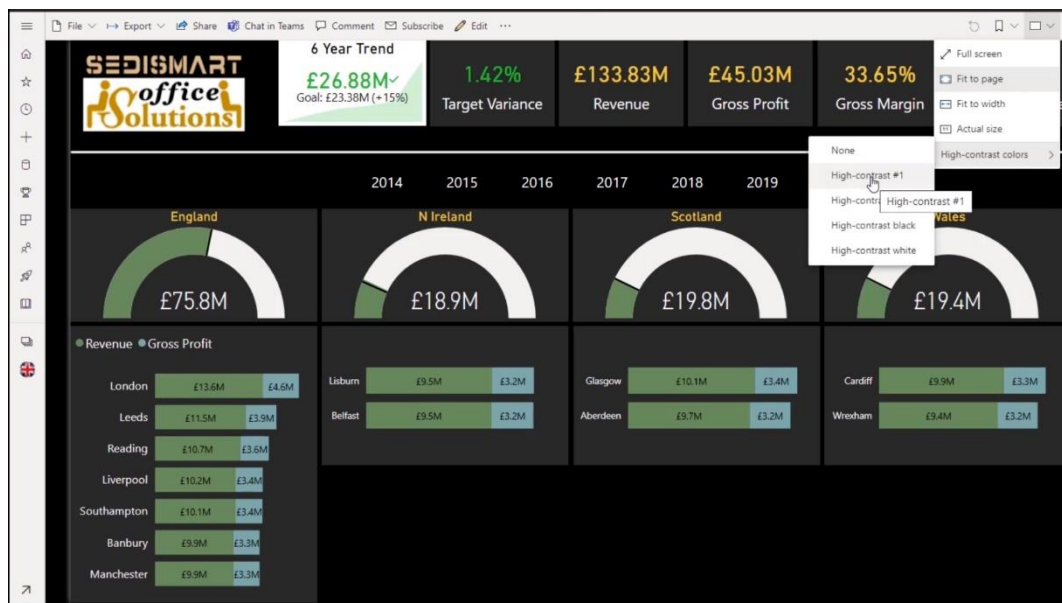
To access themes, go to the View tab, where you'll find a selection of built-in themes. Power BI includes specific themes designed to improve accessibility, such as the High Contrast and Colourblind Safe themes.



If your organisation adheres to specific accessibility standards, these can serve as useful starting points when customising your colours.

0505-making Content Accessible

These theme-based features are primarily for report developers. However, Power BI also provides accessibility options for report consumers. Users viewing a report in the Power BI service can adjust their settings via the View menu, where they'll find several high-contrast themes.



This feature is available for all reports, allowing users to personalise their experience based on their visual needs.

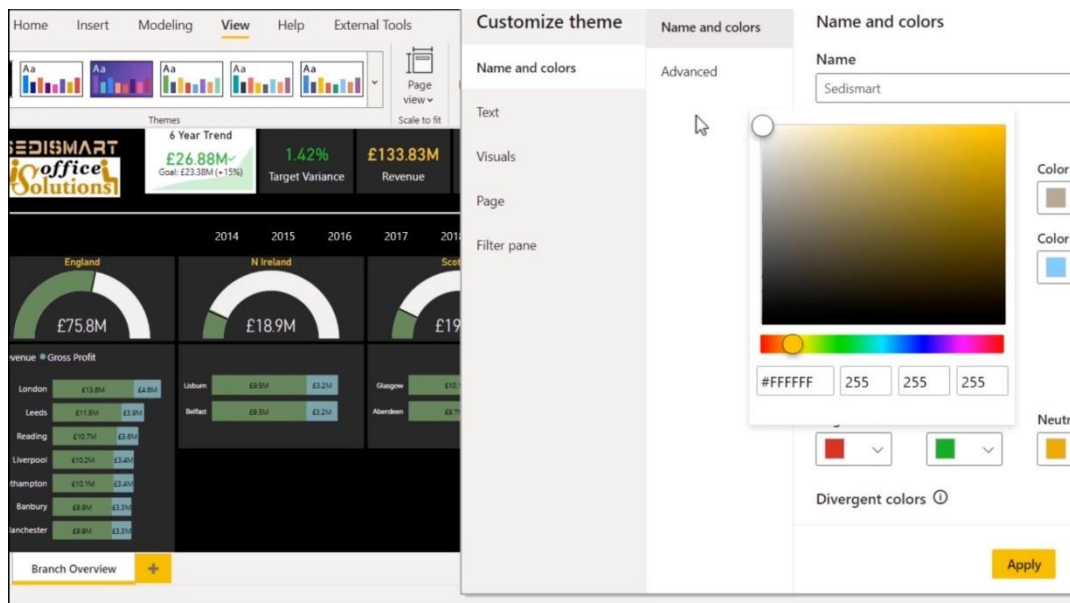


Improving Contrast for Readability

If contrast plays a crucial role in your report's design, you may want to evaluate your colour choices to ensure text remains legible. While there are online tools to test contrast scientifically, applying basic common sense is often sufficient.

For example, if data labels on a card visual appear faint against the background, you can modify them by customising the current theme. Under Customise Current Theme, check which theme colour is

applied to the text. If the text uses Theme Colour 5, changing it to white may improve contrast against a dark background. Similarly, if black text appears against a dark green or blue, making those background colours lighter will enhance readability. These adjustments ensure that contrast remains adequate without requiring complex tools.



Setting an Effective Tab Order

Another critical aspect of accessibility is tab order, which is essential for users relying on screen readers. The tab order determines the sequence in which report elements are accessed when navigating via the Tab key. Ideally, this should match the visual layout of the report, ensuring a logical reading flow.

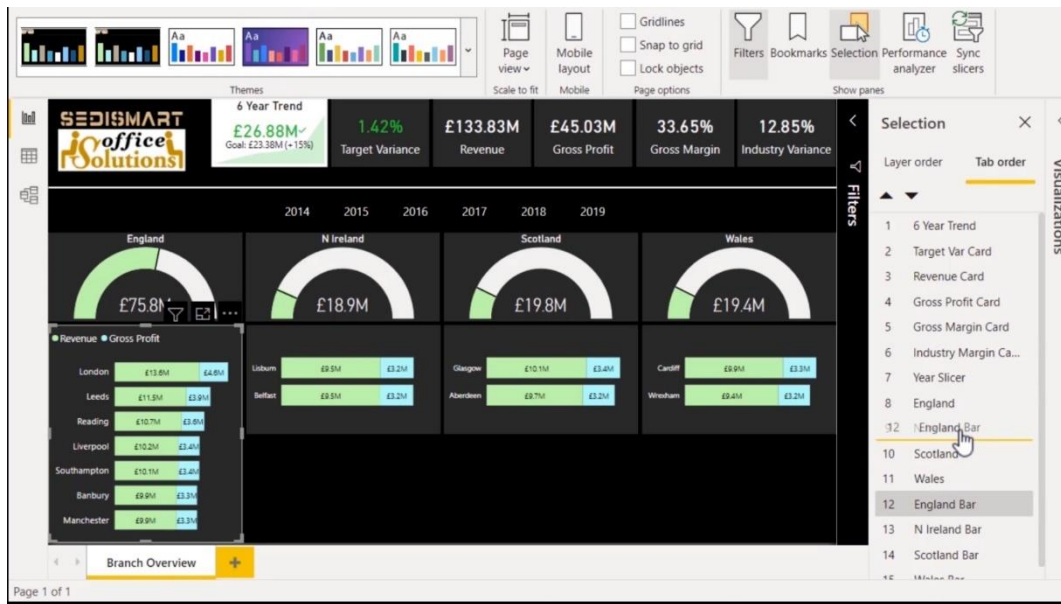
You can test the tab order in Power BI Desktop by pressing the Tab key. If navigation moves in an illogical sequence (e.g., jumping across unrelated visuals rather than progressing top to bottom), adjustments are needed.

To modify tab order:

Open the Selection Pane from the View tab.

Switch to the Tab Order view.

Ensure that all visuals have meaningful names (rather than default names). These names are read aloud by screen readers.



Drag visuals into a logical sequence, matching the expected reading order.

Once adjusted, this structure ensures a smoother experience for users navigating via assistive technology.

Using Alt Text for Screen Readers

Alt text provides additional descriptions for visuals, making reports more accessible to users who rely on screen readers. Unlike web development, where alt text is required for all images, in Power BI, you only need alt text when a visual conveys important information not otherwise communicated.

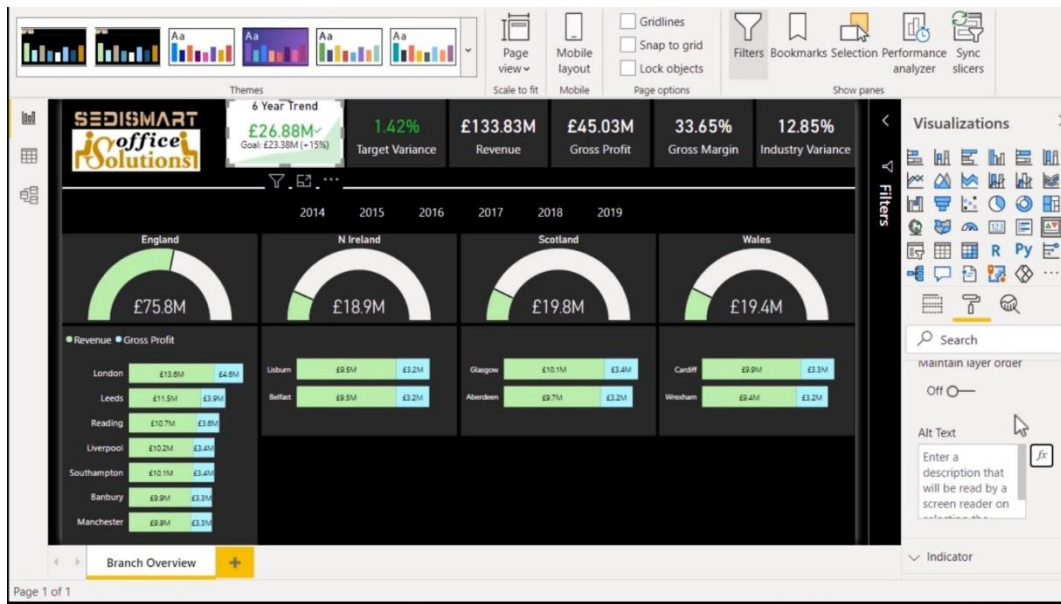
A good example is a target variance KPI, where a RAG (Red, Amber, Green) colour scheme is used. Since screen readers do not interpret colour, you need to append this information as alt text.

To add alt text:

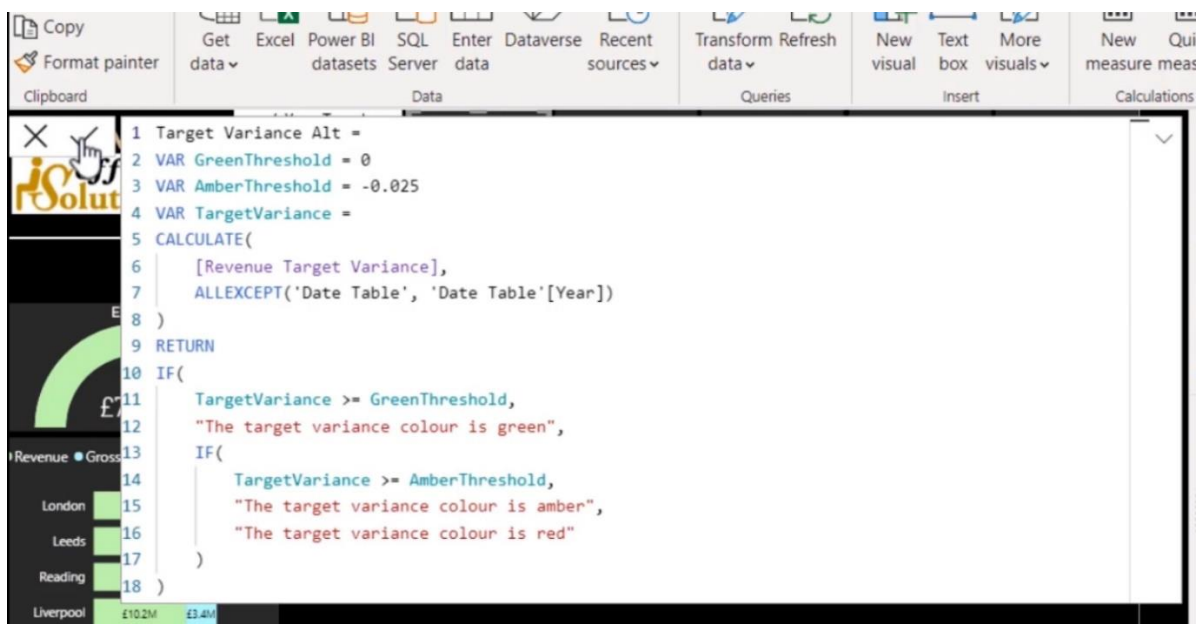
Select the visual.

Open the Format Pane, navigate to General, and locate the Alt Text field.

You can enter static text or use a dynamic measure to generate context-aware alt text.

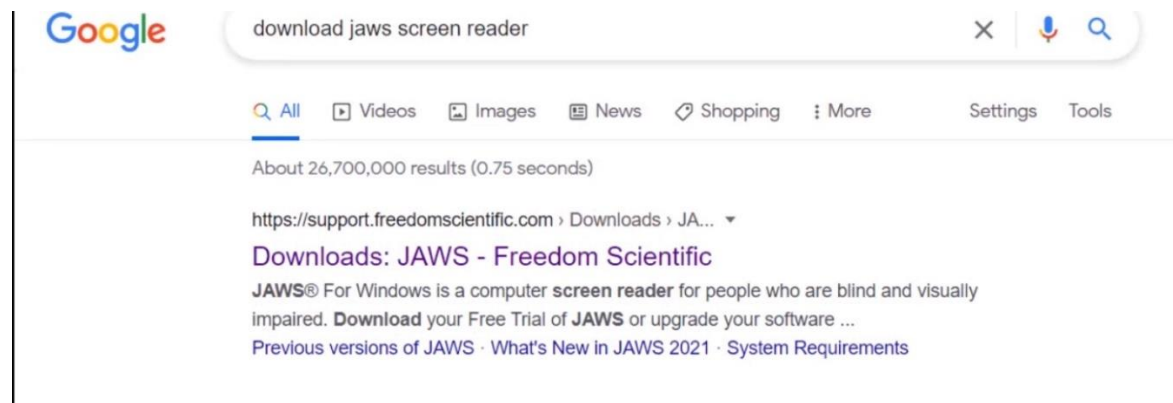


In the example shown below, a measure called Target Variance Colour determines the KPI's colour. We can copy this logic and create a corresponding Target Variance Alt measure that outputs descriptive text instead (e.g., "Target variance colour is green"). By applying this measure to the Alt Text field, screen readers will now announce the colour meaningfully.



Testing the Screen Reader Experience

To validate the accessibility of your report, you can use the JAWS screen reader software, available as a free trial version.



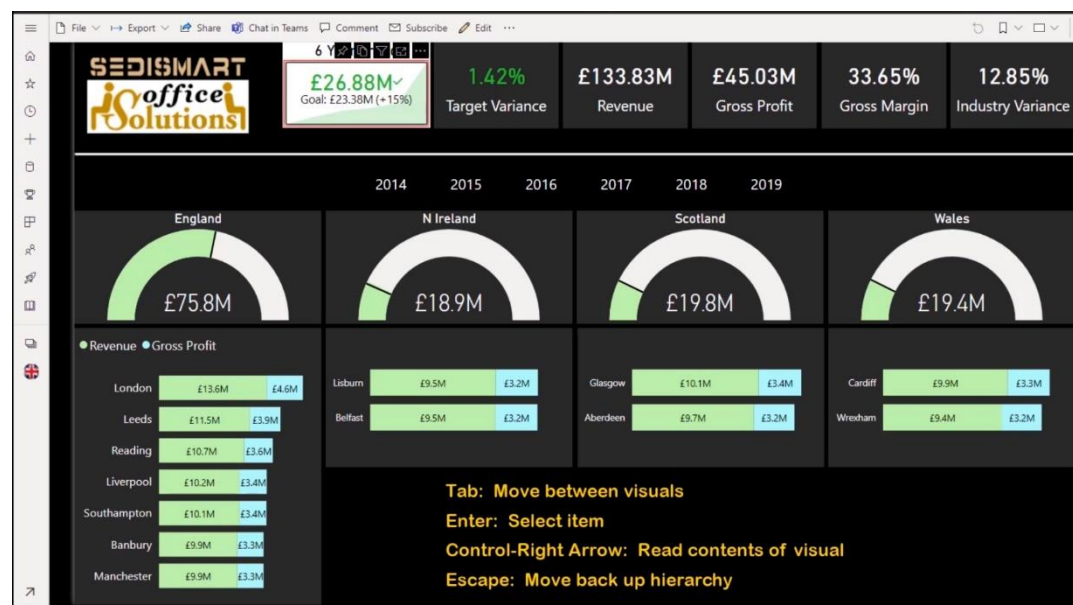
This tool allows you to experience how a visually impaired user would navigate the report.

When using JAWS:

Press Tab to navigate through different elements, ensuring the logical order is maintained.

Press Enter to enter a visual and Ctrl + Right Arrow to read its content.

Use Escape to exit a visual.



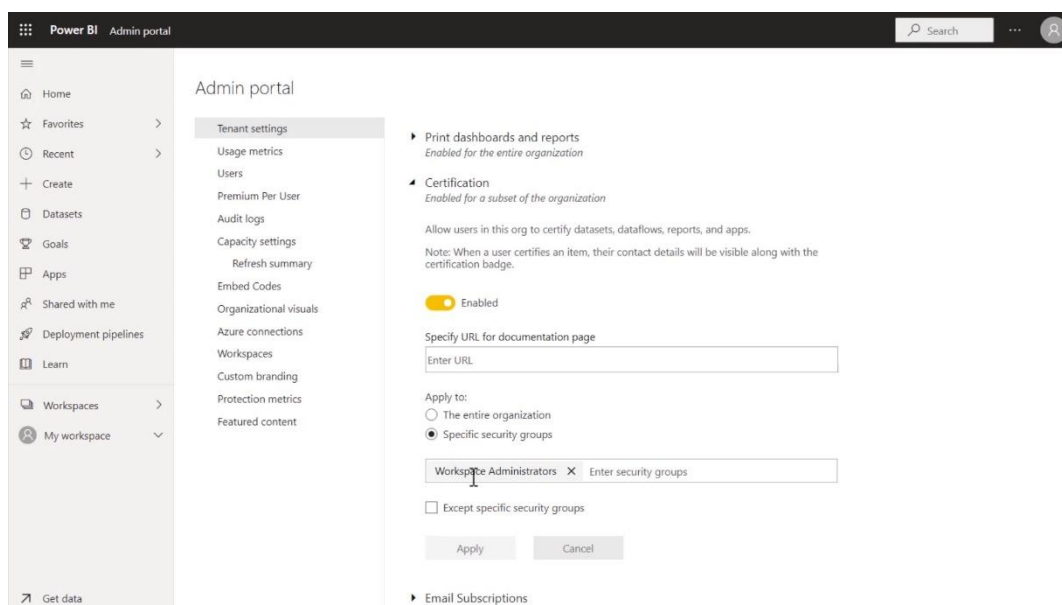
This process allows you to confirm that tab order, alt text, and contrast adjustments have been implemented effectively. By following these steps, you can ensure that your Power BI reports are accessible to a wider audience, improving usability for all users.

0506: Endorsing Content

In Power BI, there are four main types of content: apps, reports, dashboards, and Excel workbooks. However, endorsement only applies to apps and reports—dashboards and Excel workbooks cannot be endorsed.

Configuring Certification in the Admin Portal

To enable content certification, a Power BI administrator must configure settings in the Admin Portal. Under Export and Sharing, the administrator expands the Certification section. In this example, certification has been enabled for specific security groups, with the capability assigned to a group called Workspace Administrators.



Anyone in this group has permission to certify semantic models, dataflows, reports, and apps. After adding myself to this group, I now have the ability to certify content, including apps and reports.

Promotion vs. Certification

It's important to distinguish between promotion and certification:

Promotion is a self-service process available to all users.

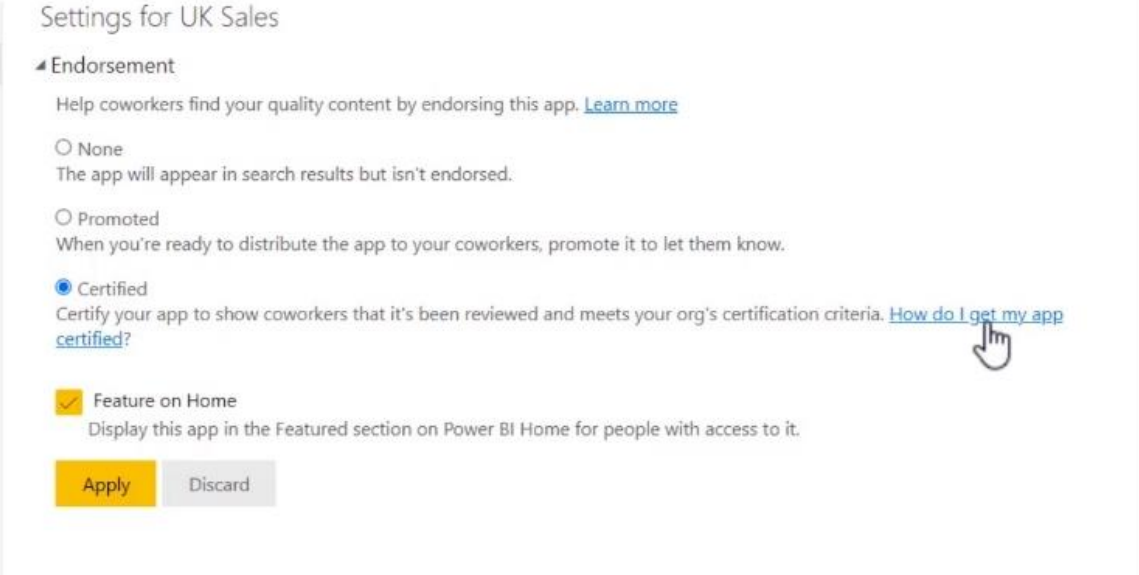
Certification requires administrator configuration and is only available to designated users within approved security groups.

Certifying an App

To certify an app, navigate to the workspace where the app was published. Open the ellipsis menu (More options) in the top-right corner and select Endorse this app.

This takes you to the App Settings page, where the Endorsement section is visible. Here, you can change the certification status:

- None (default)
- Promoted (available to all users)
- Certified (available only to workspace administrators)



Settings for UK Sales

Endorsement

Help coworkers find your quality content by endorsing this app. [Learn more](#)

☐ None
The app will appear in search results but isn't endorsed.

☐ Promoted
When you're ready to distribute the app to your coworkers, promote it to let them know.

☒ Certified
Certify your app to show coworkers that it's been reviewed and meets your org's certification criteria. [How do I get my app certified?](#)

☒ Feature on Home
Display this app in the Featured section on Power BI Home for people with access to it.

Apply Discard

Users who are not in the Workspace Administrators group will see the Certified option greyed out, along with a link explaining how to get their app certified within the organisation.

Featured Content vs. Endorsement

When selecting Certified, the option to Feature this content on the homepage is automatically suggested. However, this is an independent setting, separate from endorsement.

Featured content is controlled from the Admin Portal, under the Featured Content section, just below Certification. If this feature is enabled, certified content can be further highlighted by featuring it on users' homepages, making it more visible to those with access.

After enabling certification (and optionally, featuring the content), click Apply to finalise the endorsement process.

Certifying a Report

To certify a report, go to the Report Settings page. From the report listing, click the ellipsis menu and select Settings.

Scroll down to the Endorsement section, where the same options are available:

- None
- Promoted
- Certified

0506-endorsing Content

UK Sales
Demo

+ New Create a pipeline View Filters

All **Content** Datasets + dataflows

Name	Type	Owner	Refreshed
09-Design and configure for accessibility	Report	UK Sales	23/05/21, 09:45:27
09-Personalize visuals	Report	UK Sales	09/04/21, 18:16:19
England	Dashboard	UK Sales	—
Expenses 2014-19	Workbook	UK Sales	10/08/20, 11:12:03
Insights	Dashboard	UK Sales	—
N Ireland	Dashboard	UK Sales	—
Sales Overview	Dashboard	UK Sales	—
Sales Report	Report	UK Sales	11/07/21, 07:38:40
Scotland	Dashboard	UK Sales	—
Wales	Dashboard	UK Sales	—

Settings for Sales Report

Upload Delete

Endorsement

Help coworkers find your quality content by endorsing this report. [Learn more](#)

☐ None
The report will appear in search results but isn't endorsed.

☐ Promoted
When you're ready to distribute the report to your coworkers, promote it to let them know.

☒ **Certified**
Certify your report to show coworkers that it's been reviewed and meets your org's certification criteria. [How do I get my report certified?](#)

☒ **Feature on Home**
Display this report in the Featured section on Power BI Home for people with access to it.

Persistent filters

Don't allow end user to save filters on this report.

Save Cancel

As with apps, selecting Certified will automatically suggest enabling Feature on homepage. This can be manually disabled if desired.

Click Save to confirm the certification. The report's Endorsement column will now display Certified, making this visible to all users with access to the report. Similarly, any certified app will display a Certified badge next to its name, ensuring users can easily identify endorsed content.

UK Sales
Demo

+ New Create a pipeline View Filters Settings Access Search

All **Content** Datasets + dataflows

Name	Type	Owner	Refreshed	Next refresh	Endorsement
09-Design and configure for accessibility	Report	UK Sales	23/05/21, 09:45:27	—	—
09-Personalize visuals	Report	UK Sales	09/04/21, 18:16:19	—	—
England	Dashboard	UK Sales	—	—	—
Expenses 2014-19	Workbook	UK Sales	10/08/20, 11:12:03	—	—
Insights	Dashboard	UK Sales	—	—	—
N Ireland	Dashboard	UK Sales	—	—	—
Sales Overview	Dashboard	UK Sales	—	—	—
Sales Report	Report	UK Sales	11/07/21, 07:38:40	—	Certified
Scotland	Dashboard	UK Sales	—	—	—
Wales	Dashboard	UK Sales	—	—	—

Update app

0507: Creating Subscriptions

Subscriptions in Power BI allow users to generate interest and maintain engagement with specific items within a workspace. However, it's important to note that subscriptions operate at the item level, not at the workspace level. You can create subscriptions for yourself, for other users within your Power BI tenant, or for security groups. These subscriptions can be applied to report pages, dashboards, and paginated reports. The result is an automated email notification, sent at a specified frequency, providing direct access to the item.

Creating a Subscription

To create a subscription, you must be working within the Power BI Service. First, navigate to the item you want to subscribe to. In this example, we'll set up a subscription for the Sales Overview Dashboard.

Once you have selected the dashboard, follow these steps:

- Click the Subscribe button.
- Select Add New Subscription.

Your email address will be automatically added to the subscription. You can keep or remove it as needed.

Add additional recipients, including individual users or security groups. In this example, we'll add a user called An O'Neill.

Enter a Subject (e.g., Sales Overview).

If necessary, include a Message to provide additional context for the subscription.

Set the Start Date and End Date. If you want the subscription to continue indefinitely, leave the end date blank.

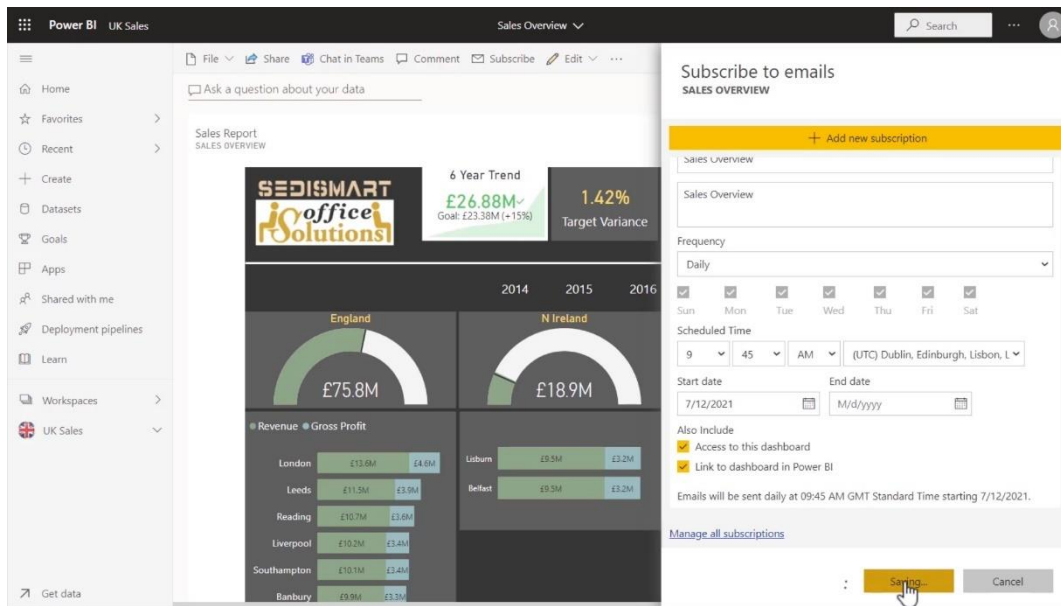
Choose the Frequency of notifications (e.g., Daily).

Specify the Scheduled Time for delivery.

Ensure that the Access to Dashboard and Dashboard Link options remain enabled so recipients can access the dashboard directly from the email.

Once all settings are configured, click Save and Close to activate the subscription.

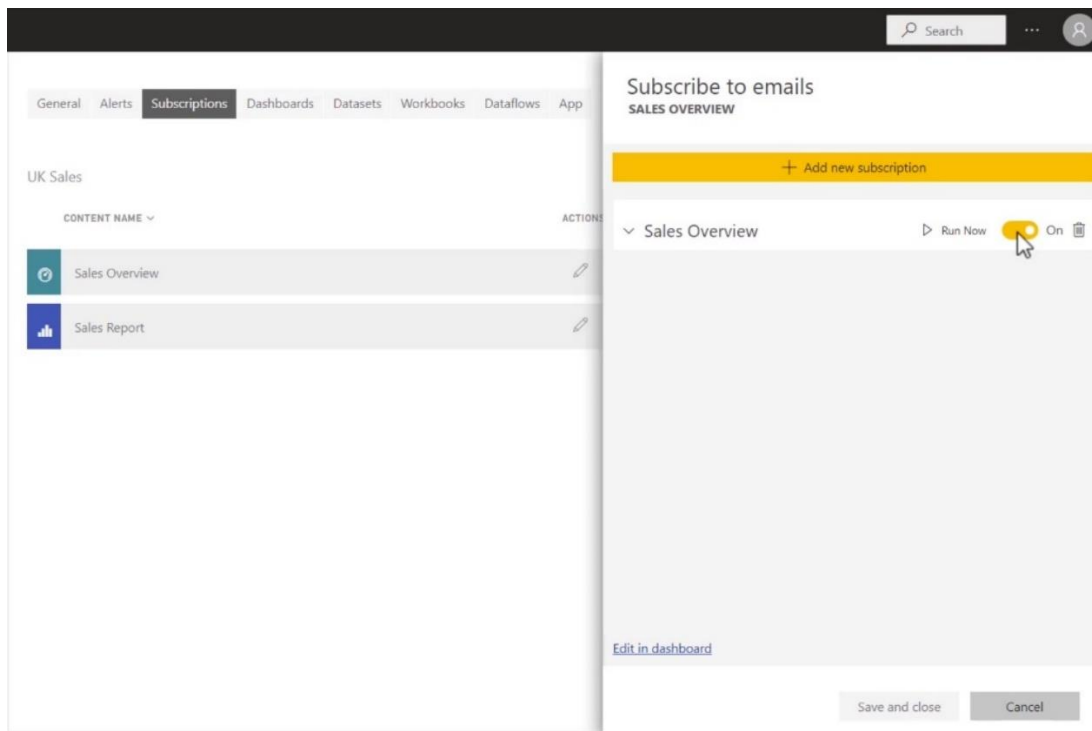
0507-creating Subscriptions



Testing a Subscription

After creating a subscription, you may want to test it immediately rather than waiting for the scheduled email. To do this:

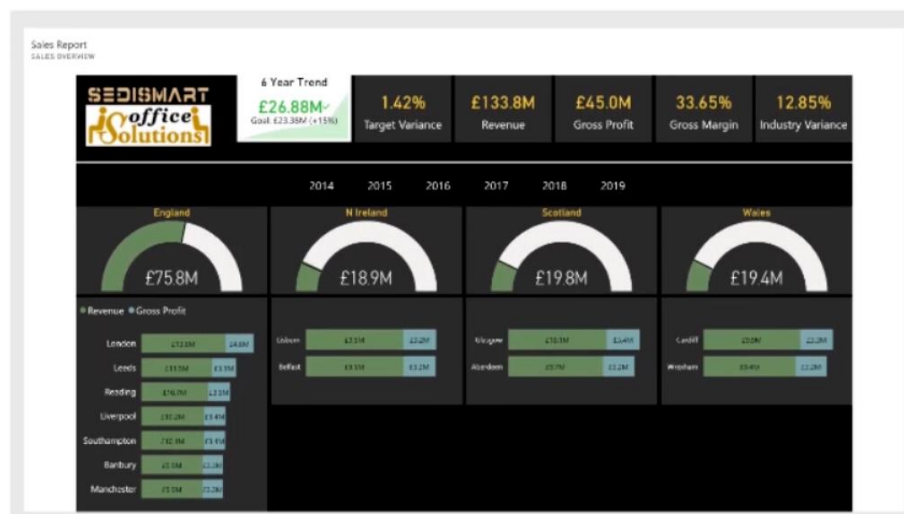
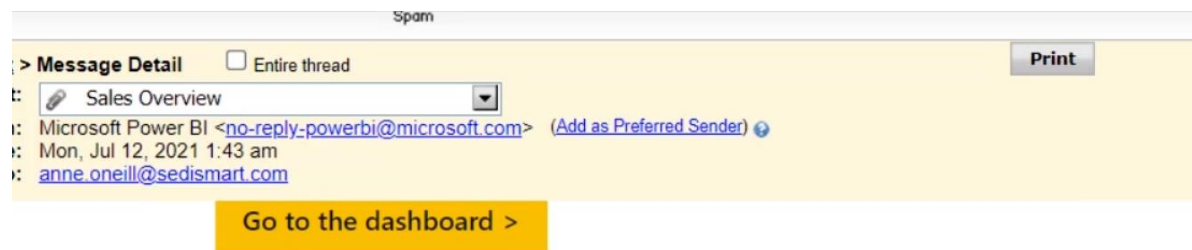
- Click the Subscribe button again.
- Locate your newly created subscription.
- Click the Run Now button next to it.



This will generate the email straight away, allowing you to verify that the subscription is working as expected.

0507-creating Subscriptions

The email received by subscribers includes a direct link to the dashboard. If the recipient is already logged in, they will be taken straight to the dashboard; otherwise, they will need to enter their login credentials.



You're receiving this email because admin@sedismart.com subscribed you to the 'Sales Overview' dashboard. The image above was generated at July 12, 2021 8:42 UTC.

Managing Subscriptions

To manage your existing subscriptions:

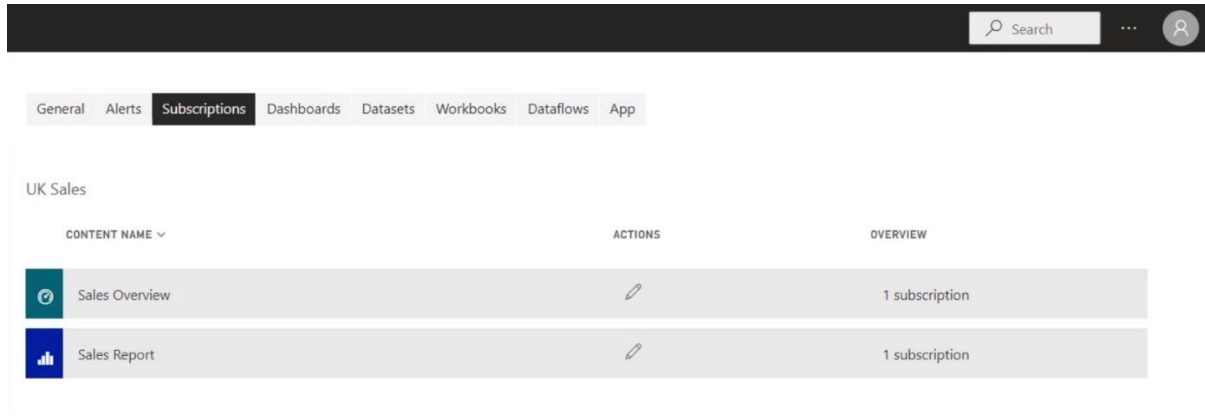
Click on the Ellipsis Menu (...).

Select Settings.

Navigate to the Subscriptions section.

Here, you will find a list of all subscriptions you have created.

0507-creating Subscriptions



The screenshot shows the Microsoft Power BI web interface. At the top, there is a dark navigation bar with a search box and a user profile icon. Below this is a horizontal menu with tabs: General, Alerts, Subscriptions (selected), Dashboards, Datasets, Workbooks, Dataflows, and App. The main content area is titled 'UK Sales'. It features a table with three columns: 'CONTENT NAME', 'ACTIONS', and 'OVERVIEW'. The table lists two items: 'Sales Overview' and 'Sales Report'. Each item has a small icon on the left, a pencil icon in the 'ACTIONS' column, and '1 subscription' in the 'OVERVIEW' column.

CONTENT NAME	ACTIONS	OVERVIEW
 Sales Overview		1 subscription
 Sales Report		1 subscription

You can:

- Edit an existing subscription – Update recipients, subject lines, messages, or scheduling options.
- Add new subscriptions to the same object.
- Delete a subscription if it is no longer needed.
- Temporarily deactivate a subscription – This is useful if you need to pause email notifications while making changes to the underlying report or dashboard.

By managing subscriptions effectively, you can ensure that users receive timely and relevant updates without unnecessary interruptions.

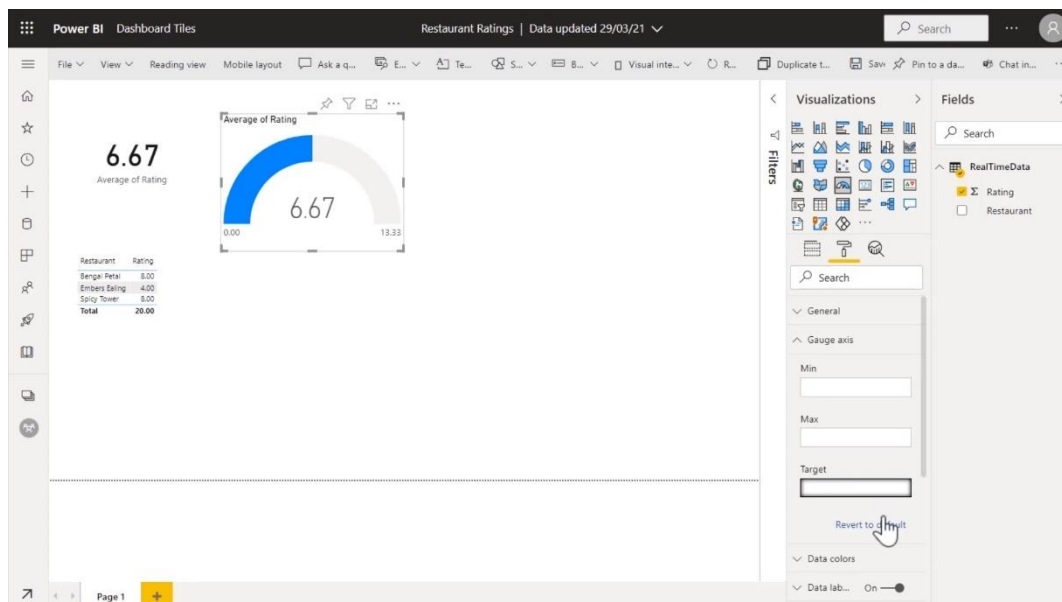
0508: Configuring Data Alerts

A dashboard alert is a notification that informs you when a key metric reaches or falls below a specified threshold. There are two key aspects to alerts: a notification that appears when you log in and an optional email notification that is sent at a chosen frequency.

Alerts are applied at the tile level and can only be set on dashboard tiles. Additionally, alerts can only be associated with three specific visual types: Card, Gauge, and KPI. Other visuals, such as tables or charts, do not support alerts.

Adding a Gauge Visual for Alerts

In our report, we already have a Card visual, but to demonstrate data alerts, we will also add a Gauge visual.



Set the value of the gauge to display the rating.

Change the aggregation from Sum to Average to reflect the correct metric.

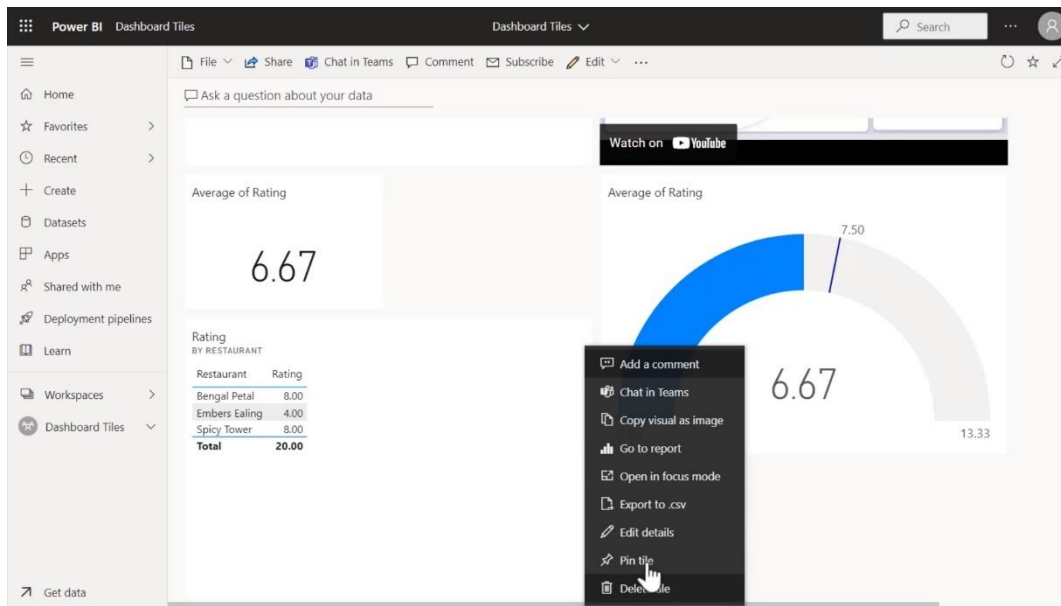
Since we do not have a predefined measure for the target, we will use a static value of 7.5.

To do this:

- Navigate to the Format section.
- Expand the Gauge Axis settings.
- Enter 7.5 as the static target value.

Once configured, pin the gauge visual to a dashboard. Then, save the changes and switch to the dashboard.

0508-configuring Data Alerts



Setting Up an Alert

With our updated dashboard, we now have two eligible visuals for alerts: the Card and the Gauge.

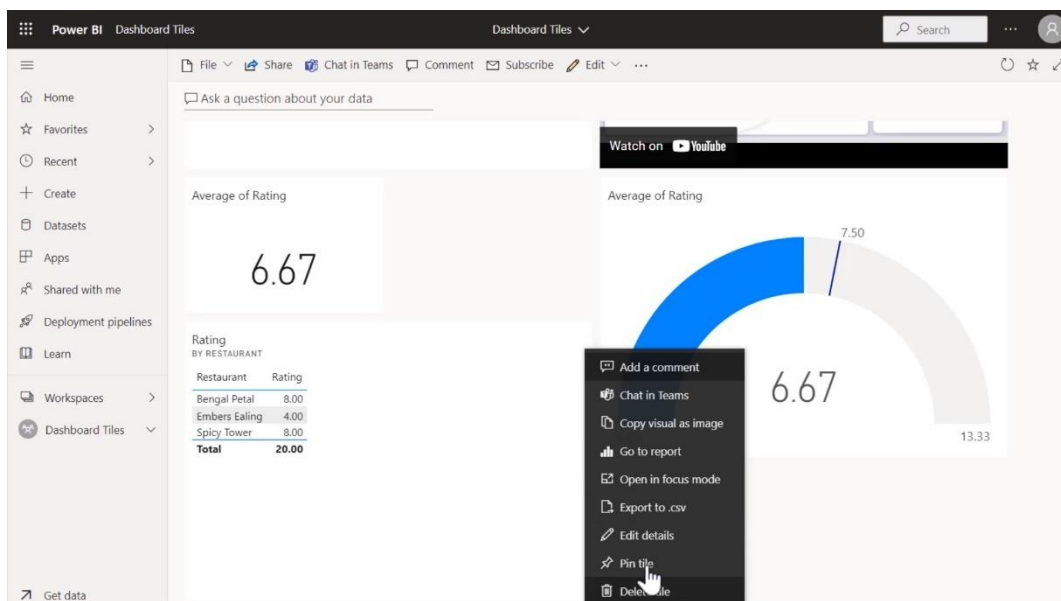
Click the ellipsis menu (:) on a supported tile.

Select Manage Alerts.

If the tile supports alerts, the option will be available. If it does not (e.g., a table visual), the option will not appear.

Let's set up an alert for the Gauge visual:

Click Manage Alerts.



Select Add Alert Rule.

0508-configuring Data Alerts

Optionally, rename the alert (for this example, we will leave the default name).

Keep the default condition as "is above" and set the threshold to 7.5.

The system automatically detects this threshold from the gauge target value.

Set the maximum notification frequency to at most once every 24 hours (default setting).

Ensure Send an email is switched on.

The screenshot shows a Power BI dashboard with a 'Manage alerts' dialog box open. The dashboard has a top navigation bar with 'File', 'Share', 'Chat in Teams', 'Comment', 'Subscribe', and 'Edit'. Below the navigation bar is a search bar 'Ask a question about your data'. The main content area is divided into two sections. The left section, titled 'Average of Rating', displays a large number '6.67' and a table of ratings by restaurant. The right section, also titled 'Average of Rating', shows a gauge chart with a blue arc and a value of '0.00'. The 'Manage alerts' dialog box is on the right side of the screen. It has a title bar 'AVERAGE OF RATING' and a subtitle 'Manage alerts'. Below the title bar is a yellow button '+ Add alert rule'. The dialog box contains the following settings: 'Set alerts rule for' is 'Average of Rating'; 'Condition' is 'Above' (selected from a dropdown); 'Threshold' is '7.5'; 'Maximum notification frequency' has two options: 'At most every 24 hours' (selected with a radio button) and 'At most once an hour' (unselected); 'Alerts are only sent if your data changes.' is checked; 'By default, you'll receive notifications on the service in the notification center.' is checked; and 'Send me email, too' is checked. At the bottom of the dialog box is a link 'Use Microsoft Power Automate to trigger additional actions' and two buttons: 'Save and close' (yellow) and 'Cancel' (grey).

Restaurant	Rating
Bengal Petal	8.00
Embers Ealing	4.00
Spicy Tower	8.00
Total	20.00

Alert Limitations and Power Automate

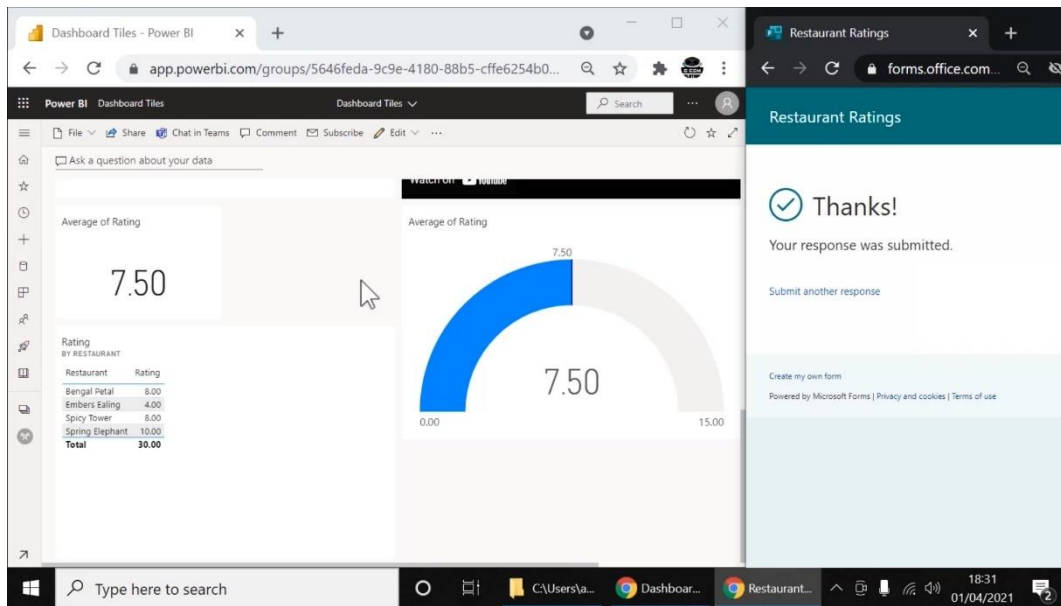
A key limitation of dashboard alerts is that you can only create alerts for yourself. You cannot create alerts on behalf of colleagues or other users.

However, if you need to notify others, you can use Power Automate to trigger additional actions when an alert is raised. A useful link within the alert settings allows you to configure Power Automate workflows directly.

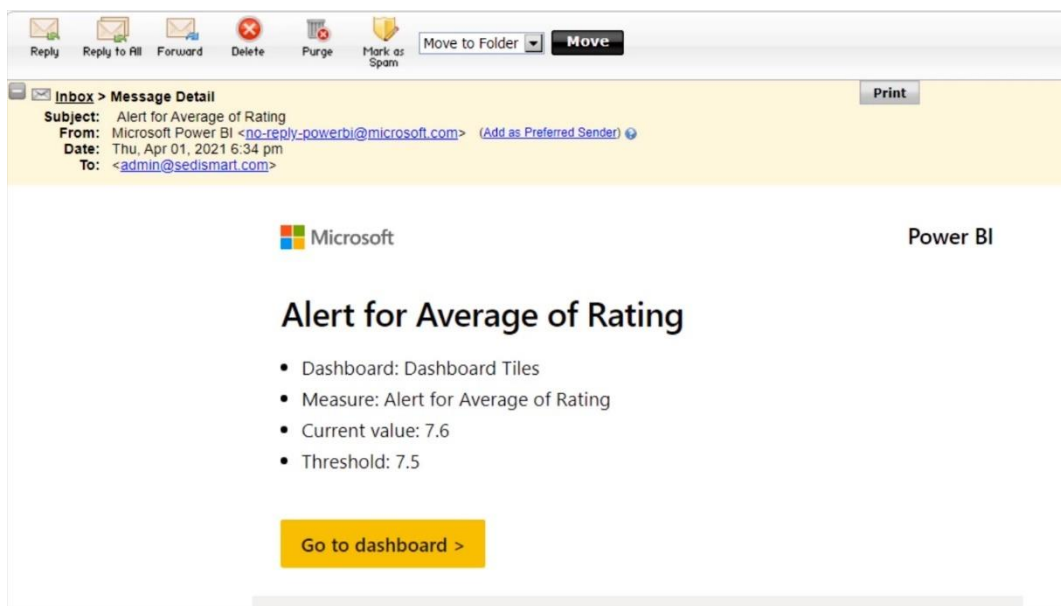
Testing the Alert

In the example shown below, we are using a Microsoft Forms form to enter a rating which will change the data and test the alert by increasing the rating:

0508-configuring Data Alerts



The threshold of 7.5 has now been reached, triggering an email.



Since we enabled email notifications, we now receive an email confirming that the threshold has been surpassed. This email also includes a direct link to the dashboard for quick access.

0509: Using Sensitivity Labels

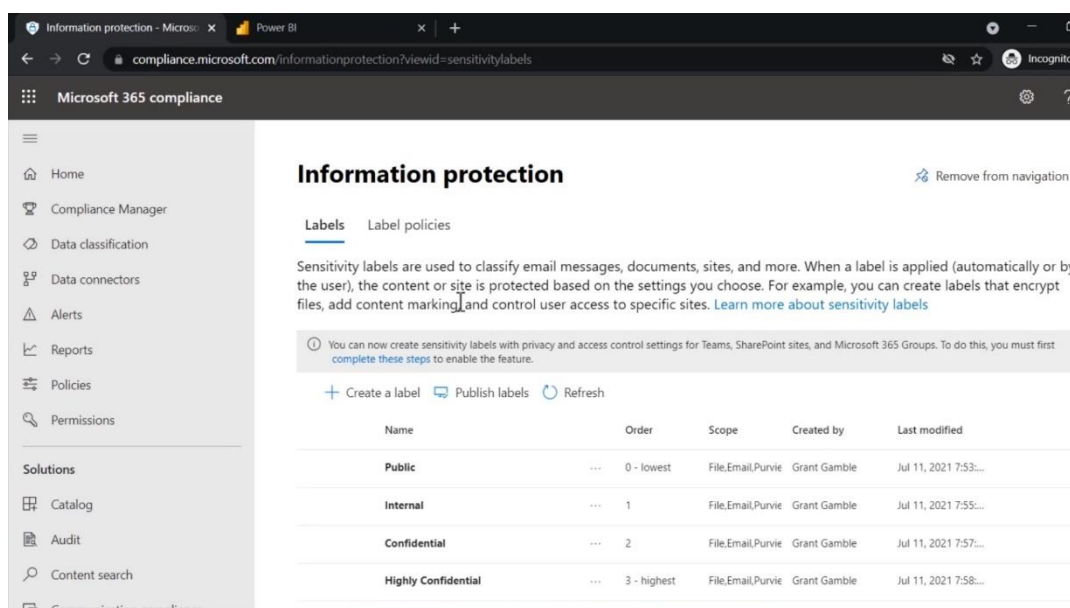
Sensitivity labels enable organizations to classify and protect sensitive data on an object-by-object basis. By applying sensitivity labels, organizations can specify the level of sensitivity for each object and enforce appropriate protections.

Setting Up Sensitivity Labels

To use sensitivity labels within Power BI, administrators must first configure them in Microsoft 365. This is not a Power BI-specific task but rather a Microsoft 365 or Office 365 function, meaning only a Global Admin can create sensitivity labels—Power BI Admins alone do not have this capability.

To create sensitivity labels, a Global Admin must go to the Microsoft 365 Security and Compliance Center at protection.microsoft.com. From there:

Navigate to Classification and select Sensitivity Labels.



Click Create a Label and follow the setup process to define the necessary labels.

In this example, four sensitivity labels have been created in ascending order of sensitivity: Public, Internal, Confidential, and Highly Confidential. Once the labels have been defined, the final step is to publish them and associate them with a label policy. In this case, the labels have been published using a policy called Corporate.

Enabling Sensitivity Labels in Power BI

Once sensitivity labels have been created and published, the next step is to enable their use within the Power BI tenant. This is a Power BI-specific setting and can be configured by either a Global Admin or a Power BI Admin.

0509-using Sensitivity Labels

To enable sensitivity labels in Power BI:

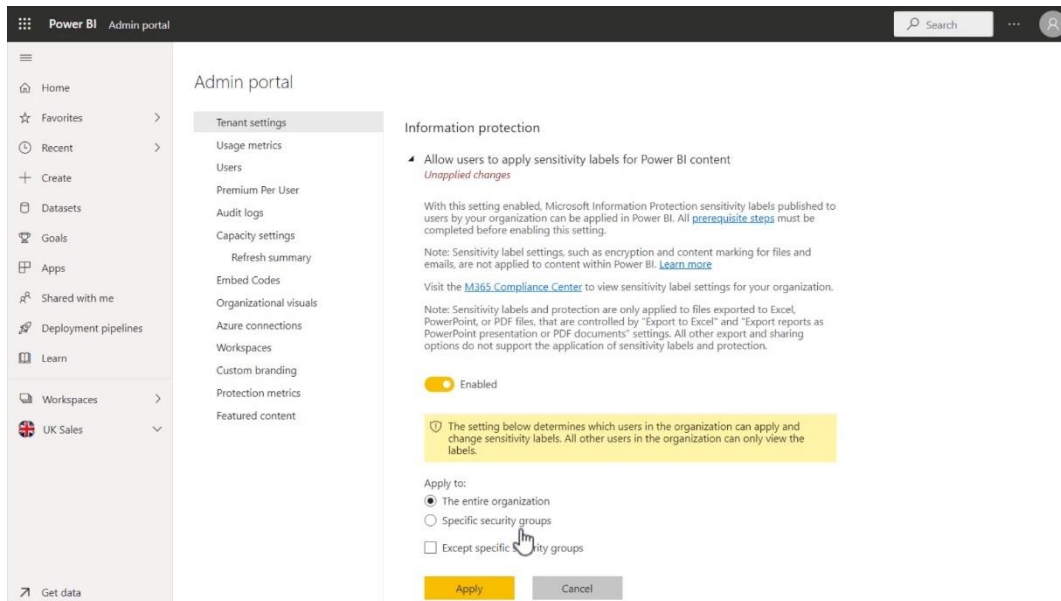
Go to Power BI Service and navigate to Settings > Admin Portal.

In the Tenant Settings, locate the Information Protection section.

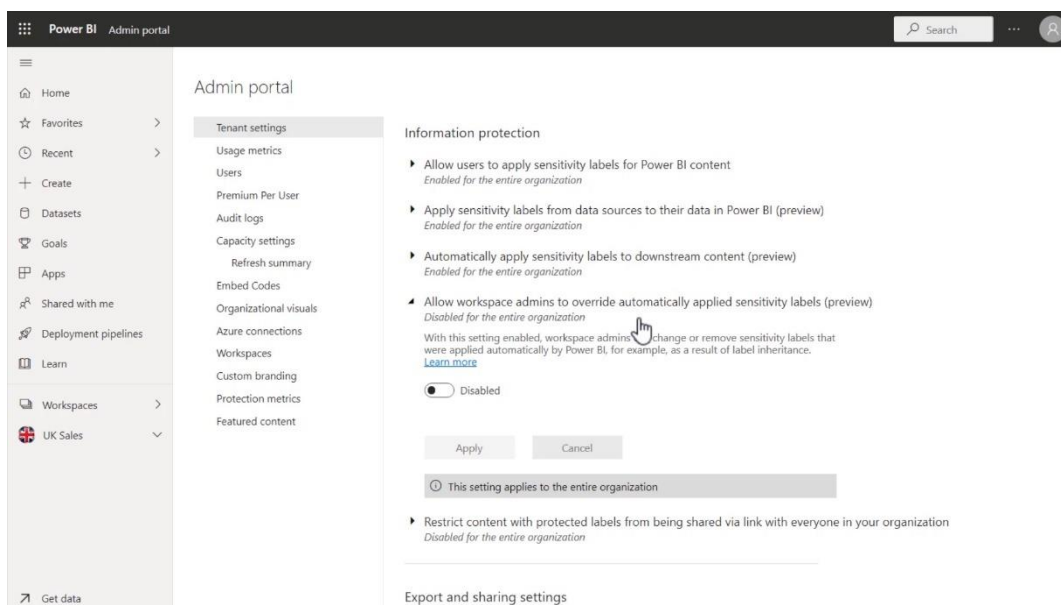
Enable the setting Allow users to apply sensitivity labels for Power BI content.

This can be applied to the entire organization or specific security groups.

Most organizations will benefit from enabling this across the entire organization.



Additional Sensitivity Label Settings



Power BI provides several additional settings related to sensitivity labels, which help refine how labels are applied and enforced:

Apply sensitivity labels from data sources to their data in Power BI:

- Enables the inheritance of labels from external data sources (e.g., SQL Server).
- Typically enabled to maintain consistency across platforms.

Automatically apply sensitivity labels to downstream content:

- Ensures that reports, dashboards, and other content inherit labels from semantic models.
- Helps maintain security and compliance across related content.

Allow workspace admins to override automatically applied sensitivity labels:

- If enabled, workspace admins can override inherited labels.
- Generally, this setting should remain disabled to prevent inconsistencies.

Restrict content with protected labels from being shared via link:

- Prevents sensitive content from being shared broadly.
- This is particularly relevant for labels that include encryption or watermarking.

Applying Sensitivity Labels in Power BI

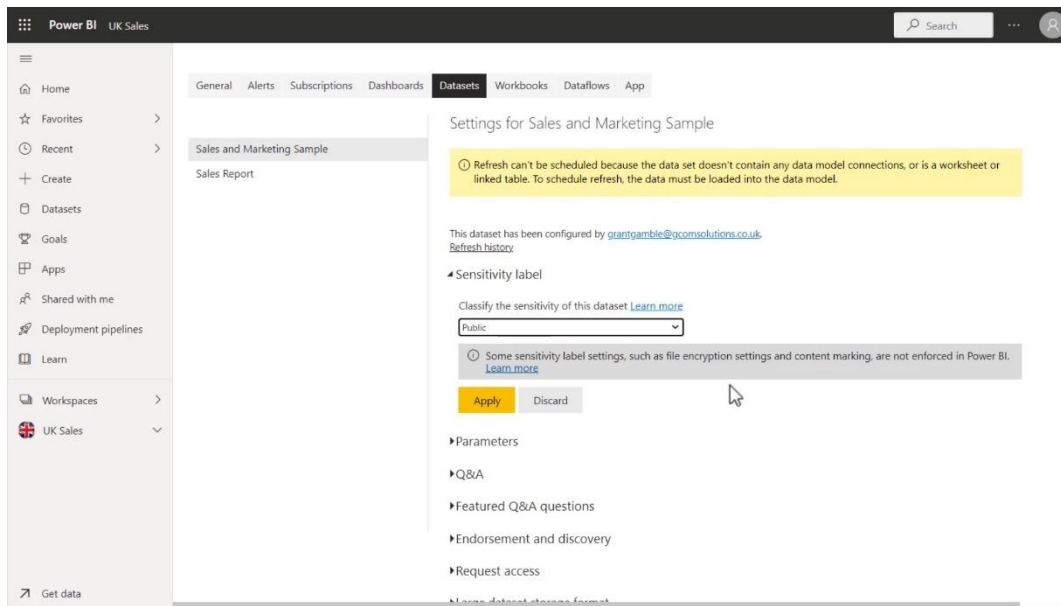
Once the administrative settings are in place, users can apply sensitivity labels to individual objects. From the workspace listing, users can:

Click the ellipsis menu (:) next to an object.

Select Settings.



Scroll to the Sensitivity Labels section and choose an appropriate label.



Using Inherited Sensitivity Labels

Instead of manually applying labels to every object, organizations can take advantage of inheritance by enabling the Automatically apply sensitivity labels to downstream content setting.

For example, consider a Sales and Marketing workspace containing:

- A semantic model
- A report
- A dashboard

If a sensitivity label is applied only to the semantic model, then:

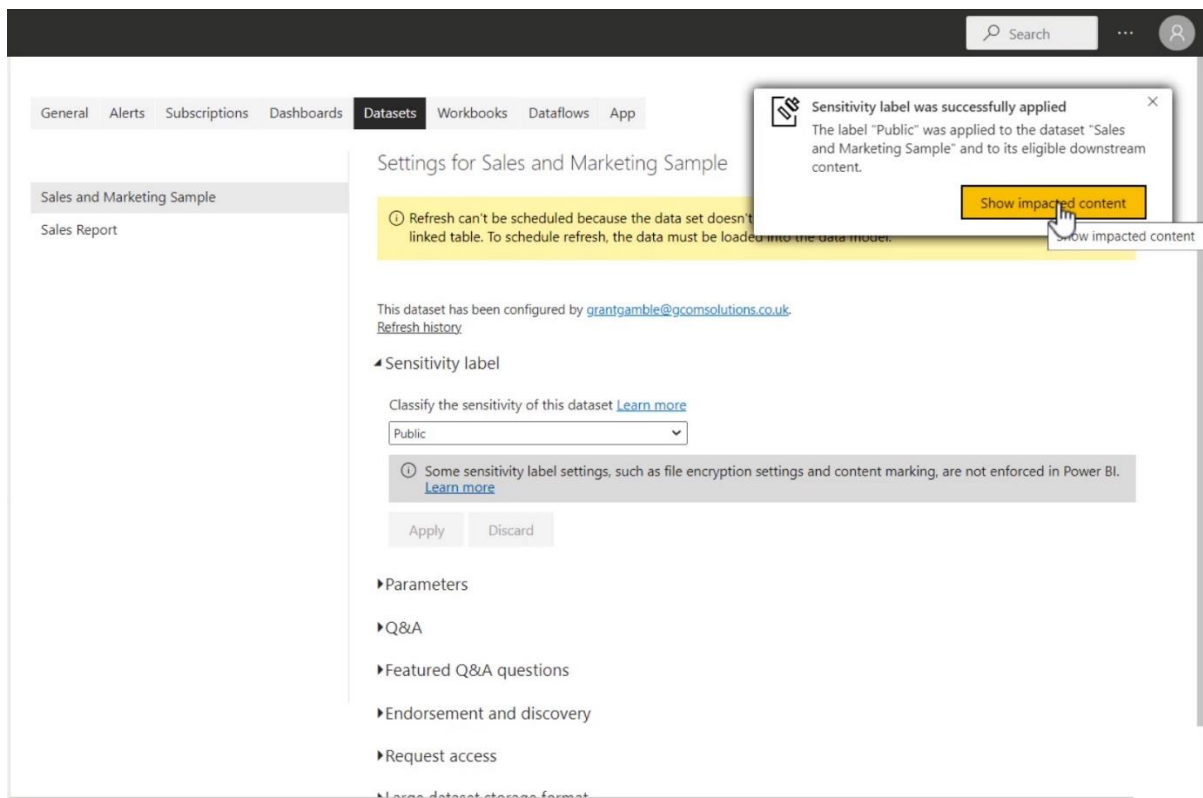
- Any reports based on that semantic model automatically inherit the label.
- Any dashboards built from those reports also inherit the label.

To demonstrate:

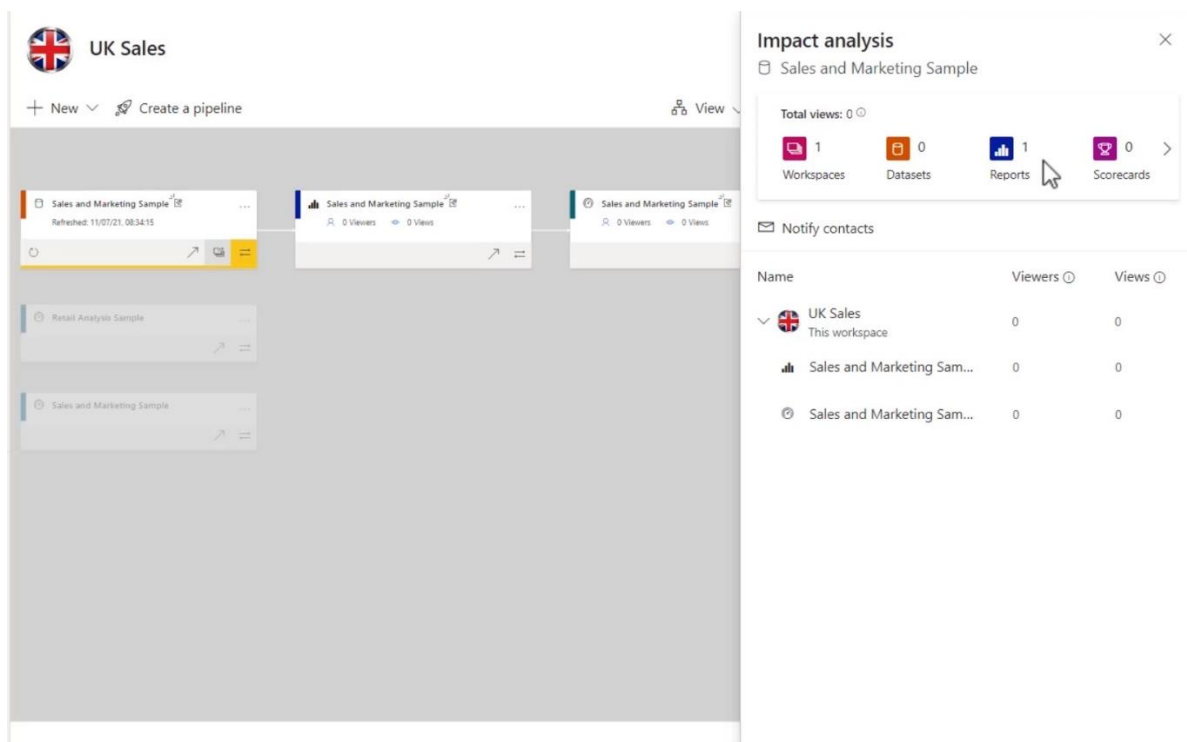
In the following example, we apply the Public sensitivity label to the Sales and Marketing semantic model.

A confirmation message appears: "Sensitivity label was successfully applied to the semantic model and its eligible downstream content."

0509-using Sensitivity Labels



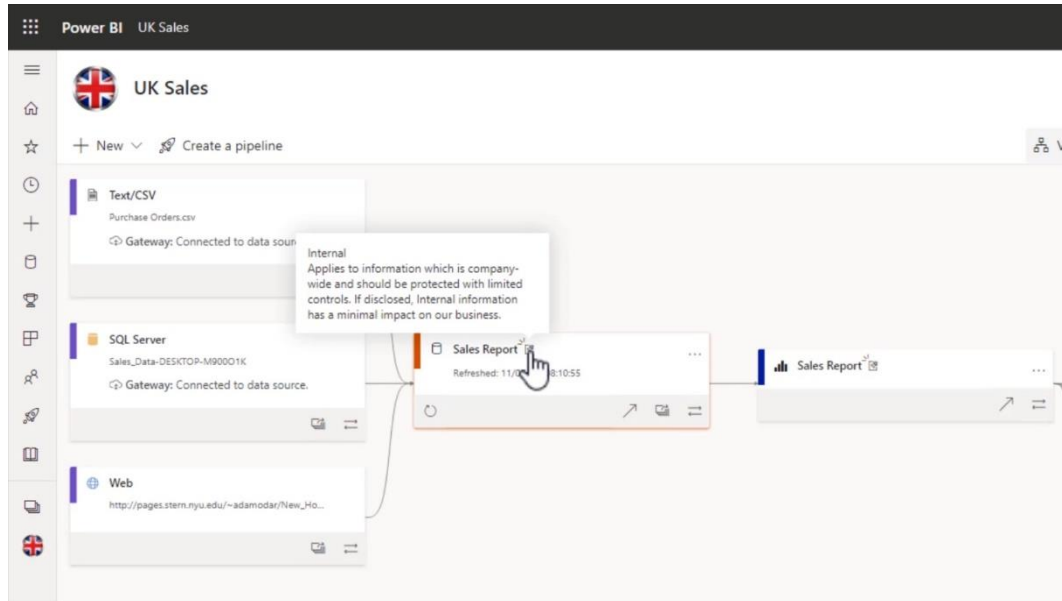
Click Show Impacted Content to view dependencies in Lineage View.



Similarly, if a semantic model is labeled Internal, any related reports and dashboards will inherit the Internal label automatically.

Viewing Sensitivity Labels in Workspaces

Once sensitivity labels are applied, they appear in the Sensitivity column within a workspace listing. Each label is accompanied by an Information (i) button, which displays a description of the label when hovered over.



The label descriptions are set by the administrator when the labels are created. Users can view these descriptions in both List View and Lineage View, ensuring clarity on data classification and protection policies.

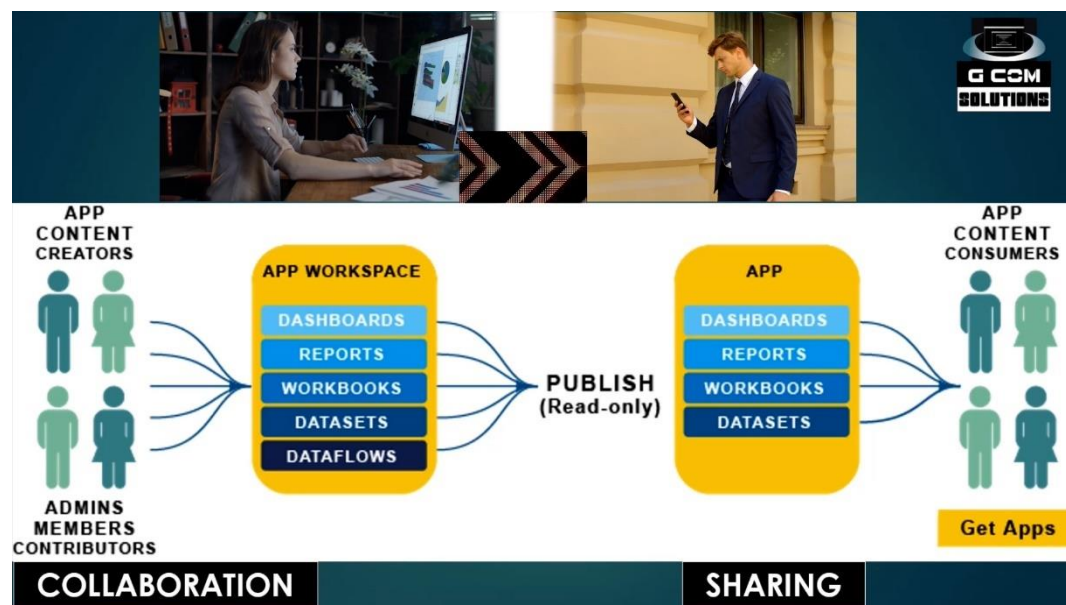
0600: Overview of Sharing Administration

In this section, we'll focus on how to share reports and dashboards effectively within your organisation. While it is possible to share individual reports and dashboards directly, this approach is not recommended. Managing shared content at an individual level quickly becomes cumbersome for both content creators and consumers. Instead, the best practice is to share content through apps.

Why Use Apps for Sharing?

Apps function like mini-websites, providing a structured way to present reports and dashboards from a single workspace. They improve manageability and ensure a more seamless experience for consumers. Importantly, only one app can be created per workspace, but not all workspace content has to be included in the app. This allows content developers to decide when to make specific items available. Whenever new content is added to an app, the app must be updated to reflect those changes.

You can think of apps as the consumer-facing version of workspaces. They allow administrators to control which groups or individuals have access and define exactly what content each group can see.



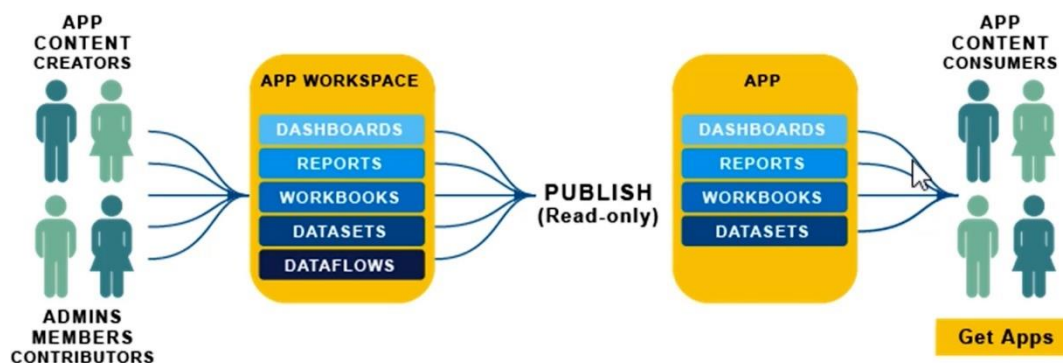
Distinguishing Between Workspace and App Access

When managing access, it is crucial to differentiate between workspace access and app access:

Workspace access should be reserved for content creators who need to build and manage reports.

App Workspace Permissions				
	Admin	Member	Contributor	Viewer
View content	Yes	Yes	Yes	Yes
Create, edit and delete content	Yes	Yes	Yes	No
Publish reports from Power BI Desktop	Yes	Yes	Yes	No
Publish/Update apps	Yes	Yes	No	No
Share individual reports and dashboards	Yes	Yes	No	No
Add members and contributors	Yes	Yes	No	No
Update and delete workspace	Yes	No	No	No
Delete users, members and other admins	Yes	No	No	No

App access is designed for content consumers who only need to view and interact with the published content.



A best practice is to work with security groups rather than individual users. This allows flexibility—some users might have workspace access for content creation, while others may only need access to the app.

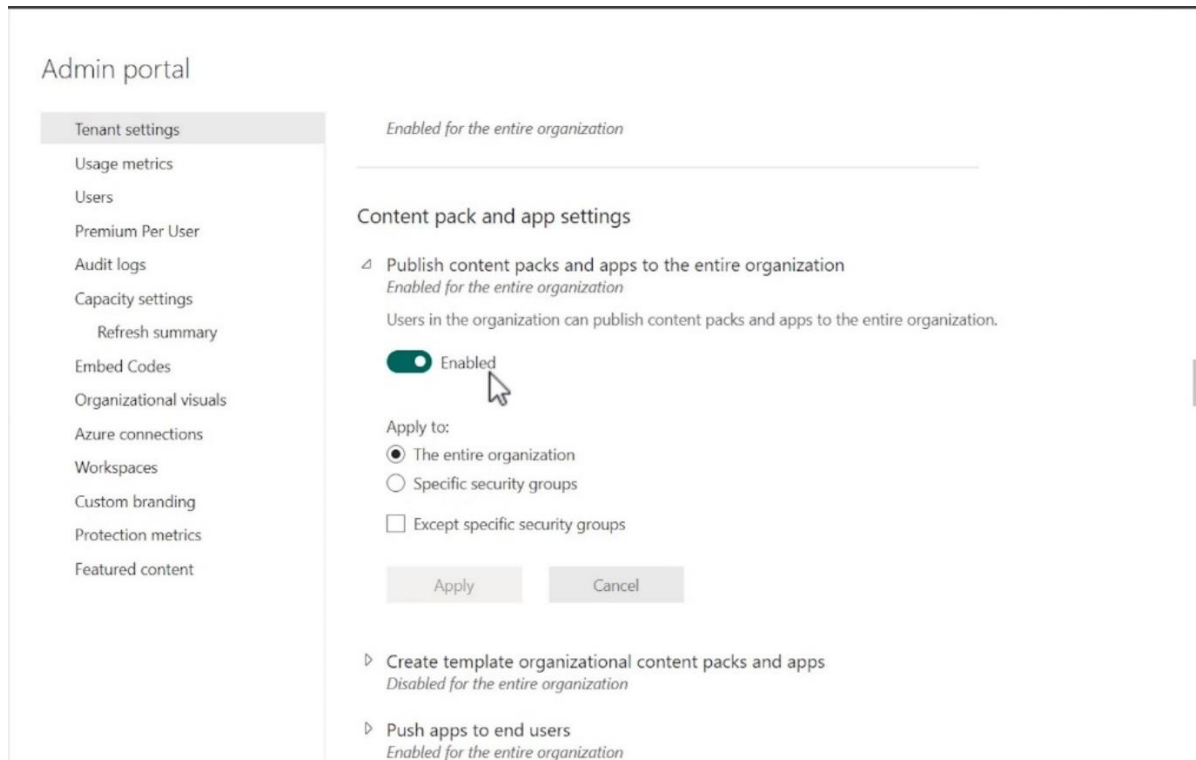
Configuring App Publishing in the Admin Portal

In the Admin portal, under Tenant settings, the relevant setting for app creation is found in the Content pack and app settings section. The key option here is Publish content packs and apps to the entire organisation.

In most cases, this setting should remain enabled for the whole organisation rather than being restricted to specific security groups.

There are two key reasons for this:

- The ability to create apps is essential for effective content sharing.
- Even with this setting enabled, only users with the Member role in a workspace can publish apps.



To maintain control, standard workspace access should be set to Contributor, allowing users to create and edit content. The Member role should be reserved for those responsible for publishing apps.

By structuring access in this way, organisations can ensure that content is shared efficiently while maintaining clarity between content creators and consumers.

0601: Publishing an App

One of the most efficient ways to share content in the Power BI service is by publishing an app workspace as an app. The app itself is essentially a read-only version of the original workspace, allowing users to consume reports and dashboards without making changes.

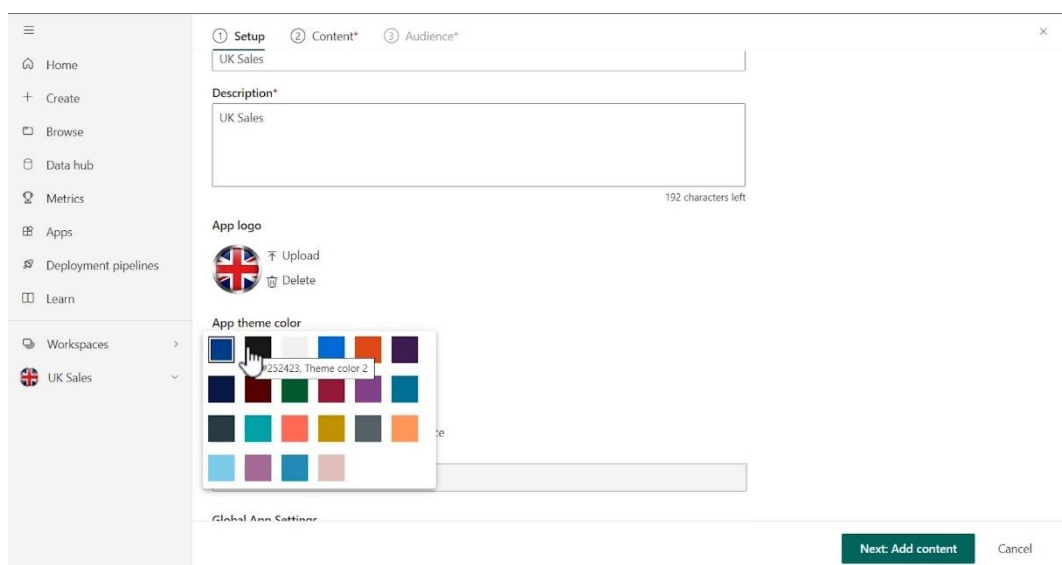
Reviewing Content Before Publishing

Before creating an app, it's important to review the content within the app workspace and decide which items should be included in the published app. Not everything in the workspace needs to be part of the final app—selecting only relevant reports and dashboards ensures a streamlined experience for end users.

Creating the App

To create an app, click on the Create App button in the top-right corner. The setup process consists of three key sections: Setup, Content, and Audience.

Setup: Configuring App Details

The screenshot shows the 'Setup' section of the Power BI app creation process. On the left is a navigation pane with options like Home, Create, Browse, Data hub, Metrics, Apps, Deployment pipelines, Learn, Workspaces, and a selected 'UK Sales' workspace. The main area has three tabs: 'Setup' (active), 'Content*', and 'Audience*'. Under 'Setup', there's a text field for 'App name' containing 'UK Sales', a 'Description*' text area also containing 'UK Sales' with a '192 characters left' indicator, an 'App logo' section showing a UK flag with 'Upload' and 'Delete' options, and an 'App theme color' palette with a color picker open showing '#232423, Theme color 2'. At the bottom right are 'Next: Add content' and 'Cancel' buttons.

In the Setup section, you configure the app's details:

- **App Name** – This is inherited from the original workspace name.
- **Description** – A clear, informative description helps users understand the app's purpose and intended audience.
- **Logo and Colour** – The logo is inherited from the workspace, but you can upload a custom one for better recognition. Similarly, choosing a colour scheme can help align the app with corporate branding.
- **Contact Information** – You can specify a security group (e.g., administrators) as the contact for any queries regarding the app.
- **Global Settings** – These include options like:

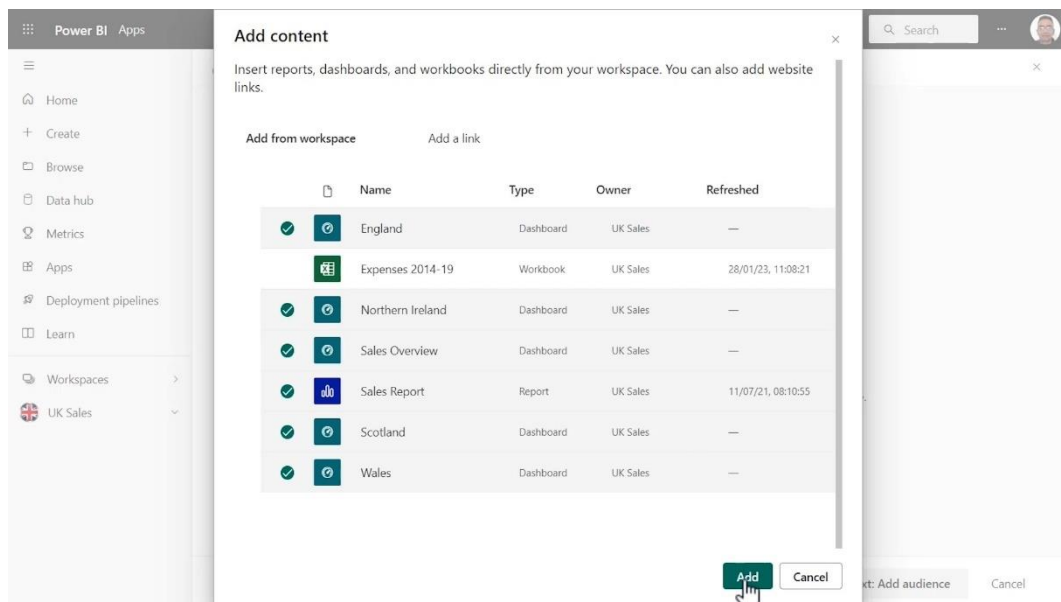
- Install this app automatically – When enabled, users do not need to manually retrieve the app; it appears in their list automatically.
- Hide app navigation pane – Generally not recommended, as the navigation pane is auto-generated and user-friendly.
- Allow users to make a copy of reports – Enabling this can lead to multiple versions of reports circulating, which may cause confusion.
- Support URL – This could be a SharePoint page or other resource providing documentation and support.

Content: Selecting Reports and Dashboards

The Content section defines which elements from the workspace will be included in the app.

Click Add Content to select specific reports, dashboards, and other items.

You do not have to include all content from the workspace—only relevant items should be added.

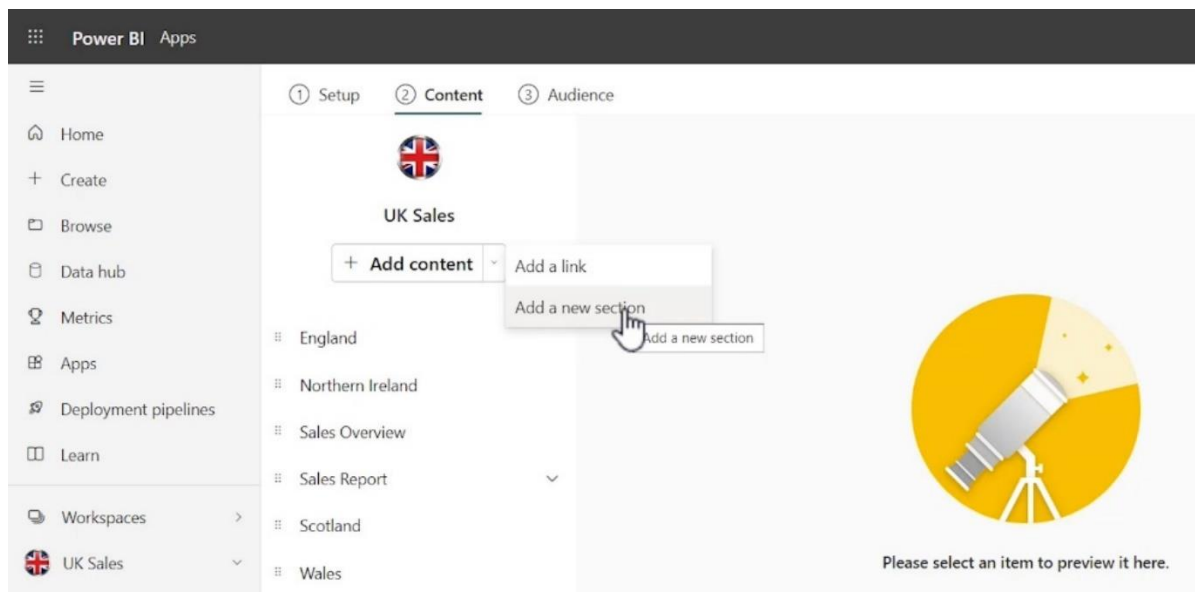


A navigation system is generated automatically, making it easy for users to find content.

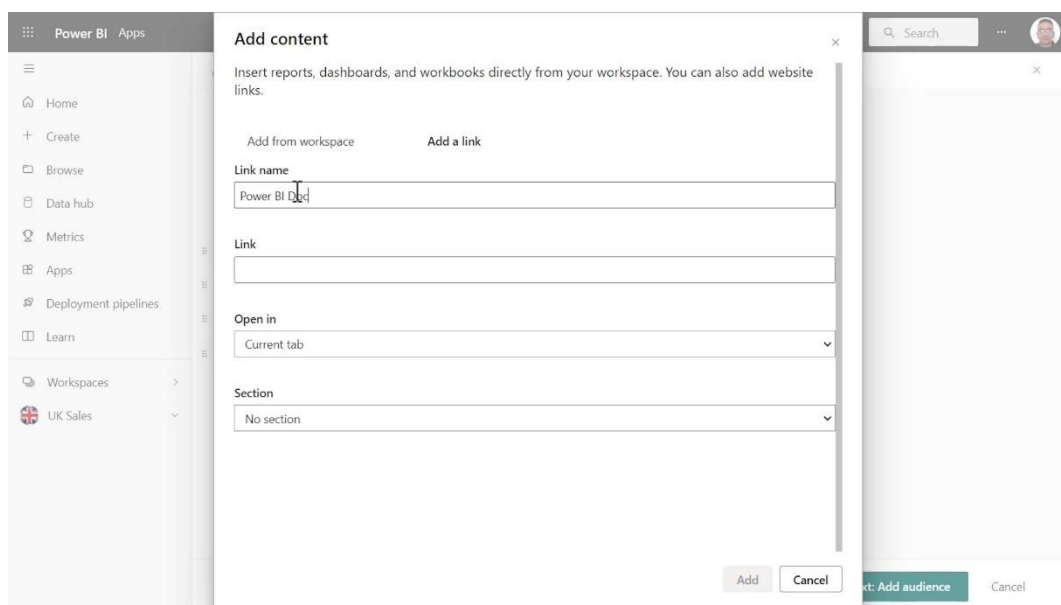
Additional customisation options include:

Sections – These act like folders to group related content. For example, you might create a section called "By Country" and organise reports within it.

0601-publishing An App



Links – You can add links to external resources, such as Power BI documentation or internal help pages. These links can be placed in a dedicated section like "Help" for easy access.



Audience: Controlling Access

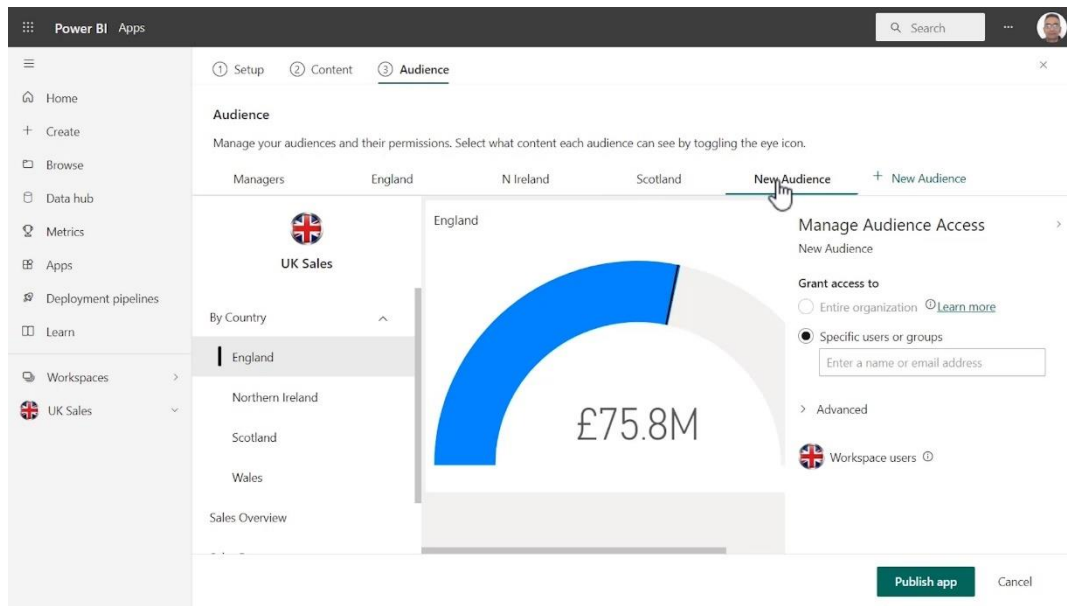
The Audience section determines which users or groups have access to the app and what content they can see.

While it is possible to share the app with individual users, using security groups is recommended for better management.

The default audience name is taken from the app name, but it can be renamed.

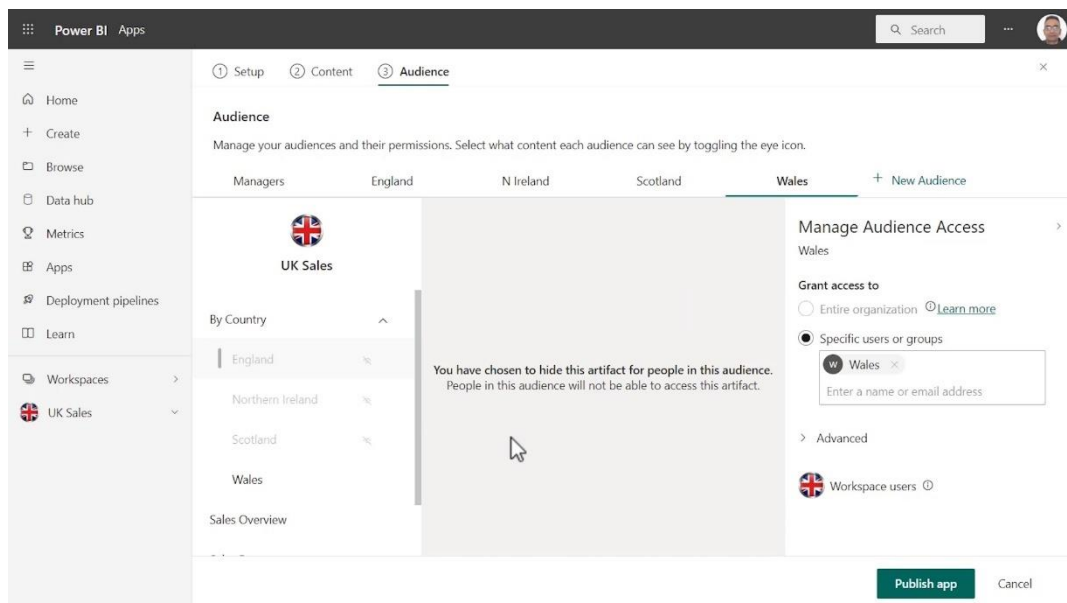
Additional audiences can be created to control access at a more granular level. For example, separate groups can be created for different regions (e.g., Managers, England, Scotland, Wales, Northern Ireland).

0601-publishing An App



For each audience, you can manage visibility settings, ensuring that users only see relevant content.

This flexible approach allows you to tailor access precisely, ensuring that different teams or departments only see the information relevant to them.

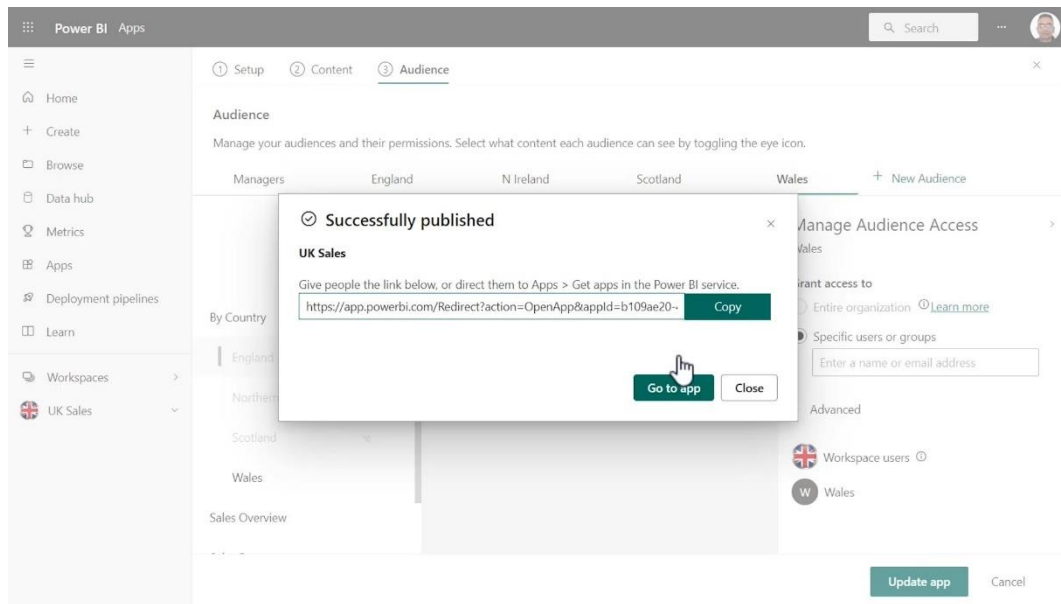


Publishing and Managing the App

Once everything is configured, click Publish App. A confirmation message appears, and clicking Publish again finalises the process.

After publishing, you can preview the consumer experience by selecting Go to App. This lets you see the app as end users will experience it.

0601-publishing An App



Although the published app itself is read-only, you can make changes by going back to the app workspace and selecting Update App. Any updates made in the workspace can then be republished to refresh the app.

Next Steps

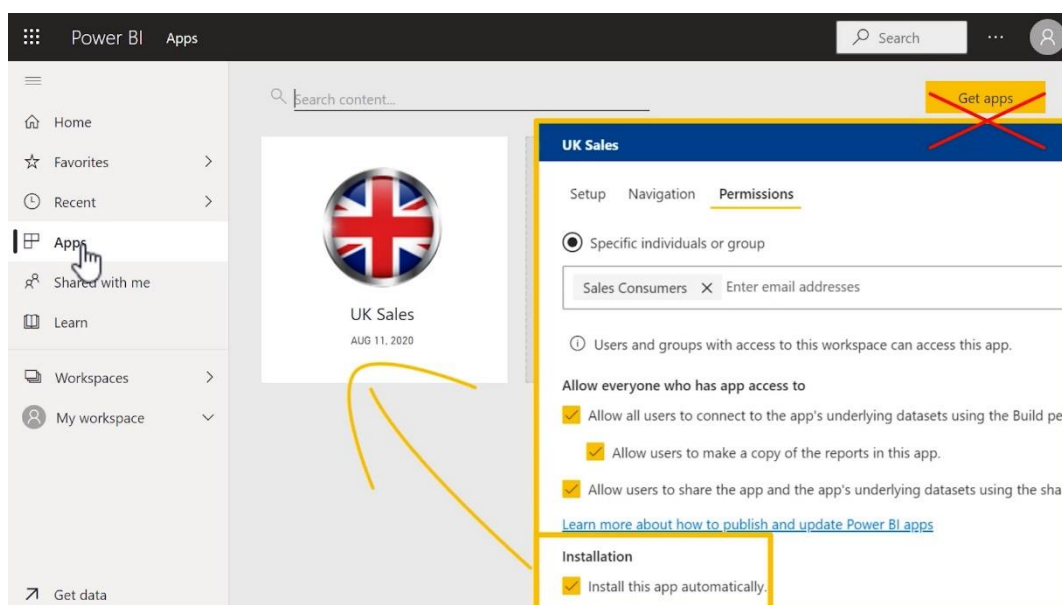
With the app successfully published, the next step is to explore how users interact with it. The following tutorial will cover the consumer experience, demonstrating how audiences navigate and engage with the app's content.

0602: The App Consumer Experience

When you share an app in Power BI, it's important to understand that you are not granting access to the underlying workspace. This distinction is crucial because while content creators work within app workspaces, consumers access the published apps without direct involvement in the workspace itself.

For example, let's consider a user who is a member of the Sales Consumer security group, which has been granted access to the UK Sales app. However, this user does not have access to the app workspace. If they navigate to Workspaces, they will find that none are available to them. Instead, to see the apps they have access to, they must use the Apps button in the navigation pane.

If the app was set to install automatically, it appears immediately in their list of available apps. If not, the user would need to click Get Apps and then navigate to the Organizational Apps tab to locate and install it manually.



Navigating the App

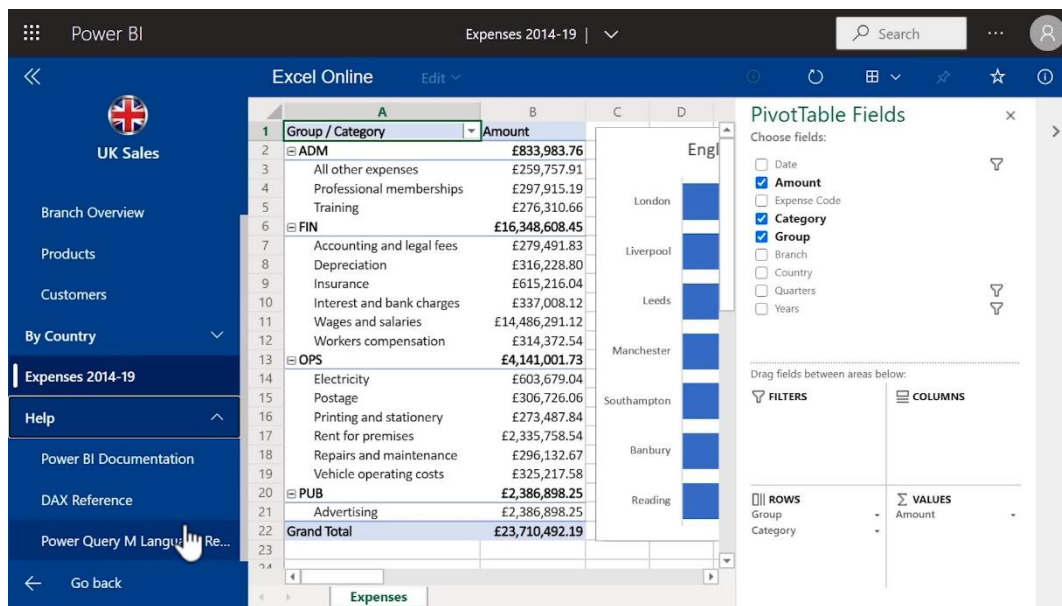
Once inside the app, the experience is structured much like a mini website. The left-hand navigation pane provides access to different sections of content. In this example:

The Sales Reports section contains links to each of the report pages, which were created automatically.

A By Country section includes links to relevant dashboards.

An Excel Workbook section provides direct access to integrated Excel files.

A Help section includes external links, all configured to open in a separate tab.



This structured approach ensures that all relevant information is conveniently accessible in a single, user-friendly container.

Publishing Options

When publishing an app, several options can be enabled that affect how users interact with the underlying data:

Allow All Users to Connect to the App's Semantic models

This option grants consumers permission to use the app's semantic model when building their own reports. By selecting View Related from the ellipsis menu, users can see the available semantic model (in this case, the Sales Report semantic model). They can then create their own reports using the full data model, including all available measures, but without the ability to modify the semantic model itself. These new reports can be saved either in My Workspace or another workspace they have access to.

UK Sales

Setup Navigation **Permissions**

Sales Consumers > grant.gamble@seadigital.com > Enter email addresses

ⓘ Users and groups with access to this workspace can access this app.

Allow everyone who has app access to

- ☒ Allow all users to connect to the app's underlying datasets using the Build permission.
- ☒ Allow users to make a copy of the reports in this app.
- ☒ Allow users to share the app and the app's underlying datasets using the share permission.

[Learn more about how to publish and update Power BI apps](#)

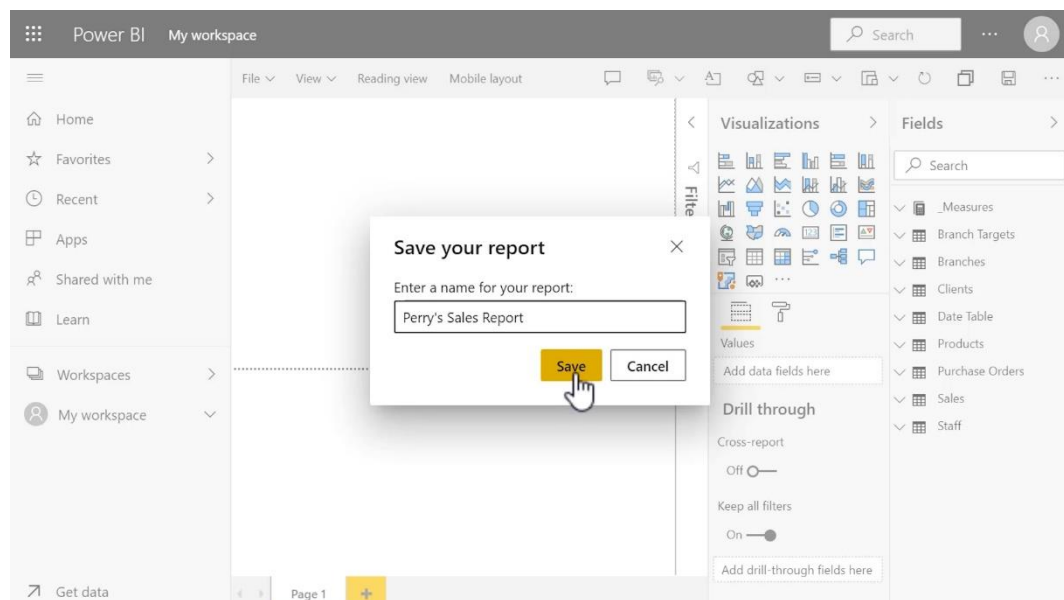
Installation

- ☒ Install this app automatically.

For instance, Perry, a consumer with no assigned workspaces, can create his own Sales Report using the shared semantic model. However, since he lacks workspace permissions, his new report is saved in My Workspace.

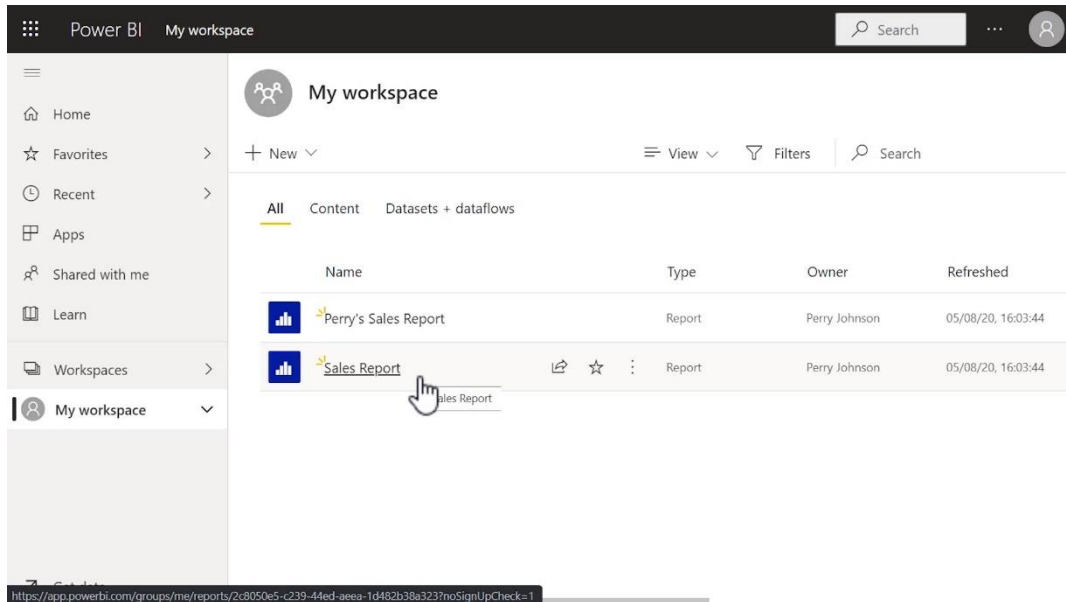
Allow Users to Make a Copy of Reports

Consumers can also duplicate reports using the Save a Copy option. This allows them to start with an existing report and modify it as needed. Perry, for example, can save a copy of the Sales Report to his My Workspace, providing him with an editable version that retains all report pages, including hidden ones. However, he remains unable to alter the underlying semantic model.



Allow Users to Share Reports

Enabling this option allows consumers to share their own versions of reports with others. Looking inside Perry's My Workspace, we now see both his blank Sales Report and his saved copy of the completed report. Because sharing is enabled, Perry can now distribute his version of the report to colleagues.



Best Practices

While these sharing and copying options might seem useful at first glance, they often lead to fragmented versions of reports circulating across the organization. This can result in multiple, inconsistent variations of what should be a single authoritative report.

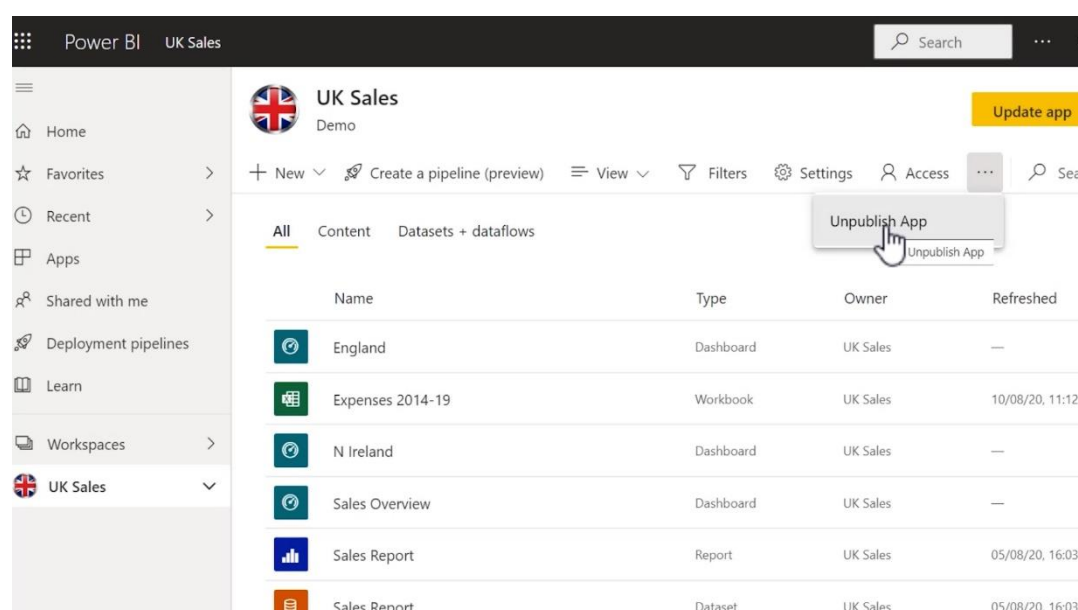
To maintain control and consistency, best practice is to disable these options unless there is a specific, justified need to allow them. Keeping reports centralized within the published app ensures a consistent and managed reporting experience for all consumers.

0603: Unpublishing an App

When you publish an app in Power BI, it becomes a distinct entity, separate from the original workspace. There are times when you may need to remove the app while keeping the workspace intact. For example, if errors or inconsistencies appear in a key report, you might choose to unpublish the app, correct the issues, and then republish. Similarly, if your process includes an approval stage, you may publish an app for a limited audience, gather feedback, and unpublish it if necessary to refine and improve it before full deployment.

Navigating to the Unpublish Option

To unpublish an app, you must be inside the app workspace, not the app itself. If you navigate to the app, you'll notice that there is no direct option for unpublishing. Instead, go to the app workspace by clicking the Edit button within the app. Once inside the workspace, open the ellipsis menu (three dots) and select Unpublish app.



Effects of Unpublishing an App

Once an app is unpublished, it no longer exists. If you check the Apps section after unpublishing, you will see that the app has disappeared. This is not just for the owner—app consumers will also lose access. For instance, if a user, such as Perry Johnson, refreshes their browser, the app will vanish from their list immediately.

Key Considerations

Unpublishing an app is a permanent action. The app itself is completely removed, along with all its settings. If you decide to republish, you will need to reconfigure everything from scratch, including navigation settings and links. None of the original publishing configurations will be retained, so it's important to be certain before proceeding.

0603-unpublishing An App

In summary, while unpublishing is sometimes necessary, it is a drastic step. If your intention is to make updates or revisions, consider whether alternative approaches, such as updating individual reports within the workspace, might be more suitable before taking this irreversible action.

0604: Updating an App

Once an app has been published, the Create App option is replaced with Update App. But when exactly do you need to update an app? Let's consider our UK Sales App as an example.

When Do You Need to Update an App?

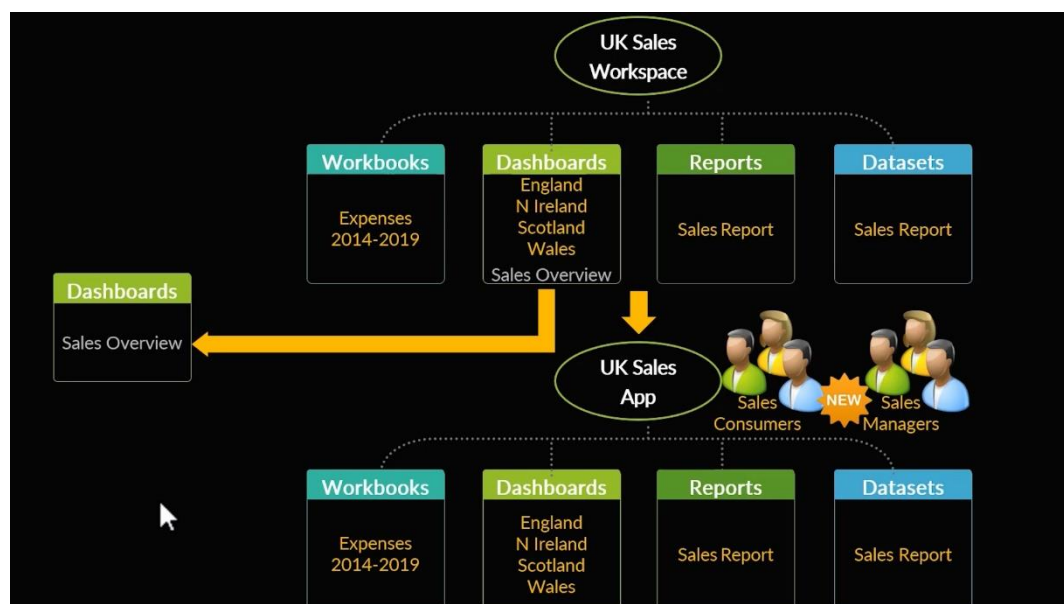
Some changes do not require an app update. For example, if the underlying semantic model powering a report is updated with new data, a simple data refresh is sufficient—there's no need to update the app itself. We'll cover data refreshing in more detail in the next section.

However, structural changes to the semantic model do require an update. If, for instance, new measures need to be added to the semantic model, an app update will be necessary.

Other examples of changes that require an app update include:

- Adding new reports
- Creating a new dashboard
- Uploading a new Excel workbook
- Modifying sharing settings

For example, if our app is currently shared with the Sales Consumers group and we now want to share it with a new group, such as Sales Managers, an app update will be required.



How to Update an App

Let's walk through the process by adding a new report to our app. In this case, we'll use one of the Power BI sample reports.

To access these:

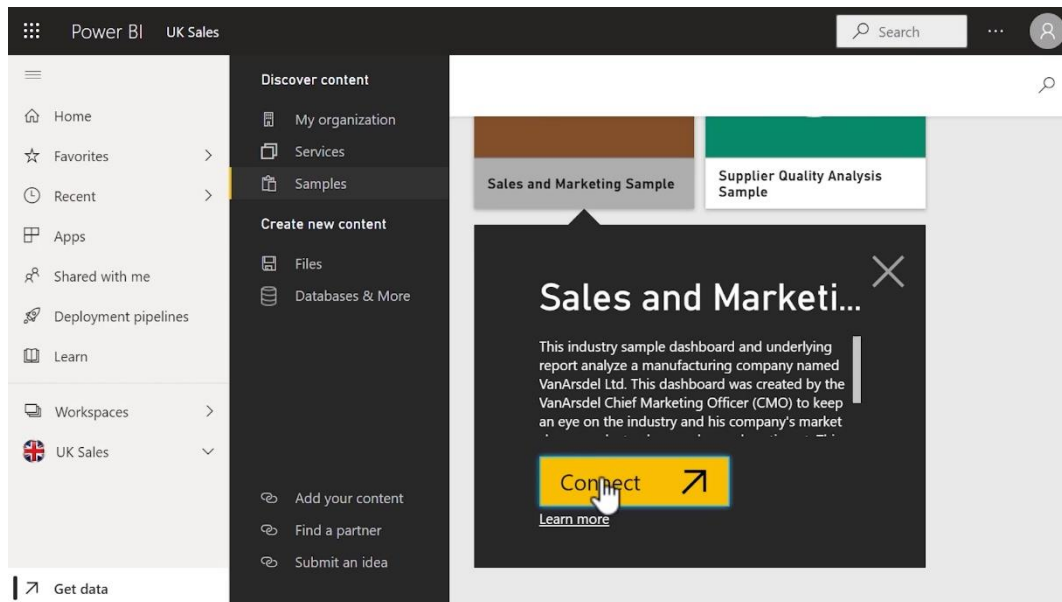
0604-updating An App

Click Get Data.

Scroll down and locate the link to Samples.

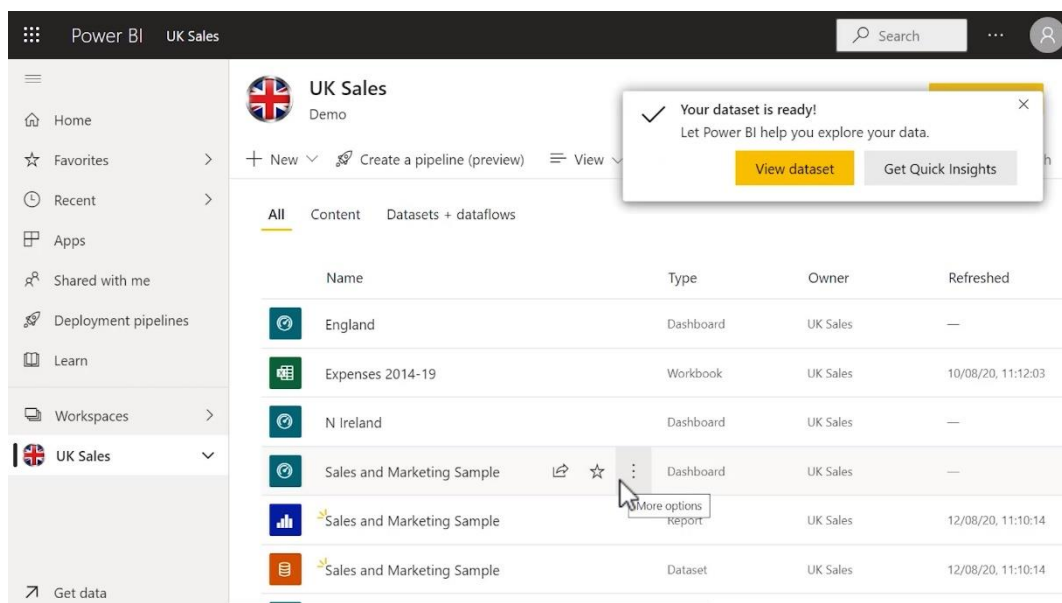
Since we're working in a Sales Workspace, we'll select the Sales and Marketing Sample.

We click Connect to import the sample.



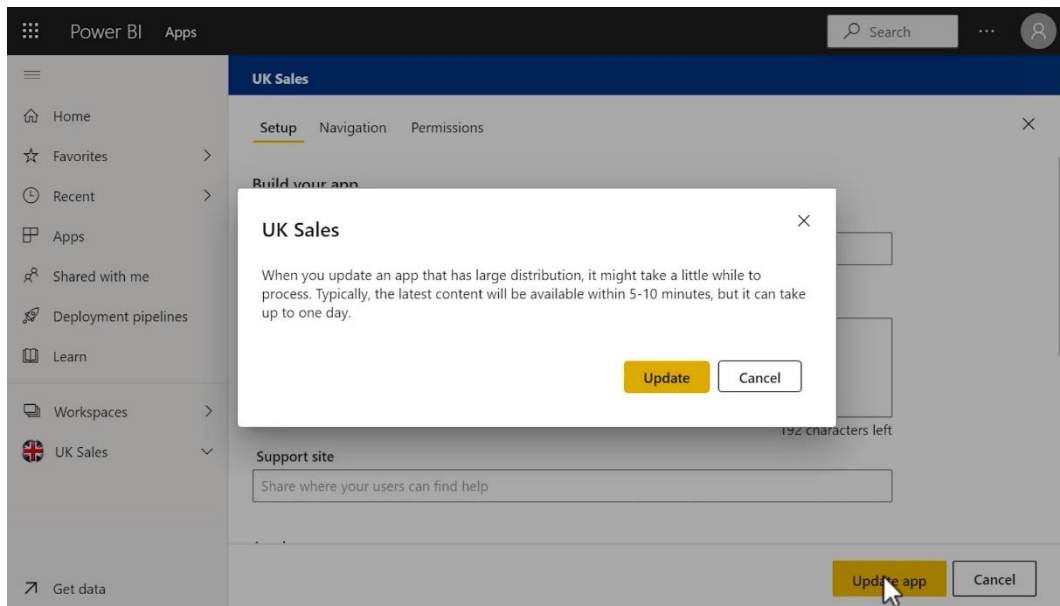
The Sales and Marketing sample includes three items:

- A dashboard
- An underlying report
- A semantic model



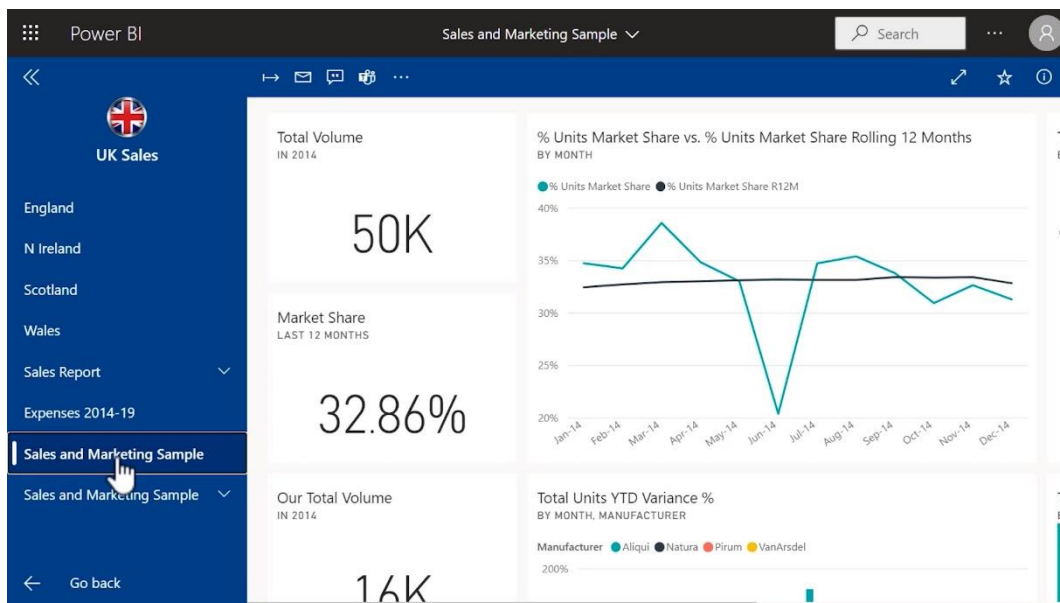
Let's say we now want the dashboard and report to be included in the app. Once the new content has been added, we proceed by clicking Update App.

0604-updating An App



Verifying the Update

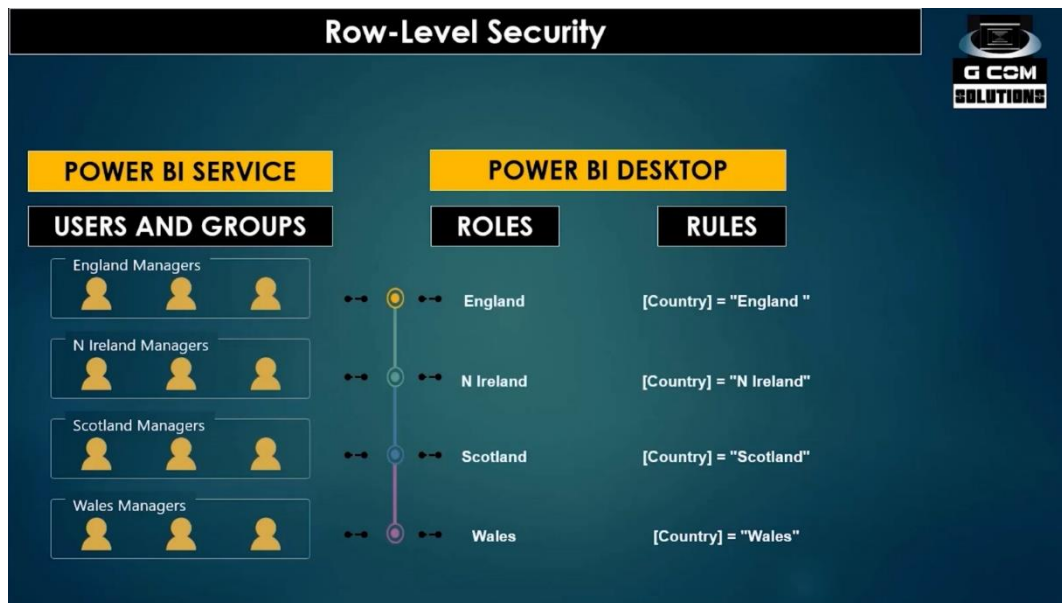
After updating the app, we can verify the changes by checking another user's account. In this case, when Perry refreshes his page by pressing F5, the updated version of the app will load, and he will see the new content.



There's no limit to how many times you can update an app, so changes can be made as often as necessary.

0605: Using Row-Level Security

Row-Level Security (RLS) in Power BI allows you to deploy a single report to multiple users while controlling what each user can see. This is achieved by assigning users to specific roles, which determine their access to data. The roles are created in Power BI Desktop, and then, in the Power BI service, individuals or groups are assigned to these roles.



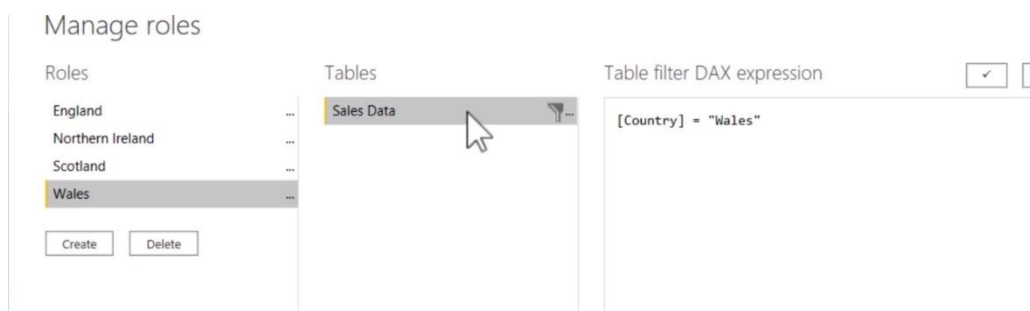
Defining Roles in Power BI Desktop

In our example, we have two main user groups: sales managers and salespeople. Sales managers should only see data relevant to their country, while salespeople should only view their own sales figures. This means they will see their country name, branch name, their own name as an account handler, and a breakdown of their sales by product range and product.

To implement RLS, we begin by creating roles in Power BI Desktop. Since we have four countries—England, Northern Ireland, Scotland, and Wales—we will create a role for each country. This is done via the Modelling tab by selecting Manage Roles.

For each role, we apply a DAX filter expression to restrict data visibility. For example, the DAX filter for Wales is:

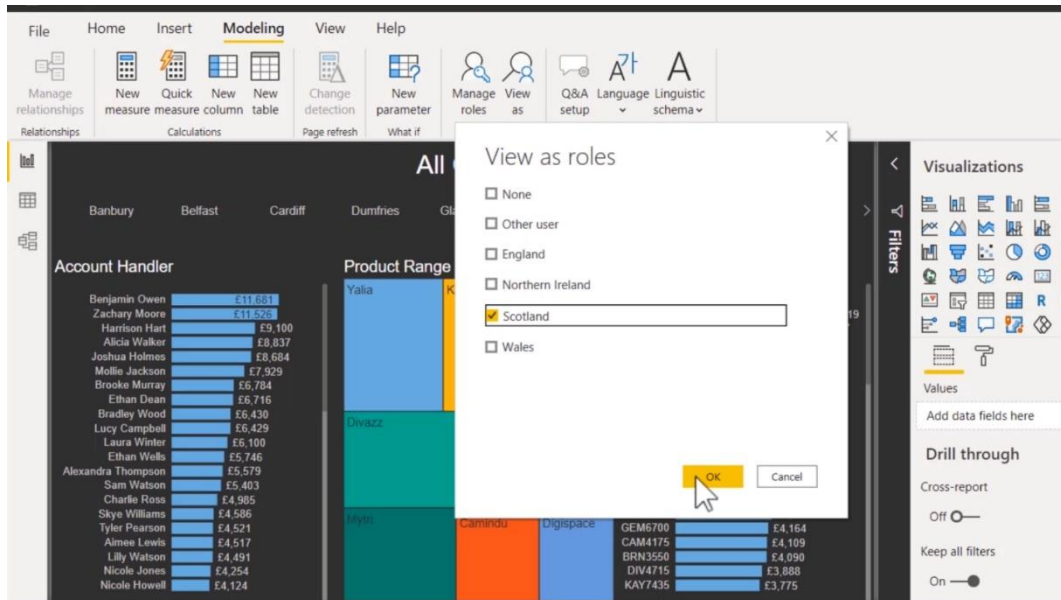
`[Country] = "Wales"`



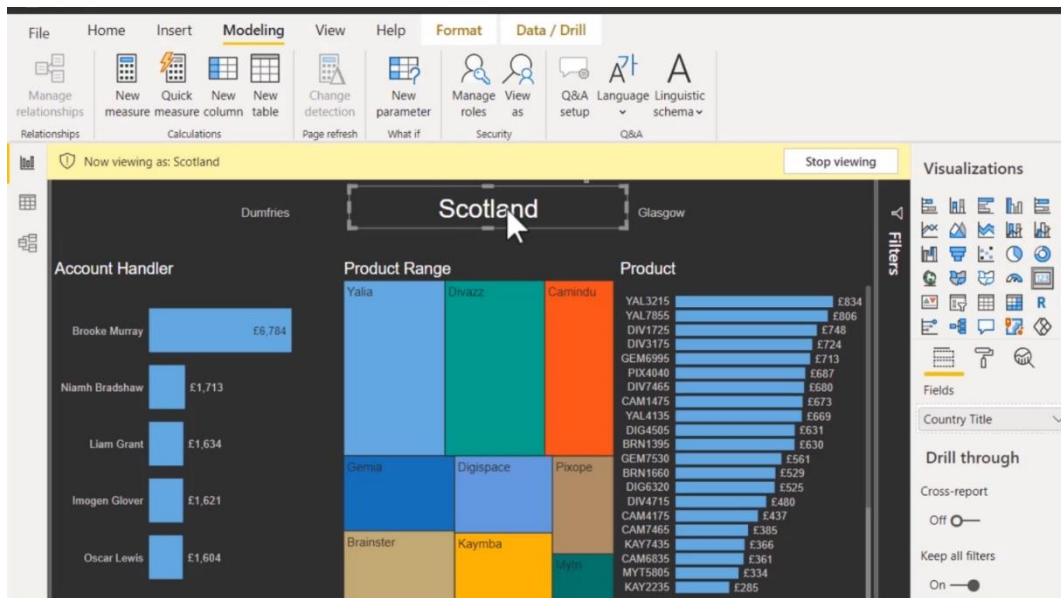
This process is repeated for each country. Once all roles are assigned their respective DAX filters, we save the roles.

Testing Roles in Power BI Desktop

Before moving to the Power BI service, we can verify that the roles are working correctly using the View As feature in the Modeling tab of the ribbon.



By selecting a role (e.g., Scotland), we should see only the branches in Scotland. Clicking on a branch will confirm that only relevant account handlers are displayed.



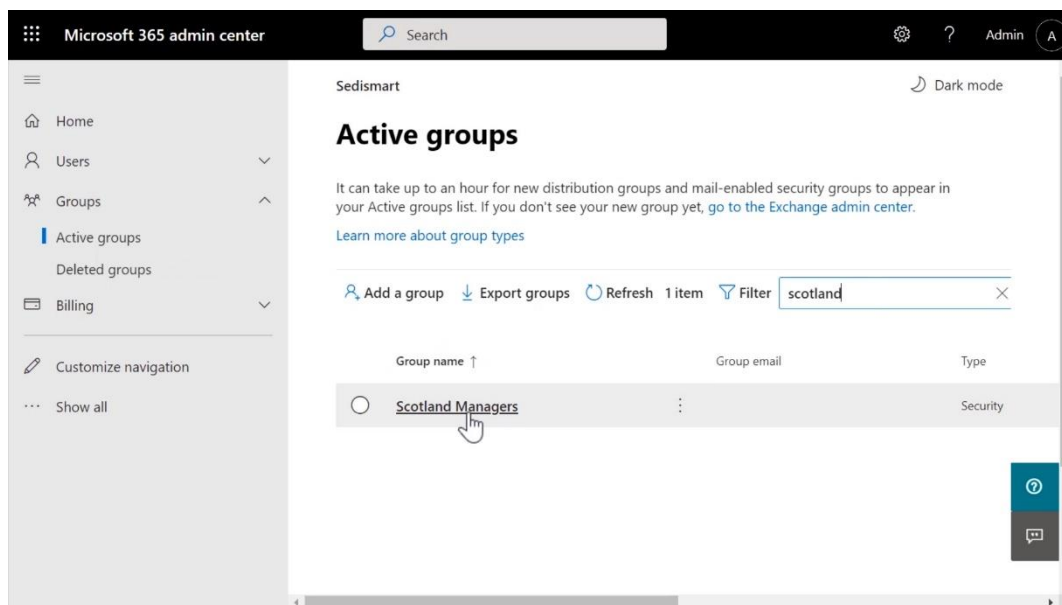
The report dynamically updates based on the selected role, with the country title using the SELECTEDVALUE DAX function to display the current country. If no country is selected, it defaults to showing all countries.

Assigning Users to Roles in Power BI Service

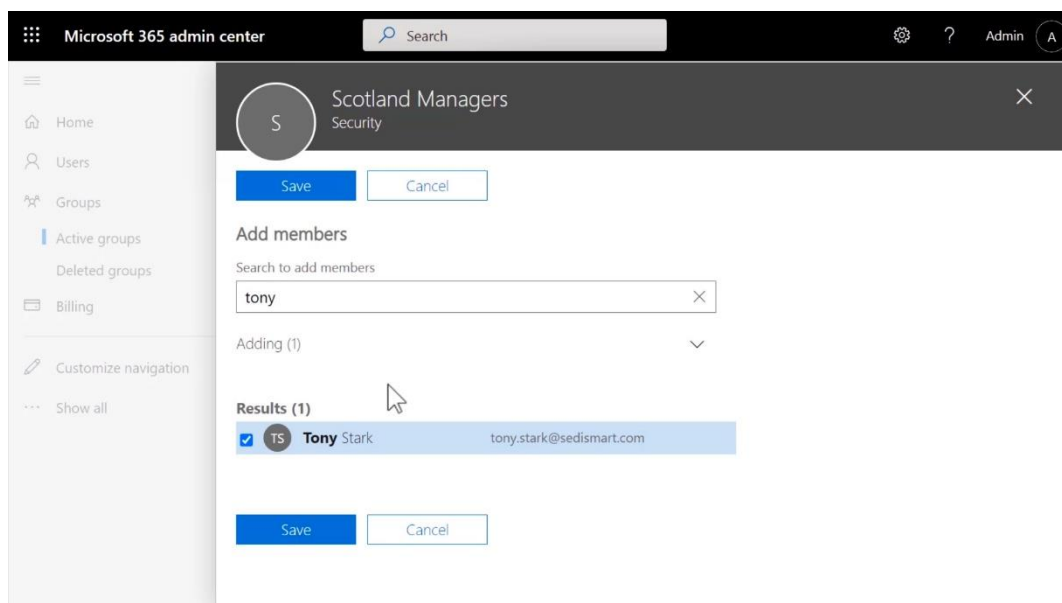
Once the roles are defined and tested, the next step is to publish the report to the Power BI service. For this example, we publish it to a workspace called UK Sales.

You can share your published report with individual users. However, sharing with security groups is a more scalable solution. To assign users to groups, we need to manage security settings. This is done by:

- Navigating to the Power BI admin portal via Settings > Admin Portal.
- Switching to the Users tab and opening the Microsoft 365 Admin Center.
- Viewing Active Groups. In this example, we have a pre-created group called Scotland Managers.



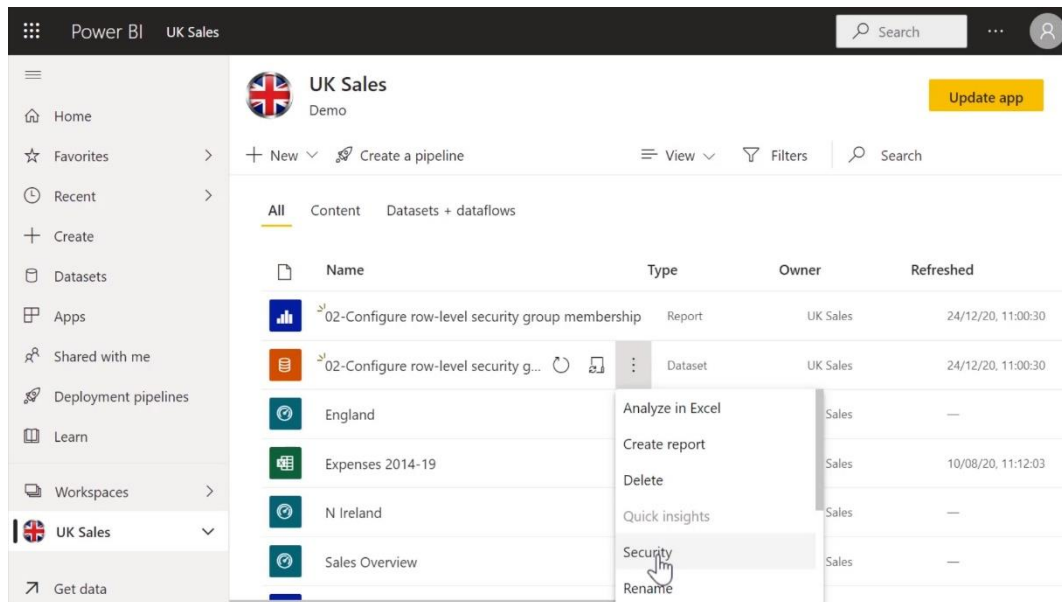
We add a user (e.g., Tony Stark) to this group by selecting Members > View All > Add Members.



Once all users are added to their respective groups, we return to the Power BI service. The semantic model linked to the published report must now be configured to enforce security rules.

Configuring Role Security in Power BI Service

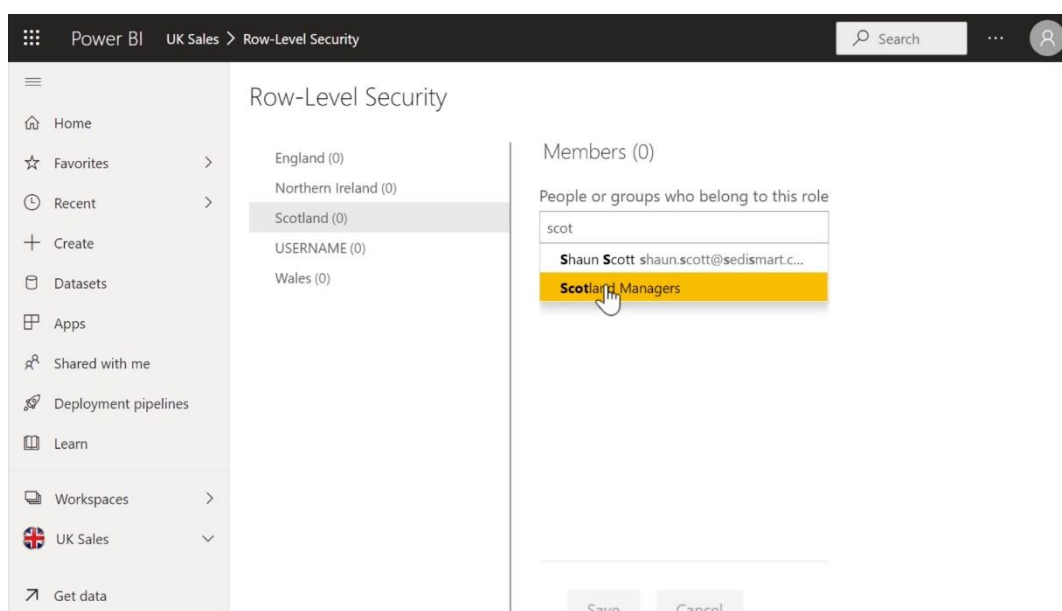
In the UK Sales workspace, we locate the semantic model associated with the report. Using the ellipsis menu (:), we select Security.



Here, we can see the roles we previously created in Power BI Desktop.

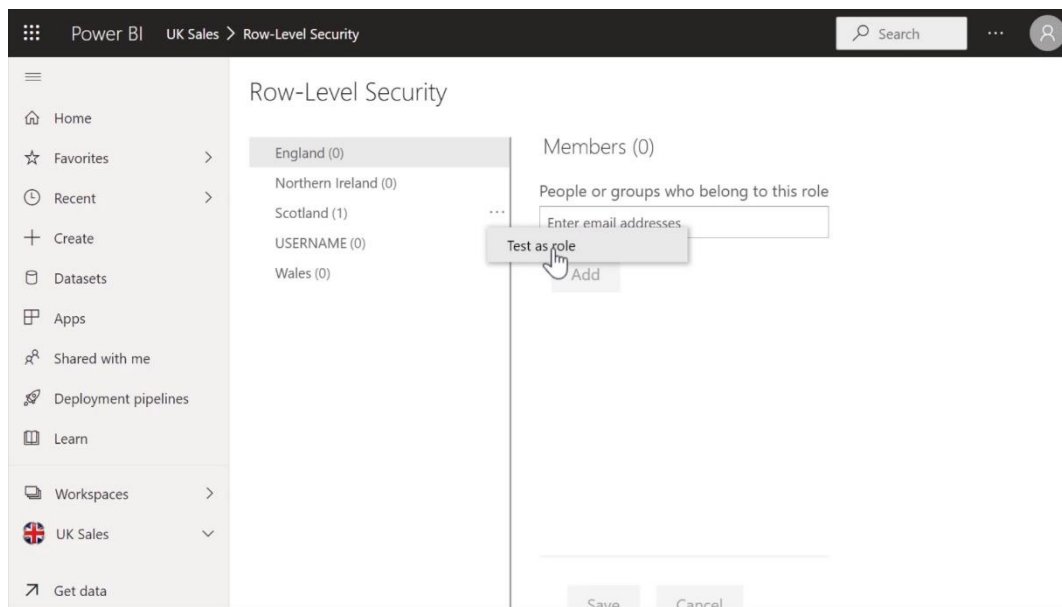
To apply security settings to the Scotland role:

- Select the Scotland role and add the Scotland Managers group.
- Click Add, then Save.

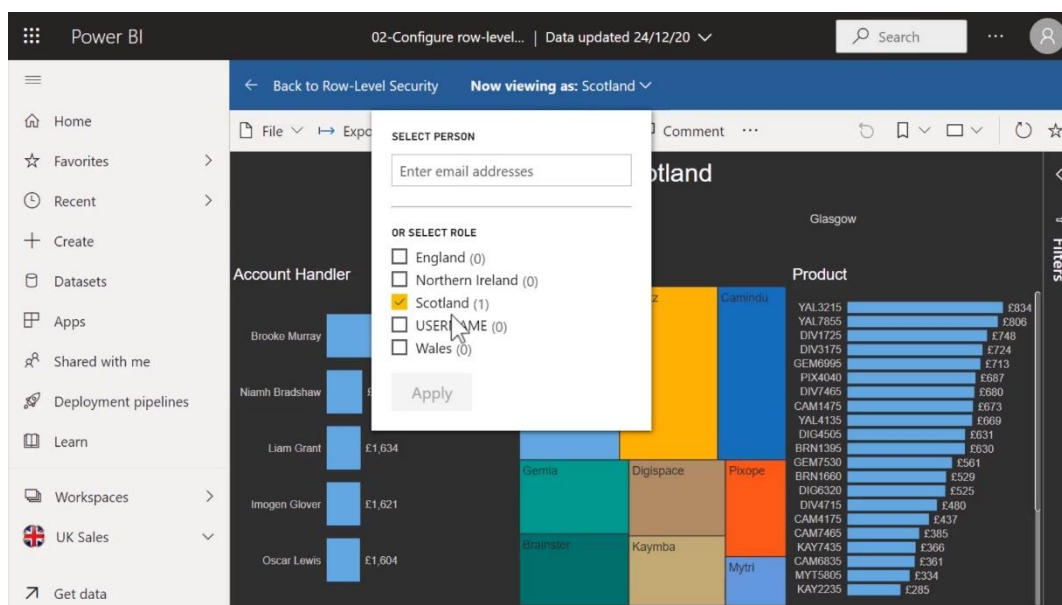


0605-using Row-level Security

To verify that the security setup is functioning correctly, we use the Test as Role option.



This simulates what a user assigned to the role will see. In this case, Scotland managers should only see branches in Scotland and their respective account handlers. If other roles had been assigned users, they could be tested in the same way.



Summary

Row-Level Security in Power BI ensures that users only see data relevant to their role. The process involves:

- Defining roles in Power BI Desktop with DAX filters.
- Testing roles locally using View As.
- Publishing the report to Power BI service.
- Assigning users or groups to roles via the Power BI admin settings.

0605-using Row-level Security

- Verifying role-based access using Test as Role.

With these steps completed, the report can be shared confidently, ensuring that each user sees only the data they are meant to access.

0606: Linking Row-Level Security to User Login

In this section, continuing our look at row-level security, we will focus on restricting salespeople—our account handlers—to viewing only their own sales figures. To achieve this, we will use their login credentials as a filter within Power BI.

Creating a Role for User-Specific Filtering

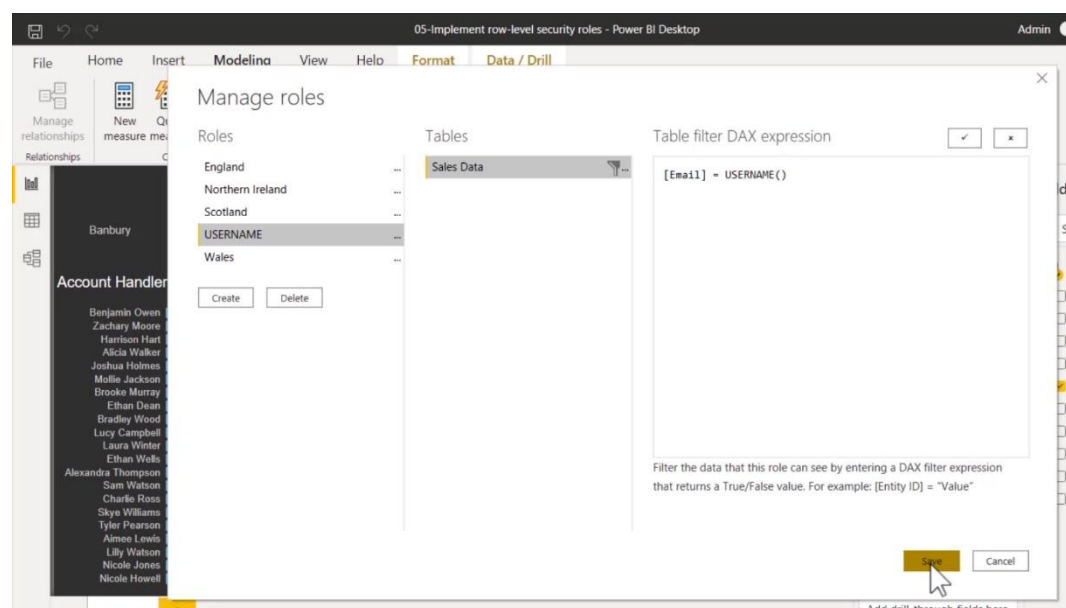
The key to implementing this restriction is leveraging the users' email addresses. If we switch to Data view, we can see that each salesperson's email address is stored in our semantic model. This is the same email address they use to log in to Power BI.

To enforce row-level security (RLS), we only need a single role: a username role. We can create this by navigating to Manage Roles under the Modeling tab. Here, we define a new role named username and apply a filter to the email column.

Instead of setting a static value (e.g., Email = "someone@example.com"), we use a DAX function:

Email = USERNAME()

Since USERNAME() is a function, we must include parentheses. Once this filter is applied, we save the role.



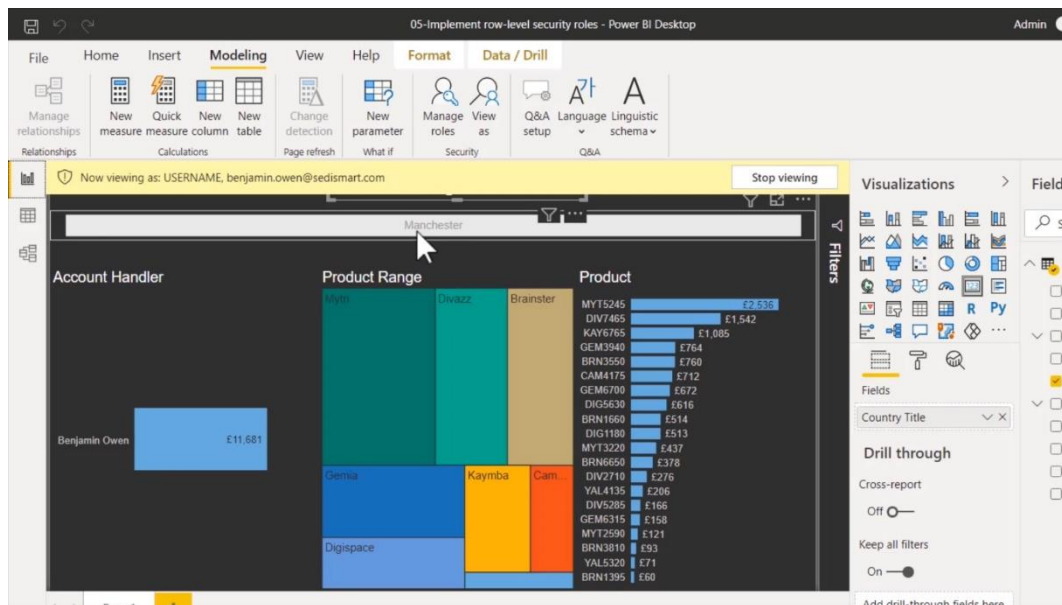
Testing Row-Level Security

To confirm that our role works correctly, we use the View As feature. Here, we select Other user, enable username, and manually enter an example email address. Let's use Benjamin Owen (benjamin.owen@savvysmart.com).

As expected, Benjamin sees only his assigned data:

0606-linking Row-level Security To User Login

- His branch (Manchester).
- His name under Account Handler.
- His personal sales figures.
- A breakdown of sales by product range and product.
- The Country field showing only England (since Manchester is in England).



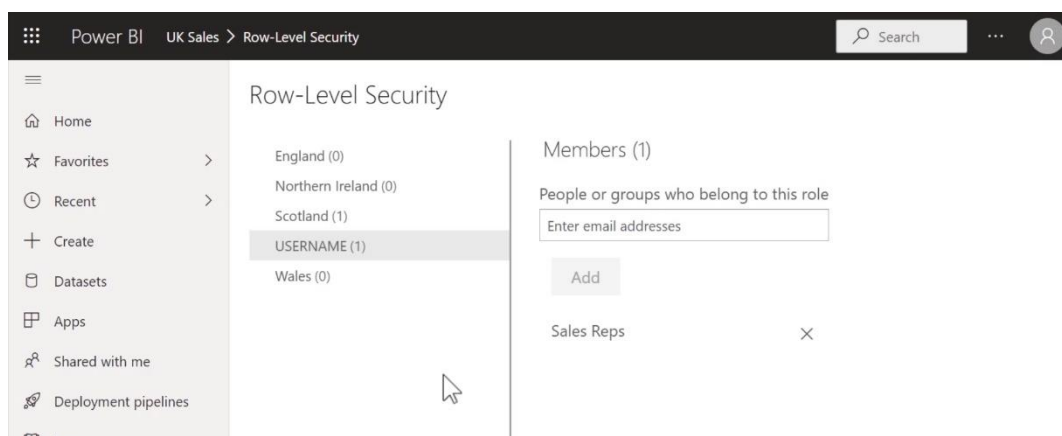
Compared to what sales managers see, Benjamin's report contains far less data—just his relevant information. This confirms that our row-level security is functioning correctly.

Assigning Users to the Role

To implement this at scale, we assign users to the username role through Power BI's Security settings.

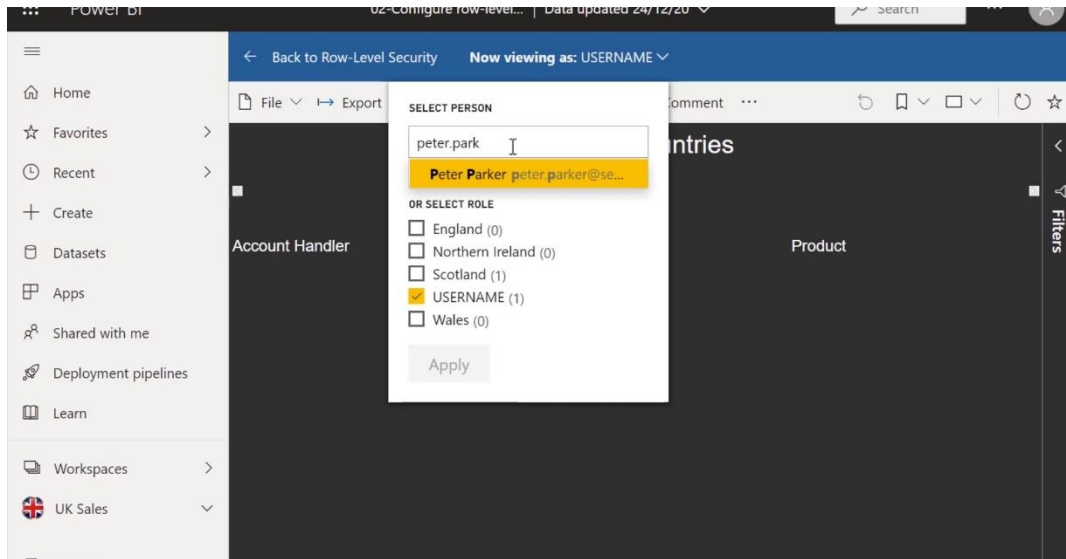
We navigate to the Admin Portal and open the Microsoft 365 Admin Center. Under Groups > Active Groups, we locate the Sales Reps group and add a test user—Peter Parker.

After adding Peter to the group, we return to Power BI and open the Security settings for our semantic model. Using the ellipsis menu (...) next to the semantic model, we select Security and add the Sales Reps group to the username role. Once added, we save the changes.



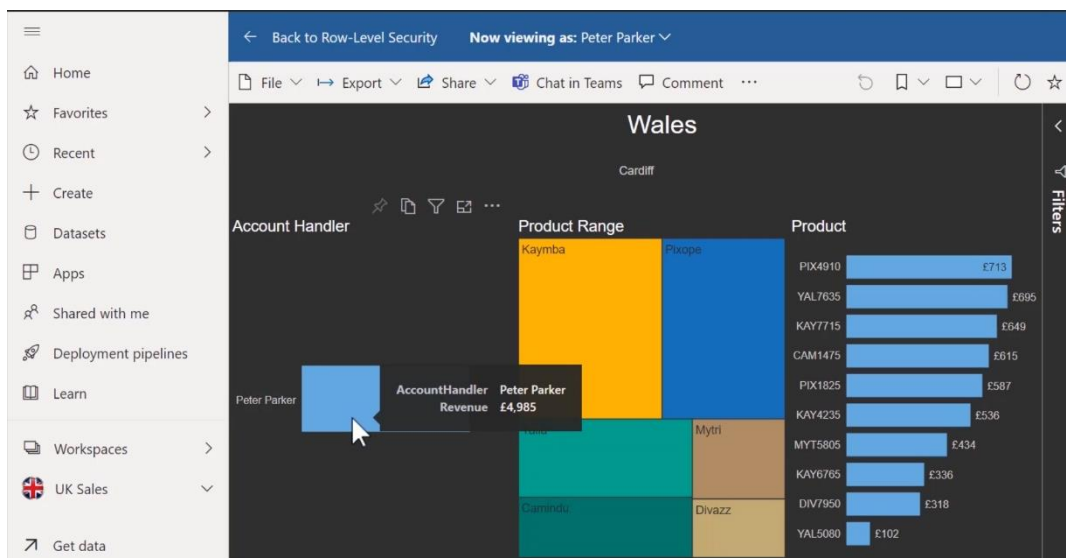
Testing the User Experience

To verify Peter's access, we use Test as Role. Initially, no data appears—this is expected since we must specify a user. We select Now viewing as username, enter Peter Parker, and click Apply.



The report updates to reflect Peter's experience. He sees:

- His assigned country and branch.
- His own name under Account Handler.
- Only his sales figures, broken down by product range and product.



This confirms that row-level security is working as intended. Each salesperson only has access to their own data, ensuring that sensitive information remains restricted while maintaining a personalized reporting experience.

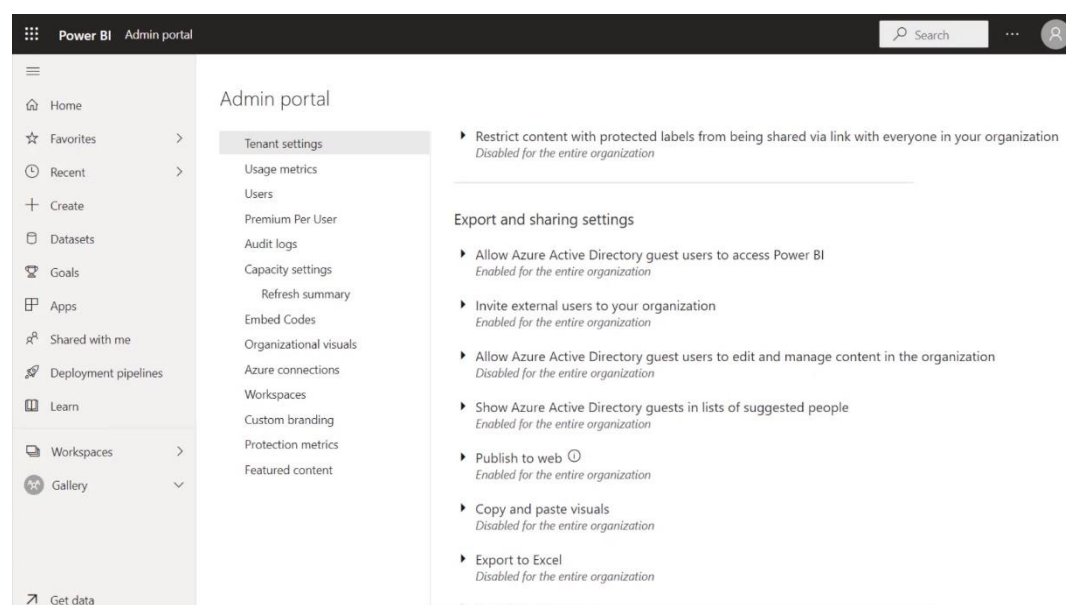
0700: Overview of Perimetric Administration

Perimetric administration in Power BI refers to managing aspects of the environment that interact with, or are influenced by, external factors. This includes decisions around sharing content with external users, publishing reports, and exporting report data.

External Guest Access

If your organisation collaborates closely with external clients, it may be beneficial to grant them access to specific Power BI reports. This can be done using external guest access settings, which allow selected users outside your organisation to view shared content.

The first four options within the Export and Sharing Settings section of the Power BI admin portal govern these external sharing capabilities. If your organisation has no need for external user access, the simplest and most secure approach is to disable all of these options.

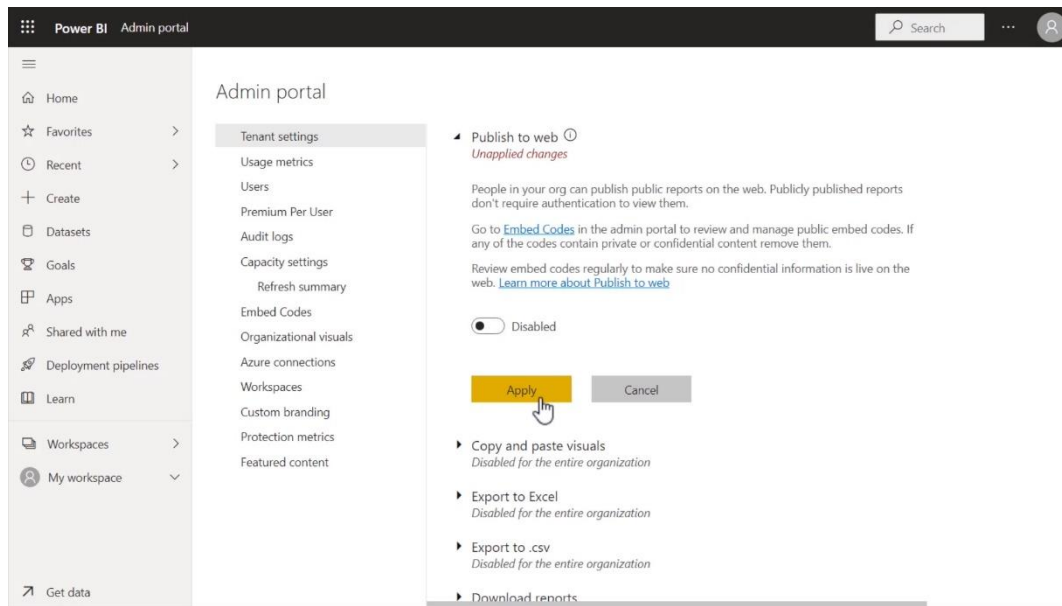


Publish to Web

One of the most critical security settings to review is the Publish to Web feature. This function allows report developers to create an embed code, providing unrestricted access to a report via any web browser—without requiring a Power BI licence or authentication.

Given the potential security risks, enabling this feature could allow a single malicious actor to expose sensitive corporate data to competitors or the public. As a best practice, this feature should always be disabled unless there is a specific, well-justified business need.

If you inherit the administration of an existing Power BI tenant, it is essential to check whether any reports have been published using this method. You can review and manage these by accessing the Embed Codes section in the admin portal, where you also have the ability to unpublish reports by deleting their embed codes.



In rare cases, such as for nonprofit organisations that need to make reports publicly available, the Publish to Web feature may be necessary. In such instances, access should be tightly controlled and granted only to a select group of users.

Exporting Report Content

Another key administrative decision is the extent to which users are permitted to export Power BI reports and report content. Power BI allows exports to formats such as PDF, PowerPoint, and image files, all of which can be managed in the Export and Sharing Settings section of the admin portal.

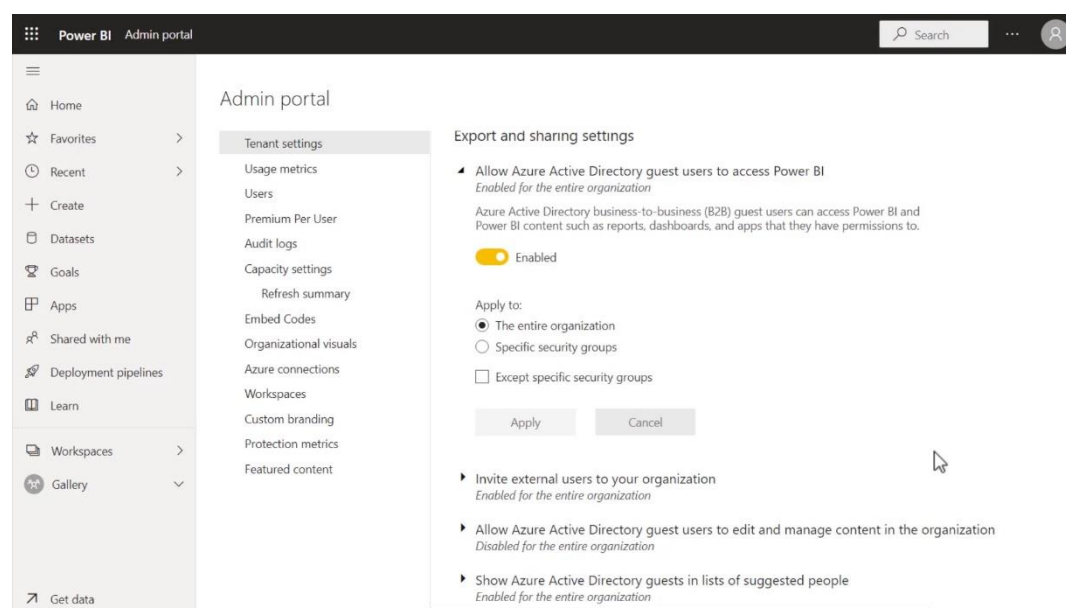
While exporting reports makes content more widely accessible—especially for those without Power BI licences—it comes with a trade-off. Once exported, reports lose their interactivity and will not reflect any updates made to the original Power BI report. However, this method can be a cost-effective way to share insights, as it eliminates the need for additional Power BI licences for report consumers.

0701: Giving Access to External Users

In the Export and Sharing section of the admin portal, you can manage how your Power BI tenant interacts with external users and how data can be shared outside your organisation. This section contains several options, with the first four specifically addressing guest users—also known as Azure Active Directory Business-to-Business (B2B) users.

Understanding Guest User Access

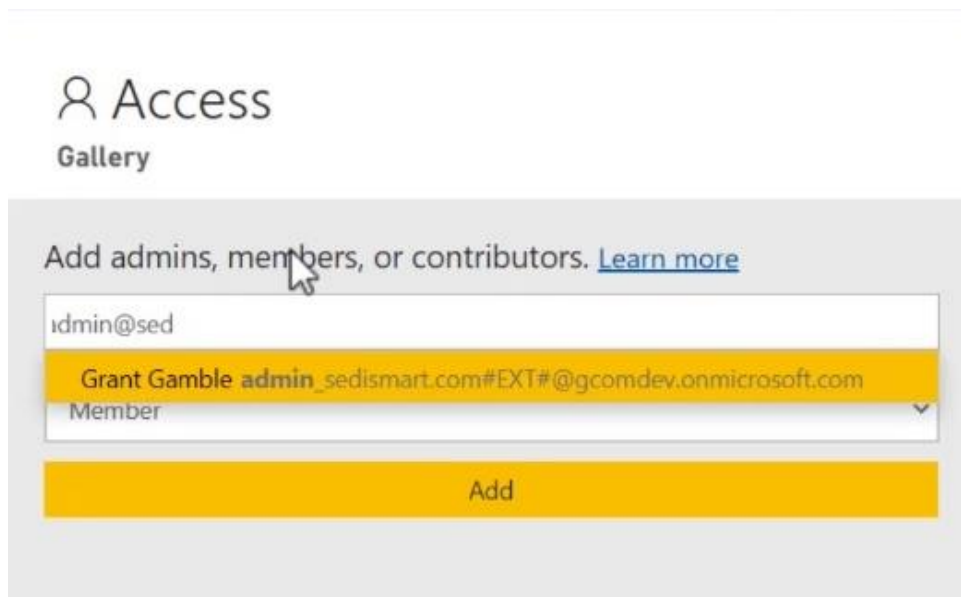
Azure Active Directory's B2B collaboration feature allows you to invite external users to work with your organisation, even if they do not have an Azure AD subscription. The first setting in this section—Allow Azure AD guest users to access Power BI—determines whether users within your Power BI tenant can share content with external B2B users. However, if this is the only option enabled, guest users must first be manually created outside of Power BI, typically via the Azure portal (portal.azure.com).



To add a guest user in Azure Active Directory:

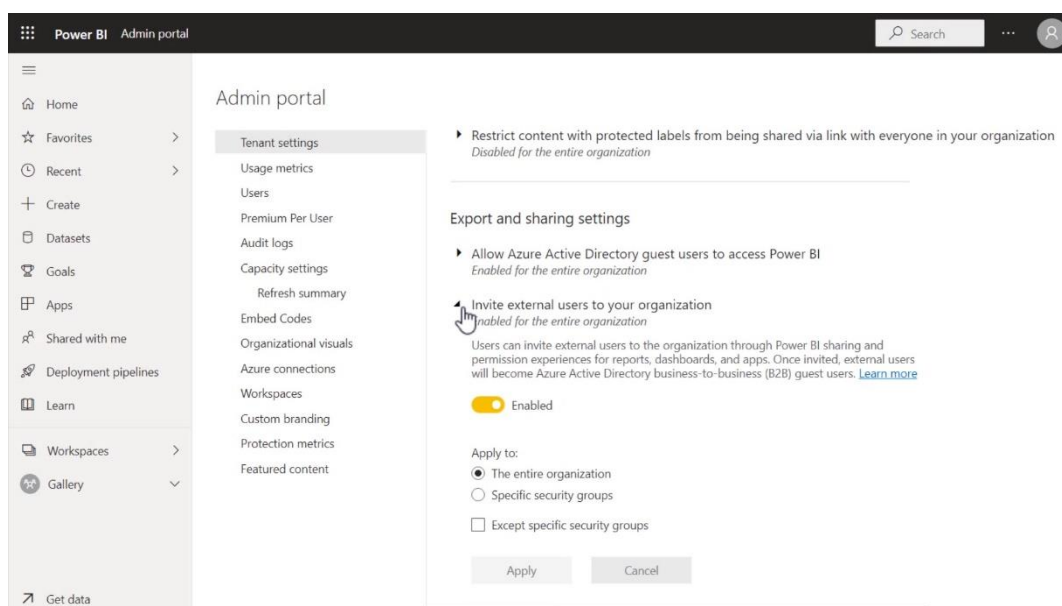
- Navigate to Azure Active Directory in the portal.
- Click on Users and select Add a new guest user.
- Choose Invite user, then enter the person's name and email address.
- Click Invite.

Once the user is created, internal users can share content with them as if they were part of the organisation. (The email shown for such users usually contains a prominent "#EXT#" flag, which makes external users easily identifiable.)



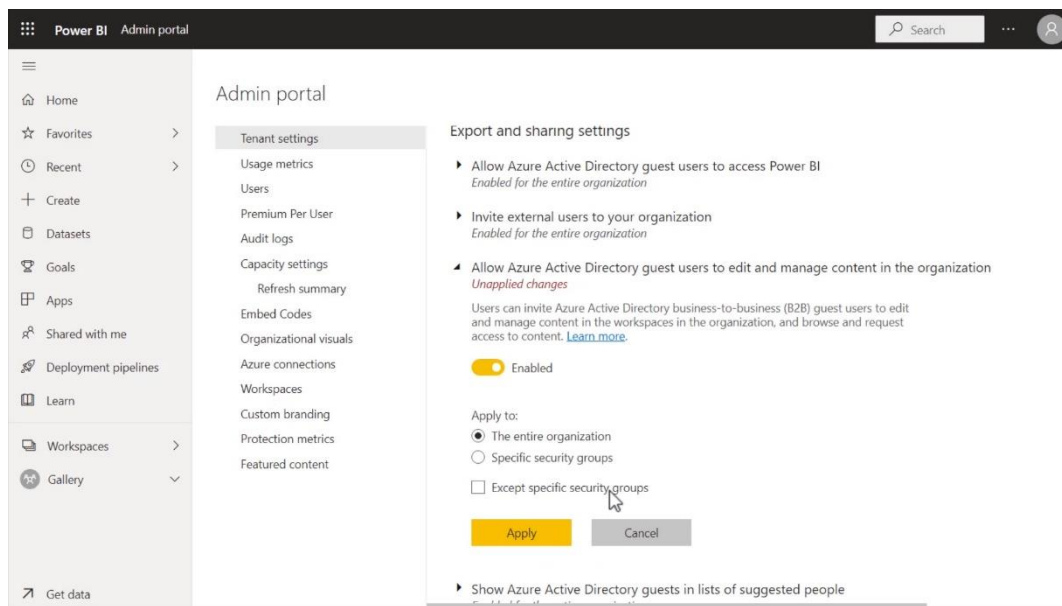
Enabling Users to Invite External Guests

The second option, Invite external users to your organisation, expands this capability. When enabled, Power BI users can invite external users themselves without requiring a separate request to the IT department. The outcome remains the same: a new Azure AD B2B guest user is created. This simplifies collaboration, especially in organisations where external partnerships are common.



Allowing Guest Users to Manage Content

By default, guest users can only view shared content. The third setting, Allow Azure AD guest users to edit and manage content in the organisation, lets external users take a more active role in Power BI workspaces. With this enabled, guest users can be assigned any of the four workspace roles, including Admin.



However, even with elevated privileges, there are some restrictions:

- Guest users cannot publish reports from Power BI Desktop into a workspace. Instead, they must create reports within the workspace itself using the Get Data command.
- They cannot build reports using shared Power BI semantic models.
- They cannot install or configure an On-Premises Data Gateway to connect to your organisation's data.
- They do not have access to the Analyse in Excel feature.

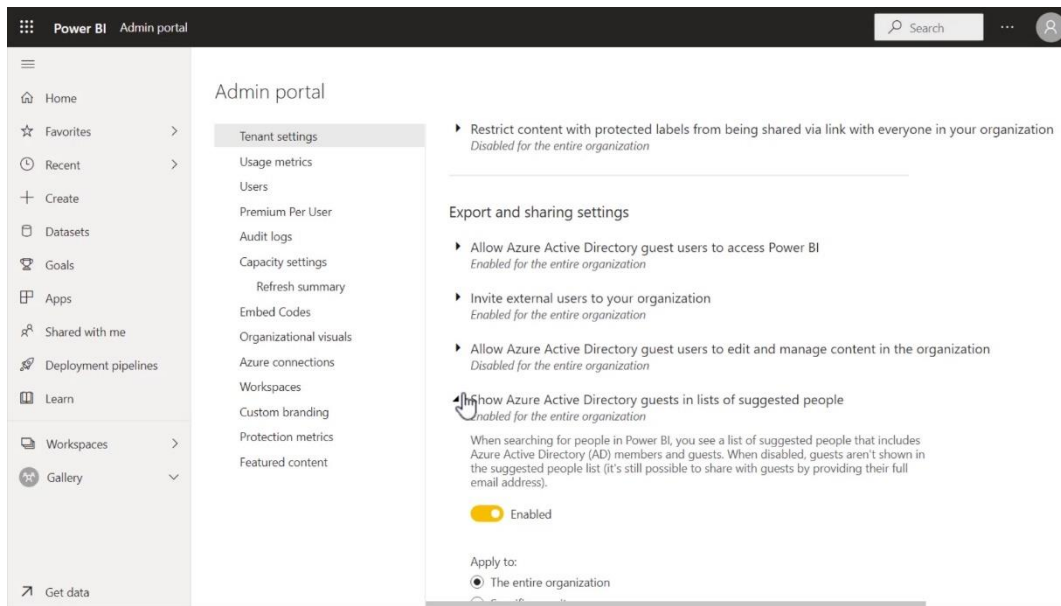
To improve the experience for external users, it's a good idea to provide them with your tenant URL. This helps them navigate to your organisation's Power BI environment. You can find the tenant URL by going to Help & Support > About Power BI, then copying and sharing the URL with the external user.

Showing Guest Users in People Picker Fields

The final setting, Show Azure AD guest users in the list of suggested people, determines whether guest users appear in auto-suggest lists when entering names in Power BI's people picker fields. If enabled, guest users will be suggested alongside internal users.

If this setting is disabled, you can still grant permissions to a guest user by manually entering their full email address. This ensures that external users can always be included in collaboration efforts, regardless of whether they appear in the suggested list.

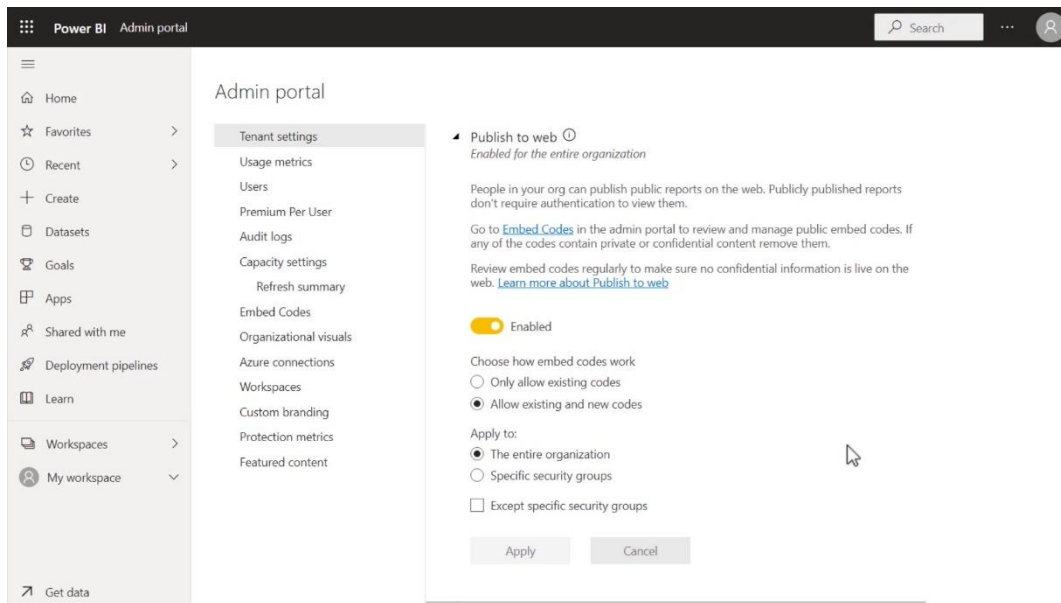
0701-giving Access To External Users



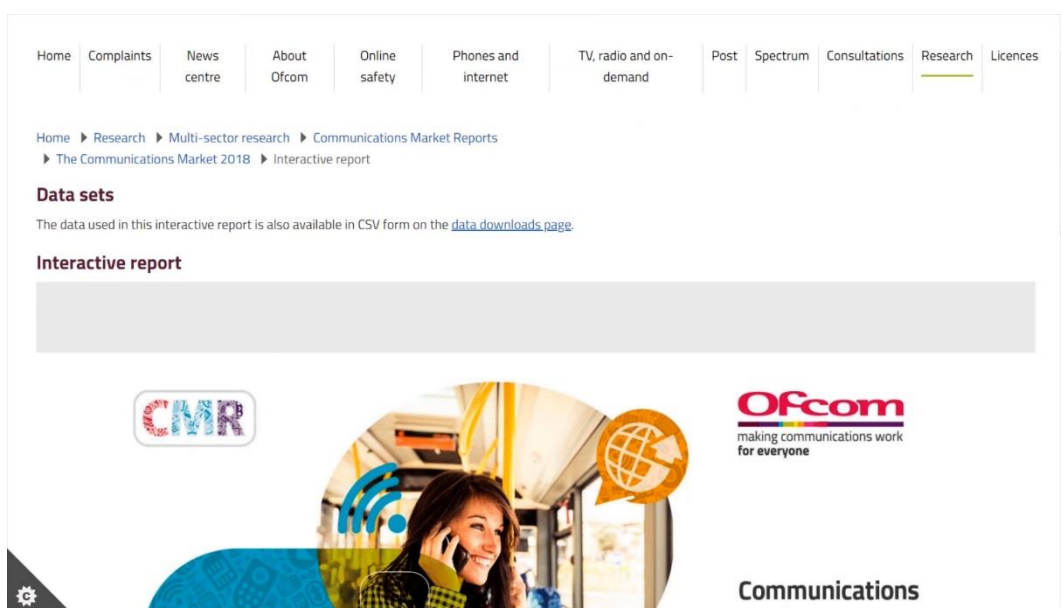
By understanding and configuring these settings, you can control the level of access granted to external users, ensuring secure and effective collaboration within Power BI.

0702: Using Publish to Web

The Publish to web feature in Power BI allows report developers to generate an embed code, which can then be shared with others or embedded within a website. Anyone with access to this embed code—or anyone visiting a webpage where it has been used—can view the report in full, without needing to sign in to Power BI or even have a Power BI licence.



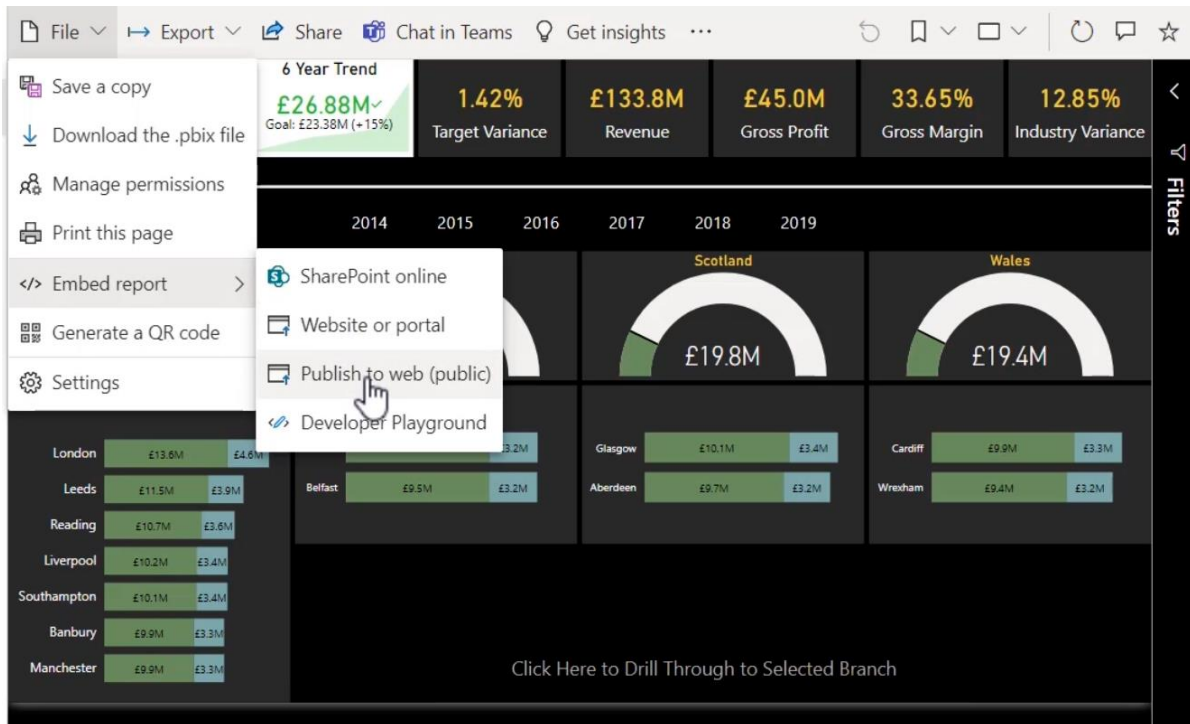
Because of this unrestricted access, many organisations will want to disable this feature entirely. However, there are specific cases where Publish to web can be beneficial—such as for nonprofit organisations that wish to share insights from public semantic models.



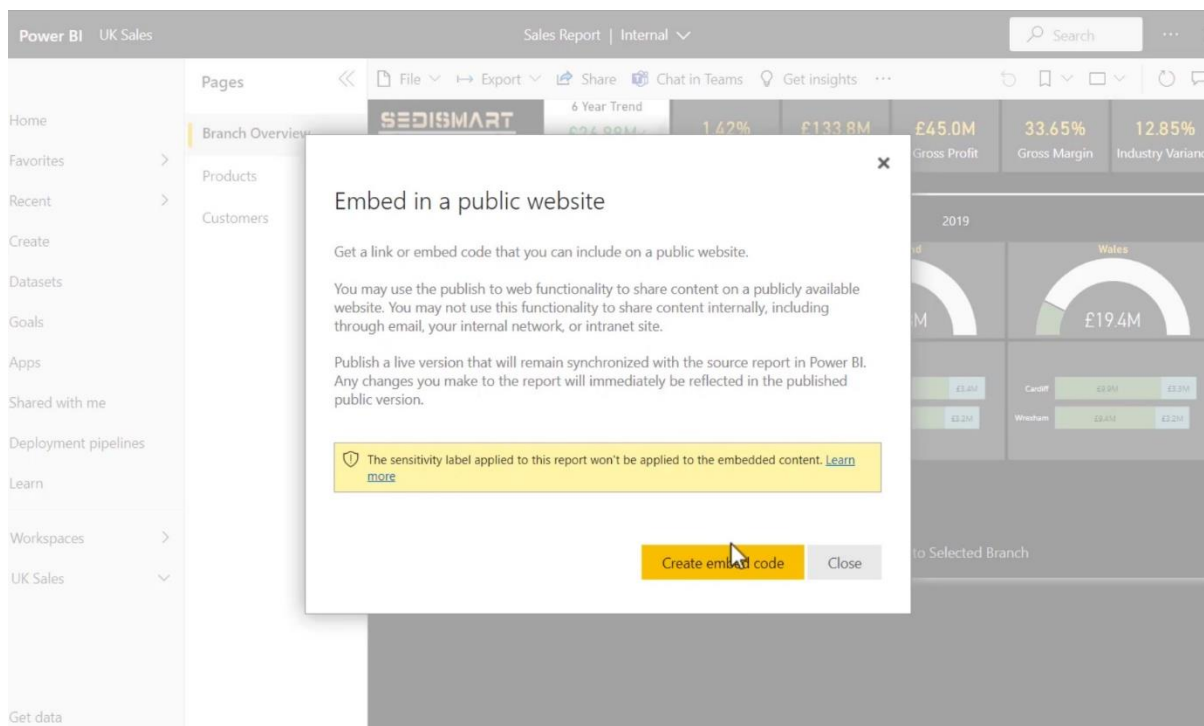
Creating an Embed Code

To publish a report using this feature:

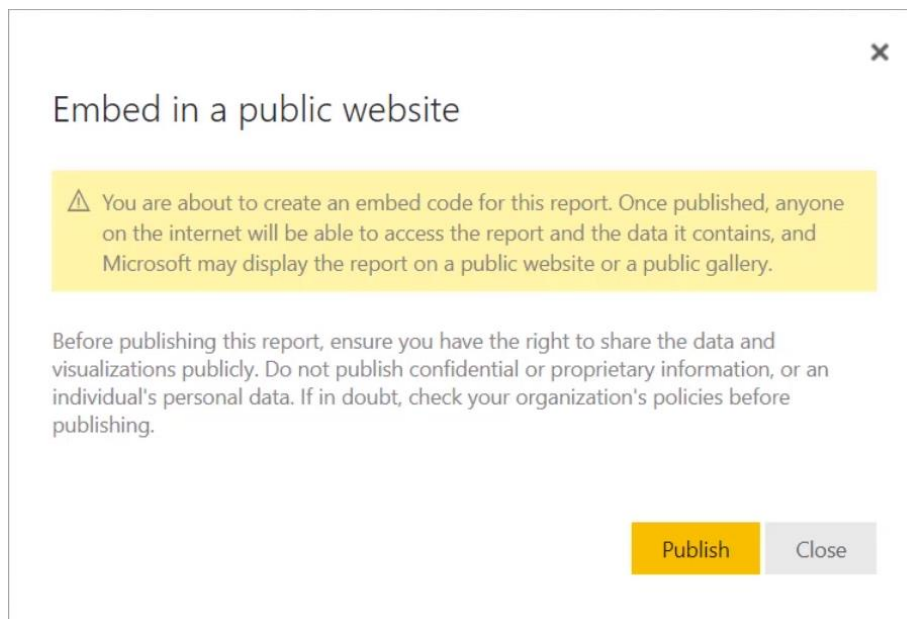
- Open the report in Power BI.
- Click File > Embed report > Publish to web.



A dialogue box will appear, explaining how the embed mechanism works. Click Create Embed Code to proceed.



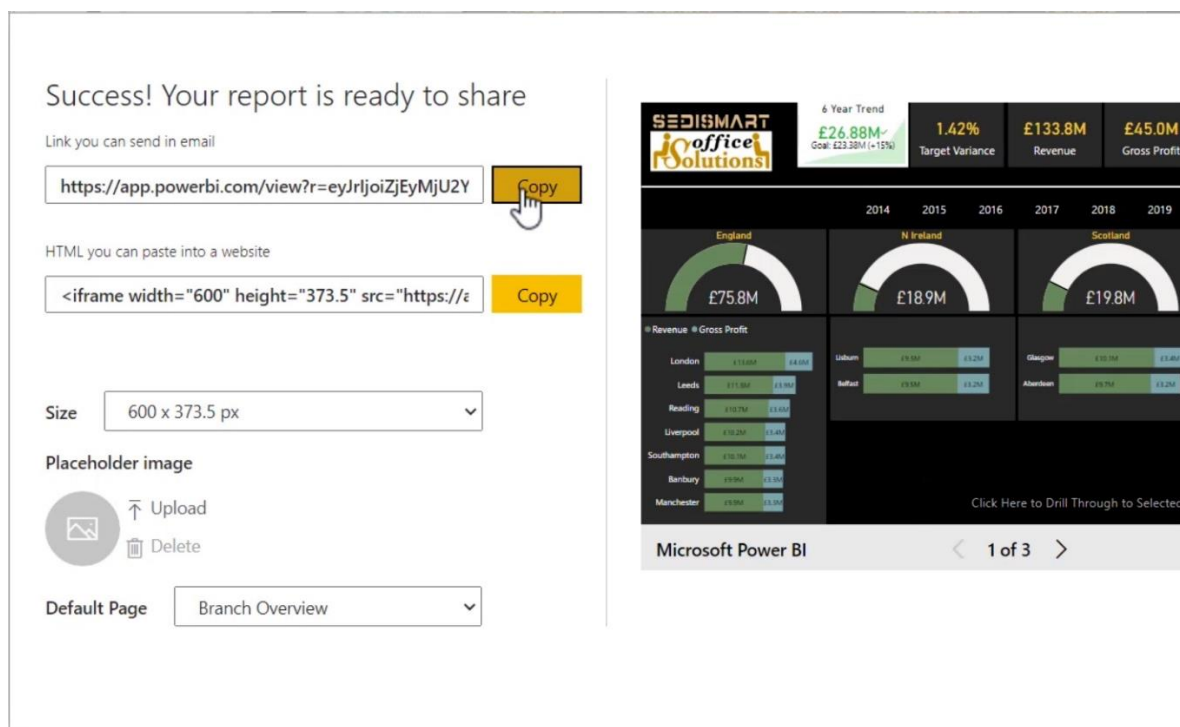
A second dialogue box will warn you that anyone with the embed code will have unrestricted access to the report. It also notes that Microsoft may display the report on a public website or gallery. If you accept this, click Publish.



At this point, Power BI generates two codes:

A direct link to the report, which can be pasted into a browser's address bar.

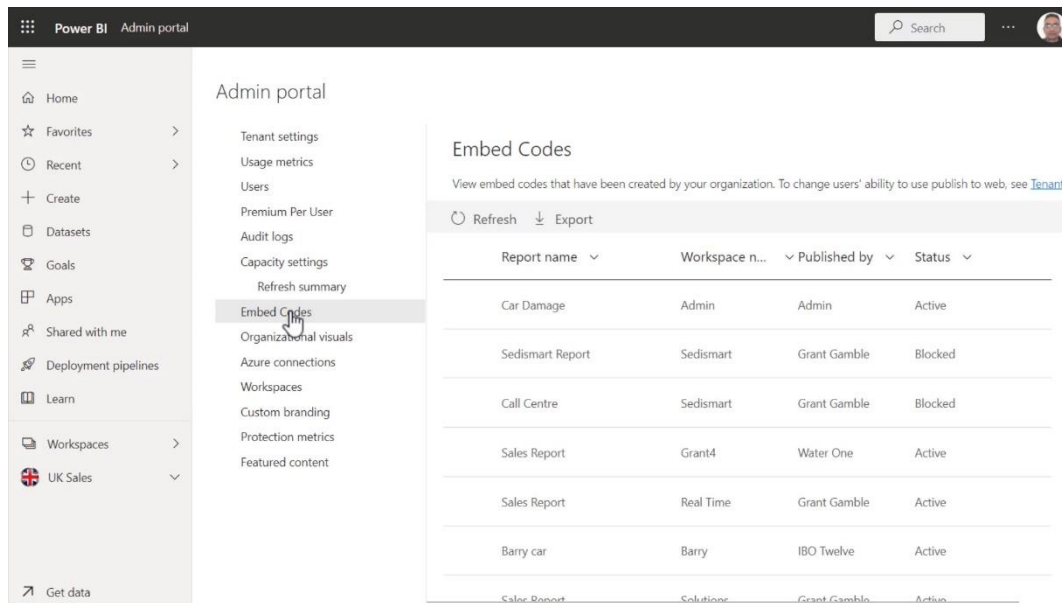
An HTML iframe code snippet, which can be used to embed the report into a webpage.



Managing and Monitoring Embed Codes

If an organisation enables Publish to web, administrators can monitor its use via the Embed codes section in the Power BI portal. Here, they can view all generated embed codes, along with details such as:

- The report name
- The workspace where the report is located
- The author
- The current status of the embed code (active or blocked)

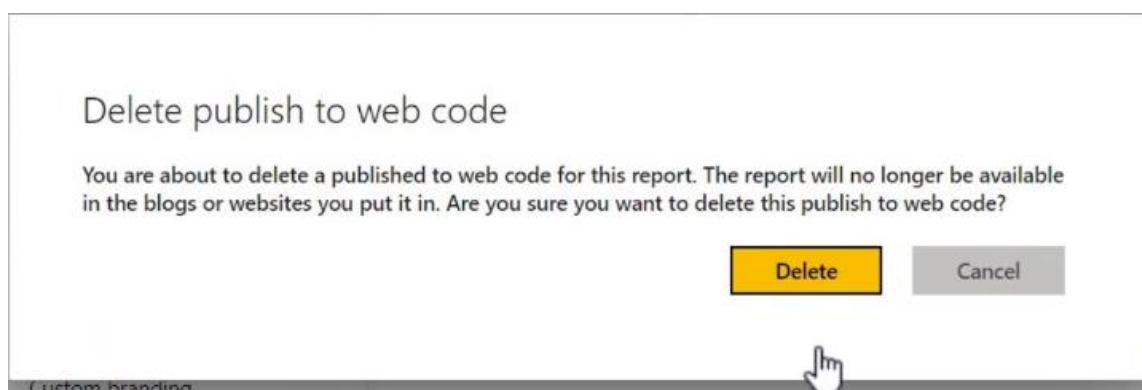


Taking Action on Embed Codes

From this portal, administrators can take two key actions:

View on web – This allows them to see the report exactly as it appears to anyone with the embed code.

Delete embed code – If an embed code is deleted, any user attempting to access the report via that URL will receive an error message stating that the content is no longer available.

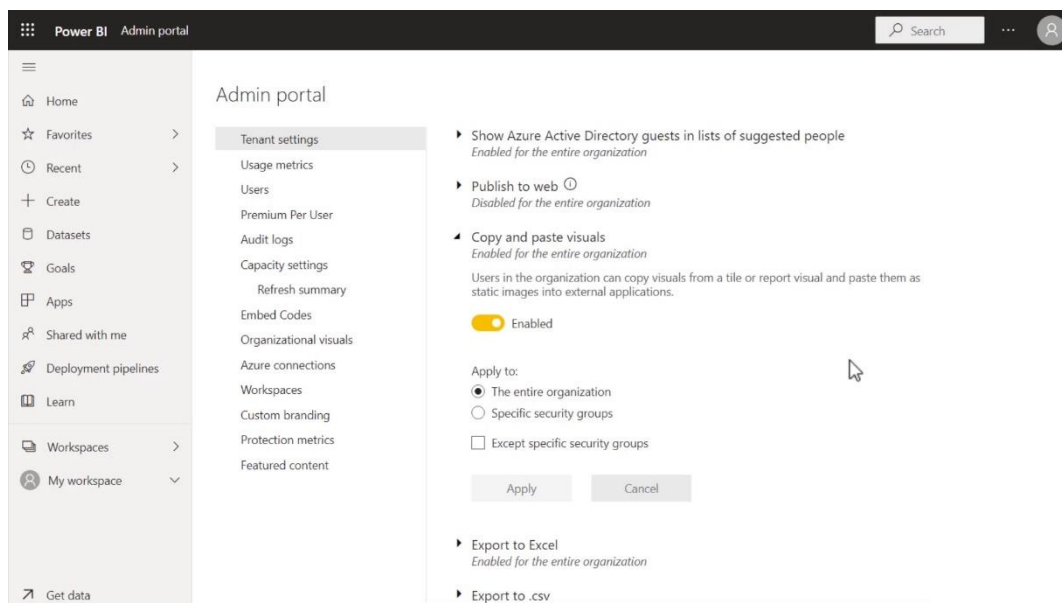


0702-using Publish To Web

Because Publish to web bypasses standard authentication and security measures, organisations should carefully consider whether to enable it, ensuring that only appropriate reports are made publicly accessible.

0703: Allowing Users to Copy Visuals

The Copy Visuals feature in Power BI enables users to copy individual visuals from reports and dashboards and paste them into other applications. This isn't just a simple image copy; Power BI also includes metadata and a link back to the original report, making it easy to trace the visual back to its source. However, for the link to work, the user must be signed in and have the necessary permissions to access the report.



How It Works

Once this feature is enabled, users will see a Copy visual as image button when they hover over a visual.



0703-allowing Users To Copy Visuals

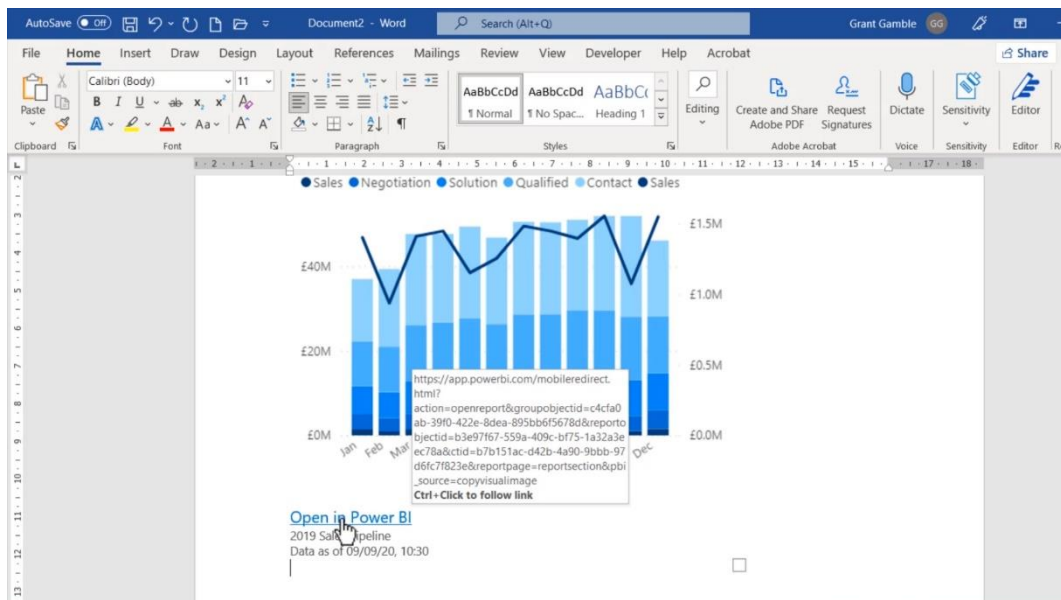
Clicking this button generates an image of the visual and places it on the clipboard. A confirmation message—Copy to clipboard—appears, and then the visual is ready to be pasted elsewhere.



Users can then navigate to another application, such as Microsoft Word, and paste the copied visual.

The pasted content includes:

- The image of the visual
- Metadata about the visual
- A link back to the original Power BI report



The embedded link allows users to return to the source report by Ctrl + clicking from within the application where the visual was pasted. Of course, accessing the report still requires the appropriate permissions.

Security Considerations

Enabling this feature does not introduce any security risks. Anyone with access to a report could already use a snipping tool or screenshot function to capture visuals manually. The Copy Visuals feature simply provides a more structured and efficient way to share visuals while maintaining traceability back to the report.

Should You Disable This Feature?

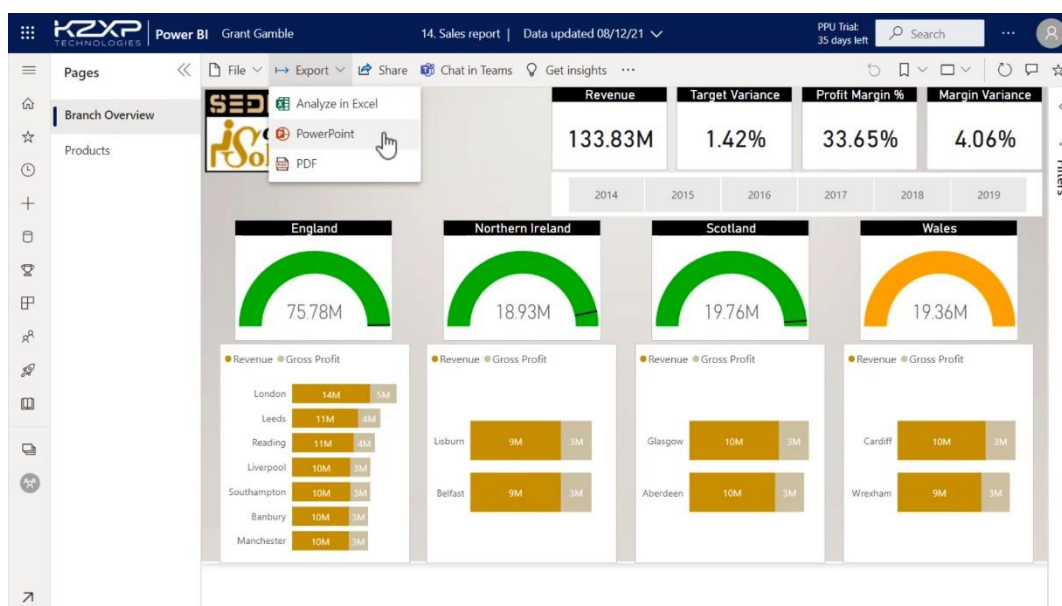
In most cases, there's little reason to disable this option. It enhances usability without compromising security, allowing users to work more efficiently while ensuring that visuals remain linked to their source reports.

0704: Allowing Users to Export Reports



In the Export and Sharing settings, Power BI provides several options for exporting reports and paginated reports into different formats. These options enable users to create report assets that can be viewed without a Power BI licence. However, while this increases accessibility, it also introduces limitations—exported reports are always static and non-interactive. Additionally, once exported, reports can be shared freely, potentially beyond the intended audience. As such, these export features are most useful for distributing carefully curated summary information to external stakeholders or internal users who do not require interactive capabilities.

Exporting Reports to PowerPoint and PDF

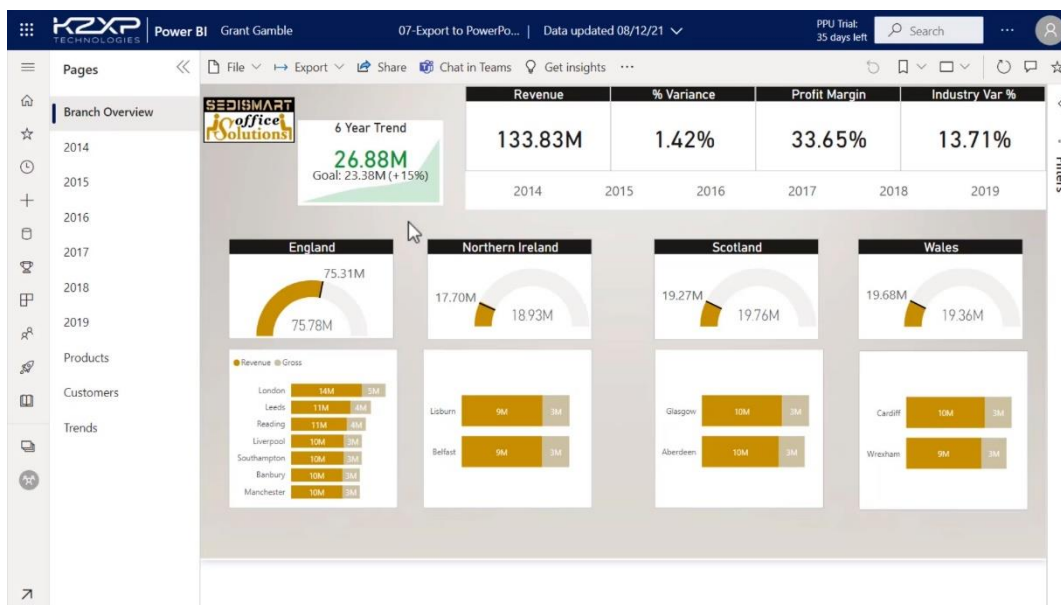


0704-allowing Users To Export Reports

Among the most commonly used export options are PowerPoint and PDF. The export to PowerPoint feature is available only in the Power BI service, whereas export to PDF can be done in both Power BI Desktop and the Power BI service. This means that even if export to PDF is disabled in the Power BI service, users can still generate PDFs from their reports in Power BI Desktop.

When preparing reports for export, it is often beneficial to create a dedicated version of the report specifically for PowerPoint or PDF export. Since interactivity is lost in the exported version, consider adding additional pages that display key insights that would normally be accessed through slicers, filters, or drill-through actions.

For example, if an interactive report contains a slicer allowing users to filter by year, the exported version could include separate pages, each pre-filtered for a specific year. These additional pages would not be needed in the interactive version but ensure that important contextual information remains accessible in the static export.



Exporting Paginated Reports

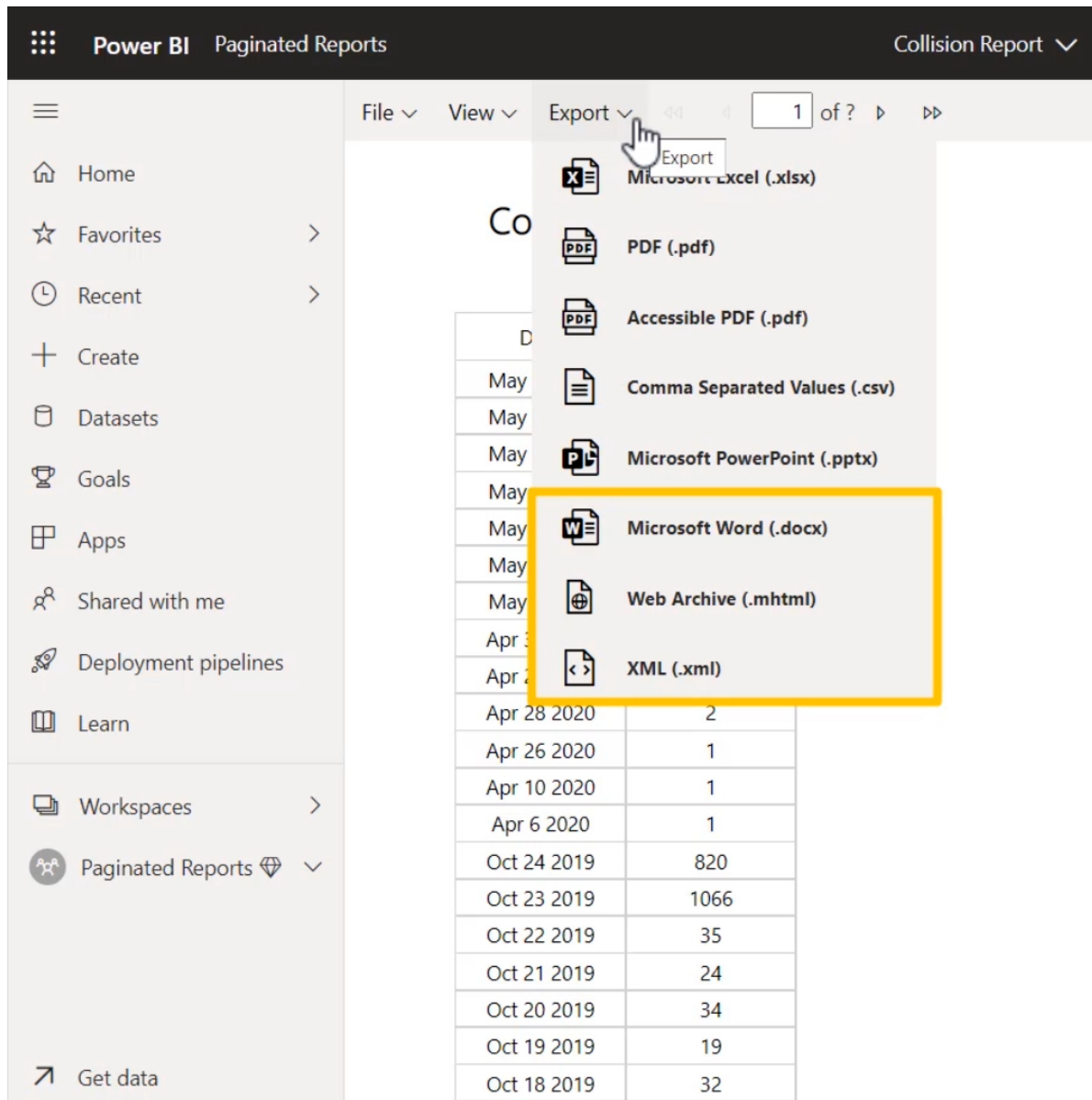
There are three export options that apply exclusively to paginated reports:

- Export to MHTML (archived web page format)
- Export to Word (.docx)
- Export to XML

Admin portal

Tenant settings	Export reports for the entire organization
Usage metrics	Export reports as PowerPoint presentations or PDF documents <i>Enabled for the entire organization</i>
Users	Export reports as MHTML documents <i>Enabled for the entire organization</i>
Premium Per User	Export reports as Word documents <i>Enabled for the entire organization</i>
Audit logs	Export reports as XML documents <i>Enabled for the entire organization</i>
Capacity settings	
Refresh summary	
Embed Codes	
Organizational visuals	

Paginated reports, which are typically more tabular and detailed than standard Power BI reports, can only be created in Premium workspaces. These export options may be useful when integrating with workflows that require structured data for further processing, such as document generation or automated reporting pipelines.



Exporting Reports as Image Files

Power BI also allows reports to be exported as PNG images, but this feature is only available via the Power BI REST API. It is not an option that can be accessed by standard users within the Power BI interface.

0704-allowing Users To Export Reports

The screenshot displays the REST API documentation for the `ExportTo` endpoint. On the left is a sidebar with a search bar and a list of API endpoints. The main content area shows the HTTP method (POST), the endpoint URL, and a table of URI parameters. Below this is a table of request body parameters. At the bottom, a 'Responses' section shows a 202 Accepted status with an 'Export' type.

Filter by title

- Export Report In Group
- Export To File**
- Export To File In Group
- Get Datasources
- Get Datasources In Group
- Get Export To File Status
- Get Export To File Status In Group
- Get File Of Export To File
- Get File Of Export To File In Group
- Get Page
- Get Page In Group
- Get Pages
- Get Pages In Group
- Get Report
- Get Report In Group
- Get Reports
- Get Reports In Group
- Rebind Report
- Rebind Report In Group
- Take Over In Group
- Update Datasources
- Update Datasources In Group
- Update Report Content
- Update Report Content In Group

> Template Apps

> Users

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HTTP Copy Try it

POST `https://api.powerbi.com/v1.0/myorg/reports/{reportId}/ExportTo`

URI Parameters

Name	In	Required	Type	Description
reportId	path	True	string uuid	The report ID

Request Body

Name	Required	Type	Description
format	True	FileFormat	The requested format for the exported file
paginatedReportConfiguration		PaginatedReportExport Configuration	The configuration used to export a paginated report
powerBIReportConfiguration		PowerBIReportExport Configuration	The configuration used to export a Power BI report

Responses

Name	Type	Description
202 Accepted	Export	Accepted

For users who need to extract visuals as images, the closest alternative is copying and pasting visuals, a setting covered earlier. However, a significant limitation of this approach is that both the report and its semantic model must be in a Premium or Embedded capacity for it to function.

Printing Reports and Dashboards

The screenshot shows the Power BI Admin portal interface. The left sidebar contains navigation links. The main content area is titled 'Admin portal' and shows 'Tenant settings'. Under 'Tenant settings', the 'Print dashboards and reports' option is highlighted. It is currently 'Enabled' for the entire organization. A yellow warning box at the bottom states that content may be printed regardless of sensitivity labels and data protection settings.

K2XP TECHNOLOGIES | Power BI | Admin portal PPU Trial: 35 days left Search ... User

Admin portal

- Tenant settings
- Usage metrics
- Users
- Premium Per User
- Audit logs
- Capacity settings
- Refresh summary
- Embed Codes
- Organizational visuals
- Azure connections
- Workspaces
- Custom branding
- Protection metrics
- Featured content

Print dashboards and reports
Enabled for the entire organization

Users in the organization can print dashboards and reports.

☒ Enabled

With this option enabled, dashboard and report content may be printed regardless of sensitivity labels and data protection settings applied to them or to their underlying dataset. [Learn more](#)

Apply to:

0704-allowing Users To Export Reports

Given that Power BI already allows export to PDF, the ability to print reports and dashboards may seem less critical. However, it provides a quick way to print the current page or specific visuals directly from the interface.

To print an entire report page or dashboard, users can navigate to the File menu and select Print this page.



To print an individual visual, they need to enter Focus mode on the desired visual and then use the Print command while in focus mode.

While printing is a fairly basic feature, it can be useful in situations where users need quick hard copies of individual elements rather than entire reports.